

EPB EXCAVATION AND CONDITIONING OF COHESIVE MIXED SOILS: CLOGGING AND FLOW EVALUATION

by

Daniela Garroux Gonçalves de Oliveira

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Abstract

When shield tunnels are being excavated along varying geological conditions, it is essential to understand the behaviour of the material being excavated, as it influences excavation speed and ground stability, in addition to other operational issues. For an Earth Pressure Balance Machine (EPBM), this requirement is increased because the excavated material, combined with the rotation of the machine screw conveyor and the thrust of the hydraulic jacks, is responsible for holding the pressure ahead of the machine. For this system to function, this material needs to maintain certain engineering characteristics. In the absence of those key target properties, additives should be mixed with the excavated soil in front of the machine or in the conveyor system, a process called ground conditioning.

There are numerous bodies of research concerning EPBM conditioning that mostly focus on either sandy or clayey soils, with relatively little published information regarding mixed ground. Terrains originating from the tropical weathering of rocks present these mixed ground characteristics. EPBM have excavated this type of ground in many areas around the world.

The aim of this thesis is to investigate and characterize this type of terrain, exploring the particularities regarding its excavation by EPB shields, including the conditioning of those materials. The focus is on cohesive mixed soil, thus restraining the topic. This research includes two main subjects: the assessments of clogging behaviour and flow behaviour of cohesive mixed soils for EPBM excavation, drawing a parallel between them whenever possible.

A laboratory routine was developed for investigating and characterising these mixed soils and their flow and clogging tendencies. Soils were assembled in the laboratory, allowing control of all the inspected elements. Several conclusions related to the clogging potential and flow behaviour could be achieved by changing certain variables: clay mineral, clay fraction, grain size of the sand fraction, water content, foam, and polymers.

This laboratory routine and some of the achieved conclusions can be also applied to materials other than cohesive mixed soils. Some considerations for future work are provided, foreseeing the need for a future correlation and validation of the assumptions here made with real tunnel drives.

Co-Authorship

This thesis was written in manuscript format according to the School of Graduate Studies regulations. Manuscript chapters use the first person “we” to reflect co-author contribution. The following chapters were published or submitted to international journals:

Chapter 3: Oliveira, D.G.G. de, Thewes, M., Diederichs, M.S., Langmaack, L., 2018. EPB Tunnelling Through Clay-Sand Mixed Soils: Proposed Methodology for Clogging Evaluation. *Geomechanics & Tunnelling* **11** (4): 375-387. doi: 10.1002/deot.201800009

Chapter 4: Oliveira, D.G.G. de, Thewes, M., Diederichs, M.S., Langmaack, L. 2018. Consistency Index and its Correlation with EPB Excavation of Mixed Clay-Sand Soils. *Geotechnical & Geological Engineering*. doi: 10.1007/s10706-018-0612-x

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I was responsible for the initial data compilation, laboratory analysis, methodology development, ATUR initial conception and writing of all manuscripts. Dr. M.S. Diederichs provided the primary funding for this research to happen, and supervision in all the testing phases, ATUR conception and the methodology development, as well as editing all manuscripts. Dr. M. Thewes provided funding for the laboratory work in Germany, including the tested material, support and supervision for all the laboratory testing and methodology development, funding and support with the assemblage of the new testing device ATUR, initial suggestion of the use of the flow table, and edited all manuscripts. Ing. L. Langmaack provided technical support for the tests with foam and polymers, suggested also the use of the flow table, and edited manuscripts of Chapters 3 and 4.

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Statement of Originality

I hereby certify that all of the work described within this thesis is the original work of the author. Any published (or unpublished) ideas and/or techniques from the work of others are fully acknowledged in accordance with the standard referencing practices.

Daniela Garroux Gonçalves de Oliveira

(September, 2018)

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List of Abbreviations

λ	Clogging Empirical Parameter ¹
λ_0	Clogging Empirical Parameter without Drops of the Beater
λ_1	Clogging Empirical Parameter after One Drop of the Beater
λ_2	Clogging Empirical Parameter after Two Drops of the Beater
λ_3	Clogging Empirical Parameter after Three Drops of the Beater
λ_7	Clogging Empirical Parameter after Seven Drops of the Beater
λ_F	Final Clogging Empirical Parameter
λ_x	Clogging Empirical Parameter after X Drops of the Beater
ρ	Bulk Density
ρ_d	Dry Density
ASTM	American Society for Testing and Materials
ATM	Atmospheric
ATUR	RUB-Queens Clay Adhesion Tool
BMS	Ball Measuring System
CB_{Factor}	Cleaning Beater and Bowl Correction Factor
CEC	Cation-Exchange Capacity
C_f	Concentration of Surfactant
CF	Clay Fraction
e	Void Ratio
EPB	Earth Pressure Balance
EPBM	Earth Pressure Balance Machine
FER	Foam Expansion Ratio

¹ In Chapter 3, λ and λ_0 are considered as synonyms, as an absence of any drop of the beater.

FIR	Foam Injection Ratio
Flow ₄₀	Flow 40 in %
Flow ₆₀	Flow 60 in %
HSSR	Hard Soil Soft Rock
I _c	Consistency Index
IGM	Interface Geomaterials
I _L	Liquidity Index
IP	Plasticity Index
m ₀	Average of Three Measurements of the Initial Specimen Diameter
m ₄₀	Average of Three Measurements of the Specimen Diameter after Forty Times Jolting
m ₆₀	Average of Three Measurements of the Specimen Diameter after Sixty Times Jolting
MEQ	Milliequivalent
MF	Mixed face
m _{Foam Solution}	Mass of Foaming Solution
MG	Mixed ground
m _{Surfactant}	Mass of Surfactant in Foaming Solution
MT	Mixing Tool
MTG	Mixed Transitional Ground
N _{SPT}	Number of SPT Blows
NP	Non-Plastic
S	Saturation Degree
SD	Standard Deviation

SPT	Standard Penetration Test
SQM	Square Meter
So	Soft
St	Stiff
TBM	Tunnel Boring Machine
TLB	Institute for Tunnelling and Construction Management
TOT	Total Mass of Tested Material
UCS	Unconfined Compressive Strength
V_{Foam}	Volume of Foam
$V_{\text{Foam Solution}}$	Volume of Foam Solution
V_s	Very Soft
V_{Soil}	Volume of Tested Soil
W	Water Content
WC	Water Content in Chapter 3
W_L	Liquid Limit
W_P	Plastic Limit
W_{POL}	Plastic Limit Calculated by the Equation from Polidori (2007)
W_s	Shrinking Limit

Chapter 1

Introduction

1.1 General Introduction

The mechanised tunnelling industry is growing worldwide. The Earth Pressure Balance - EPB shield (and its variants) is the most recurrent type of shield chosen for tunnel projects whenever encountering soft ground (Herrenknecht et al. 2011, Galli 2016). The ideal field of EPB machines has grown since its original assemblage in Japan (Maidl et al. 2012), which is a direct consequence of successful ground conditioning (Maidl 1995, Langmaack 2000, EFNARC 2005, Thewes, Budach & Bezuijen 2012). Researchers have also followed industry tendencies in trying to provide scientific support to overcome the constraints faced when excavating several different geological environments with a tunnel boring machine. There are, however, still challenges to overcome in the mechanised tunnelling field.

Mixed faces (MF), describing a setting when two or more materials with different behaviour are encountered at the same tunnel face, still present a challenging scenario (Della Valle 2001, Thewes 2004, Zhao et al. 2007, Tóth 2013, Shirlaw 2016, Comulada et al. 2016, Gong et al. 2016, Peixoto da Silva et al. 2017, Oliveira et al. 2017a). A particular type of these mixed faces is the ground that results from the tropical weathering of rocks. Oliveira and Diederichs (2016) and Oliveira et al. (2017a) used the term mixed transitional ground (MTG) for this kind of ground, also called by other terms not necessarily exclusively related to tunnel boring machines as the previous denomination, such as “HSSR - Hard Soil Soft Ground” and “IGM – Interface Geomaterials” (Viana da Fonseca and Topa Gomes 2010).

This ground is a combination of soil and rock portions. Historically, it has often been impossible to separate what is soil from what is rock, as many of its elements lie exactly at the

soil-rock boundary. The soil component, also known as residual soil, can be classified either mature or young residual soil (Futai et al. 2012), depending on its strength properties (N_{SPT} values) defined by the Standard Penetration Test-SPT (ASTM D1586 2011). During the early times of large dam projects, this type of soil was heavily investigated in tropical areas, as it was expected to exhibit a peculiar behaviour when compared to the soils from sedimentary origin (Vargas 1980, CBDB 2011, Futai et al. 2012). It can be mentioned, for example, that Karl Terzaghi and Arthur Casagrande (Figure 1-1) were both astonished by the differences of this type of ground when visiting outcrops in dam projects in Brazil (CBDB 2011, Futai et al. 2012).



Figure 1-1. Field inspection at the Itaipu dam, Brazil, from bottom to top: Arthur Casagrande, John Cabrera, Gurmukh Sarkaria and Flavio H. Lyra (CBDB 2011).

In terms of an EPB excavation, this type of ground implies having a mixed face ahead of the machine for most of the time, and on many occasions, it is not even feasible to predict the changes of the soil-rock boundary in multiple directions. In the simplest "mixed face" engineering context, sedimentary soils mainly occur in layers of pure clay-silt interspersed with layers of sand-gravel. Conversely, in a residual soil the ground is already mixed ahead of the tunnel face, with a heterogeneous and chaotic variation in its grain size distribution along the tunnel profile, as schematically illustrated by Figure 1-2.



Figure 1-2. Schematic comparison between sedimentary soil layers for a mixed face scenario, and residual ground with a mixed soil/ground scenario.

While the rock component of mixed ground poses its own engineering challenges, this research focuses on mixed soils containing sand and clay fractions, thus, a cohesive mixed soil, not including the portions of rock. It aims to characterise and assess the flow behaviour of those

materials and its tendencies for clogging. While flow behaviour and clogging² potential have been frequently addressed and researched, however, these studies have not previously included mixed clay-sand soils.

The main goal of this thesis is to define a comprehensive laboratory routine that allows for the investigation of these soils in terms of characterising the flow and clogging potential, by means of changing several variables such as: clay mineral, clay fraction, grain sizes of sand, water content, foam and polymers. The research also included a small section on the influence of mica flakes as a secondary variable in the mixed soils.

The methodology used in this research is a modified version of the method originally designed for assessing clogging in pure clay soils involving the use of a mixing machine and a flat mixing tool (Zumsteg and Puzrin 2012). In this thesis, this procedure was modified by adding an extra dropping stage with a new assembled device, called ATUR³. This device provided a tool to evaluate clogging in different clay proportions and also the effectiveness of soil conditioning. For the calibration of this new method, the universal diagram for clogging evaluation from Hollman and Thewes (2012, 2013) was considered. As in the Hollman and Thewes evaluation data from tunnel projects were used.

A flow table, originally designed for tests with concrete, was also included to evaluate the flow behaviour of those soils, which can be later correlated with EPB excavation parameters, as well as any pumpability method. The generated results were compared with tests conducted using a rheometer and slump test, providing inputs regarding those testing methodologies and the flow characteristics of cohesive mixed soils.

² Clogging is the behavior of clay soils to stick to metal surfaces, that will be detailed along the next chapters.

³ ATUR stands for *Adhäsive Tone Untersuchung RUB-Queens* (RUB-Queens Clay Adhesion Tool).

A combined laboratory routine was finalised and is described here in detail, so it can be reproduced in any other facility, including tunnel project sites. This routine could be conducted using material from a borehole investigation, for example, and would not require considerable amounts of testing material (around 1 kg), unlike the case for the slump test (requiring around 8-12 kg).

The material can be characterized in terms of its geological and geotechnical properties, in which the efficiency of soil conditioning is first verified initially and is then checked again following a second testing phase, which can be conducted along the tunnel excavation, comparing results by means of a back analysis.

It is expected that with this work a new standard laboratory routine can be established for investigating, characterising and classifying cohesive soils (mixed or not) and their interaction with an Earth Pressure Balance machine excavation. This type of standard laboratory routine would make it feasible to more reliably predict the risks, costs, excavation rates, and ultimately the success of a tunnel excavation and operation.

1.2 Thesis Motivation and Evolution

The motivation for this thesis came from both direct and indirect participation in the design and construction of the Line 5 of the São Paulo Metro. This was the second metro line excavated along the residual ground with an EPB machine in Brazil, the first one was Line 4 of the São Paulo Metro. This project encountered many difficulties, some of them related to stretches in mixed transitional ground (Comulada et al. 2016, Peixoto da Silva et al. 2017, Oliveira et al. 2017a). It then became clear that there was a gap in the knowledge and experience in tunnel drives through tropical residual ground and in this case, a gneissic residual ground.

In parallel, engineers and researchers dealing with the same type of ground in other regions of the world were consulted and it was established that the issues happening in different

locations were quite similar. Therefore, it was decided to investigate this further. Initially, data of worldwide residual soils were compiled, and are presented here in Appendix A.

Data collection was challenging due to the time constraints imposed by a typical PhD research schedule and the limited availability of confidential data from tunnel projects. Consequently, conducting tests with soils mixed in the laboratory provided a reasonable solution. In addition, working with these “artificial soils”⁴ made it possible to control specific variables, which would not be the case for data from tunnel projects, or even when dealing with natural soils.

However, testing soils assembled in laboratory implied in not considering the interaction amongst the particles of tropical soils given by the natural cementation. Surely, that has to be cited as one main disadvantage and has to be considered in future researches.

It was then decided to focus on the basic interaction of sand and clay, which are the basic components of a residual soil, making feasible to conduct this research within the given time and working frame. The influence of changes in the clay mineral type, clay percentage, sizes of sand grains, water content, and the presence or absence of additives (foam and anti-clay polymers) was investigated.

Because of the breadth of this topic, it was decided to focus mainly on the clogging behaviour of those soils, by directly assessing this using a methodology proposed by Zumsteg and Puzrin (2012), which simply comprises of mixing a certain mass of soil in a mixing machine, a HOBART⁵ mixer, and quantifying the mass of soil that was left stuck in the mixing tool, the B-

⁴ The term “artificial soil” is used here referring to soils mixed in laboratory, having primary elements, such as quartz sand or clay mineral (or mineral assembly), as a base for the soil mixture. However, these primary elements originated from a natural source and were washed and divided into a pre-selected grain size range. They were not artificially manufactured.

⁵ HOBART is a brand of food service equipment. Their mixer can be also used for tests in the cement research industry, as their equipment is more resistant than the conventional household mixing items.

flat beater. This mixing machine was chosen because it was previously used in research, and, it is available worldwide. This test has, however, frequently produced unrealistic results in the past, as detailed in Chapter 3. Therefore, a new stage was included to improve the accuracy of the results. This was accomplished by means of using a new device herein referred to as ATUR.

Then other queries around this type of ground emerged, regarding the application of certain classification systems and standardisations, such as the Atterberg limits, and, consequently, the application of the consistency index for evaluating clogging in mixed soils.

The applicability of the threshold on the grain size used for the test (< 0.425 mm) was also explored. In one set of tests, the standards defined by the ASTM D4318 (2017) were varied, as detailed in Chapter 4, proposing an alternative approach to be used for cohesive mixed soils.

The flow table was initially used in an attempt to find an alternative method for determining the liquid limit. However, the initial intention of the table application did not go through, as it was determined that soils with the same consistency index do not have similar flow values (Chapters 4 and 5). It was then understood that the flow table could provide interesting results regarding the flow behaviour of cohesive mixed soils, resulting in its inclusion in this research.

Finally, it was possible to correlate both methods, including other stages of testing, defining a combined testing methodology (Chapter 6). Additionally, this combined methodology was tested with natural soils from a site investigation of a tunnel project, however, only insights regarding these results can be delivered (Chapter 7), due to confidentiality agreements.

1.3 Thesis Goals

The main goal of this thesis is to investigate, characterise, and classify the clogging potential and flow behaviour of cohesive soils, mixed or not, by means of a new laboratory methodological routine. Considering that soils are a product of nature, and quoting Terzaghi:

“... in earthwork engineering the designer has to deal with bodies of earth with a complex structure and the properties of the material may vary from point to point” (Terzaghi 1948), the ultimate goal is to provide tools to investigate any cohesive soil regarding its expected behaviour when excavated by an Earth Pressure Balance machine, by means of a comprehensive laboratory characterising routine.

Specific goals include:

- Characterisation of cohesive mixed clay-sand soils, when changing proportions of clay fraction, clay mineral, water content, sand grain size, and additives (foam and polymers);
- Evaluation of the influence of those variables in terms of clogging potential and flow behaviour;
- Correlation of the results with other testing methods, such as rheometer and slump test, recognizing advantages and limitations of the testing procedure;
- Identification and resolution of queries regarding assumptions made for soils, verifying its application in the case of the tropical residual ground (i.e. Atterberg limits);
- Providing a comprehensive laboratory testing routine for cohesive soils mixed (or not) to be excavated by EPB shields;
- Evaluating the effectiveness and influence of soil conditioning by means of a simplified testing methodology.

1.4 Organization of Thesis

The thesis is organized such that this introductory chapter is followed by a brief background in Chapter 2, which mainly includes theoretical information not dealt in the following

parts. Chapters 3 to 6 comprise the four manuscripts, two already published and two submitted. Chapter 3 deals with the evolution of the clogging methodology, before reaching its final stage. Chapter 4 questions the applicability of the Atterberg limits and the consistency index for the tropical residual ground, correlating with EPB excavations, and introducing the application of the flow table for evaluating the flow behaviour of cohesive soils. Chapter 5 explores the testing method with the flow table, comparing results with data from tests conducted with a rheometer and slump tests. Finally, Chapter 6 combines all the methodologies, proposing a combined testing routine for the cohesive mixed soil, relating to the EPB machine excavation. Chapter 7 closes the work by means of a general and final discussion, including insights from tests with natural in situ soils and suggestions for future work, followed by the references chapter.

The thesis also includes two Appendixes, A and B. Appendix A presents the data compilation of the tropical residual soils. Appendix B presents information from all the materials tested in this thesis. It should be noted that this information was added also as an appendix to the Chapter 4 manuscript.

Chapter 2

Background

2.1 Introduction

A background is provided here with a summary of the aspects regarding the tropical residual ground, the clay minerals, the Earth Pressure Balance Machine and its interaction with the tropical residual ground. Most manuscripts included in this thesis except the last one (Chapter 6) provide a standalone theoretical background discussion. Therefore, this background chapter was written so as to avoid repetitions of the same content.

2.2 Tropical Residual Ground

Historically, the development of large cities took place mainly along sedimentary basins due to the proximity of rivers and flatter geomorphology, like São Paulo and Mexico cities, for example. Consequently, geotechnical studies have been concentrated in Quaternary, Neogene and Paleogene sediments. Nowadays, with the urban centre expansion, the geotechnical information on tropical residual soils from the outer city has become more and more important. Although tropical residual soils have been previously investigated leading to a substantial amount of laboratory and in-situ testing data, the knowledge of their engineering behaviour is not as extensive when compared to that available for the transported sedimentary soil (Oliveira et al. 2017b).

There are several references extensively exposing the particularities of the tropical residual ground, such as Deere and Paton (1971), Boyce (1985), Costa Filho and Vargas (1985), Viana da Fonseca and Topa Gomes (2010), Huat et al. (2012), Rahardjo et al. (2012), Salih

(2012), amongst others. It is important to note that not all tropical soils are residual soils, though these designations are sometimes mistakenly used interchangeably.

The diversity of materials originated from the complex process that integrates the weathering of rocks, makes it practically impossible to summarize all the aspects of all residual soil types. The resulting product, called residual ground, will immensely vary depending on the parent-rock, combined with complex and intense chemical and physical weathering processes. Consequently, one of the main characteristics of the residual soil is its heterogeneity and anisotropy. Figure 2-1 illustrates the main factors contributing to the soil formation, according to Thomas (1974).

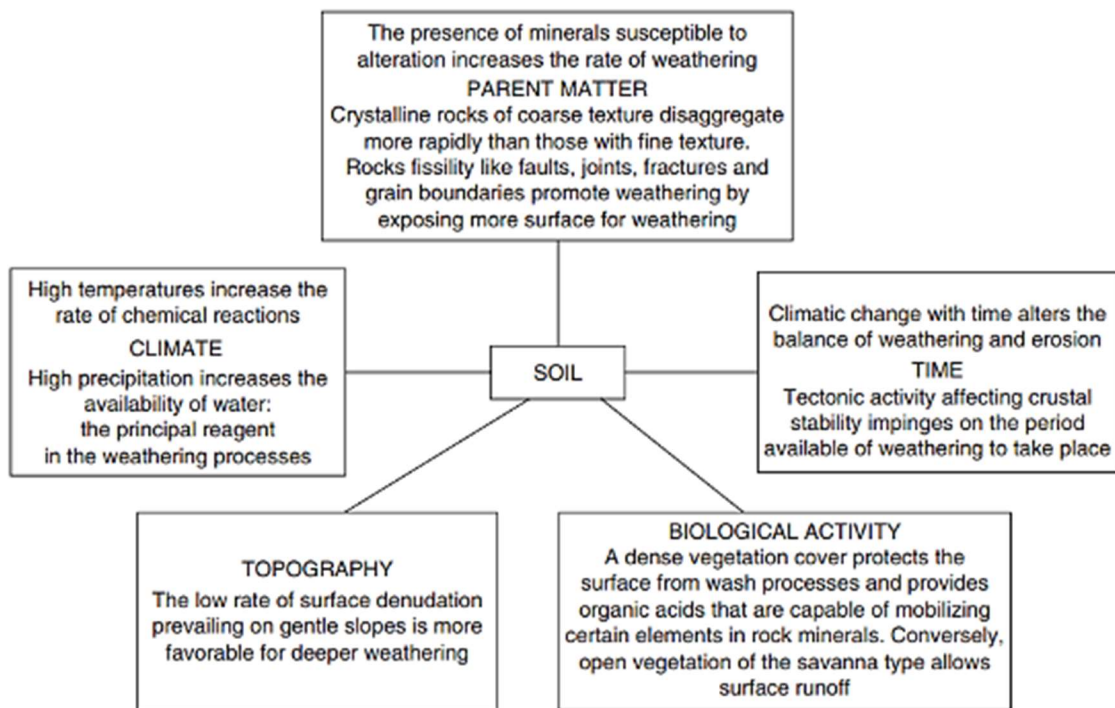


Figure 2-1. Factors responsible for soil formation (Thomas 1974).

In tropical areas, due to climate aspects such as high temperature and elevated precipitation (Futai et al. 2012), weathering – particularly chemical weathering – is extreme. This weathering leads to more expressive layers of the residual soil. Vargas (1981) mentions that the thickness of the residual soil layers in Brazil is, most probably, the highest worldwide, reaching even more than 100 meters.

Vargas (1951, 1980, 1985) mentions that tropical residual soils are fundamentally different from the sedimentary-origin soils formed at the north hemisphere, with cold climate. While in colder climate countries there is a higher contribution of illite and montmorillonite clay minerals, kaolinite is predominant in tropical countries, contributing to considerable differences in the soil behaviour. The same author makes clear that it is thus not possible to adopt the same classical soil mechanics approaches defined in the northern hemisphere, mainly conceived for transported soils, as for the massive layers of residual soil in tropical countries.

One of the most essential aspects of tropical residual soils, considering the issues with EPB drives along this terrain, is the soil/rock transition, which can be extremely irregular in this type of ground. As mentioned by Oliveira et al. (2017b), the more heterogeneous the original rock, the more irregular the soil/rock transition. The more irregular the transitional boundary, the more difficult its delimitation prior to the excavation, even with proper site investigations. Figure 2-2 from Little (1969), modified by Rahardjo et al. (2004), shows this transitional aspect of the weathering process, ranging from a 100% soil at the top to fresh rock at the bottom. It is a rather simplified scheme and the reality reveals a much more chaotic ordering of the soil/rock elements.

The grades presented in this figure are also the same ones suggested by the guidelines for rock core description, from the International Society of Rock Mechanics (ISRM 1978). Frequently, this uneven weathering aspect is influenced by the discontinuities present in the parent-rock, which would result in preferable groundwater paths, thus increasing the chemical

alterations along them. Figures 2-3 and 2-4 shows this uneven weathered limit in two different locations: for a granitic residual soil/rock in Porto and for a gneissic ground in São Paulo. It is possible to distinguish cobbles and boulders dispersed in a soil matrix.

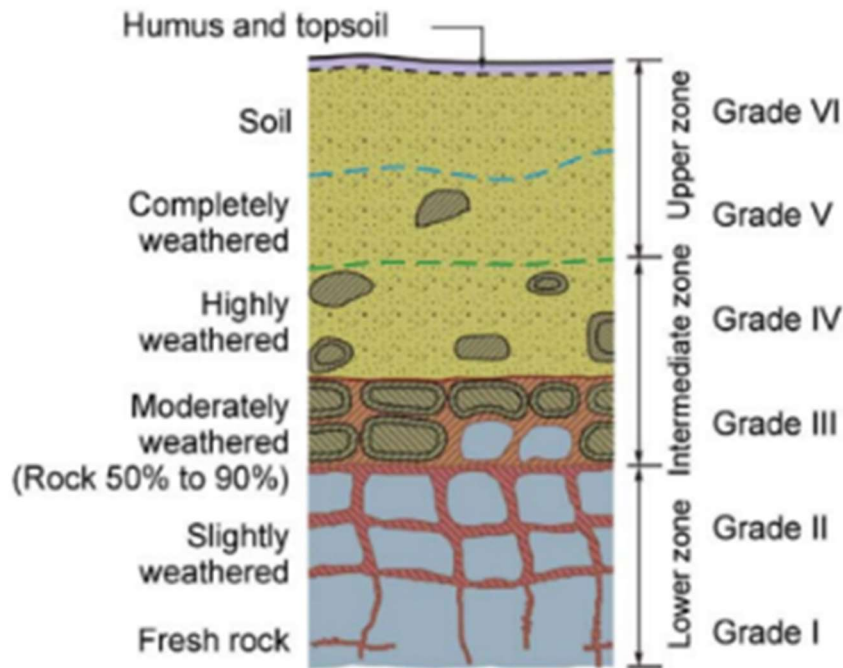


Figure 2-2. Simplified scheme of the typical weathering profile of residual soils (Little 1969, modified by Rahardjo et al. 2004).



Figure 2-3. Typical residual soil outcrop in Porto, Portugal. Vertical dimension of this picture is around 12 meters (Courtesy of Jorge Quelhas, Metro of Porto).



Figure 2-4. Typical residual outcrop in São Paulo, Brazil, northeast part of the city. Vertical dimension in this picture is around 20 meters.

This transitional soil/rock layer in a residual soil is also known as saprolite, or saprolitic ground. Above this soil/rock layer towards the ground surface, there is usually a layer of young residual soil and above that, a layer of mature residual soil. This is the current classification scheme used for the São Paulo metro projects and is illustrated here in Figure 2-5, according to the longitudinal geological-geotechnical profile presented by Oliveira et al. (2017c).

According to some authors (Futai et al. 2012, Kutner and Bjornberg 1997), the N_{SPT} values, besides other properties, could be used to divide this residual material into three different layers: the mature soil is the layer with N_{SPT} values up to 19 blows, the young residual layers above 19 blows, and the saprolite layer is the material that cannot be penetrated by percussive drilling but is not a fresh rock. The saprolite classification was also denominated as MTG by Oliveira and Diederichs (2016) in the cases of a TBM excavation, and as HSSM or IGM by Viana da Fonseca and Topa Gomes (2010).

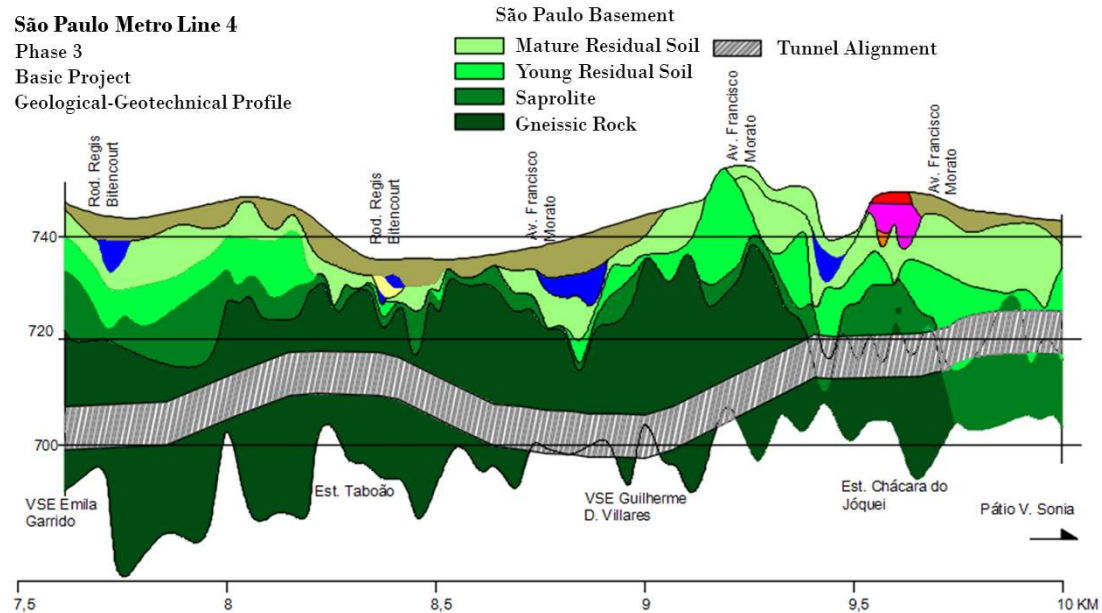


Figure 2-5. Longitudinal geological-geotechnical profile of the São Paulo Metro Line 4, Phase 3, Brazil (vertical exaggeration applied). Only the residual ground and rock layers are indicated here in green colors, the layers with remaining colors are sedimentary covers (modified from Oliveira et al. 2017c).

Moreover, as affirms Oliveira et al. (2017b), the change between horizons is gradual and their division is difficult to determine, sometimes varying according to the appraiser. These three layers do not necessarily occur simultaneously at the same site.

Figures 2-6 to 2-9 intend to illustrate the residual ground, with parts of the material being hard to classify either as a rock or as a soil. Frequently, the residual soil holds remnant structures of the parent-rock, or it is still not completely weathered.



Figure 2-6. Residual gneiss undisturbed sample.



Figure 2-7. Residual gneiss, same sample as above, but after squeezing it.

Figures 2-8 and 2-9 are photos that were taken inside a shaft along the Line 5 project of the São Paulo metro, at around 40 meters depth. In this project, during the initial site investigations, the borehole survey practically did not encounter any rock. During the construction, one of the EPB drives occurred along 300 meters of rock/weathered rock, which was not expected, damaging the machine and leading into major delays (Comulada et al. 2016, Peixoto da Silva et al. 2017, and personal communication).



Figure 2-8. Rock core, indicated by purple dashed line, in the middle of a soil matrix. Picture taken from inside shaft along Line 5 of the São Paulo Metro, Brazil.



Figure 2-9. Detail of the dissembled soil matrix from the picture in Figure 2-8 above.

Oliveira et al. (2017b) have compiled a total of 519 data points from 118 references of residual soil around the world. The complete information with all the data points is presented in Appendix A. An excel file can also be downloaded at the following link:
https://www.researchgate.net/publication/309564759_ResidualSoilCompilation_Final.

The goal of this compilation was to have a broader view of the variance in several soil parameters of the residual ground around the world. Several conclusions could be drawn from this compilation, as detailed by Oliveira et al. (2017b), but the most important aspect was the varying grain size distribution, with soils containing up to 76% clay (basaltic residual soil from

Brazil), 95% sand (granite from Brazil), and up to 55% gravel content (granites from Hong Kong and Malaysia). It was this main varying aspect leading to the primary research idea: to consider only two basic elements, clay and sand, and investigate the behaviour when changing certain variables.

There is no doubt that this is a simplified approach, as the residual ground seems much more complex than purely sand and clay having other minerals as components, as well as the cementation between the particles. Considering that there was a lack of research for mixed soils that correlates the results from EPB conditioning and excavation, it was decided to start with basic simple questions. Hopefully, other researchers can take this matter further, as mentioned in the Chapter 7.

2.3 Clay Minerals

The term clay holds several meanings, which can create many ambiguities, as mentioned by Baille (2014). It could be used to describe the size of a grain with dimensions up to 0.002 mm, as classified by soil scientists (Mukherjee 2013); it could also refer to a mineral, as a rock or soil type. The term was first formalised by Agricola in 1546, according to Karpiński and Szkodo (2015). These authors also explain the importance of this group of minerals in terms of their industrial practical applications, because of their numerous properties such as plasticity, hydration and catalytic, for example.

The term clay in this thesis is always used as the mineral term, which Baille (2014) describes as: "...a class of hydrated phyllosilicates, which are composed of a specific combination of two basic structural units, the silica tetrahedral sheet and the octahedral sheet". According to the JNC⁶, clay minerals are "phyllosilicate minerals and minerals, which impart

⁶ JNC – The joint nomenclature committee (JNC) of the Clay Minerals Society (Baille 2014).

plasticity to clay and which harden upon drying or firing”. As described by Karpiński and Szkodo (2015), these minerals are an assemblage of crystalline minerals which provide plasticity and cohesion to soils.

There are many types of clay minerals, which differ chemically, structurally, physically, and, also in their behaviour. Table 2-1 presents a synthesis of these minerals which can be divided into different classes, groups, and, even species. Here is presented a summary of the main aspects of clay minerals, focusing on both the minerals used for this research: kaolinite and bentonite⁷. The studies of clay mineralogy are vast, considering that clay minerals have a wide range of applications; here the interaction of an EPB machine excavating clay-sand soils as a natural material is explored.

Table 2-1. Classification of clay minerals and their position in the mineral world (Mukherjee 2013).

Mineral class	Mineral subclass	Mineral groups	Mineral species
SILICATE Contains SiO_4^{-4} tetrahedron as basic structural unit	Phyllosilicate SiO_4^{-4} tetrahedra are linked together to form two- dimensional sheet like structure	Kaolinite (Serpentine- kaolin)	Kaolinite, Dickite, Nacrite, Halloysite, Lizardite, Amesite, Berthierine, Cronstedtite, Nepouite, Kellyite, Fraipontite, Brindleyite
		Smectite (Montmorillonite)	Montmorillonite, Beidellite, Nontronite, Volkonskoite, Saponite, Hectorite, Sauconite, Stevensite, Swinefordite
		Illite (Interlayer- deficient)	Illite, Glauconite, Brammallite, Wonesite
		Vermiculite	Trioctahedral vermiculite, Diocahedral vermiculite
		Palygorskite- Sepiolite	Palygorskite, Sepiolite, Loughlinite, Falcondoite, Yofortierite
		Mixed Layer (Inter-stratified)	Brinrobertsite, Aliettite, Kulkeite, Rectorite, Hydrobiotite, Corrensite, Tosudite

⁷ Bentonite is mainly composed of montmorillonite (Mukherjee 2013). Here is used, to simplify, as practically synonyms.

In terms of structure, kaolinite and bentonite differ from each other significantly, as illustrated in Figure 2-10. The basic structure of clay is an alternation of silica tetrahedral sheets and aluminium or, secondarily, magnesium octahedral sheets.

Kaolinite has a 1:1 alternation structure, where the octahedral sheet is mainly composed of aluminium (gibbsite), with a strong hydrogen bonding between these sheets, and according to Baille (2014), interlayer separation does not occur with this clay mineral. The sheets of kaolinite, according to Mukherjee (2013), combine to form six-sided flakes, elongated prominently in one direction, forming the kaolinite grains.

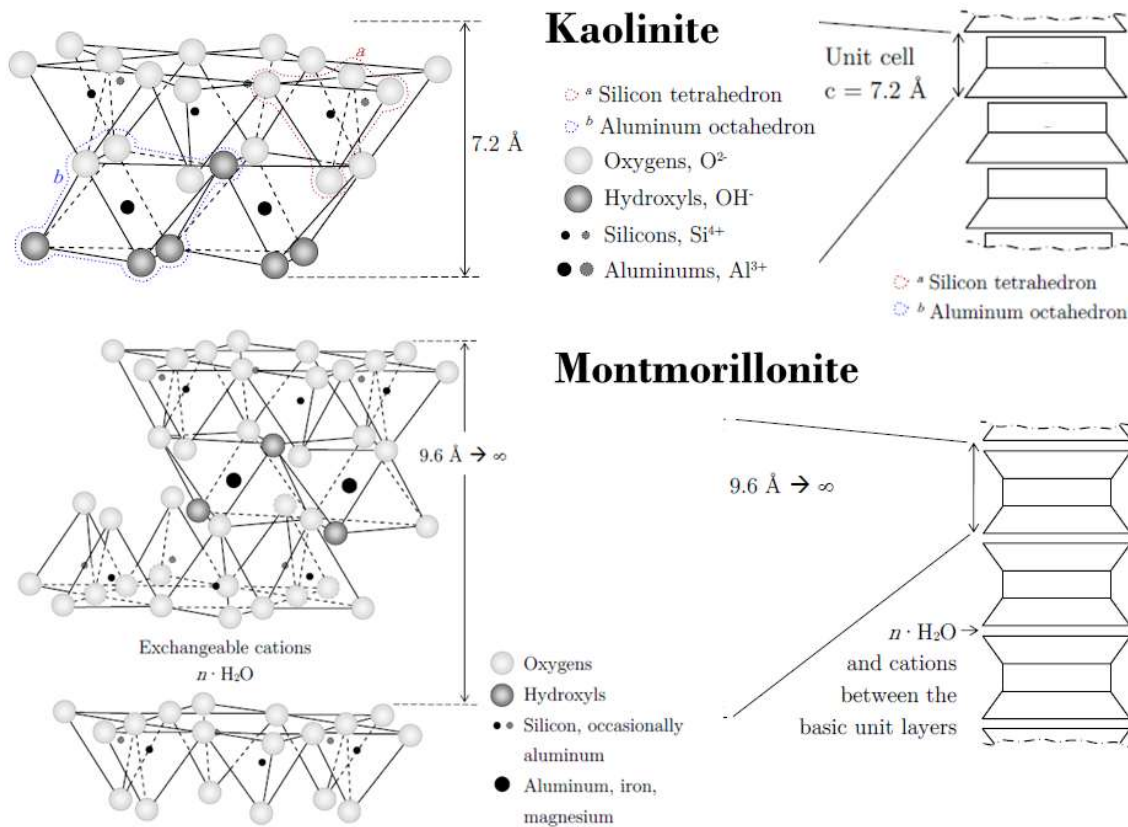


Figure 2-10. Kaolinite (top) and bentonite (montmorillonite) structure and basic sheet organization (modified from Baille 2014).

Montmorillonite has a 2:1 structure, with a silica tetrahedral sheet being squeezed between two octahedral sheets, which can be composed of aluminium, magnesium, or iron. The bonding in this mineral is weaker than for the kaolinite species, characterised by a van-der-Waals bonding allowing not only separation between the layers, but also the entering of other molecules, such as water. The inclusion of water in the molecular structure provides the swelling behaviour of this clay, as great quantities of water can fit in between the layers. The montmorillonite grain is organized in undulating mosaic sheets (Mukherjee 2013).

The specific area of bentonite is larger than of the kaolinite. Mukherjee (2013) presents the ranges of 5 to 40 sqm per gram for the specific area of kaolinite, and between 40 to 800 sqm for smectite (bentonite). In terms of cation-exchange capacity at pH 7 (CEC), the values of kaolinite are between 3 and 15 meq/100 g, whereas smectite lies between 80 and 120 meq/100 g. The CEC measures the ability of a soil to hold positively charged ions. These values for the kaolinite and bentonite clays used here as basic materials of the assembled soils can be reviewed in detail in Table B-1 (Appendix B).

The interaction of clay minerals and water at a molecular level has a great influence on its general behaviour and, depending on the mechanisms of this interface, the effects will greatly vary. This interaction does not depend solely in the characteristics of the clay, but as well as the environmental conditions, such as temperature, availability of water, and salinity (Baille 2014).

According to Konta (1995), the crystal structure and chemistry of clay minerals reproduces the state and the electrostatic charge distribution along their sheet surfaces. Their negative charge, which is concentrated along the clay surfaces, is a product of ion substitutions in their sheets, and depending on the ion substitution, the effect would enormously vary. Consequently, considering what has been discussed so far about clay minerals, soils formed by

clays from the exact same species (e.g. kaolinite soils) could behave completely different from each other, depending on other factors.

In terms of the physical aspects, the Atterberg limits are one of the most used ways to classify soils containing clay minerals, considering changes of its state by altering its water content. This subject, including limitations of the method, is discussed in detail in the Chapter 4. From the Atterberg limit, it is also possible to classify the consistency of a clayey soil, considering its liquid limit (w_L), plastic limit (w_P) and water content (w) of the soil. Figure 2-11 resumes the changes in the state of a plastic soil, with changes in the water content, regarding its consistency and liquidity limits, also correlating them with its unconfined compression strength, based on the findings of Atterberg (1911 and 1913).

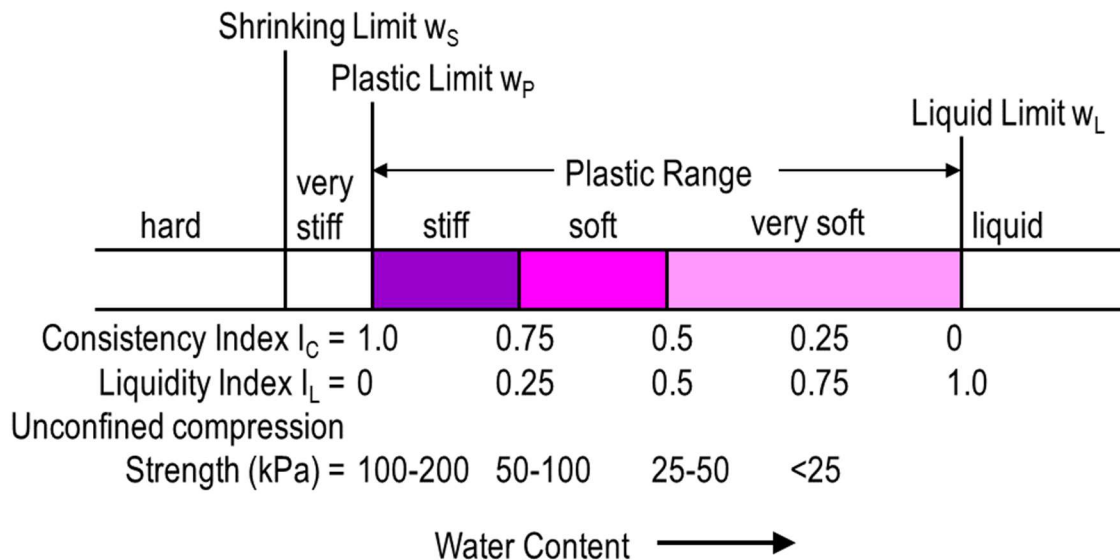


Figure 2-11. Scheme of changes in the state of a plastic soil, considering the consistency and liquidity indexes, based on the Atterberg limits, including the Uniaxial Compressive Strength - UCS values (Modified, based and translated from Atterberg 1913, Chancellor 1994, Kretschmer 1997, Witt 2017).

Figure 2-12 illustrates two different consistencies for a same soil, in this case, pure kaolinite. The consistency index is an essential parameter to characterise cohesive soils, and it considers the relation between the Atterberg limits of a soil and the water content. A low consistency implies in a softer soil and a high consistency, in a stiffer.



Figure 2-12. Same pure kaolinite sample in two different consistencies, stiff, on the left and soft, on the right.

Equation (2-1) considers the values of plastic and liquid limits (W_P and W_L , respectively), together with the water content (W), obtaining the consistency index values for a certain soil.

$$I_c = (W_L - W)/(W_L - W_P) \quad (2-1)$$

Further aspects regarding the application of the consistency index for cohesive mixed soils are presented in the manuscripts included here as Chapters 3, 4, 5 and 6. The same goes for

clogging mechanisms, which are not included here in the background, as it is already referenced in Chapters 3 and 6.

2.4 Earth Pressure Balance Machine

Mechanised tunnels are those where the tunnel excavation is performed mechanically through the use of tunnel boring machines (TBM) by means of teeth, picks or disks (ITA-AITES 2000). TBMs can be of various types, being adapted to the project needs. An Earth Pressure Balance Machine (EPB or EPBM) is a closed shield used for the excavation of soils where face support and groundwater pressure counter-effect is obtained by means of the material excavated by the cutting wheel, which serves as a support medium itself.

As detailed by Maidl et al. (2012) and EFNARC (2005, Figure 2-13), a cutterhead, shown in Figure 2-13 as ❶, fitted with tools such as disc cutters and scrapers, excavates the material, which goes inside an excavation chamber (❷). The excavation chamber, also called working chamber, is closed from the tunnel by a pressure wall (❸), and the excavated ground is mixed with mixing arms attached on the back of the cutterhead (rotors/back buckets) and on the pressure wall (stators). Sensors placed in the pressure wall control the pressure inside the chamber. A pressure-tight screw conveyor (❹) is responsible for removing the excavated soil, where the support pressure can be controlled by varying the revolution speed of the screw conveyor.

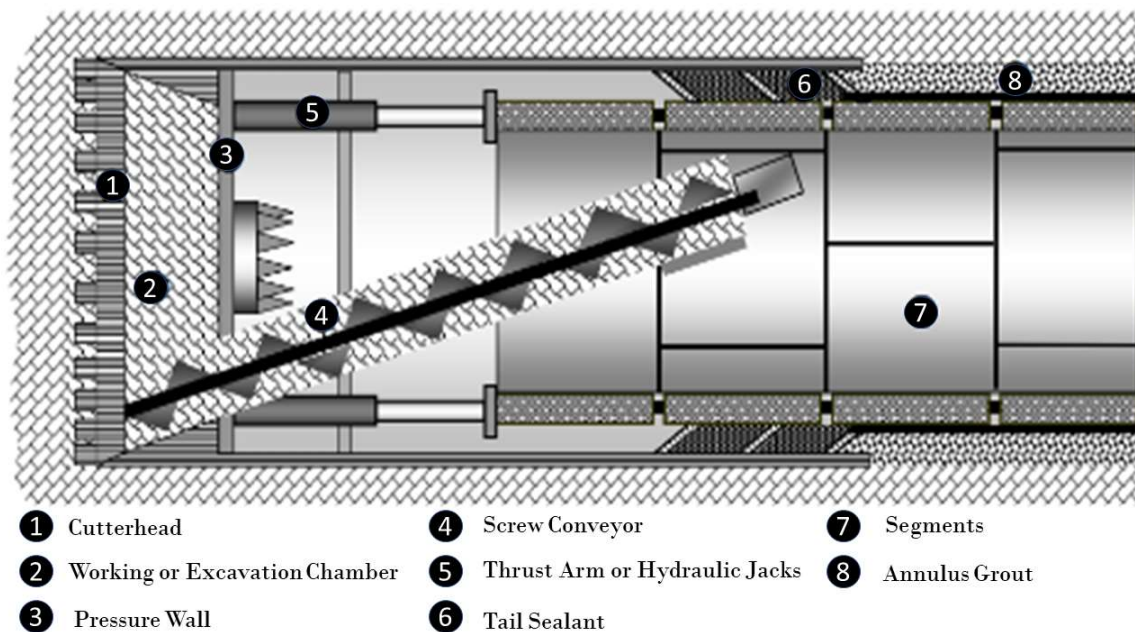


Figure 2-13. Schematic representation of Earth Pressure Balance Machine (modified from EFNARC 2005).

Thrust arms, or hydraulic jacks (⑤) push the machine forward and help to control the pressure ahead of the machine, thrusting the shield forward against the soil mass. Pre-cast concrete segments (⑦) are erected, functioning as the final tunnel lining. A grout (⑧) is injected to fill the voids created behind these segments. Last, a tail sealant (⑥) is injected at the end of the shield, working as a seal against groundwater, grout or bentonite slurry, preventing the infiltration of these elements inside the shield.

EPB machines were first commissioned in Japan in the early 1970s for tunnel drives along very cohesive soils with high plasticity (Maidl et al. 2012). Since then, the machine has extended its field of application many times by means of ground conditioning. Figure 2-14 shows the most recent field of application of these machines, including the conditioning with high-density slurries, according to Budach and Thewes (2013). There is ongoing research being conducted to extend this field even further.

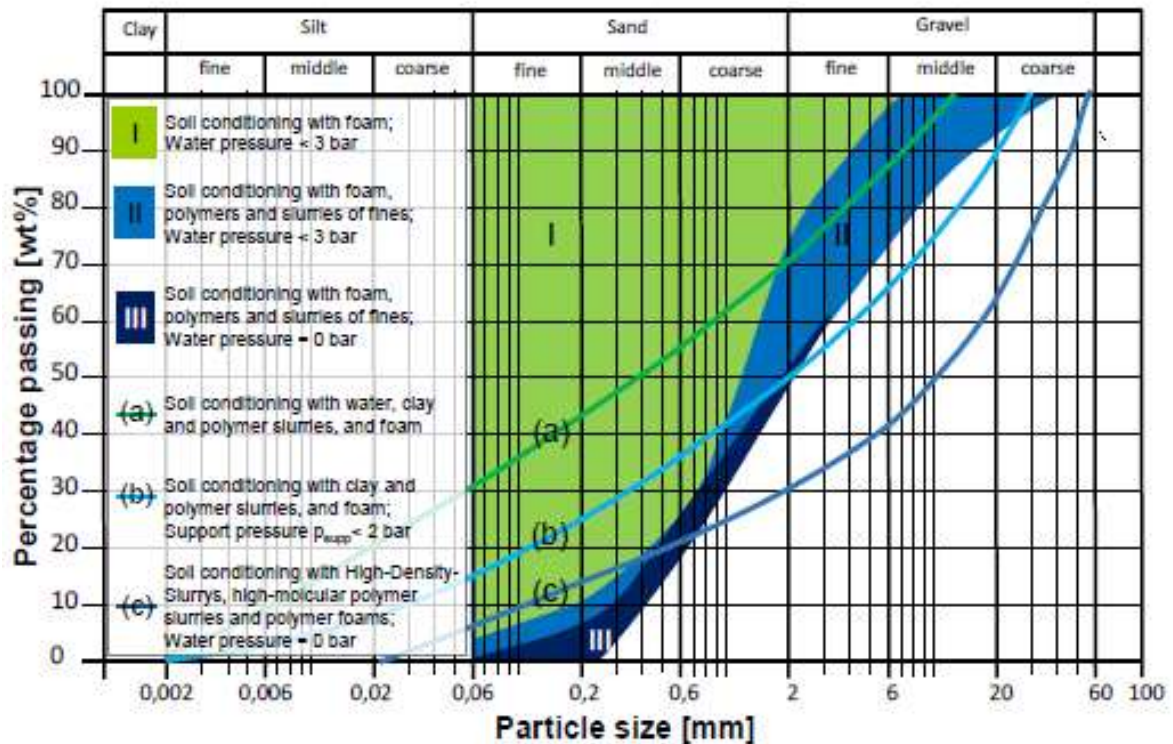


Figure 2-14. Range of application of EPB machines, with field extended by means of soil conditioning, according to Budach and Thewes (2013).

Regarding its operation, there are three possible modes feasible for the machine: closed-mode, open-mode and semi-open-mode. The closed-mode is the one used when a full support of the face is required, with loose soils and groundwater occurrence, for example. In the opposite case, for a self-supported ground such as rock or cohesive hard clays the machine can drive in open-mode, with the excavation chamber empty, with the muck being directly transported through the screw conveyor, without the need to apply face pressure. The third option is the semi-open-mode, or semi-closed, where there is only a partial filling of the chamber; this third option is usually used for mixed face conditions (Maidl et al. 2012, Galli 2016).

For an EPB operation, besides the face support, there are also the logistics of muck transportation and disposal, which can be done by means of muck cars or conveyor belts. The logistics regarding the muck disposal can be heavily constrained by the local environmental

regulations, which can be a key factor influencing the ground conditioning itself, in terms of types of permitted chemical additives, such as foam and polymers. The costs involving the transportation and disposal of the muck are always taken in consideration in a project: a liquid muck implies in higher volumes to be transported/disposed, therefore, higher costs.

Mechanised tunnelling is a very integrated process, which is something that needs to be considered when investigating the characteristics of the ground and proposing solutions for its optimum adjustments by means of conditioning. Even the availability of water on site could hold back an EPB excavation. In summary, besides the needs for proper face pressure regulation, the properties of the muck must fulfil the requirements for its transportation and later disposal.

The entire EPB logistics, face pressure or other aspects related to the operation in general, is not the focus of this thesis, but there are several comprehensive references on the subject, such as Milligan (2000), BTS (2005), Thewes (2007), Thewes et al. (2011), Maidl et al. (2012), Galli (2016), DAUB (2016), and others.

Another question to be briefly mentioned is the application of the EPB shields versus Mixshields. Mixshields are tunnel boring machines which the face support is achieved by means of an automatically controlled air cushion and a slurry suspension, happening in two chambers, separated by a submerged wall. Differently than an EPB, the muck is removed by means of pipes in a slurry circuit, where is later sorted in a separation plant above ground. The control of the face pressure is more precise than in an EPB machine (Herrenknecht Website).

Many times, due to the initial costs being lower, the EPB shields have been preferably chosen over Mixshields; in some cases, however, due to unforeseen excavation issues (including mixed ground scenarios), the final costs of a project after the occurrence of delays, settlements, or even major accidents, would have justified the initial higher costs of the Mixshield. Thewes (2007) compares both excavation types in detail, showing the “pros and cons” of each excavation

method and their typical application range. In this research, the author also includes the extended field by means of soil conditioning comparing both methods, resumed by Figure 2-15. The EPB is represented by the numbers and the Mixshield, by letters.

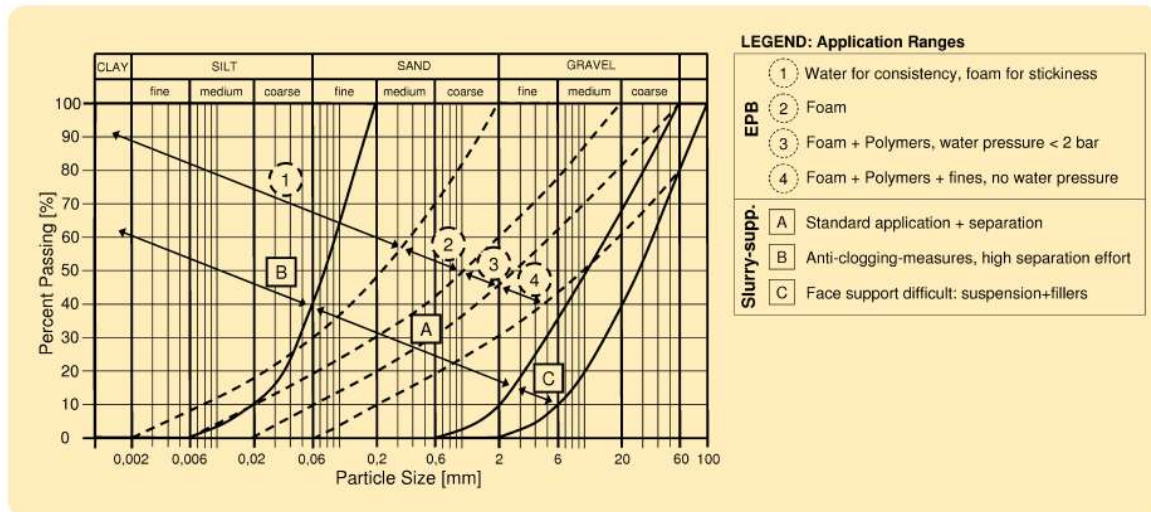


Figure 2-15. Comparison between EPB and Slurry shields application range (Thewes 2007).

In the case of a tunnel excavated using an Earth Pressure Balance Machine (EPBM), the conditioning of the ground plays a critical role in the entire excavation process. An extensive campaign of soil conditioning testing prior to a tunnel excavation can reduce risks, costs and overall improve the EPB excavation performance.

Currently, in most projects, the soil conditioning assessment is conducted during construction on a trial-and-error basis and is often done to mitigate issues during excavation in a retroactive manner. Often, this assessment is even handled only by the chemical suppliers, which could be considered to be a biased evaluation.

The gain from a preliminary soil conditioning testing phase was demonstrated by Ye et al. (2016), in a case studied for two single tunnels implemented for the Nanchang Metro Line 1, in China. The authors ran a comparison between a new conditioning scheme, based on an

assessment prior to the excavation, with another trial-and-error approach. The trial-and-error procedure resulted in project delays and difficulties in attaining the right workability of the excavated material inside the EPBM excavation chamber.

Issues with trial-and-error soil conditioning approaches become even more evident when challenging ground is being excavated, such as in mixed ground terrains. Consequently, soil conditioning is one of the most important elements for a successful EPB operation. Several aspects concerning the soil conditioning of mixed cohesive soils, as well as laboratory tests, are detailed in Chapters 5 and 6.

2.5 EPB Interaction with Mixed Transitional Ground

Mixed ground conditions present a challenging scenario for tunnel boring machines. Several tunnel delays, accidents, wear, and low performance of the machine are directly related to the excavation of this type of ground. Mixed face conditions can be found practically anywhere in the world, but it is mainly in urban tunnelling environments where these conditions cannot be avoided, mostly due alignment constraints associated with urban settings.

Even though all the cited authors (Della Valle 2001, Thewes 2004, Zhao et al. 2007, Tóth 2013, Galli & Thewes 2014, Shirlaw 2015, Peixoto da Silva et al. 2016, Comulada et al. 2016), talk about mixed ground conditions, not all of them relates to the mixed transitional ground, as defined by Oliveira and Diederichs (2016) and Oliveira et al. (2017a).

It is commonly understood that mixed face conditions contain two or more materials that present a completely different physical ground behaviour from each other, such as very distinct UCS (Uniaxial Compressive Strength), cohesion, consistency and stiffness values, weathering differences, or any other geological-geotechnical characteristics which would imply relevant different performance for the excavation and/or stability process. A transitional ground is a by-

product of the weathering alterations of the same rock type, a geological material formed in the exact same geological scenario.

A mixed transitional ground (MTG) would have both a mixed face and a transitional ground. For example, a tunnel face that contains portions of a mature residual soil in contact with a saprolite, or a young residual soil as a surrounding matrix for some boulders of a fresh parent rock, would be considered a MTG.

The accounted issues, regarding these conditions when crossed by an EPB excavation, can be resumed as follows:

- Ground loss, settlements and sinkholes;
- Slow rates of tunnelling;
- Rapid tool wear, especially disc cutters and screw conveyor;
- Damage to tools, mixing arms, and other parts of the TBM;
- Very frequent and long interventions;
- Clogging;
- Blowouts;
- High temperatures inside the excavation chamber, preventing interventions.

Several factors could be cited as the leading roles for the above related issues. The first one, already mentioned in the item 2.2, is the erratic rock top, due to the irregular weathering grading, together with paleo-topography of the basement rock. This makes tunnel alignment definition difficult, especially in urban areas where there is more restriction of space.

The second factor is related to the ratio of soil/rock at the tunnel face, with two main scenarios (illustrated in Figure 2-16A and B) which would lead to a worsened situation, as resumed by Thewes (2004). Figure 2-16A shows a closed-face EPB with a lower ratio of rock at the bottom and a loose soil at the top, which in the specific case of an MTG, could also be called

a portion of soft ground (such as the mature residual soil) and a portion of hard ground (saprolite or fresh rock, for example).

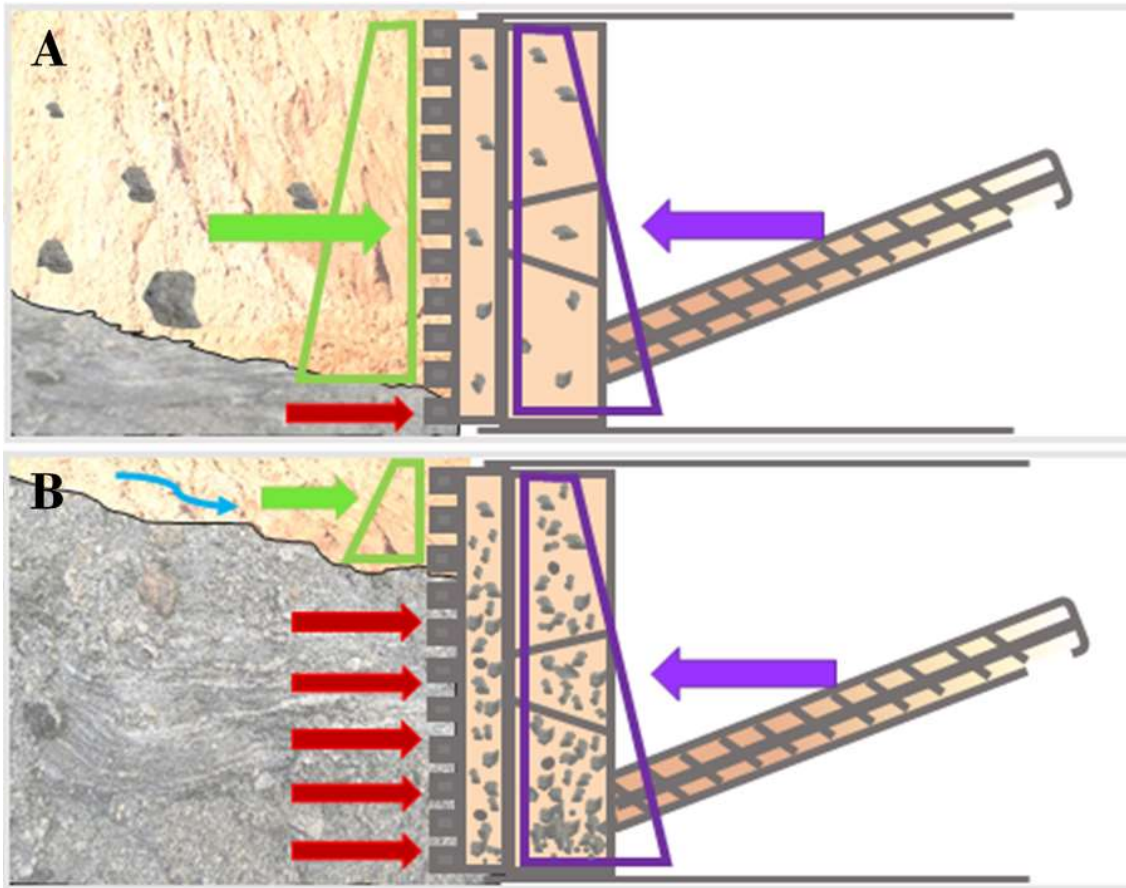


Figure 2-16. Two schematic scenarios of the mixed face: on A there is a small portion of hard ground covered by a larger portion of soft ground; on B, the opposite, a larger portion of hard ground covered by a small portion of soft ground and, additionally, ground water presence (modified from Thewes 2004).

The main issue for this situation is concerning the high impact load which the disc cutters undertake, while transitioning from a soft ground to a competent rock material, leading to the damage of the tools (Figures 2-17 and 2-18) and the constant need for changing them along machine interventions (Thewes 2004). The main difference between the MTG scenario compared to a sedimentary one is the soft/hard ground boundaries, which can have an undefined and chaotic

boundary organization in the MTG case, with a wider variety in the soil/rock positioning arrangements.

This high impact results in greater rates of wear and damage, as can be observed in Figure 2-17, from the Line 5 São Paulo subway project, and Figure 2-18, with flat discs from projects in Singapore. This issue would also culminate in other problems, affecting the machine, as clearly explained by Comulada et al. (2016): “The tool contact force is concentrated in the areas of the face with hard rock, leading to a risk of overloading the disc cutters, therefore, in a mixed face condition, it is imperative to limit the thrust force in order to avoid tool overload, which further reduces the attainable advance speed”.



Figure 2-17. Damage on the disc cutter (around 43 cm of diameter) due to mixed ground conditions in a saprolitic/residual soil boundary, Line 5, São Paulo, Brazil (Comulada et al. 2016).



Figure 2-18. Flat disc cutters (around 30 cm of diameter) damaged along mixed transitional face, in Singapore (Shirlaw 2015).

Another issue that tool wear would bring are the frequent and long interventions, mentioned and quantified in detail by Shirlaw (2015). More interventions would directly mean a lower utilization of the machine, with low performance and increasing delays.

When considering the opposite from the previous situation, which is a higher amount of rock covered by a low amount of soil and groundwater presence (Figure 2-16B), the main issue would be the inability to maintain the face pressure.

Because of the high amount of the excavated hard material (rock chips, or saprolite chips), the muck would have a high permeability and not enough cohesive and plastic material. Therefore, the desired workability would not be attained, and it would not be feasible to function

as a medium to maintain the pressure ahead of the tunnel face. This is especially an issue when excavating in the EPB closed-mode, required for cases of loose ground and groundwater presence. This would, consequently, result in ground losses, settlements and in some cases, sinkholes.

This described scenario has been encountered in the Line 5, São Paulo, Brazil, where, in some areas, the ground losses exceeded 5%, resulting in local instabilities which needed to be later filled with grouting (Peixoto da Silva et al. 2016).

The last aspect in the mixed transitional ground scenario is related to the clogging potential in clay materials, which happens when a clay mineral tends to adhere to steel. Thewes (1999, 2005) has detailed the clogging process in TBMs and affirms that if this issue is not efficiently solved, it can result in a drastic performance reduction of the machine, leading into constant and long interventions. According to Shirlaw (2015, 2016), Comulada et al. (2016) and Peixoto da Silva et al. (2016) clogging has been one of the main issues in EPB drives in Singapore and in São Paulo, Brazil.

Thewes (1999, 2005) mentions that the clogging tendency strongly depends on the content of the swelling clay mineral (smectite group) and the soil consistency. However, in Line 5, São Paulo, Brazil, there was no swelling clay encountered along the residual ground (only kaolinite clay), and clogging was still an issue. Figures 2-19 and 2-20 show a ripper (cutterhead tool) clogged in the Line 5 project in São Paulo, Brazil, and a disc cutter facing clogging issues in Singapore.



Figure 2-19. Rippers (around 20 cm) at the cutterhead clogged with clayey soil (Comulada et al. 2016).



Figure 2-20. Clogged disc cutter (around 30 cm of diameter), Singapore (Shirlaw 2015).

Clogging aspects and its evaluation will be dealt with in detail in Chapters 3, 5 and 6.

Considering the data obtained from the soil compilation presented in Oliveira et al. (2017b), the samples from Singapore and São Paulo were compared using the evaluation universal diagram of clogging potential for EPB shields, as defined by Hollman & Thewes (2012), illustrated in Figure 2-21. This diagram is explained in detail in the Chapters 3, 4 and 6.

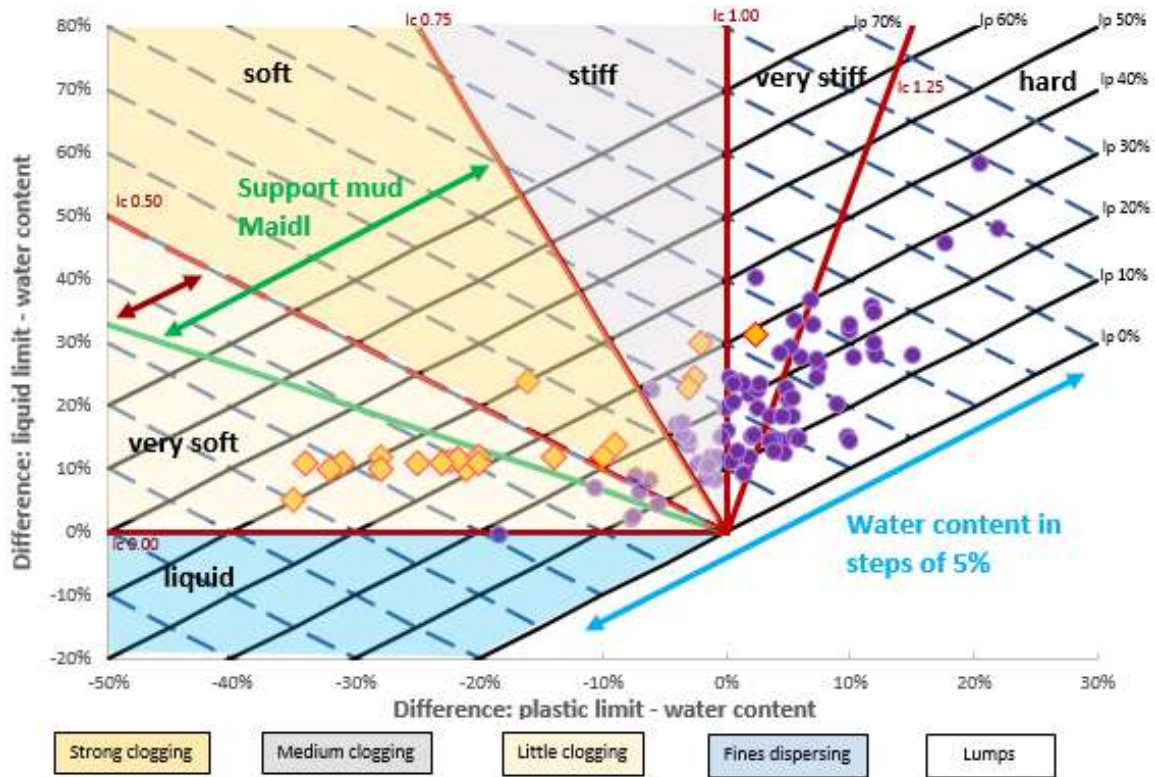


Figure 2-21. Evaluation diagram for clogging potential. The green double arrow indicates the typical consistency range for the support medium (Maidl et al. 2012). The brown double arrow indicates the partial area less susceptible to clogging (Hollman & Thewes 2012). Plotted data includes samples from Singapore (yellow diamond) and São Paulo, Brazil (purple circles) (Modified from Hollman & Thewes 2012, published in Oliveira et al. 2017b).

This diagram allows a comparison between the samples from these two distinct locations, considering its W_p , W_L and W values. While São Paulo samples would need an increase of water

content and/or foam to achieve the ideal support mud as defined by Maidl et al. (2012), the samples from Singapore would be too soft and would need additive to attain the right consistency, bringing both of them to a strong clogging potential area.

In practical terms, to attain this optimum field of the support medium, water, foam conditionings, additives, or soil fines, must be added, bringing the soil to this “perfect condition”. When considering the frequency with which a face condition can change in a mixed transitional ground environment, and the limitations of site investigation in predicting the exact changes along the tunnel alignment, the reality of tunnel projects with EPB machines along an MTG presents a unique set of challenges.

Chapter 3

EPB Tunnelling Through Clay-Sand Mixed Soils: Proposed Methodology for Clogging Evaluation

Abstract

The clogging of Tunnel Boring Machine (TBM) tools by soils has long been investigated, owing to the numerous difficulties arising in shield tunnelling as a result. Its occurrence leads to operation delays owing to the frequent and lengthy interventions required to unblock the soil stuck to the excavation tools and screw conveyor. Several authors have proposed laboratory tests for evaluating the clogging potential. One of these methods is the empirical stickiness evaluation, whereby a mixer and flat beater are used to define a clogging evaluation parameter. Following an extended test campaign using artificial soils with different clay contents and minerals, it became clear that this method was not adequate to provide reliable information regarding the tendency of a previously untested soil to clog in a tunnel drive. A new device was then implemented, which adds to the first method a kinetic energy impulse via dropping of the beater from a certain height. This combination of methods could provide a reasonable approximation of the potential for clogging to occur along Earth Pressure Balance Machine (EPB) tunnel drives. This paper presents the results of the proposed combined methodology for clogging evaluation, as well as the research evolution that led to the addition of the beater dropping stage.

3.1 Introduction

The clogging of tools by soil was initially investigated for agriculture purposes over a century ago, as it is a particularly troublesome soil characteristic. Early investigations into soil stickiness led to one of the most widely used standard soil classifications, the Atterberg limits, which defined different soil states according to their changes in water content (Atterberg 1911). Later, Casagrande (1932) refined this classification for its application to the soil mechanics field.

Clogging in tunnel boring machines, as defined by Thewes (1999, 2004), occurs through a combination of four single mechanisms: adhesion of clay particles to metal surfaces, bridging of clay particles over a metal opening, cohesion of particles, and insolubility of clay minerals. The occurrence of clogging in earth pressure balance machines (EPBMs) may cause several issues such as high torque requirements of the machine, drastic performance reductions, and lowering of advance rates. These issues result in lengthy and frequent interventions for cleaning the clogged tools, as well as additional excavation costs (Thewes 1999, 2004, Sass and Burbaum 2009, Feinendegen et al. 2011, Zumsteg and Puzrin 2012, Hollman and Thewes 2012, 2013, Hollman et al. 2014, Thewes and Hollman 2014, 2016, Peila et al. 2016, Basmenj et al. 2016, Khabbazi et al. 2017).

The evaluation of the clogging potential and possible mitigation measures is therefore essential prior to excavation of a shield tunnel. Appropriate machine design (namely location of the foam injection system, opening ratio of the excavation interface and other particularities of the machine) and effective soil conditioning play a significant role in reducing clogging issues.

Soil conditioning is one of the most important elements of EPBM excavation, as highlighted by Maidl (1995), Langmaack (2000), Merritt (2004) and EFNARC (2005), for example. The tunnel face stability following excavation by EPBM relies on the support pressure transferred from the excavated soil inside the chamber to the tunnel face. This is best achieved

with a soil that is conditioned to a paste with soft plastic consistency and low permeability (Herrenknecht et al. 2011, Galli and Thewes 2014). Numerous additives such as water, foams, polymers, dispersants and slurries with fines, including bentonite, are used to condition the soil, with the goal of achieving the expected soil characteristics.

A considerable number of research studies have dealt with clogging in EPBMs; among these are: Sass and Burbaum (2009), Spagnoli et al. (2010), Feinendegen et al. (2011), Zumsteg and Puzrin (2012), Hollman and Thewes (2013), Hollman (2014), Thewes and Hollman (2014, 2016), Basmenj et al. (2016), Khabbazi et al. (2017). Nevertheless, the influence of clay grain size and mineralogy in a mixed clay-sand material on soil conditioning, as well as on clogging occurrence, has not yet been subjected to extensive investigation.

This paper presents the research evolution that led to the inclusion of an extra step in the methodology proposed by Zumsteg and Puzrin (2012), in order to assess the clogging potential by means of a HOBART mixer. Tests were conducted with pure clay and mixed sand-clay soils, assembled in the laboratory, adding only water as a conditioner and changing the clay contents and minerals. The test results also disclose certain preliminary information regarding the influence of clay/clay mineral content in a mixed clay-sand soil. Soils assembled in the laboratory were selected to ensure reproducibility, allowing for evaluation of the new combined methodology. The results were compared to the universal clogging evaluation diagram from Hollman and Thewes (2012), indicating closer fitting between this empirical diagram based on real tunnel drives along pure clay soils and the results obtained using the proposed combined methodology.

3.2 Clogging Mechanisms and Evaluation

Issues with clogging have long been addressed for tunnel drives in clayey soils. Jancsecz (1991) demonstrated that clay adhesion to metals exhibits a strong correlation between soil

mechanical parameters, and increased adhesion occurs with increased soil plasticity, or consequently, a decreased activity index, as well as being influenced by the undrained shear resistance of the material. Clogging mechanisms were explained in detail by Thewes (1999), who defined four mechanisms, as mentioned in the introduction: adhesion, bridging, cohesion and not dissolving. Of these four mechanisms, adhesion of the particles to the metal surface is the most significant factor leading to clogging, and this adhesion can be classified as tangential or normal.

Alberto-Hernandez et al. (2017) offered a review on the occurrence of clogging in tunnel boring machines, reminding readers that a standardized method for evaluating clogging potential does not yet exist, particularly for mechanised excavations. They divide the clogging evaluation methods into two main approaches, the first being an empirical evaluation based on clogging occurrence in tunnel drives.

Several empirical charts have been proposed to evaluate clogging, such as those by Geodata (1995) and Thewes and Burger (2004), and Feinendegen (2010) for clogging evaluation in Mixshield specifically, as well as a universal chart by Hollman and Thewes (2012).

The universal chart developed by Hollman and Thewes (2012), for instance, considers the plastic and liquid soil limits, the changes in water content and consistency index (I_c), correlating that with distinct clogging potential ranges, from absent (lumps or fines dispersing) to strong clogging. Data from several projects with clogging occurrences were compiled for the design of this graph. However, one of the limitations of this correlation was not considering changes in the clay percentage, for instance, in the case of mixed soils. Also, the used data are from soils with maximum values for plasticity index around 70%, not being considered data from highly plastic soils.

The second approach to evaluating clogging is the direct assessment of particle adhesion to a metal surface, measured with different laboratory devices. Thewes (1999) constructed a

separation test device to determine the clay adhesion when a piston is pulled away vertically from a soil sample. Several tests were conducted, and the results demonstrated that the normal adhesion is directly related to the presence of clay minerals, with greater normal adhesion reported for samples containing swelling clays (such as smectite).

Feindendegen et al. (2011) proposed a classification scheme to determine the clogging potential of different clays based on the “cone pull-out test”, defining an adherence value based on the energy required to pull a metallic cone out of the soil following compaction by a standard proctor device. The results were then plotted against the soil consistency. Because the cone is rotated prior to being pulled out, the adhesion measurements are assumed to have both a normal and a shear component.

Zumsteg and Puzrin (2012) used a HOBART mixer with a B-flat beater, with the intention of qualitatively evaluating the soil clogging potential. This HOBART mixer machine is used in concrete laboratories and the bakery industry and is available worldwide. A 20-L capacity HOBART mixer was used for the first tests, and later changed to a smaller, 5-L version (Zumsteg et al. 2016).

This simple method consists of weighing the B-flat beater with any sticky soil on it, following rotation inside a mixing bowl, and comparing this with the total soil mixture weight. The mixer is rotated at speed level 1 at 100 rpm for 3 min, which, according to Zumsteg and Puzrin (2012) and Zumsteg (2014), is sufficient to “achieve mixture homogeneity at a larger scale and a steady state for the amount (mass) of clay sticking to the mixing paddle”, although it does not provide a uniform water distribution at the particle scale.

From this comparison between the mass of the material stuck in the beater and the total material mass, it is possible to obtain an empirical stickiness ratio (λ), as provided by Equation 3-1:

$$\lambda = G_{MT}/G_{TOT} \quad (3-1)$$

Where G_{MT} refers to the mass of the material stuck in the beater and G_{TOT} is the total soil mass.

Zumsteg et al. (2013) also compared the results obtained for the clogging evaluation parameter λ to the diagram proposed by Thewes (1999), to be applied in the clogging evaluation of Mixshields. The authors defined different clogging potential levels corresponding to varying values for the parameter λ : low ($\lambda < 0.2$), medium (λ between 0.2 and 0.4) and high ($\lambda > 0.4$).

Zumsteg and Puzrin (2012) developed an additional device to evaluate soil clogging potentials: the plate shear test. Using this device, the tangential adhesion was measured by rotating a plate tangentially to the soil sample, providing a parameter for evaluating the sliding resistance of the material against the steel surface, which, according to the authors, has a greater influence on the clogging potential than normal adhesion resistance. Numerous soil mixtures were tested, together with chemical additives, as explained by Zumsteg and Puzrin (2012, 2013) and Zumsteg et al. (2016).

Peila et al. (2016) conducted laboratory tests on three different clayey natural soils and explained that the samples were tested not only as an initially powdered fraction, but as pre-moulded lumps, as this would provide a closer approximation to the reality of the excavation of an EPB machine. According to the authors, it is possible that excavated soil may not be reduced into a completely crushed form, but rather may detach in lumps of variable sizes, depending on the machine cutterhead openings, which are usually between 10-30 mm in diameter. In this case, the conditioning remains only around the surface of the lumps and is not completely homogenized. Peila et al. (2013) also concluded that tests should always be carried out using only

water as the conditioning agent in order to understand the mixture behaviour fully and for comparison with the conditioned material results.

Khabbazi et al. (2017) compared the results from a piston pull-out test using 35 clayey soils in two distinct consistency indexes with the four main four evaluation charts, achieving a high match with the graph from Thewes and Burger (2004), which was not observed with the other charts.

These direct measurement methods to evaluate clogging potential did not succeed, yet, to be directly validated with real tunnel drives, or to have a good correlation with the clogging diagram from Hollman and Thewes (2012), the diagram currently applied in the tunnel industry. These methods did not consider the changes in the clay fraction, failing to evaluate mixed soils.

3.3 Materials and Methods

The tested samples were soils that were reconstituted in a laboratory, so that all of the variables could be fully controlled. Although that natural soils differ from reconstituted laboratory samples, the intention here was to validate methodologies under a controlled scenario, avoiding the local variability found in natural materials.

It was decided to focus on two different clay minerals, which would represent the two plasticity extremes: Na-bentonite (highly plastic) and kaolinite (very low plasticity). A larger number of tests were conducted with kaolinite, due to the ease and stability of dealing with this material.

Furthermore, several tests were conducted with mica flakes. The initial intention was to recreate a gneissic residual soil from the São Paulo Basin in Brazil, which is a typical tropical ground that presented numerous issues while being excavated by an EPBM (Comulada et al. 2016, Peixoto da Silva et al. 2017). Kaolinite is a typical clay mineral for tropical residual ground. Mica is also an important component, leading to the inclusion of a short chapter on this

research. Therefore, another reason that kaolinite was selected as the main tested clay mineral was so that certain correlations between tropical residual ground and clogging occurrence could later be drawn.

The HOBART mixer testing methodology was continually modified throughout the research, until it was acknowledged that an additional stage was required, as detailed in item 3.4.2, leading to the assemblage of a new device, namely the *Adhäsive Tone Untersuchung* RUB/Queens – RUB/Queens Clay Adhesion Tool (ATUR). The description illustrated in item 3.3.3 is the final methodological sequence, which achieved superior results overall; however, modification of this methodology is still under investigation and improvement.

3.3.1 Reconstituted Soils

Table 3-1 presents the main characteristics of each tested soil component, including the types of kaolinite (always cited as K) and bentonite (always cited as B). Figure 3-1 illustrates the grain size curve for the sand components used in this research. There were three types of grain size distribution for the sand fraction: “Sfine”, “Fine-Medium Sand” and “12”. All three of them are quite similar in terms of grain size distribution. Table 3-2 presents a summary of all the tested artificial soils, including their Atterberg limit values.

Table 3-1 Main characteristics of each component of constituted soils.

Product	Supplier	Spec. grav.	Chemical comp.	Specific Surface (m ² /g)	Cation exc. capacity (meq./100 g)	pH	Moisture (%)
Silica sand ^{*1}	Bell & Mackenzie	2.65	SiO ₂	-	-	6-7	-
Kaolinite - EPK Kaolin ^{*1}	Edgar Minerals	2.65	Al ₂ O ₃ , 2SiO ₂ , 2H ₂ O	28.52	4.5	5.8	Max. 1
Na-Bentonite - Volclay® 325 mesh ^{*1}	American Colloid Company	2.35	(Na,Ca)0.33 (Al1.67Mg0.33) Si ₄ O ₁₀ (OH) ₂ nH ₂ O	0.09-1.8	>70	8.5-10.5	Max. 12
Phlogopite Mica PD900 ^{*1}	LKAB	2.8	KMg ₃ (AlSi ₃ O ₁₀) (OH,F) ₂	0.3	No appreciable CEC	9	Max. 1
Silica sand ^{*2}	Euroquarz	2.65	SiO ₂	-	-	6.5-7	-
Kaolinite GHK KS 80 ^{*2}	Georg H. Luh GMBH	2.62	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃	18.2	-	4.5-9.5	0.6-2
Active Na-Bentonite IBECO B1 ^{*2}	Imerys Civil Engineering	2.65	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃ , MgO, CaO, K ₂ O, Na ₂ O	600-800	70±10	10	11±3

*1 – Initial tests presented in Item 3.4.1; *2 – Final tests presented in Item 3.4.2

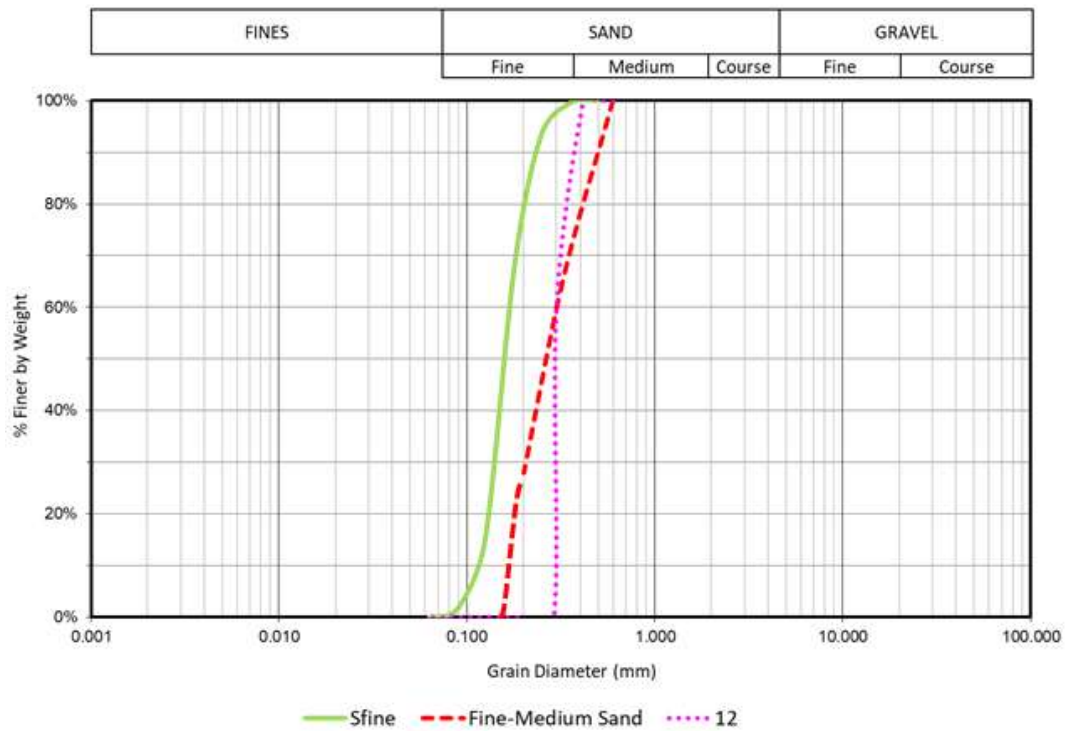


Figure 3-1. Grain size for sand portion of the soil mixture presented in Table 3-2.

Table 3-2. Tested samples with respective proportions of minerals and Atterberg limits.

Sample ID	Sand type (Fig. 1)	Sand (%)	Kaolinite (%)	Bentonite (%)	Mica (%)	W _L (%)	W _P (%)	IP (%)
FMK20	Fine-medium	80	20	-	-	24	15	9
FMK25	Fine-medium	75	25	-	-	26	16	10
FMK35	Fine-medium	65	35	-	-	29	18	11
FMK40	Fine-medium	60	40	-	-	32	19	13
12K3	12	70	30	-	-	22	13	9
12B1	12	80	-	20	-	90	24	65
12B2	12	60	-	40	-	194	36	158
12KM1	12	25	60	-	15	41	24	17
12KM2	12	10	60	-	30	47	31	16
Ksfine30	Sfine	70	30	-	-	22	16	6
Ksfine50	Sfine	50	50	-	-	32	21	11
Ksfine70	Sfine	30	70	-	-	44	29	16
K100	Sfine	0	100	-	-	62	40	22

3.3.2 HOBART Mixer Methodology – Original Version

The tests for evaluating the clogging potential were conducted using a HOBART mixer with a 5-L capacity, as shown in Figure 3-2, together with a B-flat beater, as suggested in Zumsteg and Puzrin (2012) and Zumsteg et al. (2016).



Figure 3-2. HOBART N50 mixer (around 40 cm height) and B-flat beater (around 20 cm height).

For each sample, the soil component was initially weighed with a 0.1-g precision scale, and subsequently mixed according to the proportions defined in Table 3-2. Deionised water was added to achieve the initial water content, which was defined for each sample based on its consistency, usually starting with consistencies no higher than 1 to avoid damaging the machine

with hard soils. There was a total waiting period of 24 h prior to starting the tests, to allow time for the clay minerals to react, or swell, enabling improved moisture homogenization. The material was maintained in a closed container to avoid moisture loss.

In total, six to twelve tests were carried out for each sample, with the water content being increased every time until the consistency reached a level close to zero. For each increase in moisture content in the same mixture combination, the B-flat beater and total soil sample were weighed, providing the parameter λ_0 .

Following each test, for every water content increase, the moisture content was verified following the methodology defined by ASTM D2216 (2010), thereby verifying whether the mixture was moisturized homogenously. The moisture contents of samples, as measured from the beater and mixing bowl, were compared. As these samples often did not exhibit equal moisture contents, the original methodology had to be modified; the speed of the mixer was increased from 139 rpm (speed level 1) to 285 rpm (speed level 2) for at least a couple of seconds until the mixture was homogenized, and then returned to speed level 1 for 3 min.

3.3.3 Combined Methodology – HOBART and ATUR

Following several tests, as originally conceived by Zumsteg and Puzrin (2012), it was realized that for certain consistencies, any vibration or drop in the B-flat beater would be sufficient to totally or partially remove the soil stuck in the beater, as illustrated in the results of item 3.4.1. Therefore, a trial was initiated to determine different means of dropping the beater from a certain height. Initially, this was achieved by dropping the beater firstly on top of a plastic bowl, and then on top of a Casagrande cup base, which has a standardized material and size specification (ASTM D4318 2017), leading to the first results demonstrated in item 3.4.1.

Every time the beater hit the surface, the tool was weighed again, initially resulting in three additional measures of the empirical clogging evaluation parameter λ , denoted by λ_1 , λ_2 and

λ_3 , representing the values achieved after 1, 2 and 3 blows, respectively. The impact that the beater with the soil suffered by dropping the beater into a plastic bowl was lower than when dropping it onto the Casagrande cup base; therefore, due to the lack of a steady standard, the preliminary results are merely illustrative.

Finally, a device was assembled to provide the B-flat beater with a free fall from the same height (37 cm) along the same axis. Figure 3-3 illustrates this new assembled device, the denominated ATUR, which stands for Adhäsive Tone Untersuchung RUB-Queens (RUB-Queens Clay Adhesion Tool), from research cooperation between Queen's University in Canada and Ruhr Universität Bochum in Germany.

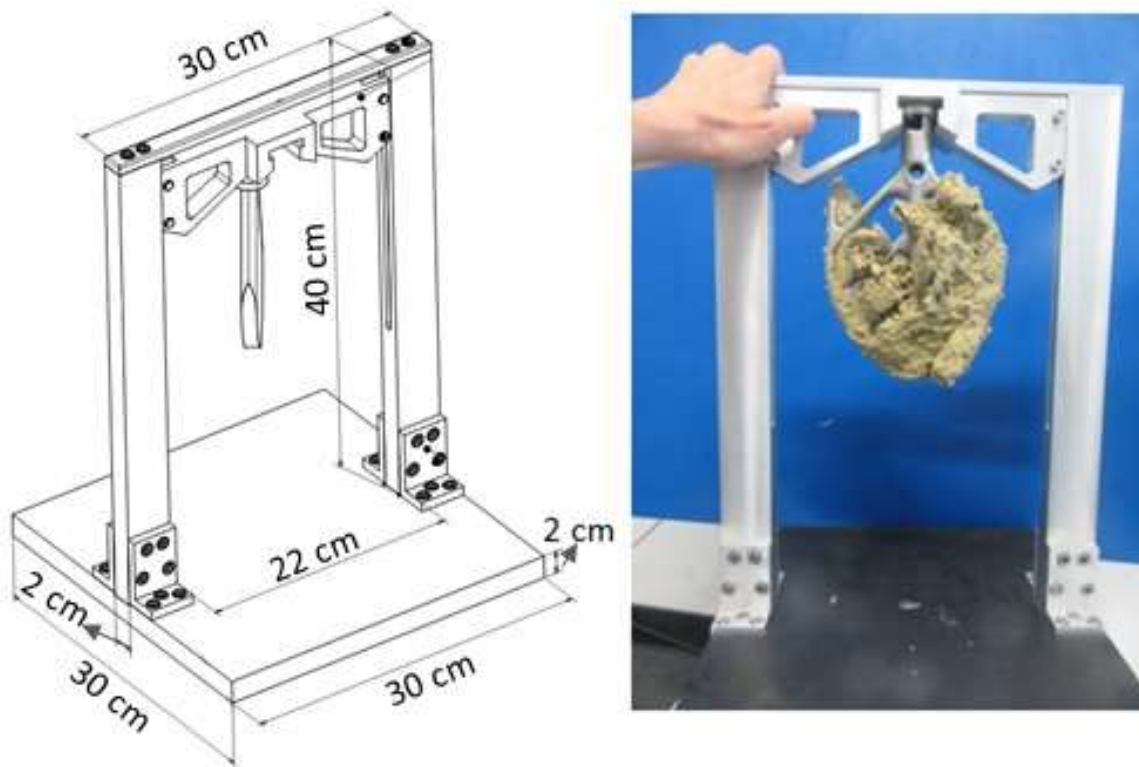


Figure 3-3. ATUR device: drawing with dimensions on the left and B-Flat beater with soil stuck to it on the right. The device has a height of 40 cm, as showing in the drawing.

Later, a calibration was conducted considering mainly pure kaolinite samples for different consistency indexes, as shown in item 3.4.2. The final methodology sequence is schematically presented in Figure 3-4. The sequence may undergo certain changes in the future, especially considering that a total of seven drops (λ_7) was necessary to achieve similar results to three drops (λ_3) when using the Casagrande cup base, which has a higher material stiffness.

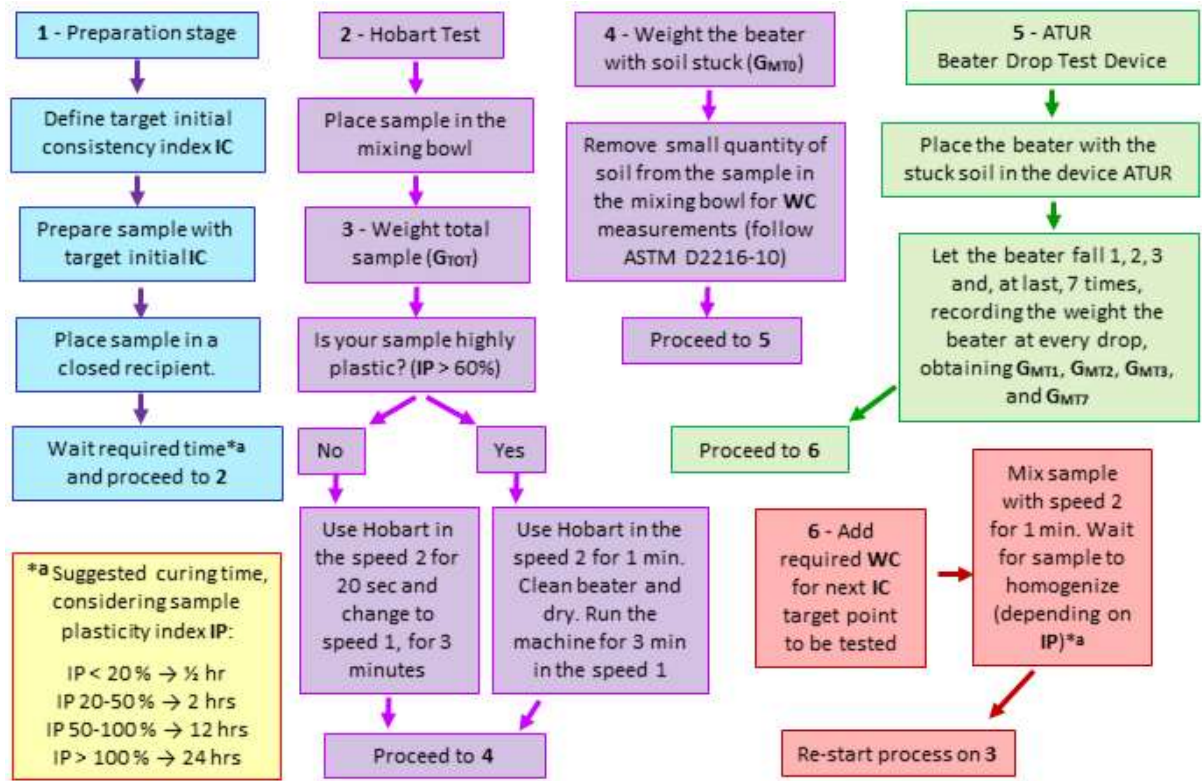


Figure 3-4. Schematic sequence for combined methodology for clogging evaluation with HOBART and ATUR, considering the following parameters: I_C (consistency index); IP (plasticity index); WC (water content); and the clogging evaluation parameters: G_{TOT} (total mass of sample); G_{MT0} (total mass of soil stuck in the mixing tool); $G_{MT1,2,3,7}$ (mass of soil after dropping the beater 1, 2, 3 and 7 times, respectively). The curing time is considered the time that the sample is left to rest in a closed container, to achieve moisture homogenisation.

3.4 Results and Discussion

3.4.1 Initial Results – Preliminary Phase

Following the HOBART testing methodology, a total of 35 samples were tested during the preliminary testing phase. Firstly, the results for the mixtures with a fine-medium sand mixed with kaolinite are illustrated, compared to the original method of simply weighing the B-flat beater, using the initial dropping version without a reliable method for dropping the beater. Figure 3-5 presents the results of the empirical clogging parameter λ for samples with increasing clay contents. The EPB consistency range (between 0.4 and 0.75) is illustrated with a grey square in the images.

It should be noted that the λ values tend to increase with increasing consistency beyond 0.8, when the clay fraction increases from 20 to 25%; these values continue to increase as the clay content increases. For these initial tests, although the new dropping adaptation was still crude, completely distinct λ_0 and λ_2 values were reported at the 25% and 35% clay content levels along the EPB consistency range.

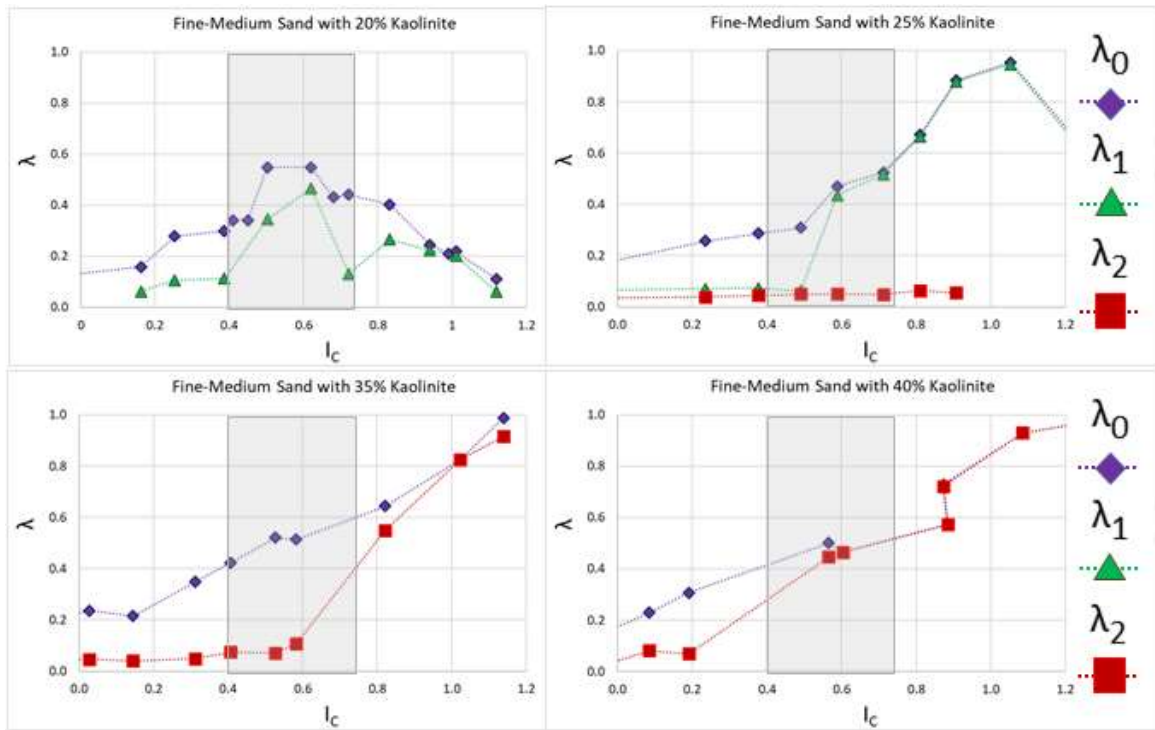


Figure 3-5. Fine-medium sand mixed with 20%, 25%, 35% and 45% kaolinite. In the x-axis is plotted the consistency index of the tested samples and in the y-axis, the clogging parameter λ .

Figure 3-6 presents the results of bentonite clay with sand (grain size denominated 12 in Figure 3-1), comparing two samples containing 20% and 40% bentonite, respectively, each mixed with sand of sieve number 50. Following several drops, the soil was totally detached from the beater for consistencies higher than 0.8, which is similar to the results obtained for samples containing kaolinite. This observation was more pronounced for the sample containing 20% bentonite than for that with 40% bentonite.

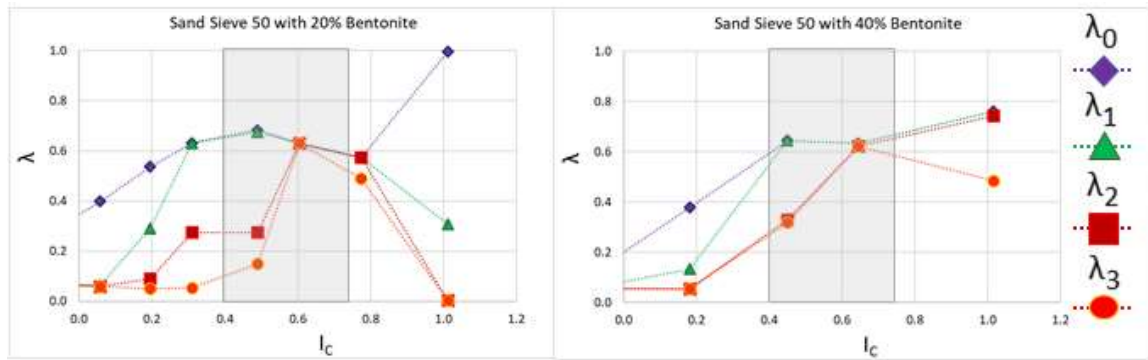


Figure 3-6. Comparison of samples 12B_1 and 12B_2 with sand sieve 50 and bentonite.

Following several tests, particularly for low plasticity samples with a relevant sand contribution, the majority of the soil stuck in the beater would be easily removed with any vibration or mixing tool impact. Likewise, for certain consistency ranges and soil combinations, most of the soil would remain stuck, even once the beater was abruptly jolted. Figure 3-7 illustrates four cases for two different clay mineral samples.

On the top left is an example of a sample with 30% kaolinite mixed with fine sand and on the top right, a mixture of 20% bentonite with fine sand, both with a consistency index of 1 (equal to the plastic limit). In this case, without any beater impact, the sample would be classified as high clogging ($\lambda=0.9$ and $\lambda=1.0$ for the kaolinite and bentonite samples, respectively). However, after always dropping the sample from the same height, once (λ_1), twice (λ_2), or even three times (λ_3), there was a significant decrease in the soil stuck in the beater, leading to an absence of clogging. Nevertheless, for lower consistencies, as seen in the bottom of the figure, a significant amount of soil was present in the beater, even after several jolting of the mixing tool.

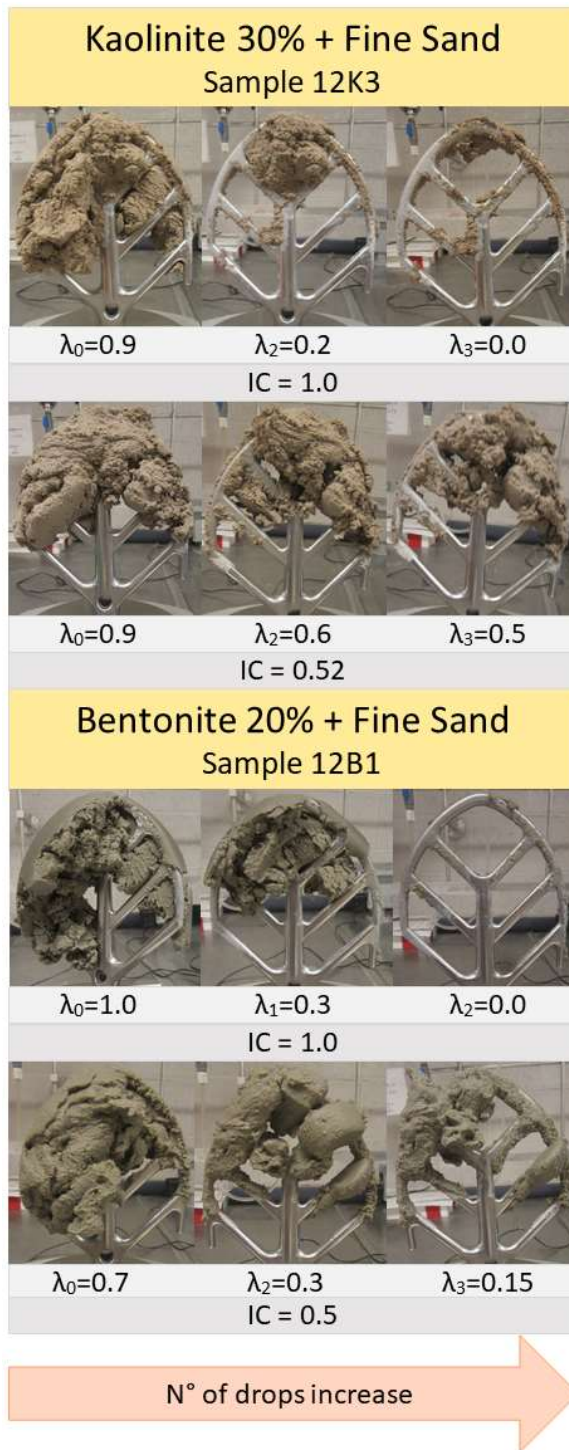


Figure 3-7. Two tested samples with distinct consistencies (I_C): on top, a kaolinite mixture and at the bottom, a bentonite one. The yellow arrow indicates the direction that increases the number of drops of the beater, providing different values for λ (λ_0 = zero drops; λ_1 = one drop, and so on).

This same behaviour was observed during several tests: material was frequently being removed from the beater with the extra blow stage for stiff to very stiff soils higher than the EPB consistency range, and for very soft soils lower than the EPB consistency range. In these cases, the samples did not exhibit the same high clogging results from the original λ value. Mostly, for both clay cases of clay, kaolinite and bentonite, it was observed that, within the ranges where the adhesion of the clay to the metal surface was significant, the beater dropping did not really affect the value of parameter λ . This observation occurred mainly for soft soils with consistency values of 0.5 and 0.8, characterised by the diagram of Hollman and Thewes (2012) as a strong clogging area.

For stiffer consistencies, detachment occurred in lumps, while for softer consistencies, it occurred as a sticky and liquid paste, illustrating the strong connection between the consistency indexes and clogging tendency. Likewise, it is demonstrated that this original methodology may exhibit a superior fit for samples with higher clay contents or pure clay materials, and for a certain consistency range, mainly between 0.5 and 0.8. As the consistency values increase or decrease beyond this range, the additional stage of dropping the beater could provide more accurate results for the clogging potential, compared to the graph from Hollman and Thewes (2012).

Figure 3-8 presents all the samples plotted in the graph of Hollman and Thewes (2012). In the top graph, the samples have not received any jolting, and on the bottom are plotted the results after three beater drops from the same height. There is improved fitting between the graphed clogging potential ranges and the empirical parameter λ_3 , using three drops of the beater, compared to the sampling of stuck soil without applying any kinetic energy (λ_0). However, too many discrepancies can be detected and the lack of a solid procedure for dropping the beater certainly influenced the results.

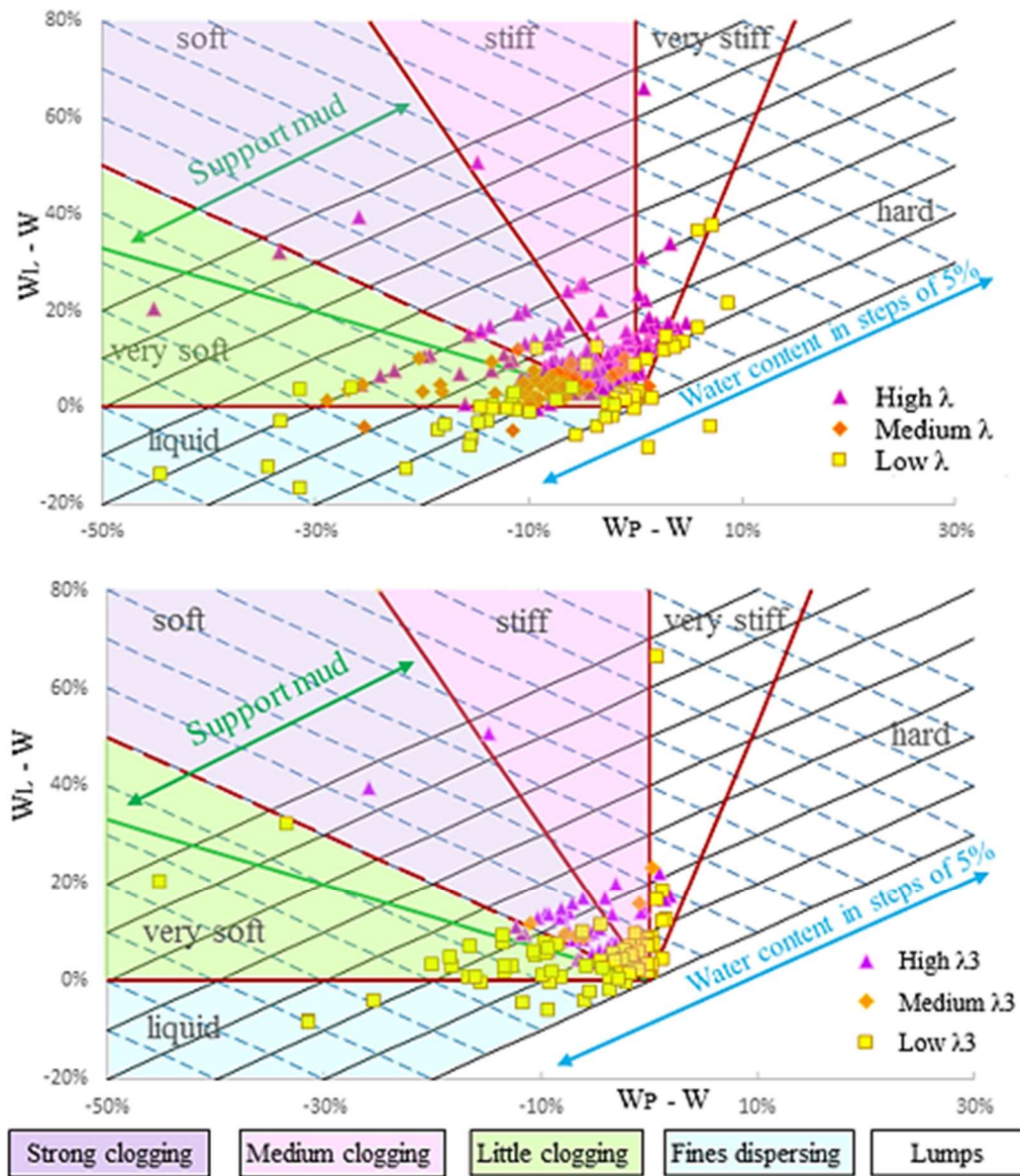


Figure 3-8. Results from preliminary testing phase plotted in the clogging potential universal graph (modified from Hollman and Thewes 2012). The top chart represents the samples without any beater dropping, as originally defined by Zumsteg and Puzrin (2012), while the bottom chart presents the modified version after three beater drops.

Furthermore, several tests were performed including mica flakes, as previously mentioned, in order to represent the typical residual soil originating from a gneissic rock, which is rich in this mineral. The results are presented in Figure 3-9, and the λ values are quite similar at

consistency values below 0.7. For higher mica flake contents, λ decreases slightly at consistency values above 0.7. The values of λ_1 , λ_2 and λ_3 decrease more evidently with higher mica flake contents.

Presumably, the mica flakes functioned as a lubricant, detaching the material from the metal surface and thereby diminishing its adherence. This phenomenon was even more evident when the beater was dropped, which can be related to the low binding forces between the mica flake layers.

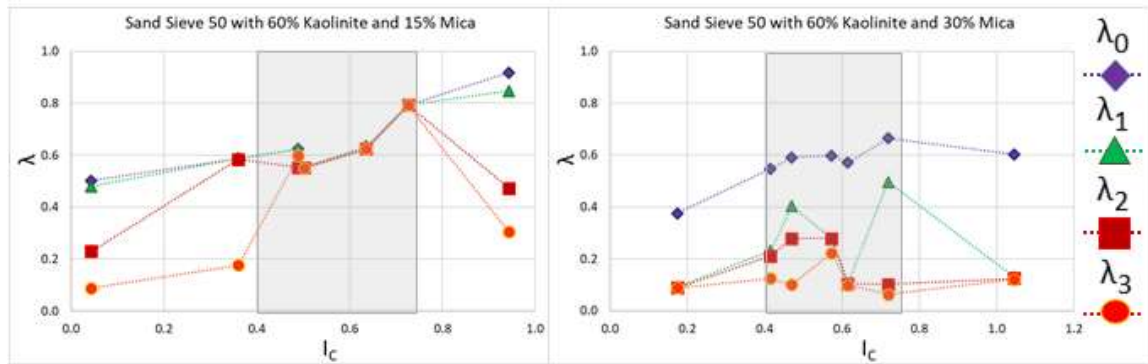


Figure 3-9. Comparison between samples 12KM_1 and 12KM_2, with fixed 60% kaolinite, and 15% and 30% mica flakes (phlogopite), respectively.

These results could even be compared to the EPBM excavation along a gneissic residual soil, such as that of the São Paulo Metro, Line 5, Lot 3, Brazil (Peixoto da Silva 2017 and personal communication). At this site, stretches with higher biotite content (originally from the dark gneissic bands) would not present clogging issues, even with the presence of clay minerals, while excavated sections with clay but without mica flakes (light bands from the gneissic parent rock) exhibited clogging. However, there was a higher incidence of face instability along portions with higher mica content, which was less apparent along stretches with less mica flakes and higher clay content.

3.4.2 Combined Methodology – Calibration and Initial Results

The following tests demonstrate the initial results obtained using the combined methodology of the HOBART mixer together with the dropping stage conducted with the new device, ATUR. The concept for the dropping stage is schematized in Figure 3-10. A clayey material would possess adhesion stress, maintaining the sample attached to the metal surface (top of figure) along a certain surface area, which would depend on certain variables, such as consistency and adhesion, as already investigated by numerous authors (Thewes 1999, Feinendegen et al. 2011, Zumsteg 2014, Basmenj et al. 2016, Khabbazi et al. 2017).

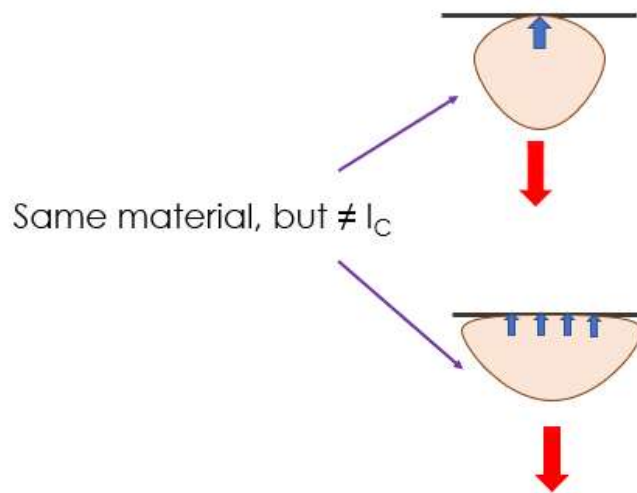


Figure 3-10. Scheme of drop stage added to HOBART mixing methodology. A soil sample (represented by the shape in pink colour) with two different consistency indexes exerts a certain adhesion force (represented by the blue arrow), which attaches this sample to a metal surface (black line). Depending on the clogging potential and responding to a counterbalance kinetic energy (red arrow), this soil may or may not detach, separating from the metal surface.

If a certain counterbalance of kinetic energy is exerted, as in the case of the laboratory tests, the drop of the beater, or in a real EPB tunnel drive, the machine cutterhead rotation and soil flow inside the excavation chamber (therefore, not static), the clogged material would either

remain stuck or not, depending on its clogging potential. This can be verified during laboratory tests, and later correlated with the empirical chart from Hollman and Thewes (2012), which is based on real tunnel drive data.

When clogging is pronounced, the counterbalance of kinetic energy (red arrow in Figure 3-10) would not be sufficient to detach the sample from the surface. Ideally, this counterbalance kinetic energy should be calibrated with a large amount of tunnel data, which could undoubtedly increase this calibration refinement.

Thereafter, tests were conducted using the combined methodology, initially only dropping the beater up to three times, with the results illustrated in Figure 3-11. The samples were mixed with very fine sand, denoted as “sfine” in Figure 3-11. From these three graphs (30, 50 and 70% kaolinite) it can be observed that at higher ($I_C > 0.75$) and lower ($I_C < 0.4$) consistencies, the differences between λ_0 and λ_3 are greater, precisely because clogging should not be expected at these consistencies. This demonstrates that the empirical stickiness parameter λ_0 does not reflect the expected clogging potential.

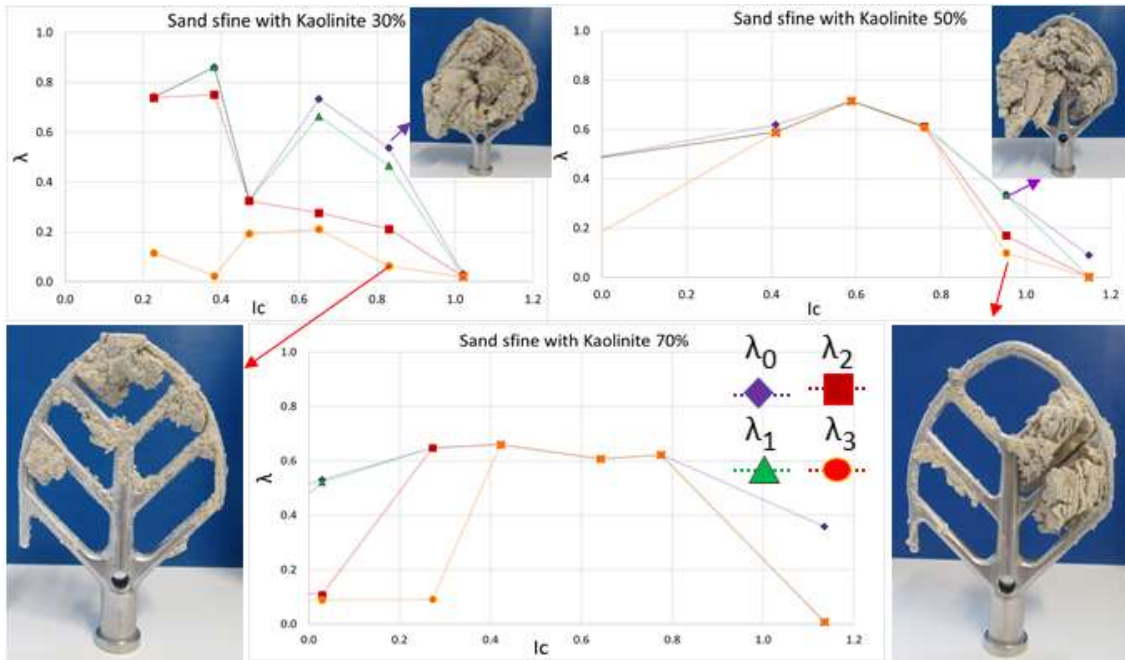


Figure 3-11. Results from three different samples, containing 30%, 50% and 70% kaolinite, showing four different empirical stickiness values of λ .

The final stage involved calibrating the number of drops of the beater with the ATUR device. Pure kaolinite soils were tested, and the results were crossed with the clogging potential, as defined in the universal chart of Hollman and Thewes (2012). The target consistency values were defined considering the limits among distinct clogging potentials for this chart (Figure 3-12). The results are displayed in Table 3-3, where the cells are coloured according to the clogging potential (yellow – low; orange – medium; red – high). The limits defined for these ranges were adjusted to better fit the ranges from the graph of Hollman and Thewes (2012), with λ_x values (x representing any number of drops, including none as 0): < 0.2 = little clogging potential; between 0.20 and 0.46 = medium clogging potential and higher than 0.46 = strong clogging potential. These values differ slightly from those defined by Zumsteg and Puzrin (2012).

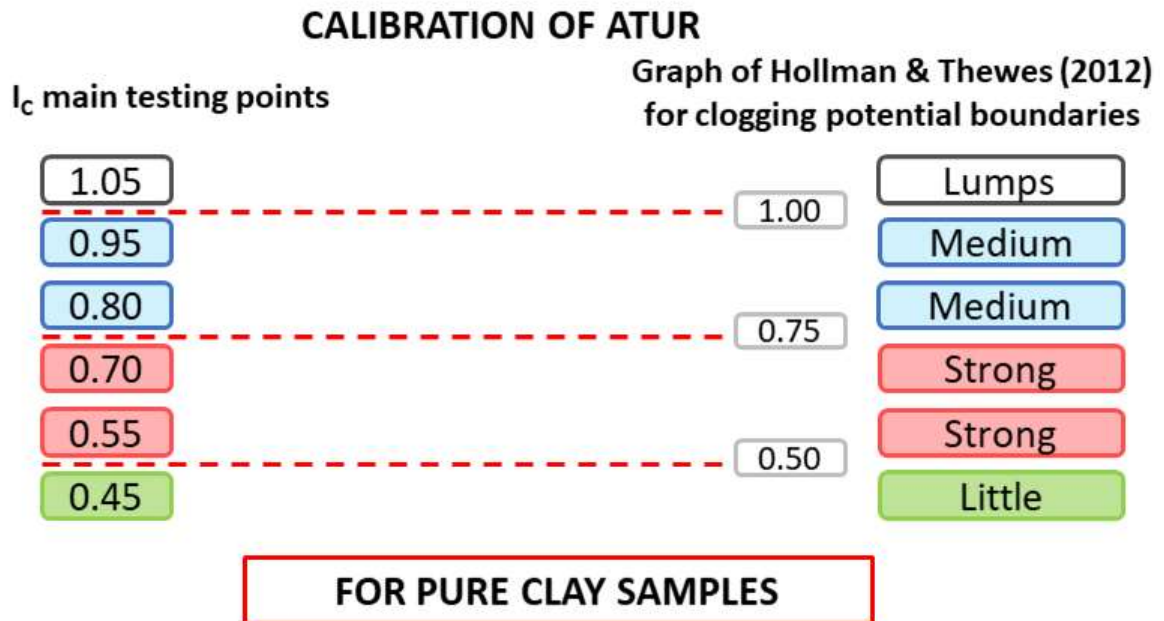


Figure 3-12. Main testing target points for the initial ATUR calibration, considering clogging potential boundaries as defined by chart of Hollman and Thewes 2012, with limits presented on the right.

Table 3-3. Results for initial calibration of HOBART + ATUR values compared with clogging potential ranges of Hollman and Thewes 2012 (indicated as H&T in the table). The colours in the cells are as follows: red for high clogging potential, orange for medium clogging potential, yellow for little clogging potential and white for occurrence of lumps.

TEST ID	TEST N°	Water content	IC	λ_0	λ_1	λ_2	λ_3	λ_7	H&T
Kpure_T2	1	40.5	1.0	0.62	0.62	0.62	0.62	0.06	Lumps
	2	42.2	0.9	0.65	0.65	0.65	0.39	0.39	Medium
	3	47.2	0.7	0.60	0.60	0.60	0.60	0.60	Strong
	4	49.8	0.6	0.56	0.56	0.56	0.56	0.56	Strong
	5	54.1	0.4	0.64	0.64	0.64	0.29	0.09	Little
	6	56.8	0.2	0.54	0.54	0.53	0.30	0.13	Little
Kpure_T3	1	40.2	1.0	0.62	0.62	0.62	0.27	0.08	Lumps
	2	42.4	0.9	0.56	0.56	0.56	0.56	0.41	Medium
	3	45.9	0.7	0.67	0.67	0.53	0.53	0.48	Strong
	4	48.8	0.6	0.60	0.60	0.60	0.60	0.37	Strong
	5	51.8	0.5	0.64	0.64	0.64	0.64	0.64	Strong
	6	53.3	0.4	0.66	0.66	0.65	0.65	0.13	Little
Kpure_T4	1	39.8	1.0	0.63	0.62	0.61	0.58	0.07	Lumps
	2	41.5	0.9	0.59	0.59	0.59	0.59	0.56	Medium
	3	44.8	0.8	0.67	0.67	0.67	0.67	0.62	Medium
	4	47.1	0.7	0.66	0.66	0.66	0.66	0.66	Strong
	5	50.6	0.5	0.71	0.71	0.71	0.71	0.62	Strong
	6	53.5	0.4	0.61	0.61	0.61	0.60	0.10	Little
	7	61.6	0.0	0.53	0.52	0.16	0.12	0.06	Little
Kpure_T5	1	39.9	1.0	0.64	0.64	0.64	0.64	0.09	Lumps
	2	41.3	0.9	0.67	0.64	0.61	0.58	0.46	Medium
	3	45.1	0.8	0.60	0.60	0.60	0.60	0.58	Medium
	4	47.8	0.6	0.66	0.66	0.66	0.66	0.66	Strong
	5	50.9	0.5	0.72	0.72	0.72	0.72	0.35	Strong
	6	53.9	0.4	0.68	0.68	0.67	0.67	0.09	Little
	7	61.1	0.0	0.52	0.51	0.22	0.13	0.05	Little

As can be observed from Table 3-3, following seven beater drops using the ATUR device, it was possible to obtain very close results to those defined by Hollman and Thewes (2012) with the universal chart. If the results without any beater dropping are considered (λ_0) for all the tested consistency indexes, the clogging would be high, and the results would not correspond with those expected from this chart. Furthermore, kaolinite is a very low plasticity clay and should not be expected to exhibit such extensive clogging for this wide range of consistency values. Thus, these values certainly need to be adjusted with additional obtained data and crossed with tunnel drives.

Certain values of λ_7 (marked in bold and italics in the table) still do not entirely fit the chart of Hollman and Thewes (2012); nonetheless, the results should be closer to the reality of clogging occurrence in EPBMs than those of the original empirical stickiness evaluation using the HOBART machine alone, as proposed by Zumsteg and Puzrin (2012).

At present, the addition of an extra final stage is also under consideration, characterising the ease of removing the soil from the beater and mixing bowl. Although this would provide an imprecise parameter, it may aid in distinguishing between low potential clogging and a lump, as in the second case where the soil is easily detached from both the beater and bowl by hands and with the use of water only, while the first option, namely a large amount of soil, sticks to the fingers, and soap is required for the cleaning process. Furthermore, this could later be correlated with the ease/difficulty of realising interventions, as a soil that is challenging to clean and remove from the metal parts of the machine would imply lengthier interventions.

3.5 Conclusions

The process of clogging during EPBM excavation of clay soils is complex and influenced by several variables. These variables are not only directly connected to the soil or underground water properties, but also interrelated with the conditioning additives used during the process, the machine itself, including the design of the cutterhead openings, positioning and shape of excavation tools, locations of foam injectors, foam generator, machine torque, and other details, with the further addition of high temperatures and pressure. It is a challenging task to combine all these variables into a holistic model. With limited work previously conducted in this research area particularly that can be directly applied to EPB operations, issues of clogging tend to lead logically to a trial-and-error approach during actual tunnel excavation.

A preliminary investigation, consisting mainly of a preliminary clogging assessment prior to excavation, may offer qualitative characterisation of the soil and its expected behaviour when

properly conditioned, which could possibly mitigate future clogging issues. Complex and expensive laboratory methods or devices are undoubtedly necessary for understanding the clogging process, particularly concerning soil adhesion. However, this combined methodology offers as a main advantage practicability and ease, with the aim of reaching a higher number of researchers as well as tunnel professionals, thereby providing the ability to assess clogging potential easily, and enable comparison with other data samples.

It has been demonstrated that data produced using the HOBART mixer on its own does not reflect the clogging potential for certain consistency ranges or lower clay contents in the case of mixed clay-sand soils. The opposite effect was illustrated using the combined methodology with the addition of the extra dropping stage, which can achieve satisfactory results.

Undoubtedly, the greatest advantage of this methodology is the very simple and practical procedure involved, along with the fact that a HOBART mixer is a device that can easily be acquired worldwide.

Furthermore, the combined methodology provides the possibility of evaluating the efficiency of soil conditioners, such as foam and polymers. In this study, the authors evaluated the effects of mica flakes in the soil, which were considered as part of the soil. It could even be speculated that for scenarios with high mica contents, as in the example of excavation along a weathering ground product of a mica-schist, clogging issues are not expected to occur, even with high clay contents. Moreover, the results could indicate that by adding mica flakes in the front of the machine, if that were physically and economically feasible, could imply a decrease in clogging issues.

Device calibration can be performed with pure clay materials, changing their consistency values as demonstrated in section 3.4.2. Later, a standardised base can be defined, as was carried out with the Casagrande cup base (ASTM D4318 2017). With the continuation of this research

and further calibration with real tunnel drives, it is expected that a reasonable and simple clogging assessment methodology can be achieved, which can be correlated to other proposed clogging evaluation methods.

3.6 Acknowledgments

Support for this research was received from the Natural Science and Engineering Research Council (NSERC), Canada and funding from Queen's University. The authors would like to acknowledge the support of the Collaborative Research Center SFB 837, "Interaction Modelling in Mechanised Tunnelling", funded by the German Research Foundation (DFG). We are also grateful to the technical staff at Ellis Hall, from the Geotechnical Laboratory of the Civil Department at Queen's University, for providing laboratory space, and to the laboratory team at TLB, at Ruhr University Bochum. Acknowledgments are also extended to the LKAB minerals, represented by Mr. Magnus Johansson, who kindly provided the phlogopite flakes. Special thanks to Mrs. Yvonne Ueberholz from Ruhr University Bochum, for the device assemblage, and Dr. Mario Galli from PORR Deutschland GmbH, for assistance with the scientific device name.

Chapter 4

Consistency index and its correlation with EPB excavation of mixed clay-sand soils

Abstract

The behavioural properties of excavated ground have significant influence on the excavation process performed by an Earth Pressure Balance Machine (EPBM), as they are among the main factors responsible for maintaining the pressure ahead of the face, which affects face stability. Therefore, understanding the characteristics of the excavated material along with its flow behaviour is essential for a successful EPB tunnel drive. In scenarios involving the excavation of fine-grained soils containing clay minerals, the consistency index has been widely used as a guideline to define the ideal state of the excavated material. However, there are certain restrictions for the use of this index, the first of which are the Atterberg limits. These limits become more restrictive when mixed soils are involved. This study presents a brief review of the application of the consistency index and Atterberg limits in order to predict the performance of an EPB excavation. This study presents the results of a laboratory testing campaign with artificially mixed clay-sand soils by using a flow table as a preliminary flow assessment of cohesive soils.

4.1 Introduction

The Earth Pressure Balance Machine (EPBM) was first commissioned by the Japanese in the early 1970s (Maidl et al. 2012) and is currently the most frequently applied tunnel boring machine in soft ground (Herrenknecht et al. 2011). An EPBM is a closed shield used for the excavation of soft ground where face support and groundwater pressure control are obtained by means of the material excavated by the cutting wheel, which serves as a support medium itself. To obtain an ideal paste of excavated material, it may be necessary to add water or other additives (foam, polymers, and slurries with fines) to bring the mixture, also called “muck”, to a satisfactory workability and permeability. This muck should maintain the required face pressure, as well as other excavation requirements, such as its extraction and transport. This process is called soil conditioning, and it is considered to be one of the most important parts of the EPB excavation process (Maidl 1995, Langmaack 2000, EFNARC 2005, Thewes et al. 2012).

In cases involving soils with fine particles, including clay minerals, muck workability is addressed in terms of consistency through a correlation between the plasticity index and the water content. Maidl (1995) stated that in an EPB excavation, whenever face support is required (unstable ground; groundwater occurrence), the consistency of the soil should range from pasty to soft (consistency should be between 0.4 and 0.75), and the soil should feature a low permeability (Herrenknecht et al. 2011, Galli and Thewes 2014, Galli 2016).

In terms of an EPB excavation, concerns related to the state of the muck go not only to the excavation process, but also to its removal and disposal. Liquid muck would produce complications regarding its transportation along conveyor belts or muck cars, which would cause a significant increase in the extracted volume and later disposal (Herrenknecht et al. 2011, Maidl et al. 2012). Consequently, the “ideal” characteristics of this excavated and conditioned material

is a balance between all the parts that this element plays in the EPB operation: excavation, support medium, removal, and later disposal.

Several authors (Casagrande 1958, Sivapullaiah & Sridharan 1985, Nagaraj et al. 2012, Claveau-Mallet et al. 2012, Mishra et al. 2012, Haigh et al. 2013, Haigh 2016) mentioned the numerous limitations of the Atterberg limits. Considering that the consistency index is based on these limits, their limitations would certainly have an influence on the delimitation of the consistencies of the “ideal muck”, or any other parameter that would consider the Atterberg limits. Those limitations are mentioned in this study, in addition to a description of how this standard method would not always provide a realistic consistency index.

This is accomplished by providing several examples of laboratory tests with artificially mixed clay-sand soils with three different clay minerals and different clay-sand proportions. The standard methodology of the Atterberg limits was modified including bigger grains, always comparing with the standardized results, proving that by changing the methodology more reasonable results could be obtained, which would represent closely the real characteristics of a mixed soil.

In addition, the flow of mixed clay-sand soils was analysed using a flow table originally designed for laboratory tests on concrete. This method of testing soil samples with soil conditioning agents was already proposed in EFNARC (2005), mainly to test the plasticising effect of certain conditions for non-cohesive soil. Here, the method has been modified to better suit cohesive soils. Several conclusions could be drawn concerning the flow behaviour of the tested soils. Mainly showing that cohesive soils holding the exact same consistency index, do not necessarily have similar flow behaviour.

4.2 Theoretical Considerations

In the first decade of the 20th century, Atterberg determined the amount of water content at which soil would change from a liquid to a plastic condition (liquid limit – W_L), as well as from a plastic condition to a solid condition (plastic limit – W_P) (Atterberg 1911), by studying different soil states. Later, Terzaghi (1926a) realized that understanding this soil property was essential to understanding the overall behaviour of the soil, and that a more efficient method should be developed to allow for reproducibility by other operators. Subsequently, Casagrande (1932) designed a more reliable method to define the plastic and liquid limits.

Casagrande (1958), after realizing issues with the standardization of his method around the world, suggested a standard specification for the Casagrande cup and grooving tool to measure the liquid limit, which is still used today and standardised by the ASTM D4318 (2017). Casagrande (1958) advised that his method for obtaining the liquid limit should be replaced by a more rational test that is based on the shear strength of the material. In some countries, this sort of test has already been implemented—it is known as the fall cone test, and it is used as an alternative method to attain the liquid limit (Houlsby 1982). However, Casagrande's method is still the most commonly applied test to obtain the Atterberg limits of soils, even after several attempts have been made to replace it (Sivapullaiah & Sridharan 1985, Mishra et al. 2012, Nagaraj et al. 2012, Haigh 2012, Haigh 2016).

Many authors have investigated the application of this method and pointed out its limitations for determining liquid and plastic limits; several examples of these limitations are listed as follows:

Liquid Limit

- The method has limited reproducibility for identical soil. There is a coefficient of variation (standard variation/mean) of up to 8% due to differences in the volume and

mass of the clay placed in the cup, the grooving tool used, the hardness of the Casagrande base, and the fall height adjustment (Clauveau-Mallet et al. 2012, Haigh 2012, Haigh 2016). Also, essential to mention, that even when presenting a good reproducibility, for same soils, variations in its coefficient may occur due to its natural variability.

- There are difficulties associated with utilizing this method within low-plasticity soils. The soil mass would rather slide rather than flow, which makes it difficult to cut a proper groove. (Sivapullaiah & Sridharan 1985, Mishra et al. 2012).
- Different results are obtained depending on the type of the standardised base used, either soft or hard, according to different specifications, which depends on the national design standard (British Standard, ASTM, French code) (Haigh 2016).
- The above result does not correlate with any change in the soil behaviour, and the results are more dependent on the device rather than on a physical characteristic of a certain soil state (Haigh 2016).
- The stiffness of the bench upon which the test is performed could induce different results (Casagrande 1958).

Plastic Limit

- The brittle failure that determines the plastic limit while rolling the thread is related to either air entry or cavitation in the clay when the water ceases to behave as a continuum and capillary suction starts to prevail; therefore, the point which this failure occurs is not related to a constant strength, as initially considered (Haigh et al. 2013).
- The test relies on the judgment of the operator, which would imply that different results could be obtained when different operators are involved (Sherwood 1970; Nagaraj et al. 2012; Haigh, Vardanega & Bolton 2013).

In addition to the aforementioned issues, it is important to note that these limits are traditionally defined only for fine-grained soils, the threshold of which is soil that can pass through a sieve number of 40. This means that only the portion of the soil containing grains below 0.425 mm (fine-sand) would be considered in the test. In the case of mixed soils, for a typical residual ground sample with a well-graded distribution, medium and coarse grains should be added to the evaluation. However, it is not well-defined how this should be accomplished.

There have been several attempts to define correlations concerning the influence of different proportions of clay fractions and clay minerals in the Atterberg limit values. Seed et al. (1964) verified that with clay content above 40%, there would be a linear relationship between the plasticity index and clay content. This was true for studies on kaolinite, illite, and montmorillonite. However, this was not always the case for scenarios involving lower clay content soil, and the liquid limit would be mainly influenced by its clay content (as clay grain size – grains smaller than 0.002 mm) and by the clay mineral.

After detailed investigations on bentonite and kaolinite mixed with different sizes and shapes of sand, Sivapullaiah & Sridharan (1985) concluded that the shape of the sand would not influence the liquid limit; however, this was not true for its grain size. They also suggested a procedure to obtain the liquid limit for low-plasticity soils by mixing them with soils of higher plasticity and assuming a linear relationship.

Polidori (2007) found a correlation between different clay fractions that applied mainly to platy clays and excluded non-platy clays (such as halloysite, allophane, and attapulgite mixed with sand) and its resulting limits, which, except for the materials with low clay content, would follow a linear relationship. A relationship between the plastic limit (W_P), liquid limit (W_L), and the clay fraction (CF) was defined by an empirical equation (Equation 4-1):

$$W_p = 0.04 W_{L(\%)} + (0.26 CF_{(\%)} + 10) \quad (4-1)$$

After an extended literature review, Nagaraj et al. (2012) concluded that even though the limits defined by Atterberg (1911) are essential for understanding the behaviour of fine-grained soils, it is not feasible to expect similar correlations for the undrained shear strength of different soil in a same state, as initially presumed by Casagrande (1932; 1958).

The consistency index (I_c) is derived from the Atterberg limits and indicates the firmness of soil and the changes in water content that allow it to vary from the following states: liquid, very soft, soft, stiff, very stiff, and hard (Terzaghi 1926b). At a consistency index of zero (0), soil is equivalent to its liquid limit, and at a consistency index of one (1), soil is equivalent to the plastic limit. The consistency index can be calculated with the following equation (Equation 4-2):

$$I_c = (W_L - W)/(W_L - W_p) \quad (4-2)$$

Where W_L refers to the liquid limit, W refers to the water content, and W_p refers to the plastic limit.

In the case of EPB excavation and soil conditioning, the consistency index of both the soil and the conditioned material is an important parameter. This consistency parameter has been used to define the ideal state of a cohesive material to function as a suitable support medium. The consistency index is also applied as one of the main parameters for the evaluation of clogging, which occurs when clayey soils stick to the metallic parts of the machine (Thewes 1999, Thewes 2004, Hollman & Thewes 2012, 2013).

Considering the consistency index of a soil along with its water content, Hollman & Thewes (2012) defined an empirical evaluation for the occurrence of clogging, which led to the creation of a clogging chart, which is shown in Figure 4-1.

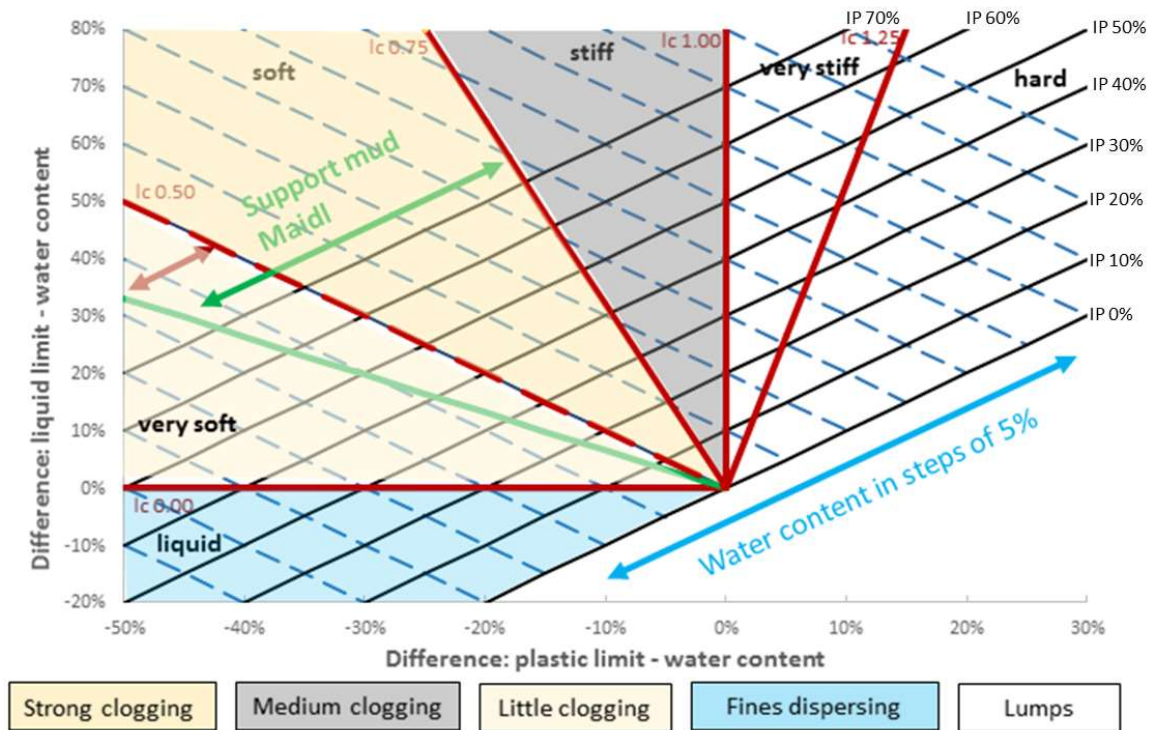


Figure 4-1. Evaluation diagram for clogging potential, taking into consideration the consistency index of the soil and changes in the water content (Hollman and Thewes 2012).

The green double arrow indicates the ideal consistency range of the support medium of the material inside the excavation chamber of an EPB, as defined by Maidl (1995), and the brown double arrow indicates the partial area that would be less vulnerable to clogging. Therefore, both the Atterberg limits and the consistency index are substantial parameters for the EPB excavation and soil conditioning. If there are limitations to the Atterberg limits, they are going to be extended to the consistency index.

4.3 Methodology

All the tested soils were assembled in the laboratory by mixing quartz sand and three different clay mixtures. A total of 71 soil mixtures were reconstituted for this study, and they contain different grain sizes and proportions of sand with bentonite and kaolinite clays. A clay mixture called Friedland, which contains montmorillonite and illite, along with other minerals as specified in Appendix B, are included in these 71 soil mixtures. For all of the mixtures, the Atterberg limits were obtained and 153 tests on the flow table were performed. Details for all of the mixtures are included in Appendix B, including their Atterberg limits and the measured and calculated plastic limits. The calculated limits were obtained by using an equation from Polidori (2007) for comparison purposes.

The samples were assembled using six different grades of quartz sand, as shown in Figure 4-2. All of the samples containing grains that were coarser than that recommended by the Atterberg limits standard, ASTM D4318 (2017), even when retained in the 40-mesh sieve (medium and coarse sand, grains bigger than 0.425 mm), therefore, the whole material, were included in the tests. Aside from this modification, all of the remaining Atterberg limit testing followed the recommendations of ASTM D4318 (2017). The method used to determine the liquid limit was multipoint method A, where 3–4 different points were obtained with the Casagrande cup.

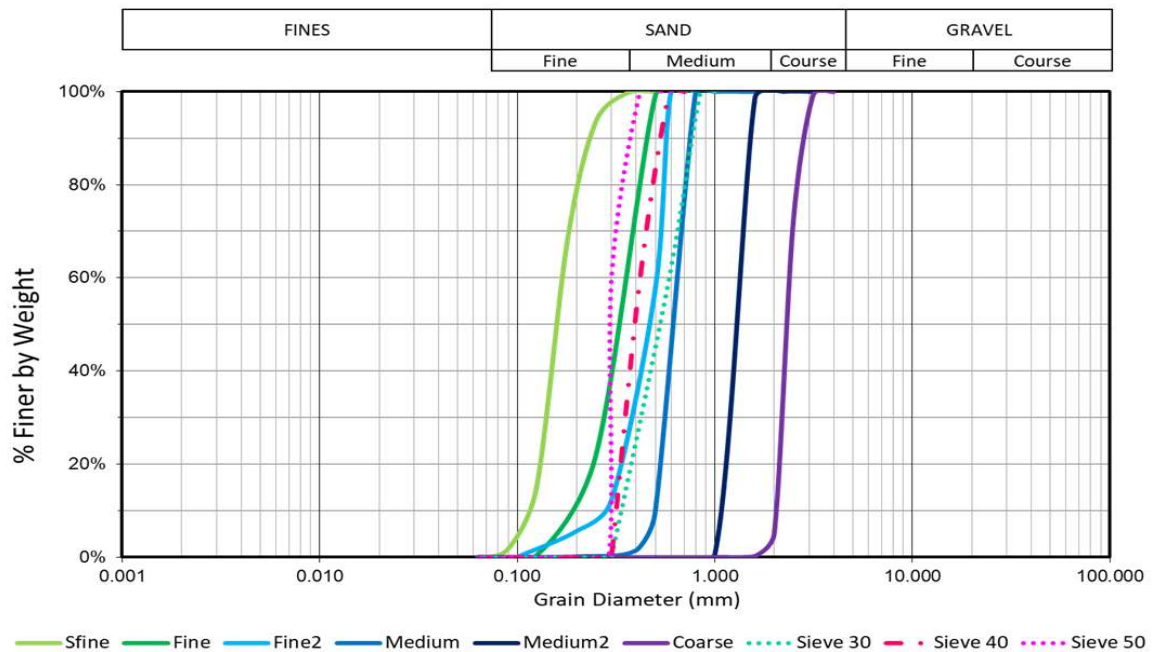


Figure 4-2. Grain size distribution of all six different sand sizes used in most of the tests.

The flow table, also known as the Hagerman flow table, was originally introduced in Germany in the 1930s, and is a test that the concrete industry uses to analyse the flow of concrete. It involves measuring the spread of material after it is subjected to jolting (Tattersall 1991). It is standardised by ASTM C230/C230M (2014) and by ASTM C1437 (2015).

EFNARC (2005) recommended the use of a flow table to test the plasticising effect of foam agents (with or without polymers) on sandy samples. According to the specifications, it is suggested that the test was to be first performed with water followed by soil conditioning additives. The table should be jolted 15 times and the flow of the sample will be measured by comparing the initial diameter with the final diameter, after the dropping stage is completed.

The original procedure was modified for the tests with clay-sand soils. The suggestion of dropping only 15 times (EFNARC, 2005) was not enough to establish a comparison between the samples containing clay minerals and the tested consistency indexes.

Figure 4-3 illustrates the setting and details for the testing of clay-sand mixed soils on the flow table. All of the tests were conducted on a glass surface that was fitted to a metal plate, which helped with the cleanup after each test (Figure 4-3a). Each soil sample was prepared at least 24 h before the intended test, and a high-speed mixer was used to ensure there was uniform mixing of the sample, especially in terms of moisture homogeneity. The metal cone was initially greased with a fine coat of vegetable oil (olive oil) to prevent the sample from sticking in the cone. This test could also be conducted without the oil coating to have an initial evaluation of the clogging potential by observing the amount of soil that became stuck in the cone after lifting it (Figure 4-3c).

It was essential to make sure that there were no air bubbles in the soil while placing the material inside the cone. To accomplish that, the soil was first placed in the corners of the cone by applying pressure (Figure 4-3b). After half of the cone was filled, up-and-down movements were repeatedly performed using with a spoon (around 20 times) to spread all the soil inside the cone. The remainder of the casing was covered by repeating the up-and-down movements with a spoon to ensure uniform filling, and excess material was removed with a spatula. Then, the cone was lifted, and the initial diameter was measured with a caliper. The value obtained is m_0 .

The table was jolted 40 times and the diameter was measured three times to obtain an average (Figure 4-3e), which was used to obtain the m_{40} parameter. Initially, the test was continued and measured again after 20 more drops were added, making 60 drops in total. This allowed us to obtain a third parameter, m_{60} . The prospect of dropping the table 60 times seemed to create a test that would be too long, and 40 jolts were already considered sufficient to allow for a comparison to be made comparison between different samples.

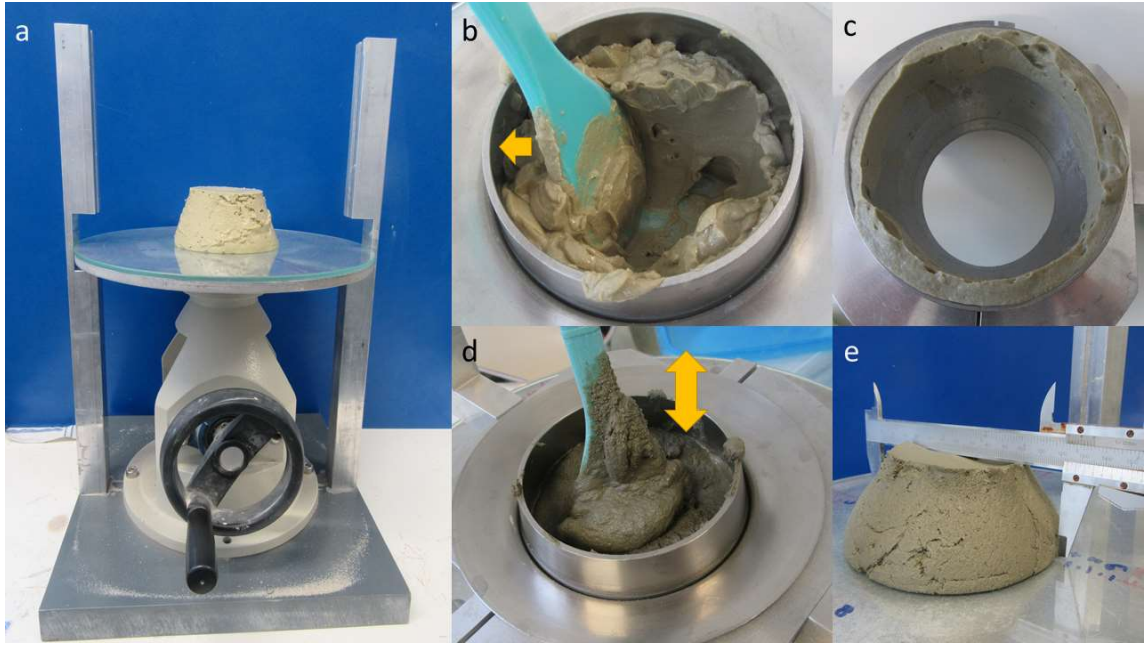


Figure 4-3. Flow table setup for tests with mixed clay-sand soils: a) general setting with one sample; b) initially filling the metal cone, pressing the material against the corners, and avoiding bubbles of air inside the specimen; c) without coating the cone with oil, some soil can remain in the frame if it is sticky; d) shaking the soil with up-and-down movements to ensure full distribution inside the cone; e) measuring the specimen with a caliper after jolting.

The following equation was used obtain the results for the flow (Equation 4-3):

$$Flow_{40}(\%) = ((m_{40} - m_0)/m_0) \times 100 \quad (4-3)$$

Where m_0 refers to the average of three measurements taken for the initial specimen diameter, and m_{40} refers to the average of three measurements of the specimen diameter taken after 40 jolts. The same equation can be used for $Flow_{60}$, where m_{40} will be replaced by m_{60} .

All the samples were weighted while wet and after being dried in an oven for at least 48 h at 100 °C. The bulk and dry densities were calculated after measuring the sample volume using the metal cone, as well as the void ratio and saturation degree. The water content of each test was checked using the standard of moisture content determination from ASTM D2216 (2010), and the

consistency index of each sample was calculated. At least two tests for each consistency index of each sample mixture were obtained.

4.4 Test Results and Discussions

The results of all the tests are presented and discussed in this section.

4.4.1 Mixed Clay-Sand Soils: Atterberg Limits

Figure 4-4 shows the results for the Atterberg limit tests for samples mixed with kaolinite and sands with sieve values of 30, 40, and 50.

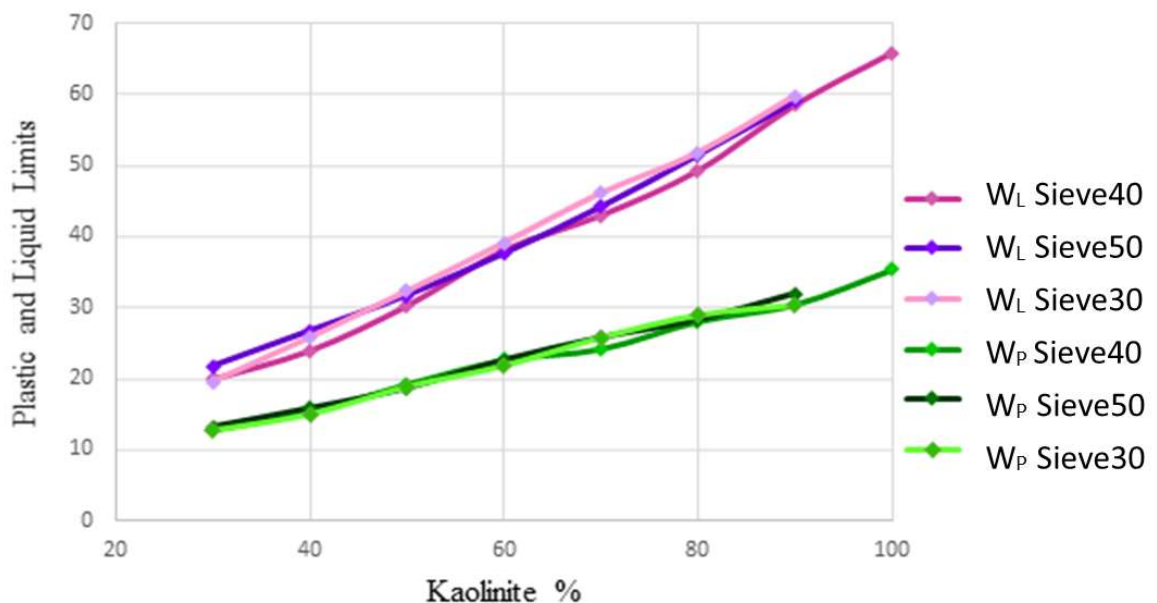


Figure 4-4. Liquid and plastic limits for three different grain sizes sand with kaolinite.

It was not possible to observe significant differences in the results between these particular grain size ranges (with minor variations); however, the results did follow a clear linear trend, even for clay contents as low as 30%. It was possible to successfully proceed with the liquid and plastic limit methodologies when including grain sizes slightly above the threshold

defined by the Atterberg limit standardization (material passing through a 20-mesh sieve and retained at a 30- or 40-mesh sieve).

However, the limitations of those tests with bigger grains than the standard threshold became clear when the tests were performed with other samples. Figure 4-5 shows the Atterberg limits obtained for bentonite-sand mixtures, and Figure 4-6 shows that for the Friedland clay-sand samples. The Casagrande cup and the thread rolling methods became more difficult to perform as the grain size increased and the clay percentage decreased. The main issues with determining the liquid limit were to properly cut the groove and, with bigger grains, it was clear that the soil was sliding rather than flowing.

The main limitation for the plastic limit was to roll the thread with coarser grains, even with high clay content. Many times, it was noted that the thread was crumbling due to the size of a sand grain, and not necessarily because it achieved the plastic limit point, providing an unrealistic value.

The limitation of the plastic limit started with 2 mm grains (limit between medium and coarse), approximately, because the samples with coarse sand (even with a high clay content of 70%) could not provide reliable results. It would be safe to assume that the plastic limit could be easily performed for grain sizes up to 1 mm in size. The liquid limit could be extended to grains up to around 2 mm; Figure 4-7 illustrates those above-mentioned limitations.

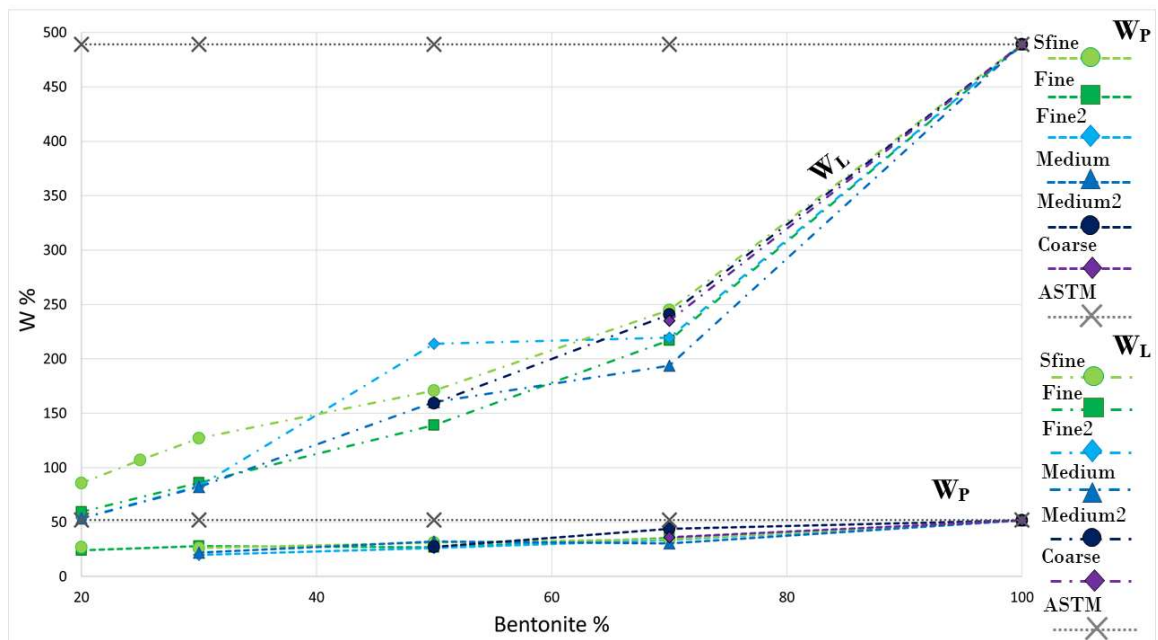


Figure 4-5. Atterberg limit values for mixed bentonite clay-sand soils (W_L – Liquid Limit; W_P – Plastic Limit). The grey dashed lines represent the plastic limits based on the ASTM standard specifications, and only take into consideration the portion of soil with grains that passed through the 40-mesh sieve; therefore, a pure bentonite sample.

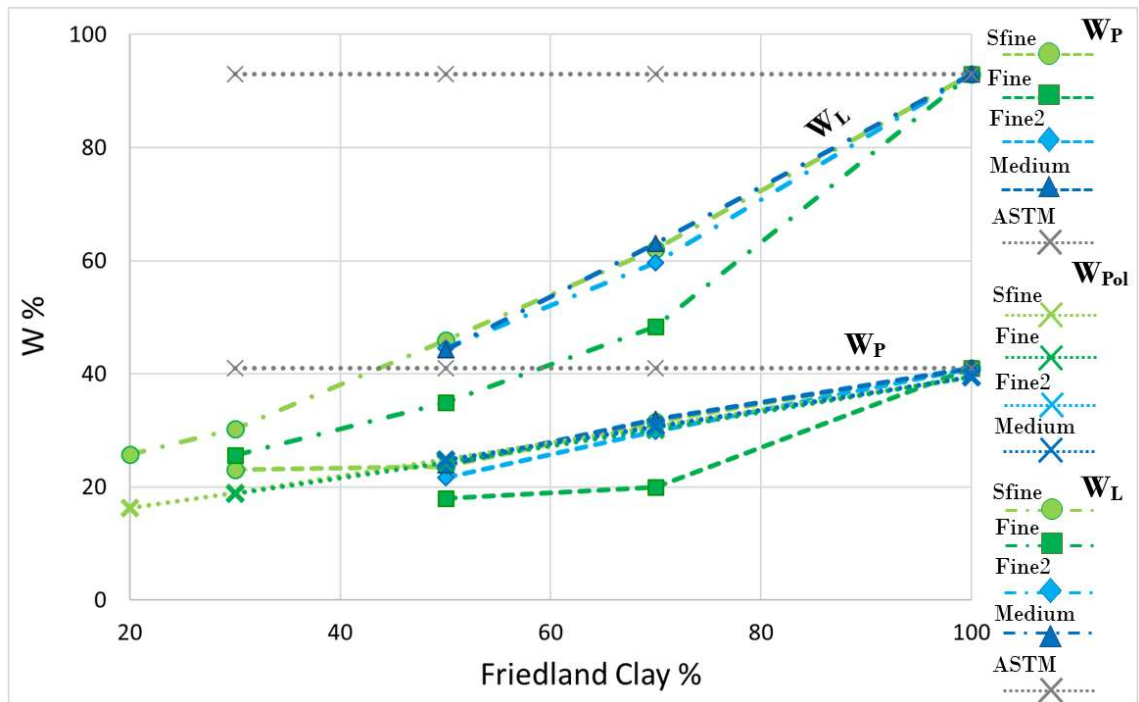


Figure 4-6. Atterberg limit values for mixed Friedland clay-sand soils (W_L – Liquid Limit; W_P – Plastic Limit). The dashed grey lines represent the limits considering the ASTM specifications (therefore, only considering the Bentonite portion), and the dashed coloured lines represent the plastic limit values (W_{POL}) calculated with the equation provided by Polidori (2007).



Figure 4-7. Physical limitations of performing the plastic (a) and the liquid (b) limits on samples with coarser grains than the assumed threshold of the method (grains smaller than 0.425 mm). The bigger grains in the thread would induce its cracking and these grains would make it hard to cut a groove with the proper tool.

In Figures 4-5 and 4-6, looking only for the “sfine” and “fine” lines for the liquid limit, it is possible to affirm that the water content needed to achieve a liquid limit is lower for coarser grains, but that does not seem to be the case for other samples. Nevertheless, even with all the limitations, it was still possible to attain a linear relationship, as already mentioned by other authors. By modifying the rule of the 40-mesh sieve of the standard ASTM D4318 (2017), it was possible to obtain more realistic values for the Atterberg limits.

In the same figures, the grey dashed line represents the limits for the pure clay samples. In the case of this clay mixed with a medium-grain sand; for instance, if only the fine particles are considered (clay component), the W_P and W_L values would be much higher than the modified ones for any clay content below 100%. Consequently, this would directly affect the values of the consistency index, as shown in Figure 4-8, where the left sample (a) was mixed considering the

Atterberg limits as defined by the standards. This led to an extremely liquid mixture for an I_C of zero (liquid limit). The sample on the right (b) was used with the modified version of the standard, including coarser grains. Visually, it is evident that the modified consistency values are a better fit than the values that were obtained by following the standard methodology.

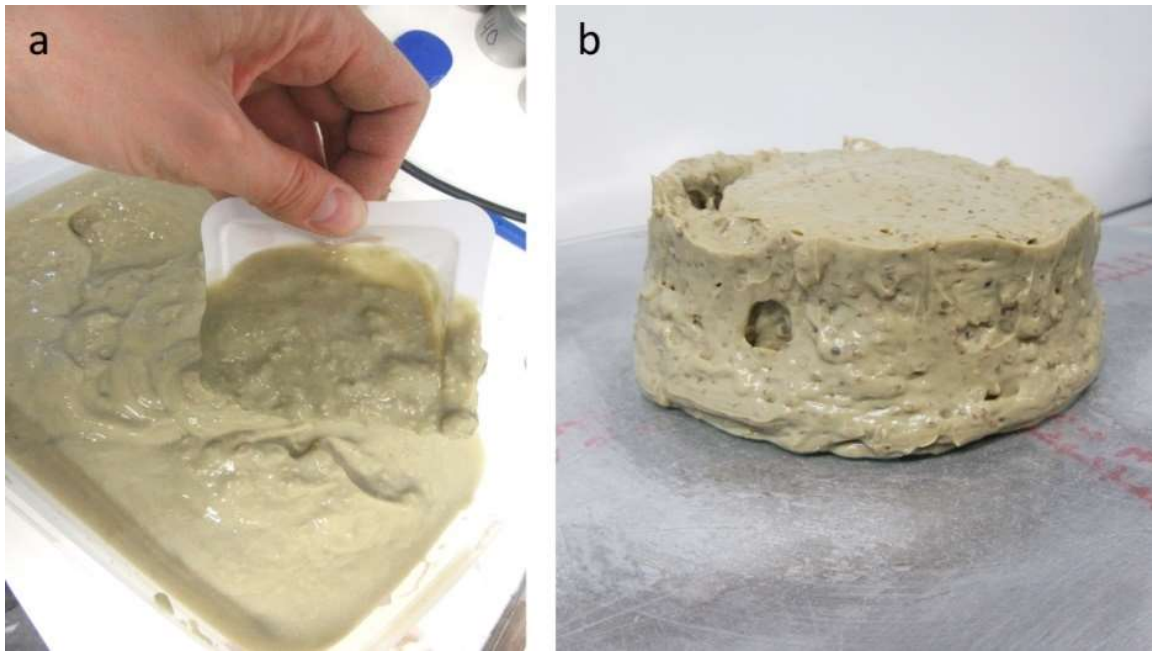


Figure 4-8. Two mixed clay-sand samples expected to have the following value: $I_C = 0$: a) following the standard and not including the grain sizes bigger than 0.425 mm; b) including all the grain sizes.

The measured results were also compared to the calculated values by considering the equation defined by Polidori (2007) for the plastic limit. This allowed us to obtain results that were consistent. Figure 4-9 shows these comparative results for the bentonite clay-sand samples. These can be compared with Figure 4-6 for the Friedland clay-sand results.

For the Friedland clay, two samples, “sfine” and “medium”, show very good correlations, even for low clay content, but for the “fine” sample, the correlations are not as close for the measured and calculated plastic limits. For the bentonite clay, the correlations seem better at

above 40% clay content, with exception of the “medium” and “medium2” samples, whereas at 70% clay content, the measured values seem to be quite far from the calculated limits. This may be emblematic of the difficulties associated with rolling the thread with bigger grains.

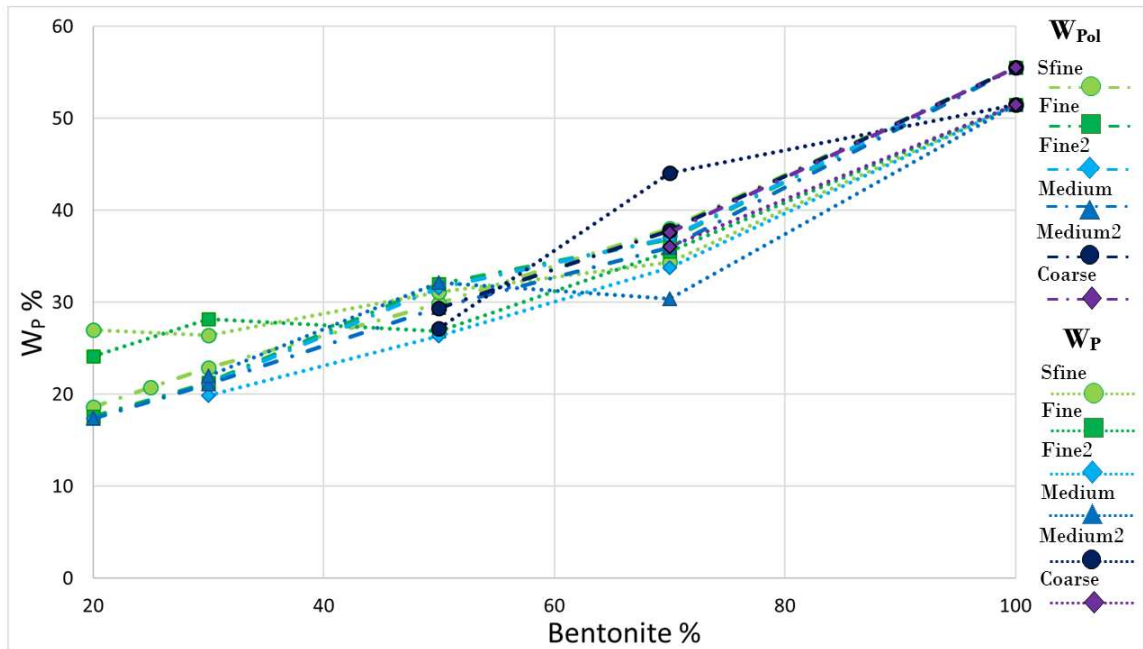


Figure 4-9. Detail of the plastic limit values measured (dotted lines) and calculated (dashed lines) for the bentonite clay-sand samples.

4.4.2 Flow Table

Several tests (minimum of three repeated tests) with different samples, each at three different consistency index values, were conducted using the flow table. Initially, this was conducted with the intention of searching for a replacement for the liquid limit test, assuming that a sample with similar consistency would flow in the same way. However, this did not occur. Nevertheless, it was possible to achieve certain insights concerning the flow of those mixtures, which could be most likely correlated with the flow inside the excavation chamber and screw conveyor of an EPB machine.

Figure 4-10 exemplifies a test with three different clay mineral/mixtures (bentonite, kaolinite and Friedland), for a consistency index of zero, therefore, at their liquid limit state, and for different proportions of very fine sand (“sfine”, see Figure 4-2 for grain size distribution). For this I_C , jolting was conducted 40 times ($Flow_{40}$) and 60 times ($Flow_{60}$), and it can be observed that they resulted in very similar flow. The average percentage value is represented by the circle (kaolinite), diamond (Friedland), and triangle (bentonite), including the standard deviation values.

The different behaviour observed between the different clay mineral mixtures is evident in Figure 4-11. For the Friedland samples, the flow increases almost linearly as the clay content increases. There is a slight downward curve from 30–50% of clay content for that mixture, but for the bentonite and kaolinite mixtures, this downward curve is quite pronounced. Initially, at low clay content, the flow decreases with the increase in clay content. The flow is at its maximum and turning point at around 65% for the bentonite, and at 87% for the kaolinite. After these turning points, the flow increases with the increase in clay content, which is more perceptible for the bentonite clay. It should be taken into consideration that the Friedland mixture is not a pure clay mineral, which could be affecting the results.

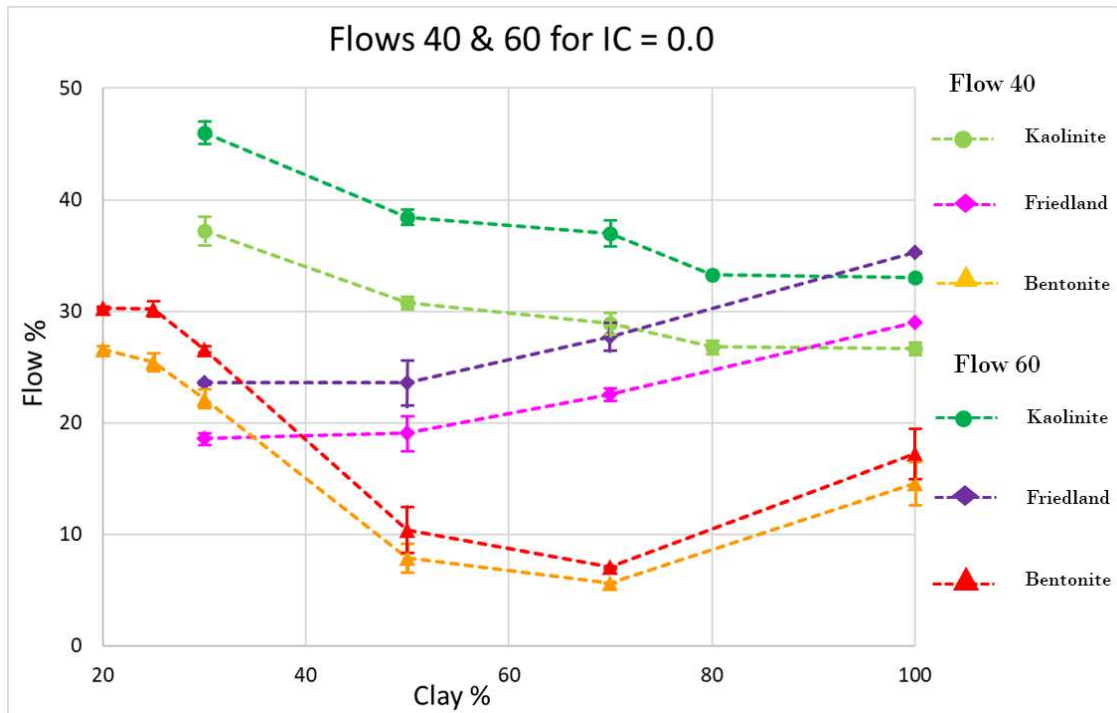


Figure 4-10. Flow₄₀ and Flow₆₀ for different clay proportions, for three different clays and $I_C = 0.0$.

Figure 4-11 also shows other consistency indexes (-0.1 and -0.2), and it is possible to affirm that lower consistencies would imply higher flows, as the mixture is more liquid-like and flows easily; this is logically consistent. Figures 4-10 and 4-11 can be compared with Figure 4-12, which shows the physical indexes of those samples at a consistency index of 0.0—their liquid limit value. In terms of densities (ρ – bulk; ρ_d – dry), both values decrease as the clay content increases. However, the void ratio (e) and the saturation degree (S) values increase as the clay fraction increases.

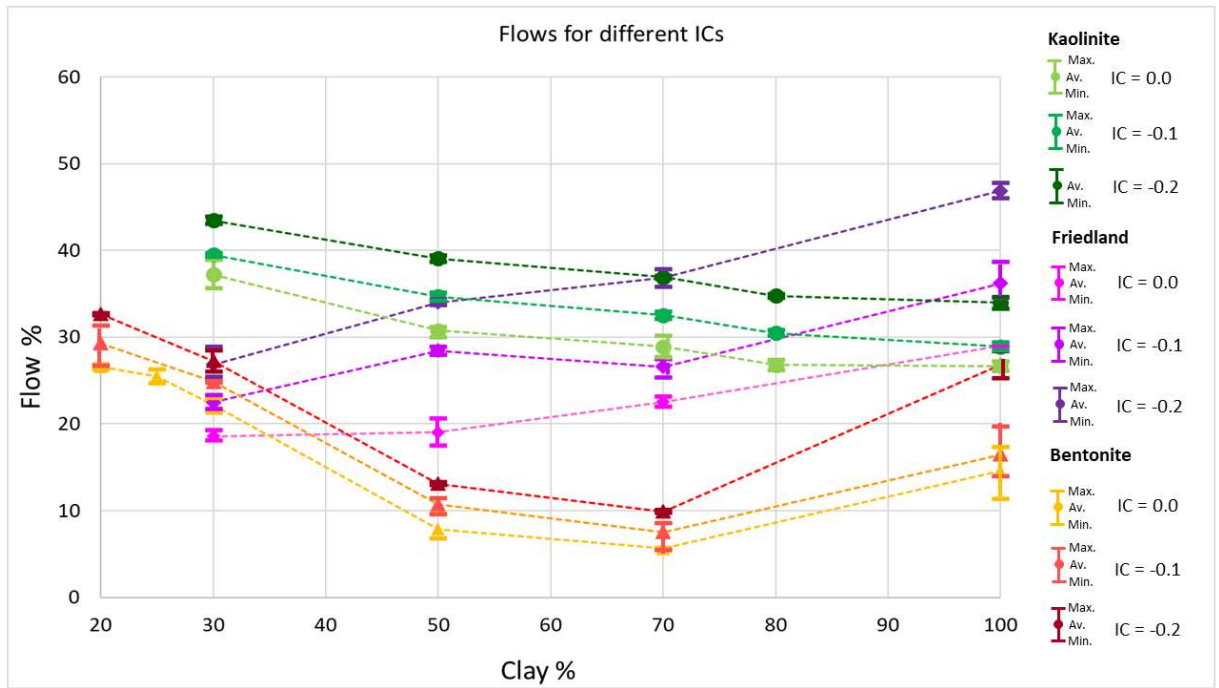


Figure 4-11. Flow₄₀ for different clay-sand mixtures for three different I_C : 0.0, -0.1, and -0.2.

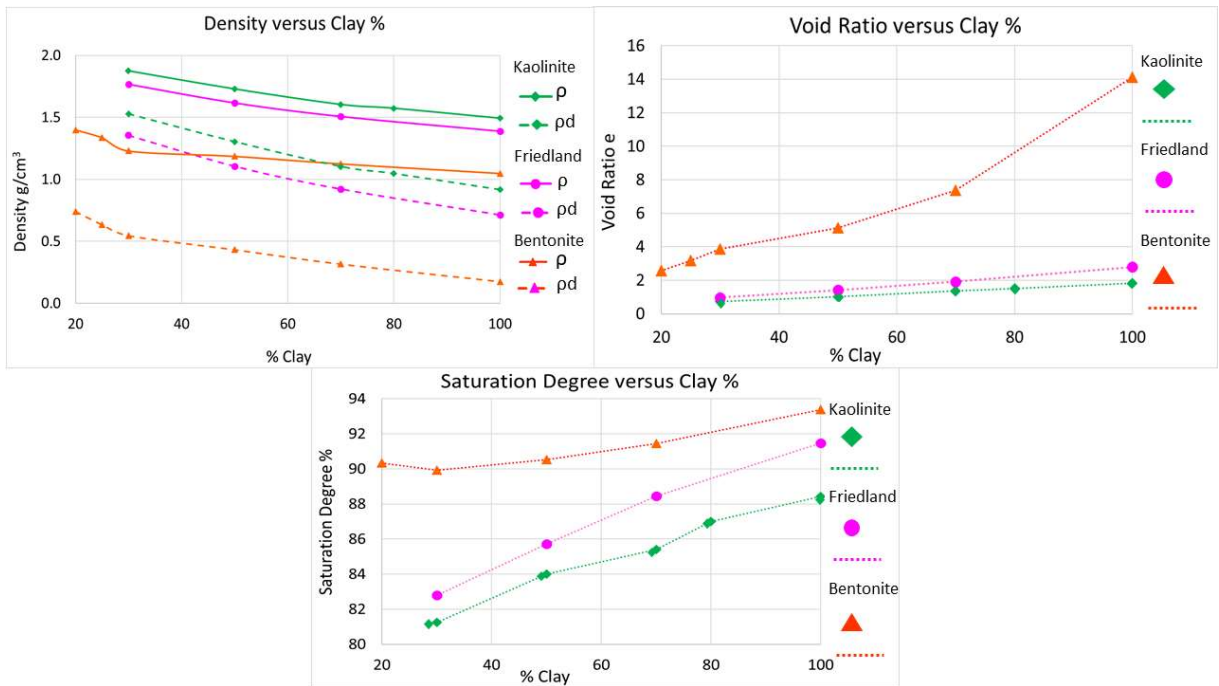


Figure 4-12. Physical indexes of the samples with $I_C = 0.0$, with bulk (ρ) and dry (ρ_d) densities, void ratio (e), and saturation degree (S).

Finally, Figure 4-13 proposes a schematic model for the clay-sand interaction and its flow, on a grain-size level, before and after jolting. The top of the figure shows the loose state before any compaction or impact, and the bottom of the figure shows the compacted state after jolting procedure.

For low clay content, the sand grains touch each other, and the clay flakes only fill the voids between the sand grains. The clay flakes would have no influence in the compaction with the jolting stage, happening only the readjustment of the sand grains into a more compacted packing of grains. With the increase in clay content, the sand grains are “floating” in the clay matrix, and after the jolting impact the clay flakes will not allow as much rearrangement of the sand grains, when compared to the lower clay content scenario. This would result into a lower flow of the entire mixture, lower even than the case of pure clay samples. Therefore, there is a threshold of clay-sand content for the lowest flow possible, and this value will differ depending on the clay mineral.

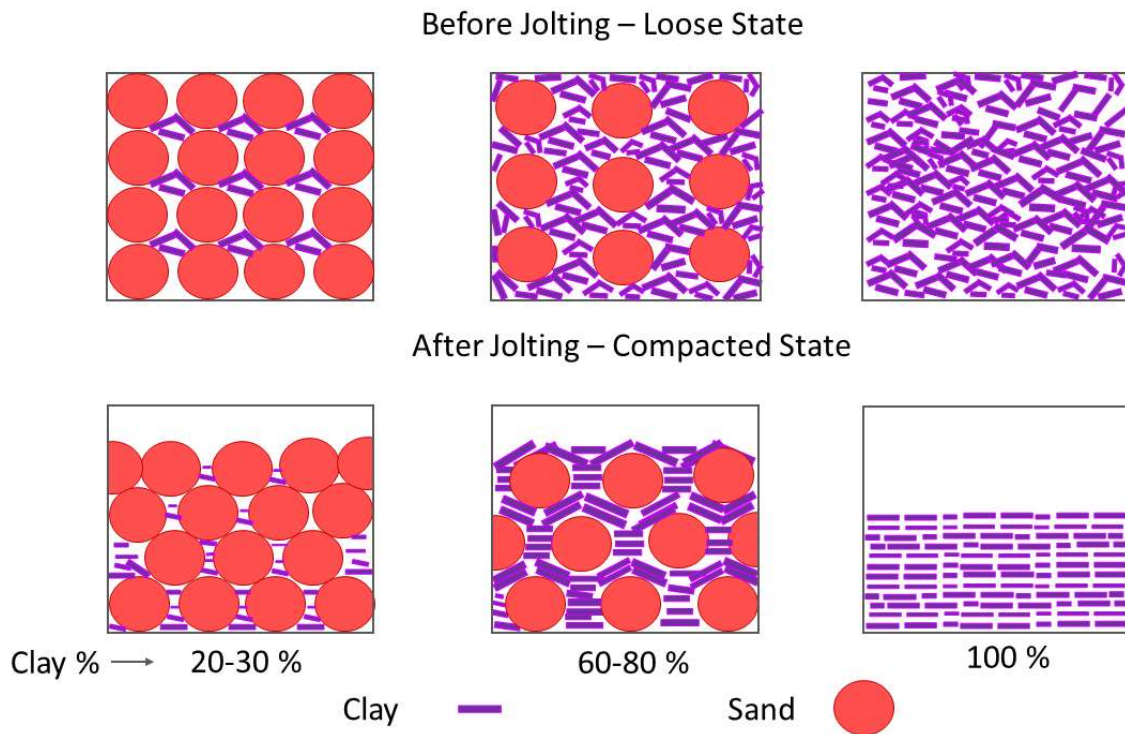


Figure 4-13. Proposed simplified arrangement for the microstructure of soils with different percentages of clay and sand content, before and after jolting.

4.5 Final Considerations

4.5.1 Tunnelling into Mixed Face: A Simple Case of Sand and Clay

Most of the research on EPB soil conditioning was conducted either for pure clay material (Mair et al. 2003, Merritt 2004, Spagnoli et al. 2011a, b, Zumsteg et al. 2012, Picchio and Boscaro 2013, Zumsteg 2014, Peila et al. 2016) or pure sand material (Vinai 2006, Borio 2010, Budach 2012, Galli 2016). Some questions concerning the soil conditioning and the behaviour of a mixed soil when excavated by an EPB machine lead into the following query: Is the current application of the Atterberg limit and the resulting consistency index sufficient to characterise those mixed soils? This question has resulted in testing with modified Atterberg limits, including tests with coarser grains.

This inquiry can be illustrated with a simple example (Figure 4-14), which shows two different scenarios with distinct combinations between two materials: a medium-grain sand (in beige), with particle sizes above 0.425 mm, and pure clay (in green). In terms of the Atterberg limits, only the clay fraction would be considered for obtaining the plastic and liquid limit values and for providing same consistency index values.

Even though it is mentioned in the standard methodology (ASTM D4318 2017) that the coarser fraction of the soil should be considered to evaluate the properties of this material, there are no clear instructions on how that must be considered, or in which occasions this really must be considered. And, as it was shown so far, this lack of caution could lead into unrealistic property characterisation of the excavated material, which could influence the optimal consistency of the muck and the soil conditioning process.

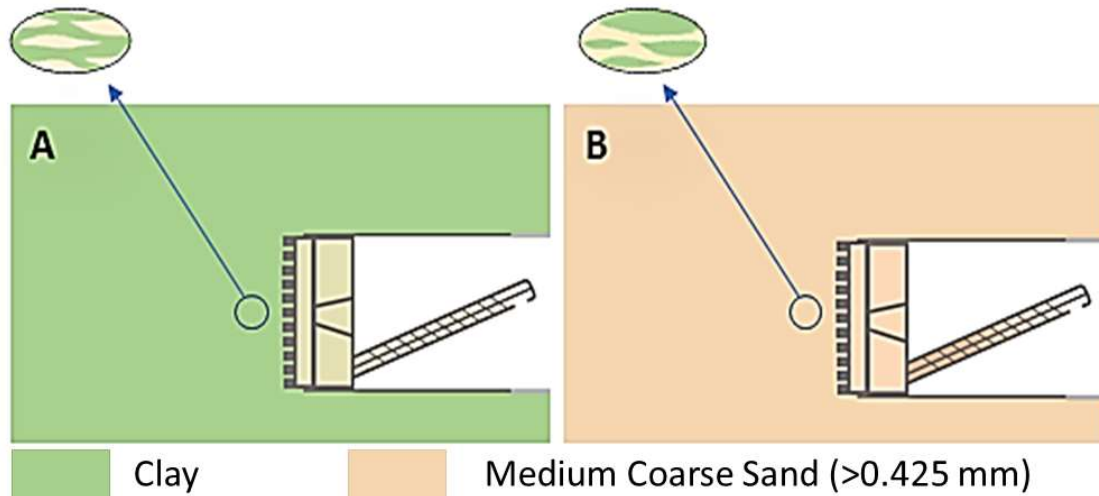


Figure 4-14. Comparison between two scenarios of interspersed lenses with same consistency index.

In terms of finding the ideal EPB consistency and classifying the soil from liquid to hard, it might be necessary to include larger grain sizes, or consider other alternatives for obtaining a

consistency index that would be closer to the material reality. For mixed soils, we suggested that they be excavated with an EPB machine and that the Atterberg limits be obtained. However, these limits should be obtained in a way that deviates from the standard to include coarser grains, at least for the liquid limit, and that the equation defined by Polidori (2007) be used to calculate the plastic limit for platy clays.

A second alternative, especially for soil fractions bigger than 2 mm (coarse), would be to replace the coarser grains of the original soil with laboratory fine sand and calculate the Atterberg limits for it. The real value should be closer to this new modified value than to the one considering only the fine particles of the natural soil, especially if the errors already assumed for the Atterberg limits are taken into consideration. This procedure is similar to the one suggested by Sivapullaiah & Sridharan (1985) for low-plasticity soils but utilize laboratory fine sand to replace the coarser grains.

These two alternatives are based on the fact that there is not much variation in the liquid and plastic limit values for different grain sizes, as seen in Figures 4-4, 4-5 and 4-6, especially when there are so many limitations within these methods to obtain the liquid and plastic limits. They might not be completely precise, but they still provide a valuable evaluation of soil properties.

Mixed clay-sand soils are found in several geological settings, such as in sedimentary alluvial-fluvial deposits, or lagoon-beach environments, where there is a high frequency of clay-sand interspersed layers. Mixed clay-sand soils are also found in tropical residual grounds, where the soil is usually well-graded. There are areas containing high clay content, or other portions containing gravel, and all areas constantly change in size in the vertical and horizontal directions. Therefore, this subject is not a single scenario; it needs to be addressed and solutions should be proposed.

4.5.2 Flow Table as a Future Replacement for Slump Tests for Cohesive Soils

The flow table has been mainly used to analyse the flow of concrete mixtures, or, as suggested by EFNARC (2005), for sandy samples. The slump test, which was originally applied in the concrete industry, is already part of the current routine for soil conditioning evaluation for EPB laboratory research and industry to evaluate the workability of the soil conditioned soil (see Vinai et al. 2008, Budach & Thewes 2015).

The main application for both tests so far has been low or non-cohesive soils. For instance, conducting slump tests with a certain percentage of clay might present a certain challenge in terms of the applying the mixing procedure needed to assure total homogeneity of the sample. In addition, its application to cohesive soils has not yet been investigated.

A slump test requires a high quantity of material, around 8–12 kg, depending on the material composition. The first issue with that is the handling of the method itself, which requires bigger mixers to assure the homogenization of the sample, and a bigger foam generator (in the case of evaluating conditioned soils), then the higher volume of material to be disposed afterwards. The other problem is to provide such a quantity of material before the machine starts to excavate; for instance, to initially evaluate the efficiency of the soil conditioning (polymers, water, foam and so on). Usually, before the excavation, there is only borehole material available from the initial site investigation phase, which is not enough to run tests with the slump device.

The flow table; however, requires between 400–600 g of material only, which would be feasible to acquire from the borehole site investigation. Besides that, there are other advantages with handling procedure and disposal for a lower material quantity to be tested. This device also provides more dynamic flow information, which is not provided by a slump test, which is a more static evaluation (Galli 2016). During the jolting stage, it could be possible to obtain more refined mixture flow behaviour information, of the material conditioned or not. It could also, in future, be

correlated with the flow inside the excavation chamber or screw conveyor of an EPB machine. This device could even be installed in tunnel boring machines to properly evaluate the muck while the excavation is taking place.

Undoubtedly, more tests should be conducted with this device. These additional tests could compare the results from slump tests with rheometer results to analyse the flow behaviour of soils and conditioned soils. In addition, the flow table device could offer an easier and more efficient solution to analyse the behaviour of conditioned materials for EPB excavation than the slump test, especially for cases of mixed soils with contribution of clay particles.

4.6 Conclusions

Tests with several artificially mixed clay-sand soils were conducted to provide conclusions regarding their Atterberg limits. These limits directly affect the consistency index values of those samples, which is necessary to characterise the material to be excavated by an EPBM. A new test routine was also proposed by using the flow table for mixed clay-sand soils, which allowed us to obtain the flow of the tested samples. Based on our experiments, we can draw the following conclusions:

1. Atterberg limit methodology can produce errors and may not be very precise.

Therefore, Atterberg methodology should be used with caution, especially in the case of mixed soils and, even more so for low-plasticity mixed soils.

2. To obtain these Atterberg limits with a modified version of the standard for the Atterberg limits can be more reasonable in terms of characterizing the excavated material (or to be excavated) for an EPB operation, especially, to achieve a more realistic consistency index.
3. This modification of the Atterberg limits can be performed by including the bigger grains, up to 2 mm; or, in case of coarser grains in the soil (above 2 mm, and

depending on the clay content), these coarser particles can be replaced by artificial quartz fine sand. Then, the liquid limit value would be approximated from the measured value of this new natural/artificial sample, which is a more realistic approximation than only considering the fine particles.

4. For coarser grains and low clay content, the equations from Polidori (2007) would provide a better fit for the plastic limit than trying to obtain those values directly from the plastic limit rolling-thread procedure.
5. The flow table could deliver valuable data regarding the flow behaviour of soils, conditioned or not, and perhaps be correlated with the flow inside the EPB excavation chamber and screw conveyor. This would optimize the EPB excavation process, especially for cases involving mixed soils.
6. This flow table could be a better replacement for the slump tests. It requires less material, features an easier handling procedure, and may offer better information about the flow behaviour of conditioned soil mixtures.
7. More tests need to be conducted with the flow table. These tests may include more clay minerals and comparisons between slump test results and rheometer results in order to provide a better idea of how this flow table could replace/complement the slump test.

4.7 Acknowledgements

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Chapter 5

Flow Behaviour of Mixed Clay-Sand Soils for EPB Machine Excavation

Abstract

Earth pressure balance (EPB) tunnel boring machines are shield TBMs that rely on their own excavated material as a support medium to maintain the support pressure at the face. This material also needs to have the necessary properties to be transported and, finally, disposed of. Whenever the natural material does not fulfil the necessary requirements, additives like water, foam, polymers, and fines, must be added, modifying the excavated ground to the desired conditions. The rheological properties of any excavated material, together with any additives, must be investigated and understood, as they will influence the flow behaviour of this conditioned material, directly affecting the machine operation and tunnel logistics. While studies assessing the flowability related to the EPB excavation of sand or clay soils are available, there is a lack of information on mixed soils. Moreover, in nature, a particular type of mixed soils occurs that is typical of tropical weathering, for which very limited geotechnical data exists in the context of its TBM-related behaviour. This paper presents the results from a testing campaign with mixed clay–sand samples, aiming to reproduce a simplified tropical weathered mixed soil, investigating its flow behaviour when changing certain controlled variables: clay–sand proportions, clay mineral, size of the clastic grain mixed with the clay, water content, and additives (foam and polymers). Results from the tests conducted with a flow table, a slump test, and a rheometer device were compared, providing insights about the flow behaviour of the tested samples and its interaction with an EPB machine.

5.1 Introduction

The flow properties of an excavated material, mixed with any conditioning (muck), will influence certain excavation parameters along a tunnel boring machine (TBM) soft ground excavation process. The flow behaviour of the muck affects the cutterhead and the screw conveyor torque, together with the material flow inside the pressure chamber and along the screw conveyor, thus influencing the pressure balance inside the pressure chamber. Additionally, it affects the removal, transportation, and disposal of this excavated, and conditioned, material.

The removal can be done, for example, by conveyor belts or muck cars along with a muck pumping system, which transports the muck through pipes under high pressure. While a soft material would be desirable to hold a proper flow inside the excavation chamber, a stiffer consistency, which means less water content and less volume of the material, would be ideal as a final disposal product, reducing the costs of muck transportation and disposal.

This research investigates the flow behaviour of cohesive soils, with focus on mixed soils. Mixed soils occur in several geological scenarios worldwide, but one of the best examples of their occurrence is the by-product of the tropical weathering of rocks, which results in residual soils of different alteration degrees (Deere and Patton 1971, Vargas 1980, Rahardjo et al. 2004, Viana da Fonseca and Topa Gomes 2010, Futai et al. 2012, Huat et al. 2012, Oliveira et al. 2017a). The particularity of this type of ground is its variability and heterogeneity, vertically and horizontally, as the chemical and physical weathering process does not affect the rock mass homogeneously, but rather following pre-existent discontinuities along the rock mass. The resulting material varies not only in grain size distribution, but also in all of its geological–geotechnical properties.

Concerning EPB excavation, those properties have substantial implications in the excavation of this soil. This material is already mixed in-situ, presenting a wide range of grain

sizes, from fines to boulders, in varying proportions. The challenges for excavating this tropical weathered ground with shield machines, especially related to the mixed face portions, have been cited by many authors (Shirlaw et al. 2000, Della Valle 2001, Zhao et al. 2007, Viana da Fonseca and Topa Gomes 2010, Shirlaw 2015, Comulada et al. 2016, Shirlaw 2016, Oliveira and Diederichs 2016, Peixoto da Silva et al. 2017, Oliveira et al. 2017a, b).

Due to the complexity and worldwide variability of these tropically weathered soils, it was decided to investigate a simplified combination of the two main elements that compose this soil: clay and quartz sand. Mixed clay–sand soils, prepared in laboratory, having all the variables controlled and allowing reproducibility of results, were tested with a flow table, a rheometer, and a slump cone. The influence on the flow behaviour of the change in clay minerals, clay fractions, sand grain sizes, consistency indexes (therefore, water content), and some additives such as foam and polymers, was observed. The chosen clay minerals were kaolinite and bentonite, so that extreme plasticity spectra could be investigated. An emphasis was given to the kaolinite samples, as this is the clay mineral occurring in tropical weathered soils, as a by-product of the mineral feldspar present in acid rocks, such as granite and gneiss (Vargas 1980, Huat et al. 2012).

Results from tests with these three devices were compared and conclusions about the flow behaviour when changing variables were achieved. Implications in terms of flow regarding these mixed soils are drawn, suggesting the use of the flow table as a laboratory method to investigate cohesive soils. Limitations of each method were also pointed out and recommendations for future in-situ tests with real tunnel drives are provided. The focus of this research was the tropical weathering soil; however, most of the conclusions and the laboratory method can be applied in other geological scenarios.

5.2 Theoretical Considerations

A brief summary regarding the main characteristics and challenges while excavating a mixed transitional ground, along with previous experiences with laboratory assessments of the flow behaviour of conditioned soils applied to EPB excavation, is provided.

Oliveira and Diederichs (2016) proposed the term “mixed transitional ground” for a certain type of ground when excavated by a shield machine. A mixed transitional ground (or face, to emphasise the scale of the TBM) is an assembly of different materials belonging to the same geological scenario and formation. A tropical weathered ground is the best example of a mixed transitional ground, a product of exactly the same rock, same formation environment, but with different weathering degrees (Deere and Patton 1971, Vargas 1980, Rahardjo et al. 2004, Viana da Fonseca and Topa Gomes 2010, Futai et al. 2012, Huat et al. 2012, Oliveira et al. 2017a).

Figure 5-1 illustrates some examples of this mixed transitional ground.

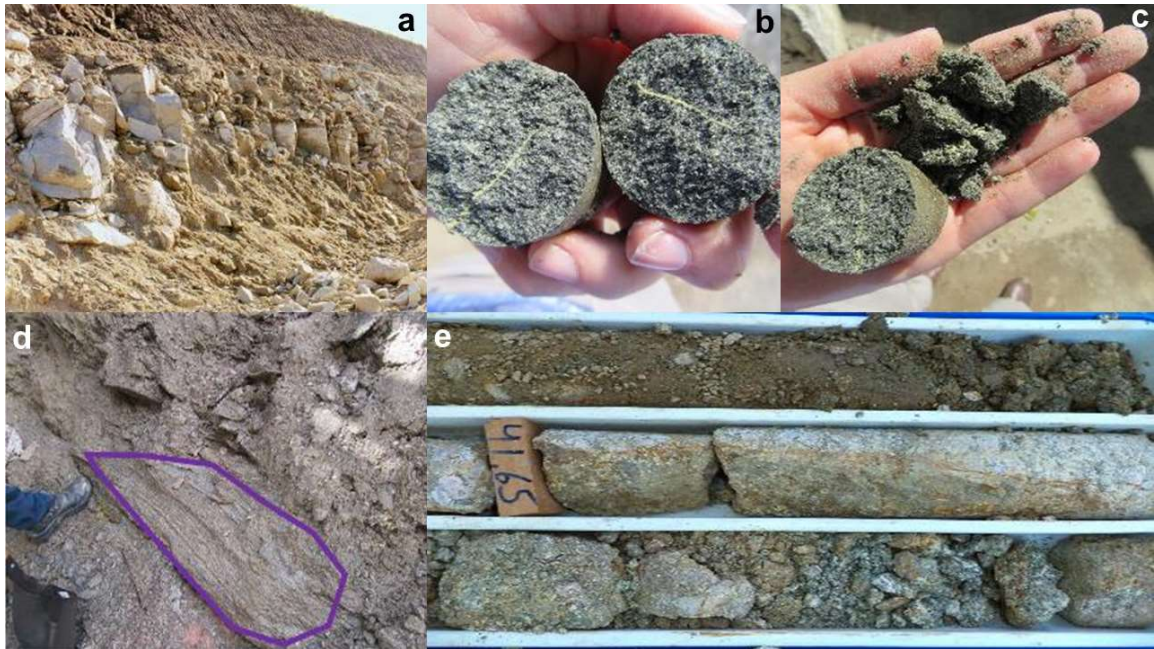


Figure 5-1. Examples of mixed transitional ground: (a) – outcrop in Porto, Portugal, showing rock cores inserted in a soil matrix, the residual soil, height of 20 meters; (b) and (c) – pieces of borehole samples of a gneissic rock, which looks like a rock (b), but when squeezed is easily disintegrated (c); (d) – bottom of a shaft along a subway line in São Paulo, Brazil, with gneissic rock (circled in purple) surrounded by the same type of ground but totally weathered; (e) – borehole core showing how this variability of soil/rock can occur even in depths of over 40 m.

To counterpoise the above scenario, an example that cannot be included in this term (mixed transitional ground), as it is considered simply a mixed ground, would be the sedimentary layers of a fluvial–alluvial setting, with interspersed layers of clay and sand. Even though these layers could belong to the same geological unit, they were deposited in completely different regimes of energy flow, as the clay layers needs total absence of movement to be deposited, while sand settles with higher levels of energy.

The term “mixed” implies significant differences in the geotechnical behaviour, such as relevant variances in consistency, strength, permeability, and stiffness. A saprolite in contact with a mature residual soil, occurring at the same time in the face of a TBM, would be a mixed transitional ground (Oliveira and Diederichs 2016, Oliveira et al. 2017a, b).

This research is mainly focusing in the soil component of the mixed transitional ground, the residual soil. A compilation of hundreds of references on this type of ground, including over five hundred compiled data sets, was presented by Oliveira et al. (2017b), which makes it possible to have a better picture of how variable is this type of ground worldwide.

Several authors have discussed the main issues when excavating this type of ground with TBMs (Shirlaw et al. 2000, Della Valle 2001, Babendererde et al. 2005, Zhao et al. 2007, Viana da Fonseca and Topa Gomes 2010, Shirlaw 2015, 2016, Comulada et al. 2016, Oliveira and Diederichs 2016, Oliveira et al. 2017a, b, Peixoto da Silva et al. 2017), including problems with loss of face pressure, damage on the excavation tools, and clogging occurrence, all of them leading into long and frequent interventions, settlements, and sinkholes.

In terms of the geotechnical characterisation of this material, Oliveira et al. (2018b) commented on the limitations of defining its consistency index (Equation 5-1) using the standardised method to calculate the Atterberg limits (ASTM D4318 2017), which consider only grains below 0.425 mm. The consistency index IC is given by:

$$Ic = (W_L - W)/(W_L - W_p) \quad (5-1)$$

Where W_L refers to the liquid limit (%), W refers to the water content (%), and W_p refers to the plastic limit (%).

It was demonstrated this exclusion of grains bigger than 0.425 mm would lead into unrealistic characterisation of the consistency of certain materials. The standardised method from ASTM D4318 (2017) does mention that the larger grains should be taken into consideration; however, it does not specify how this should be considered. Oliveira et al. (2018b) proposed a

modification of the standardised method for the case of mixed ground scenarios, presenting solutions to achieve a more realistic consistency of the material.

While excavating this type of ground, clogging owing to the clay fraction of the soil is one of the issues faced (Shirlaw 2015, 2016, Comulada et al. 2016, Oliveira and Diederichs 2016, Peixoto da Silva et al. 2017, Oliveira et al. 2017b). Many authors characterised the clogging mechanisms and correlated them with the excavation of clayey ground with TBMs (Thewes 1999, 2004, Sass and Burbaum 2009, Feinendegen et al. 2011, Zumsteg and Puzrin 2012, Hollman and Thewes 2012, 2013, Basmenj et al. 2016, Khabbazi et al. 2017). Most of the studies have focused on pure clay sedimentary soils or rocks disintegrated to soil by the shield excavation. The influence of the clay fraction in a mixed soil, however, was not included in the discussions or the proposed solutions, although it is an essential matter in this type of mixed transitional ground.

Peila et al. (2016) mentioned that a TBM excavates a stiff clayey soil as lumps. These lumps, however, will only be conditioned with foam, or other additives, in their surface, without reaching their core. Because of this fact, the authors suggest conducting the laboratory tests starting with lumps, and not with an initial clay powder, which would be unrealistic. It is essential, therefore, to recreate as much as possible the real condition of a tunnel excavation.

In the case of the mixed transitional ground, and especially focusing on the soil matrix, the clay particles are already mixed with silt, sand, or even bigger elements in the soil, which is an entirely different scenario than that of a sedimentary ground. Figure 5-2 summarises this scenario, comparing it with a typical sedimentary clay–sand mixed ground.

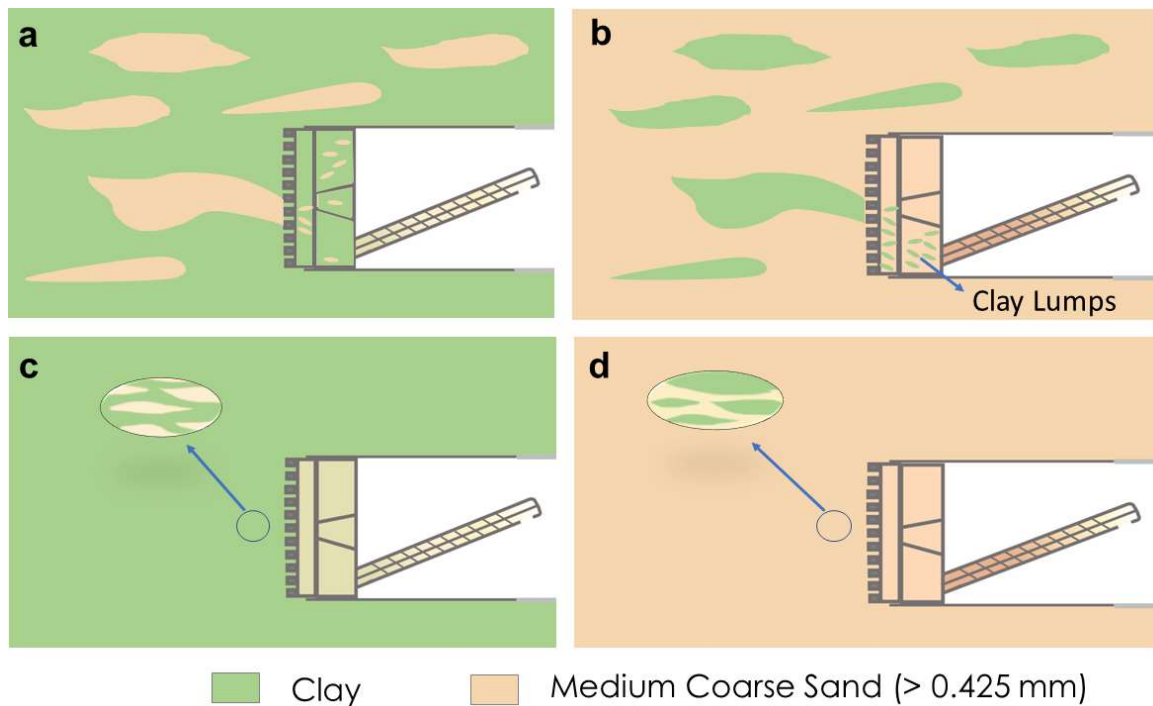


Figure 5-2. Four different scenarios of mixed clay–sand soils: (a) and (b) correspond to a sedimentary ground, where clay and sand layers are interspersed; however, as they are not really mixed in a small scale, they will be excavated as separate entities; (c) and (d) depict a residual ground, where clay and sand are closely mixed even before the machine start excavating.

The sketches at the top, (a) and (b), are from a typical clay–sand sedimentary environment. When the machine starts cutting the soil, it will remove either sand portions (a) or lumps of clay (b). In both cases, sand and clay are not entirely mixed, and lumps are “swimming” on a sand matrix, or vice-versa. It does not make sense to consider the total mass in terms of consistency index, because both unities will remain separate, even inside the excavation chamber.

For the cases at the bottom, (c) and (d), of a typical residual soil, the clay and sand are naturally mixed in-situ and there is not observable separation of this material. Then, in this case, the entire material should be taken into consideration to calculate the Atterberg limits, and consequently, the consistency index.

Therefore, for characterising any material and its excavation with an EPB machine it is necessary to visualise how the machine will be excavating the ground and what will be happening while the material flows inside the chamber, through the screw conveyor, conveyor belt (or muck cars, or muck pumping), and on to later disposal.

When characterising any type of ground to be excavated by an EPB machine, the influence of its consistency, flowability, hydraulic conductivity, and compressibility (Herrenknecht et al. 2011, Galli and Thewes 2014, Galli 2016) should be taken into consideration. To properly function as a support medium, the excavated material together with any additives needs to exhibit certain characteristics such as a suitable flow behaviour. In the case of a cohesive soil, this flow behaviour is translated into its consistency index (I_c), which, according to Maidl (1995), is suggested to be between 0.4 and 0.75. Regarding cohesionless soils, this flow could be assessed by means of the slump test. A suitable flow behaviour for a certain material would be achieved for slump heights between 10 and 20 cm (Vinai 2006, Vinai et al. 2008, Budach 2012, Budach and Thewes 2013, Galli and Thewes 2014, Budach and Thewes 2015). However, which rule should be applied when there is a mixed soil with only 20% or 30% of clay content?

EFNARC (2005), considering the absence of a standardised procedure to characterise the excavated material and its interaction with any additives for EPB excavation, suggested laboratory tests for the foam and conditioned soils. Most of these tests come from the concrete industry, such as the slump test and the flow table. Regarding the flow table, however, the suggestion for this test is limited to the verification of the plasticising properties of conditioned sands, applying 15 drops of the flow table and measuring how much the sample spreads after lifting the metal cone in relation with its initial diameter. Oliveira et al. (2018b) proposed a modification of this method to better suit cohesive soils, increasing the number of drops to 40,

which is adapted to verify the spread for cohesive soils. The authors presented preliminary results regarding the flow of samples of sand mixed with kaolinite or bentonite, finding that for the same consistency index the flow is different for the two different clays, or clay–sand mixtures.

Galli (2016) dealt with the rheology of conditioned sands, with the primary focus on understanding its flow behaviour in EPB machines by comparing the results between slump tests and rheometers. According to the author, flow is a non-static phenomenon and the slump test provides a single parameter, the equilibrium-state at rest, not providing a complete understanding of the flow behaviour. By means of various rheological tests, it is then possible to characterise the flow behaviour of different soil–water foam mixtures, concluding that it is very dependent on the shearing conditions. The tests included a ball measuring system (BMS), where a sphere eccentrically rotates on a circular path in a sample cup, measuring a shear stress for changes in the shear rates. More details on the system will be given in Section 3.4, and it can be accessed in Galli (2016).

Considering that some of the tests included foam, it is necessary to present the definition of the main foam parameters, such as foam concentration (C_f), foam expansion ratio (FER), and foam injection ratio (FIR). According to EFNARC (2005):

Foam concentration (C_f) – Equation 5-2:

$$C_f (\%) = 100 \times (m_{Surfactant} / m_{Foam\ Solution}) \quad (5-2)$$

Where:

$m_{Surfactant}$	mass of surfactant in foaming solution
$m_{Foam\ Solution}$	mass of foaming solution

Foam expansion ratio (FER) – Equation 5-3:

$$FER = V_{Foam} / V_{Foam\ Solution} \quad (5-3)$$

Where:

V_{Foam} volume of foam

$V_{Foam\ Solution}$ volume of foaming solution

Foam injection ratio (FIR) – Equation 5-4:

$$FIR (\%) = 100 \times (V_{Foam} / V_{Soil}) \quad (5-4)$$

Where:

V_{Foam} volume of foam

V_{Soil} volume of tested soil

5.3 Material and Methods

5.3.1 Tested Materials

The samples were reconstituted to simulate a simplified residual soil, focusing mainly on two main components, quartz sand and clay fraction, represented by two different clay minerals, kaolinite and bentonite. The tested samples were prepared in the laboratory. These three materials used as the base for the sample preparation are described in Table 5-1. The main purpose of the soils prepared in the laboratory is to assure reproducibility and total control of all the variables.

Table 5-1. Main characteristics of each component of the constituted soils.

Product Specification	Supplier	Spec. Grav.	Chemical Comp.	Specific Surface (m ² /g)	Cation Exc. Capacity (meq./100 g)	pH	Moisture (%)
Silica Sand	Euroquarz	2.65	SiO ₂	-	-	6.5-7	-
Kaolinite GHL KS 80*2	Georg H. Luh GMBH	2.62	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃	18.2	-	4.5-9.5	0.6-2
Active Bentonite IBECO B1	Imerys Civil Engineering	2.65	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃ , MgO, CaO, K ₂ O, Na ₂ O	600-800	70±10	10	11±3

For the sand fraction, silica sands of five different grain sizes were used, as shown in Figure 5-3. A total of 33 samples were prepared with different clay fractions, different sand grain sizes, and different clay minerals (either bentonite or kaolinite). All these samples are described in Table 5-2, together with their Atterberg limit values.

For the Atterberg limit values, whenever grains larger than 0.425 mm (Sieve 40) were included, the authors applied the modified method as suggested in Oliveira et al. (2018b), including the largest grains, in order to achieve a more realistic consistency. The plastic limit of the sample Kmedium30 was calculated using the equation suggested by Polidori (2007), as described in Oliveira et al. (2018b).

The samples were mixed in closed, cleaned, and dried plastic containers; the target water content was added, mixed, and left to cure between 12 h, for low plasticity samples (kaolinite), and 24 h, for high plasticity samples (bentonite). The samples with more than 30% of bentonite were then mixed a second time with a high-speed mixer (500 1/min), and left to rest another 24 h, assuring a high level of moisture homogeneity.

For all the testing procedures, the moisture content was always checked to verify if the target consistency index was achieved, applying the methodology suggested by ASTM D2216 (2010). The main target consistency was $I_c = 0.0$ for the initial flow table tests, mainly because that is the liquid limit value of the samples and that is the ideal consistency to reproduce the flow

table tests. Initially, as detailed in Oliveira et al. (2018b), the intention was to find an alternative for the calculation of the liquid limit methodology, including larger grains, and this is the main reason why the majority of the tests were done with this consistency.

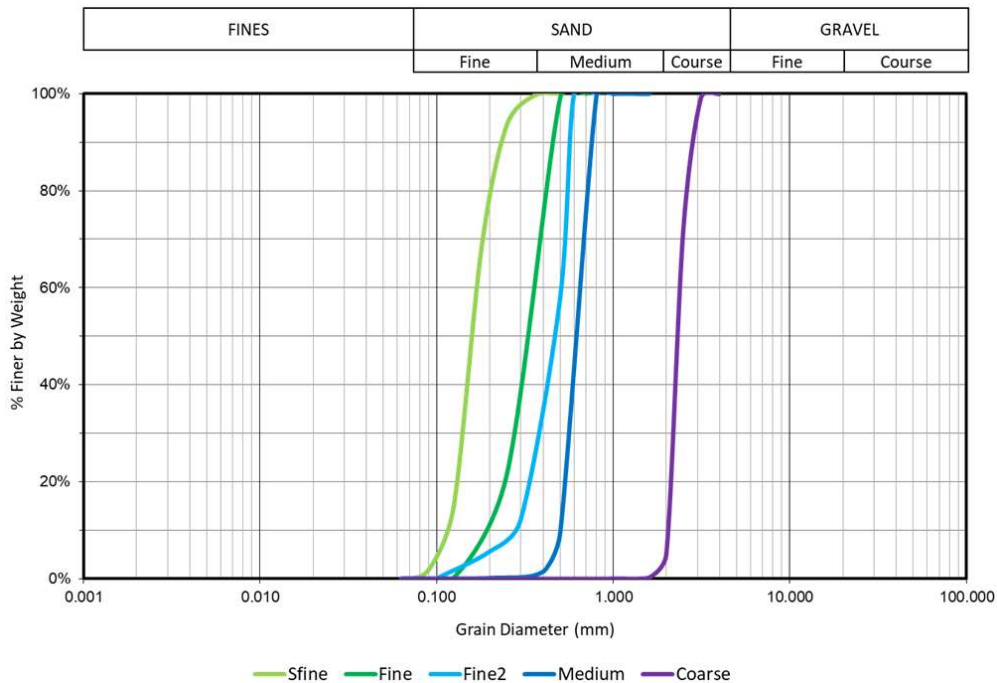


Figure 5-3. Grain size distribution for the sand portion of the soil mixture presented in Table 5-2.

As a general observation, achieving the target consistency with bentonite samples presented a higher challenge than that with the samples containing kaolinite, and this issue was solved by adding the second high-speed mixing stage. In addition, by verifying the moisture content before the test was conducted reduced the necessary testing time.

Some of the samples, such as the ones with a high fraction of bentonite, required up to five days to be prepared in order to assure fully homogeneity in the moisture content and an exact consistency index. Furthermore, each test was repeated at least three times, providing an assurance of the results and additionally the standard deviation, which is included in the results.

Table 5-2. Tested samples with the respective proportions of minerals and Atterberg limits.

Sample ID	Type of Sand	Sand (%)	Kaolinite (%)	Bentonite (%)	W _L (%)	W _P (%)	IP (%)
Ksfine10	Sfine	90	10	-	NP	NP	NP
Ksfine20	Sfine	80	20	-	NP	NP	NP
Ksfine30	Sfine	70	30	-	22	16	6
Ksfine50	Sfine	50	50	-	32	21	11
Ksfine70	Sfine	30	70	-	44	29	16
Ksfine80	Sfine	20	80	-	49	31	18
K100	-	0	100	-	62	40	22
Kfine30	Fine	70	30	-	22	18	4
Kfine50	Fine	50	50	-	32	21	11
Kfine70	Fine	30	70	-	45	28	17
Kfine2_30	Fine2	70	30	-	20	15	5
Kfine2_50	Fine2	50	50	-	32	21	11
Kfine2_70	Fine2	30	70	-	44	27	17
Kmedium30	Medium	70	30	-	20	17	3
Kmedium50	Medium	50	50	-	31	23	8
Kmedium70	Medium	30	70	-	43	27	16
Kcoarse50	Coarse	50	50	-	27	21	6
Kcoarse70	Coarse	30	70	-	41	29	12
Bsfine20	Sfine	80	-	20	86	27	59
Bsfine25	Sfine	75	-	25	107	27	80
Bsfine30	Sfine	70	-	30	127	27	100
Bsfine50	Sfine	50	-	50	171	31	140
Bsfine70	Sfine	30	-	70	245	34	211
B100	-	0	-	100	486	53	433
Bfine30	Fine	70	-	30	83	34	-
Bfine50	Fine	50	-	50	139	27	-
Bfine70	Fine	30	-	70	217	36	-
Bfine2_30	Fine2	70	-	30	83	20	-
Bfine2_50	Fine2	50	-	50	157	26	-
Bfine2_70	Fine2	30	-	70	219	34	-
Bmedium30	Medium	70	-	30	82	22	-
Bmedium50	Medium	50	-	50	160	32	-
Bmedium70	Medium	30	-	70	194	30	-

5.3.2 Additives, Foam Parameters and Foam Generator

For this study, some samples were mixed with foam and foam with anti-clay polymer, from two different suppliers. The suppliers and their respective products are not directly disclosed here. Instead, each product is cited as product A, B, or C, as listed in Table 5-3. For all the products, the concentration of the foam surfactant in water (C_f) was 3%, only because that is the concentration value suggested by most of the chemical suppliers. Further tests should be conducted with cohesive soils to verify the requirement of such concentration.

Table 5-3. Additives used in some of the tested samples.

Product Name	Description
Product A	Foam
Product B	Foam with Anti-Clay Polymer
Product C	Foam with Anti-Clay Polymer

For the tests with foam and foam with polymers, which were carried out only with the flow table using the method defined by Oliveira et al. (2018b), every sample was previously assembled with the consistency of 0.4 and brought to the consistency of 0.0 by adding the additive to be tested (products A, B, or C). For most of the samples, the target FER (Equation 5-3) was 9.5, which was the lowest reasonable FER able to be produced in the laboratory and added to the soil within enough time before the foam lost its stability and started to separate. For samples with 30% of clay, tests were also conducted with a higher FER value, of 20 as the target. The FIR (see Equation 5-4) was defined according to the necessary moisture content to reach the target consistency ($I_c = 0.0$). All the tests were conducted at atmospheric pressure.

Table 5-4 presents the parameters of the foam and tested samples. The foam was produced to achieve the target FER of each test, allowing a range of ± 0.5 of the tested FER, because of the difficulty to achieve every time exactly the same FER, given the limitations of the

foam generator. The time was measured from the production of the foam until the test was conducted, not exceeding 15 min. The higher the clay fraction was, the higher the difficulties for mixing the soil and foam. This reveals some limitations in testing cohesive soils with foam, especially for low FER values, which have a shorter stability time. The mixing was done by hand to avoid foam loss and to assure homogenisation.

Table 5-4. Parameters of the samples tested with foam and foam with anti-clay polymers.

Sample ID	Clay %	FER Target	Tested FER Range	FIR Range	Product
Ksfine30A1	30	9.5	9 - 10	38 - 42%	A
Ksfine30A2	30	20	19.3 – 19.7	82%	A
Ksfine30B1	30	9.5	9.7 - 10	40.5 - 41.5%	B
Ksfine30B2	30	20	20.5 - 22	83%	B
Ksfine30C	30	9.5	9.1 - 9.6	39 - 40.8%	C
Ksfine50A	50	9.5	9.5 - 10	61 - 65%	A
Ksfine50B	50	9.5	9 - 9.7	53.5 - 57.5%	B
Ksfine50C	50	9.5	9.6 - 10	56.8 - 59.5%	C
Ksfine70A	70	9.5	9.2 - 9.7	66 - 71%	A
Ksfine70B	70	9.5	9.3 - 9.4	66.4 - 67%	B
Ksfine70C	70	9.5	9.5 - 9.7	68.3 - 69.4%	C

The foam generator is the same described by Galli (2016) as “TLB Mini Foam Generator”. It is designed to supply pressure air/liquid up to 4 bar, having glass pearls of 5 mm in diameter inside the tube where the liquid/air is mixed. The flow regulation is done manually by means of a flowmeter (Krohne).

5.3.3 Flow Table

The flow table is a well-known test in the concrete research field, used to evaluate the flow of a certain concrete mixture. It is also known as the Hagerman, DIN, or mortar flow table. It measures the spread of the concrete after it is submitted to 15 drops of the table (Tattersall

1991). It is standardised for this use by ASTM C230/C230M (2014) and by ASTM C1437 (2015).

EFNARC (2005) recommended the application of the same table to test the plasticising effect of foam in sands. Oliveira et al. (2018b) proposed a modification of this methodology to test soil samples with clay fraction contribution, suggesting jolting the table 40 times, which allowed a better evaluation of the flow in the case of clayey samples.

Figure 5-4 shows the test set-up:

- a) The table includes a glass surface, which not only allows a better flow of the clay samples but also facilitates the cleaning after each test.
- b) It is essential to place the soil in the cone, pressing against its walls and corners, assuring that there are no empty spaces left, especially for samples with higher consistency and higher clay content.
- c) The container surface should be greased with oil; otherwise, owing to the adhesion properties of the clayey soil, it will stick to the mould (which also could be a qualitative evaluation of the clogging potential).
- d) After placing the soil in the extreme corners, as described in b, the area in the middle of the cone should be filled with soil in layers, and for each layer, the soil should be moved with up-and-down movements, using a mixing tool (e.g. a spoon), assuring complete filling of the container.
- e) After jolting the table, the spread should be measured three times with a calliper.

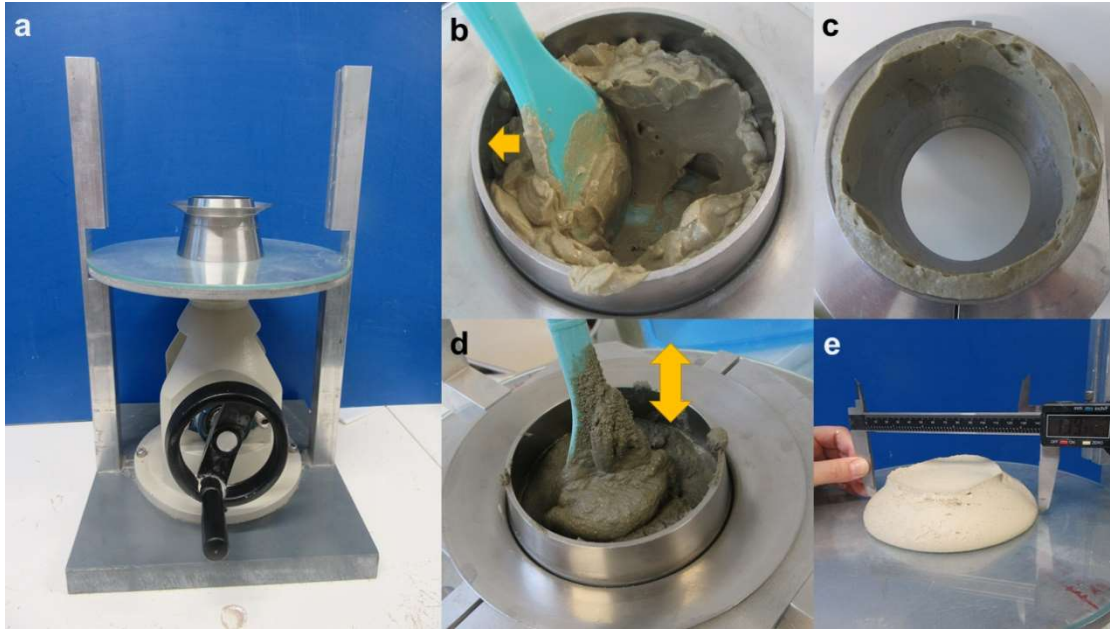


Figure 5-4. Flow Table: (a) – table set-up, including a glass surface; (b) – placing the soil initially pressing against the corners of the cone; (c) – to avoid soil stuck in the cone, an oil film should be spread in the cone; (d) – up-and-down movements should be done for every layer of soil, assuring complete spread of the sample inside the cone; (e) – after jolting of the table, the spread diameter should be measured.

Oliveira et al. (2018b) suggested jolting the table between 40 and 60 times. It was realised that 40 times was enough, considering that 60 times produced similar results but demanded additional effort in the testing methodology, especially when the table is not automatic, and the jolts must be inflicted manually. The data presented here was obtained with 40 drops of the table, and to acquire the flow results the following equation was used (Equation 5-5):

$$Flow_{40}(\%) = ((m_{40} - m_0)/m_0) \times 100 \quad (5-5)$$

Where:

m_0 – average of three measurements of the initial specimen diameter

m_{40} – average of three measurements of the specimen diameter after 40 times jolting

5.3.4 Rheometer

The rheometer used for the tests was the Anton Paar Rheolab QS, which was equipped with a BMS with 8 mm in diameter and eccentricity of 38 mm. Figure 5-45a and b illustrates the device. In Figure 5-5a, it is possible to visualise the entire device with a soil sample placed in a circular container of 500 mL, and Figure 5-5b shows the detail of the ball system of 8 mm. The ball was immersed to half the height of the container and rotated, completing a total of 6 revolutions, following a shear rate that was logarithmically increased and measuring the shear stress for several points.

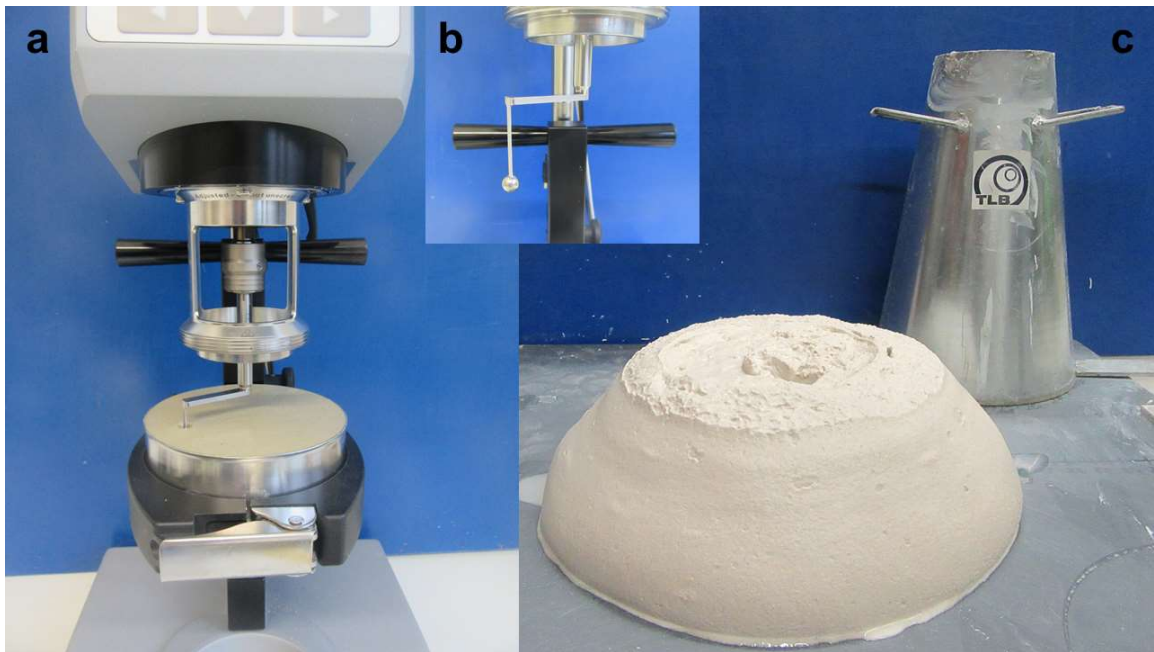


Figure 5-5. Rheometer Anton Paar Rheolab QS with the BMS (a) and (b), and the slump test with an example of the tested material showing a 20 cm slump (c).

As described by Galli (2016), Galli and Thewes (2018) and Freimann et al. (2017a, b), the first-round holds unsheared conditions, as it is the first round of the device, and the remaining 2nd to 6th rounds are the sheared ones. Unsheared (round 1) and sheared (rounds 2 to 6) conditions

are presented in separate graphs. Further details of the BMS and settings can be accessed in Galli (2016), Freimann et al. (2017a, b), and Galli and Thewes (2018).

The main limitation to use the rheometer was the consistency of the samples, especially for the samples with kaolinite and higher sand content (70% of sand, 30% of kaolinite), which presented higher yield stresses, and thus, were above the limit in which the system could work for the required testing consistency index. It was only feasible to test the samples with kaolinite mixed with “sfine” sand (Figure 5-3) when it reached the consistency index of minus 0.8, which is too liquid for normal EPB operation, especially concerning its transportation in conveyor belts and later disposal. However, if the muck is removed through a muck pumping system, it would be reasonable to investigate such liquid consistencies.

After every test with the rheometer, the same sample was then tested with the flow table. For future studies, another type of rheometer device should be used to test cohesive soils in typical EPB consistency indexes, as this device do not allow higher consistency values, which are necessary for investigating realistic EPB muck consistencies.

5.3.5 Slump Test

The slump test was also originated in the concrete industry, standardised by DIN EN 12350-2 (2009), and later adapted to the EPB soil conditioning tests to evaluate the flow behaviour and workability of conditioned soils (Quebaud et al. 1998, Leinala et al. 2000, Vinai 2006, Vinai et al. 2008, Borio 2010, Budach 2012, Budach and Thewes 2013, Peila et al. 2013, Freimann 2013, Budach and Thewes 2015, Galli 2016, Galli and Thewes 2018). The tests that are presented here were conducted using only water as conditioning agent; they do not include the part of the procedure that adds foam to the soil, and therefore, they exclude the time restrictions of the tests with foam (Vinai 2006, Peila et al. 2008, Budach 2012, Freimann 2013).

Only three samples were tested with the slump test and compared with the flow table: Ksfine10, Ksfine20, and Ksfine30. From all these three, only the last one has low plasticity, while the other two are non-plastic. A total of 13 kg of solid sample was reconstituted, mixing by hand to avoid loss of the clay fraction, which could occur if mixed with a mechanical mixer. The water was added and again mixed by hand. It took around 2 h for each sample to be properly mixed. Another reason to proceed with the manual mixing was to assure fully homogenous mixing, breaking the clay lumps by hand.

For cohesionless soils, owing to the weight of the sand grains and absence of issues with the dispersion of the clay, the mixing can be done mechanically, which would take only a few minutes. The entire procedure was physically challenging and time demanding, which also demonstrates a limitation of the slump test for cohesive soils, even for samples with low plasticity.

After the mixing stage, the sample was put into rest for at least 6 h. For testing, the sample was placed in layers inside the slump cone (Figure 5-4c), always avoiding the formation of empty spaces by immersing and moving a steel bar inside the cone, as suggested in DIN EN 12350-2 (2009). The cone was lifted and the height of the slump (difference between final and initial height) was measured. The slump flow was also measured, taken the average of two measures of the spread diameter.

The test was conducted until the target slump height values, of 10 and 20 cm, were reached, and then a sample was removed for moisture content calculations (ASTM D2216 2010). The same test was repeated three times for each sample and for each height, taking an average of the water content for the samples and calculating the consistency index for the third one (Ksfine30).

5.4 Results and Discussions

In this section, the results from the tests with the flow table, compared with the slump and rheometer tests, are shown and discussed. Figure 5-6 compares the results of flow behaviour with the flow table, for samples with different clay fractions with kaolinite (purple diamond) and bentonite (red triangle). All the samples are in the consistency of zero, at their liquid limit value. Each symbol indicates the average value and the lines above and below show the standard deviation. The bulk density (g/cm^3) of each sample is also included in the squares presented in Figure 5-6.

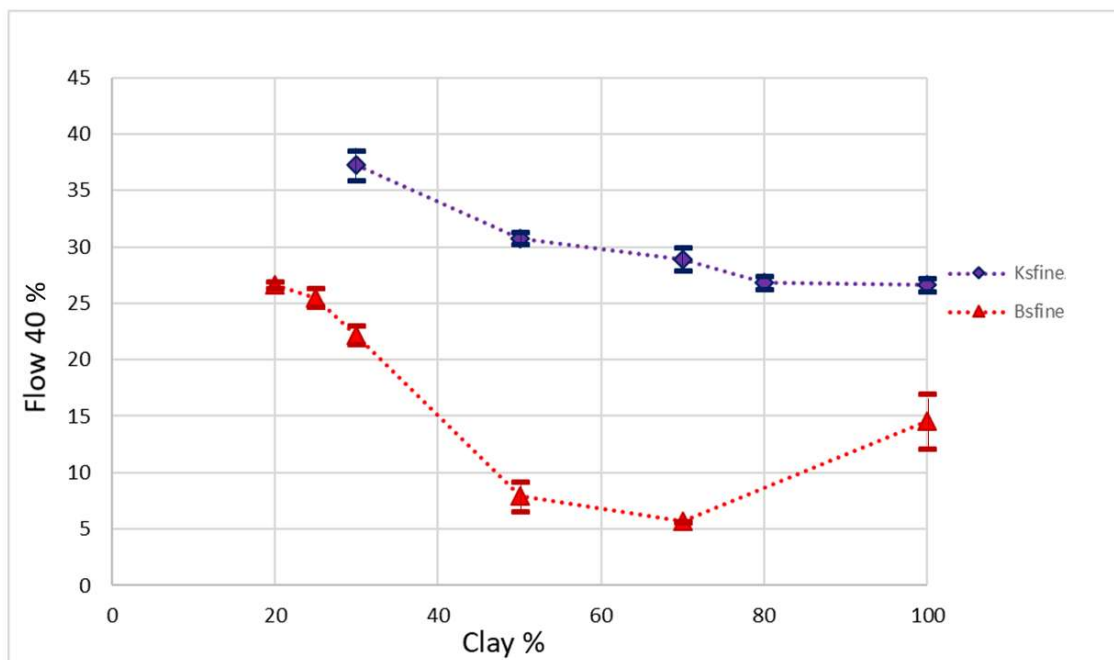


Figure 5-6. Flow₄₀ for kaolinite and bentonite mixed in different percentage with fine sand, for $I_C = 0$. The number in each square represents the bulk density of each soil in g/cm^3 .

Up to 70% of clay fraction, the samples with both clay minerals show a decrease in flow with the increase in clay fraction, with the bentonitic samples having a steeper negative slope gradient and an overall lower flow than the samples with kaolinite. After 70% of clay content, for

the samples with bentonite there is an evident increase in flow, which is not apparent in the samples with kaolinite. The last two points of kaolinite (80% and 100%) seem to have practically similar flows. Samples with 40% and 100% of bentonite would have similar flow values, for example.

Oliveira et al. (2018b) suggested that this decrease and increase in flow, more evident for bentonite, is caused by the soil skeleton, where for a low percentage of clay fraction the clay mineral is mainly filling the voids between the sand grains. With the increasing amount of clay minerals, the clay plates start working as a sheet between the sand grains, not allowing their full rearrangement after the jolting stage.

This effect is more evident for bentonite because of its swelling effect and surface area that is larger than that of the kaolinite sheets. In addition, with 100% of bentonite, the small size of the plates, the absence of sand, and the condition that the water content is much higher in its liquid limit make it possible for the sample to become more compacted but also to flow more, when compared to that with a mixture of 65–70% of bentonite. It is essential to consider the effect of friction and cohesion. For pure clay samples, only cohesion is playing a role in the strength of the material, and thus, it affects its flow. With increasing sand content, the friction is also participant in the flow behaviour of the material.

Therefore, with the same consistency index, for different soils, it occurs a different flow behaviour, even with the same clay percentages and sand grain sizes. That also should be expected to occur inside the EPB excavation chamber, through the screw conveyor, along the conveyor belts, and to final disposal. For cohesive soils, to have the same consistency index does not indicate the same flow behaviour, at least in terms of the results obtained with the flow table, which still need to be investigated further, especially by analysing samples obtained in tunnel drives.

It is important to note that each material was in its liquid limit point. This means that the open groove was closing to a similar length, after 25 blows of the Casagrande cup device (ASTM D4318 2017). Consequently, the same flow results for the same consistencies were expected; however, this did not happen. The consistency index was verified twice, before and after the test, confirming the $I_c = 0.0$. Perhaps this difference happened because of the quantity of the soil, which was larger for the flow table case than for the Casagrande device. However, that would not fully explain the behaviour of the mixed samples with bentonite. Perhaps with further tests, including undrained shear strength tests, additional conclusions could be achieved.

Figures 5-7 and 5-8 analyse the effect of increasing the grain size of the sand component in the flow behaviour, using the flow table, for kaolinite (Figure 5-7) and bentonite (Figure 5-8) samples, with increasing clay content. It is observed, in general, that a larger size of sand grain decreases even more the flow, especially with 30% of clay content, which means a greater contribution of sand, with the difference being larger for samples with 30% of bentonite. With the increase in clay to 50% and 70%, this difference is not so evident; samples with bigger grains sometimes show even higher values of flow than those with smaller grains (e.g. Ksfine70 and Kfine70 have, respectively, the lowest and highest flows). The biggest different for kaolinite is with 30% of coarse sand grains; however, this result could be affected by errors in the determination of the liquid and plastic limits for this sample, which uses a modified method as proposed in Oliveira et al. (2018b). This modified version to calculate the Atterberg values has its limitations precisely for the coarse size of grains, with around 1–2 mm of grain size.

The tests with the flow table have limitations with coarser grains, as it is more difficult to flatten the soil at the surface of the cone; bigger grains would keep moving around, destroying the top moulded surface in the soil when a scoop is used to make the surface even. The results of Kcoarse50 have been discarded because of the limitations of the test with grains larger than 2

mm, which are worse for lower percentages of clay. For coarser grains, and for gravels and bigger materials, slump tests are surely a better choice than flow table tests.

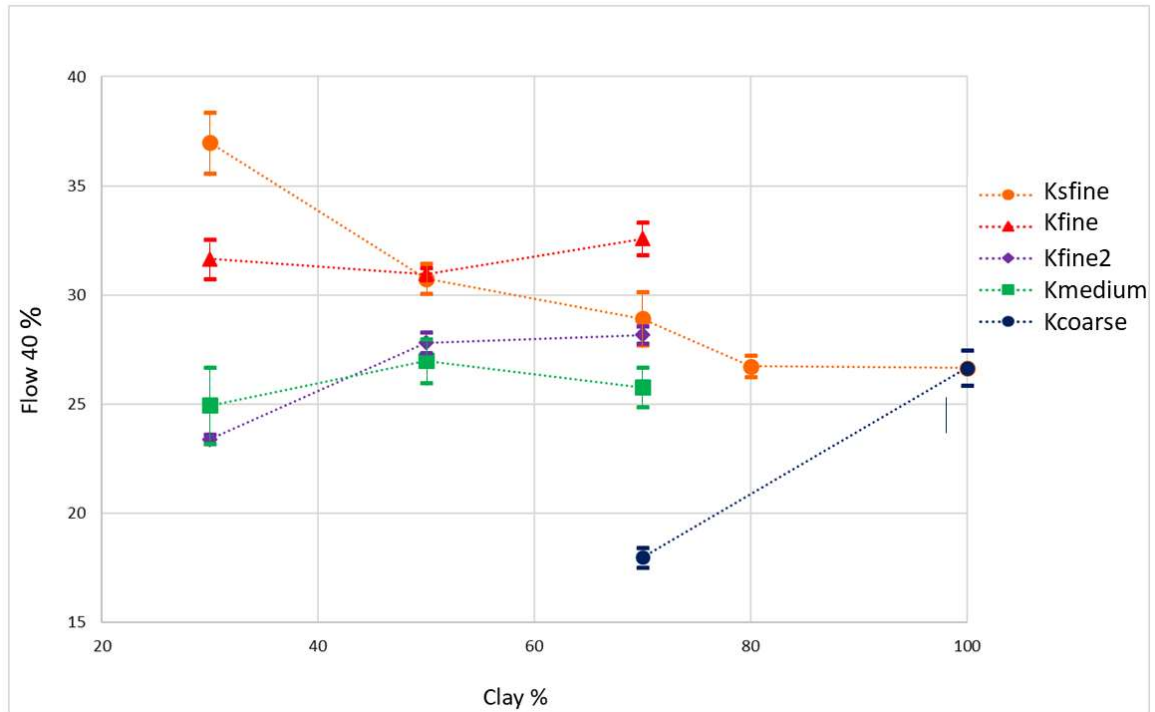


Figure 5-7. Flow₄₀ for kaolinite samples mixed with different sand grain sizes, for $I_C = 0$.

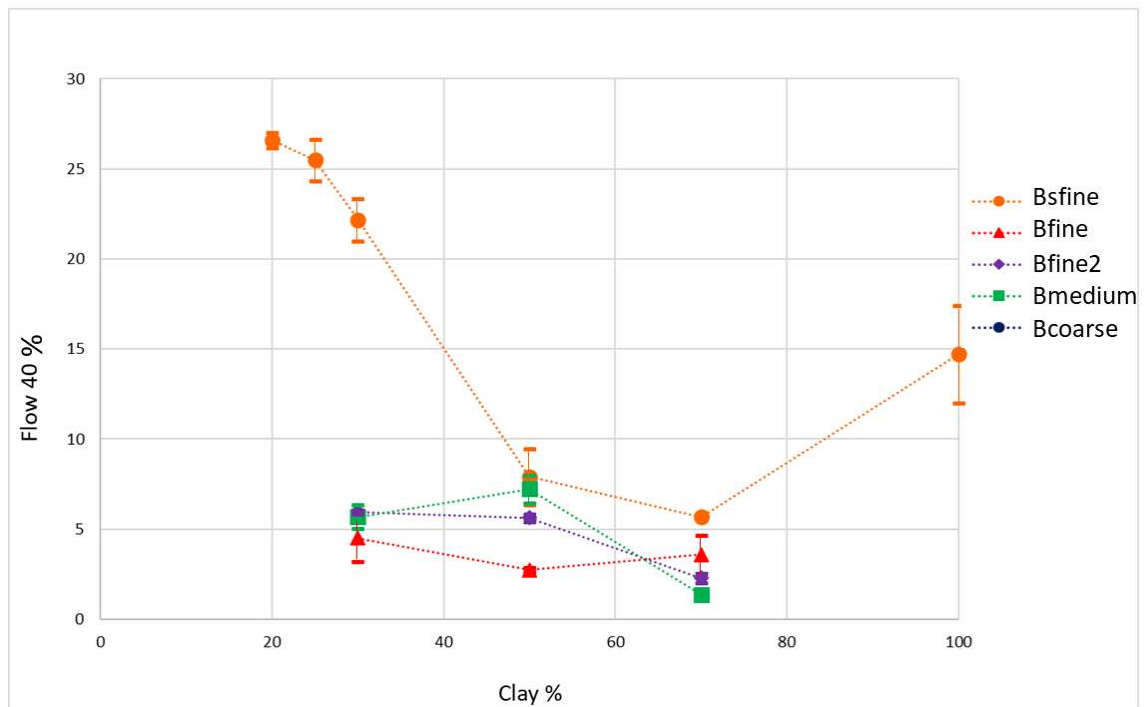


Figure 5-8. Flow₄₀ for bentonite samples mixed with different sand grain sizes, for $I_C = 0$.

Figure 5-9 presents a comparison between samples with kaolinite, in different fractions, mixed with “sfine” sand in four different consistency indexes: 0.4 (green), 0.0 (purple), -0.2 (red), and -0.8 (orange). As expected, with the decrease in the consistency index, which means a higher water content for the same plasticity index, there is an increase in the flow values. In general, for all four consistencies, the flow seems to decrease with the increase in the clay fraction, except for 50% of clay and $I_C = -0.8$. It was not possible to test samples with 100% of kaolinite with $I_C = 0.4$, as the sample with 70% was already a challenge owing to its stiffness and cohesion. Most probably, samples with bentonite could be tested with $I_C = 0.4$, but this was not done because the focus was the typical clay in the tropical ground, i.e. with kaolinite.

For 100% of kaolinite content, with $I_C = 0.4$, it would not be feasible to place the sample inside the cone and fill all the empty spaces, which shows a great limitation of this method for higher consistencies and higher clay content in a regular EPB muck removal operation. However,

for mixed soils, this method seems to be suitable, even when considering a muck pumping system for muck transportation, where lower consistencies (far below 0) are required.

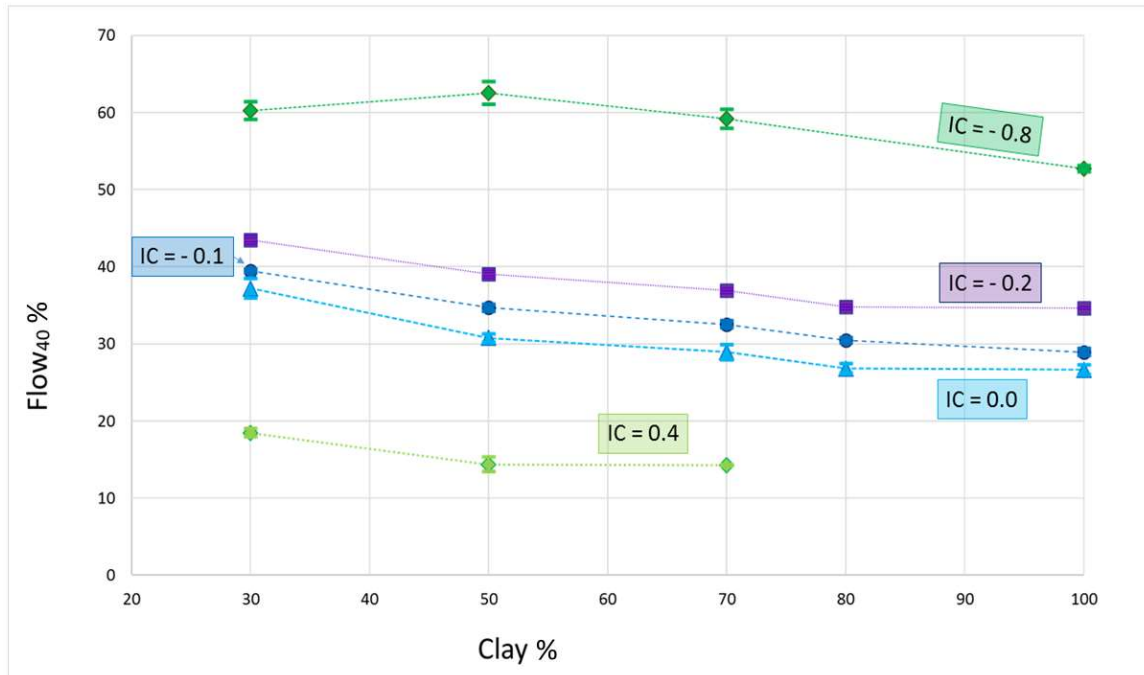


Figure 5-9. Flow₄₀ for kaolinite with fine sand in different % and different I_{Cs} .

Figure 5-10 shows a compilation of several flow tests conducted with soils prepared with bentonite, pure and mixed, and with kaolinite, pure and mixed. There is a tendency of the kaolinite samples to hold higher flow values when compared with the bentonite samples. The bentonite samples that show the highest values are the ones with lower clay content, i.e. 20, 25, and 30% of bentonite mixed with sand.

Figure 5-11 shows three different samples being squeezed or mixed. The samples in the top figures (Figure 5-11A and B) are from pure clay, bentonite, and kaolinite, respectively. The squeezed aspect looks quite similar, which was clearly noticed during the conducted tests. Both samples are around the lowest EPB ideal consistency range, with I_{Cs} of 0.43 and 0.44,

respectively. The samples below contained 30% of kaolinite, mixed with fine sand, with consistency of 0.4. These last samples are also shown in Video 5-2, which is part of the complementary material for this chapter. The mixed kaolinite samples seemed always harder than the pure clays, or than the bentonite with mixed sand, even with the consistency of 0.4.

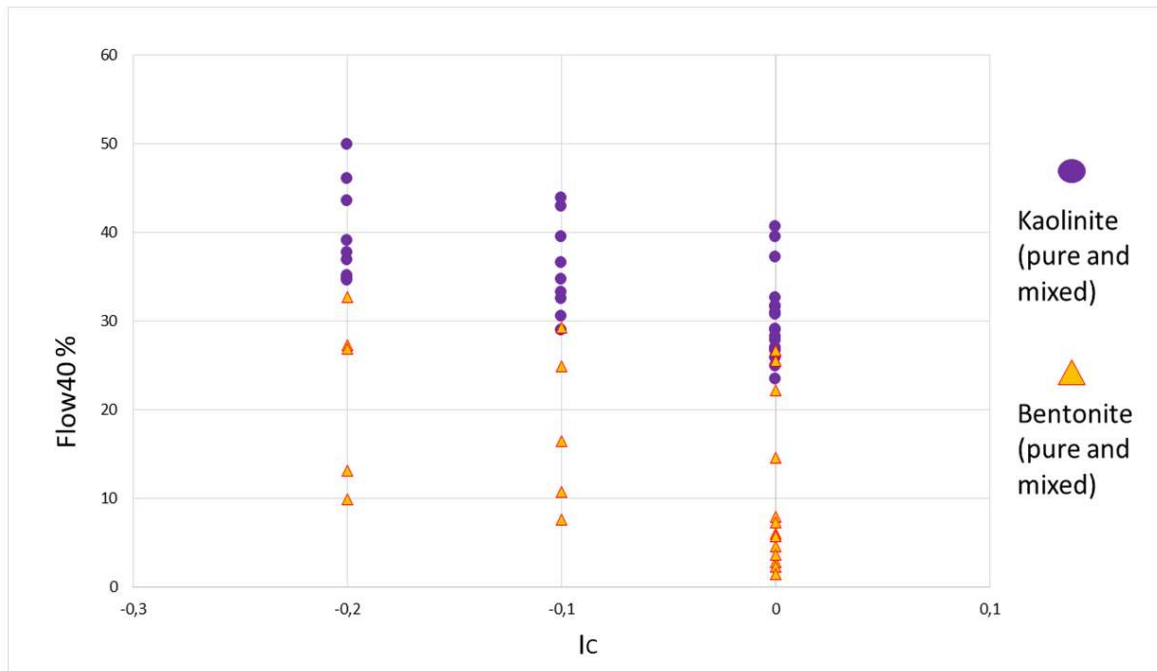


Figure 5-10. Comparison between all the tested samples with three different I_{Cs} (0.0, -0.1, and -0.2), for kaolinite samples, pure and mixed with different sand grains (purple circles), and for bentonite samples, pure and mixed with different sand grains (orange triangles).

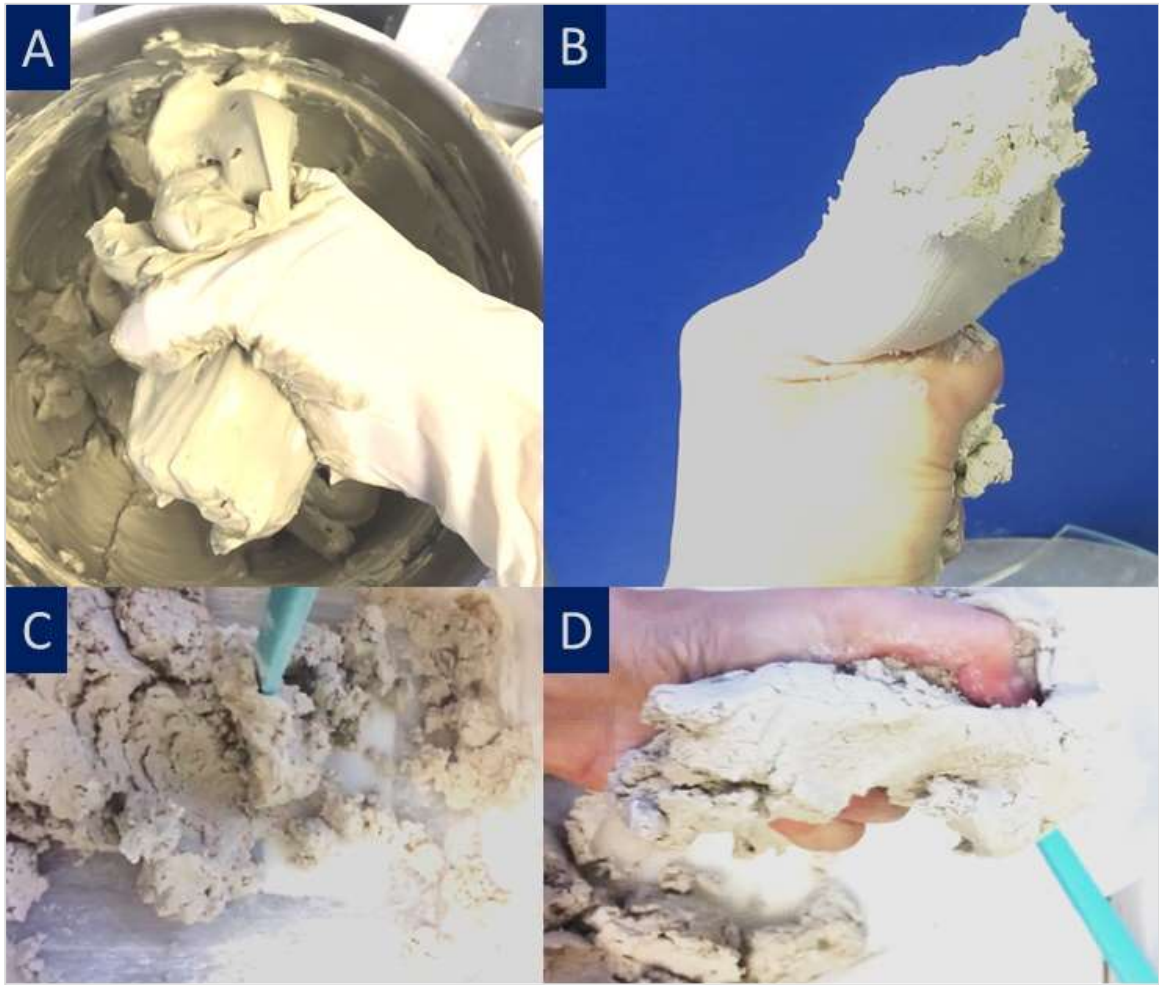


Figure 5-11. Comparison between two pure clay samples, bentonite (A) and kaolinite (B), with similar consistencies, $I_C = 0.43$ and 0.44 , respectively, showing what would be the lowest boundary of the ideal EPB range. The samples shown in (C) and (D), with $I_C = 0.4$ but with only 30% of kaolinite, are mixed with fine sand. The feeling was that the mixed sample was too dry, being difficult to even mix.

This can be seen in the tests conducted with the rheometer. Even though with a much lower consistency, all the kaolinite samples had a higher shear strength than the bentonite samples, as shown in Figure 5-12. All the samples that were tested with the BMS rheometer were built with the consistency index of -0.8 owing to the limitations of the testing device and the high yield stress values for the samples with 30% of kaolinite. This can be observed in Figure 5-12, with the orange straight line.

Surely, this is the main limitation of the test, considering that the ideal range of consistency for an EPB is between 0.4 and 0.75, as stated by Maidl (1995). However, these results could still be applied to the EPB muck pumping system, which implies more liquid consistencies.

Figure 5-12 shows that K30% presents the highest yield stress and the highest shear stress values with the increase in shear rates. The other kaolinite samples seem to have similar values of shear stress with the increase in shear rate, when compared to the bentonite samples. For the bentonite samples, the pure clay sample (purple in the graph) showed the lowest shear strength, while the samples with 50% and 70% of bentonite content had the highest shear strength, precisely those that had shown before, in Figure 5-6, the lowest flow values.

The conditions shown at the graph in Figure 5-12 were unsheared, meaning the first round of the BMS, which had higher values than the average of the second to the sixth rounds, shown on Figure 5-13, with kaolinite (top) and bentonite (bottom).

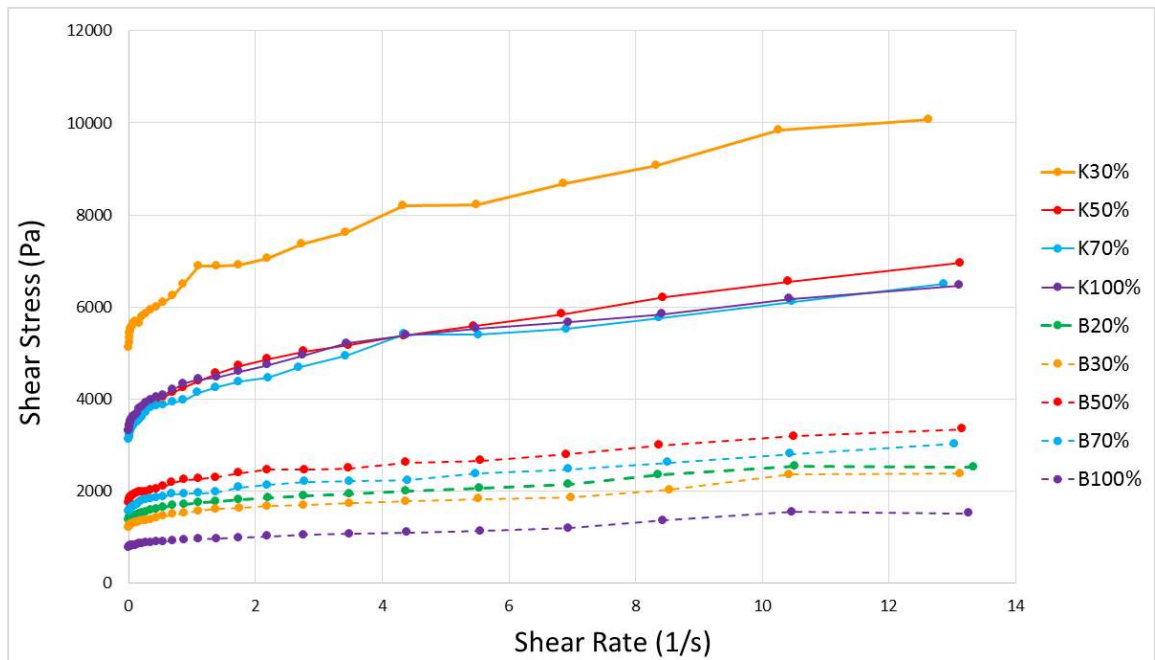


Figure 5-12. Rheometer for Kaolinite and Bentonite samples with different % of fine sand, unsheared (1st round), for $I_c = -0.8$.

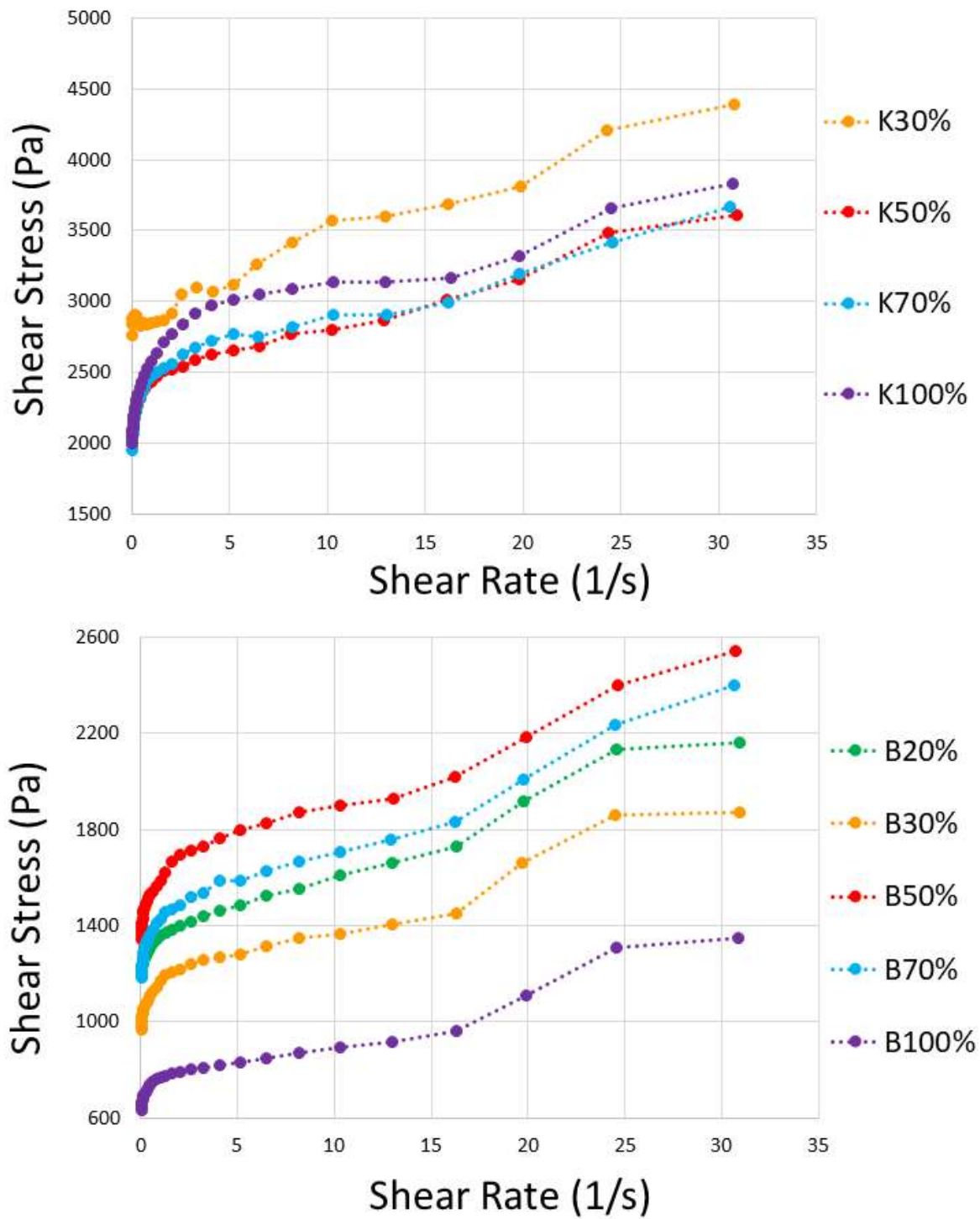


Figure 5-13. Results from round 2 to 6, sheared, with the rheometer, for kaolinite (top) and bentonite (bottom) samples, at $I_c = -0.8$.

It is interesting to observe that for the Ksfine30% there are local changes in the shear stress curve, with higher peaks showing a certain resistance to flow, requiring a higher shear stress, which could mean a local change in the shearing behaviour.

Figure 5-14 shows the variation of viscosity values versus shear rate (top) and clay fraction (bottom), for sheared conditions. As seen in the top graph in Figure 5-14, it is possible to say that viscosity decreases with the increase in shear rate, thus behaving as a non-Newtonian fluid, with shear-thinning behaviour. For kaolinite, besides K30%, all the other values are quite similar, without any significant changes. However, the differences are too low, considering the measured values of viscosity, which might not hold any significant meaning.

Tests with bigger grains could not be conducted with the rheometer (Kfine30 and Kmedium30). The samples were assembled but the rheometer would not rotate most of the time. For Kfine30 it was only possible to conduct the test in some of the rounds. Only with even lower consistencies, it would be possible to run tests with bigger grains. Nevertheless, it is possible to expect that bigger sand grains imply higher shear stress values, the same as observed in terms of flow values with the flow table.

For the bentonite samples, it could be possible to make a correlation with the flow results, especially when looking at Figure 5-14, at the bottom, showing the clay fraction. The highest values of viscosity, approximately 50% and 70%, are also the lowest flow values, as observed in Figure 5-6. Flow tests were also conducted with the bentonite samples tested in the rheometer, thus holding a consistency of -0.8 , and the same results as those of samples with consistency of 0 were obtained.

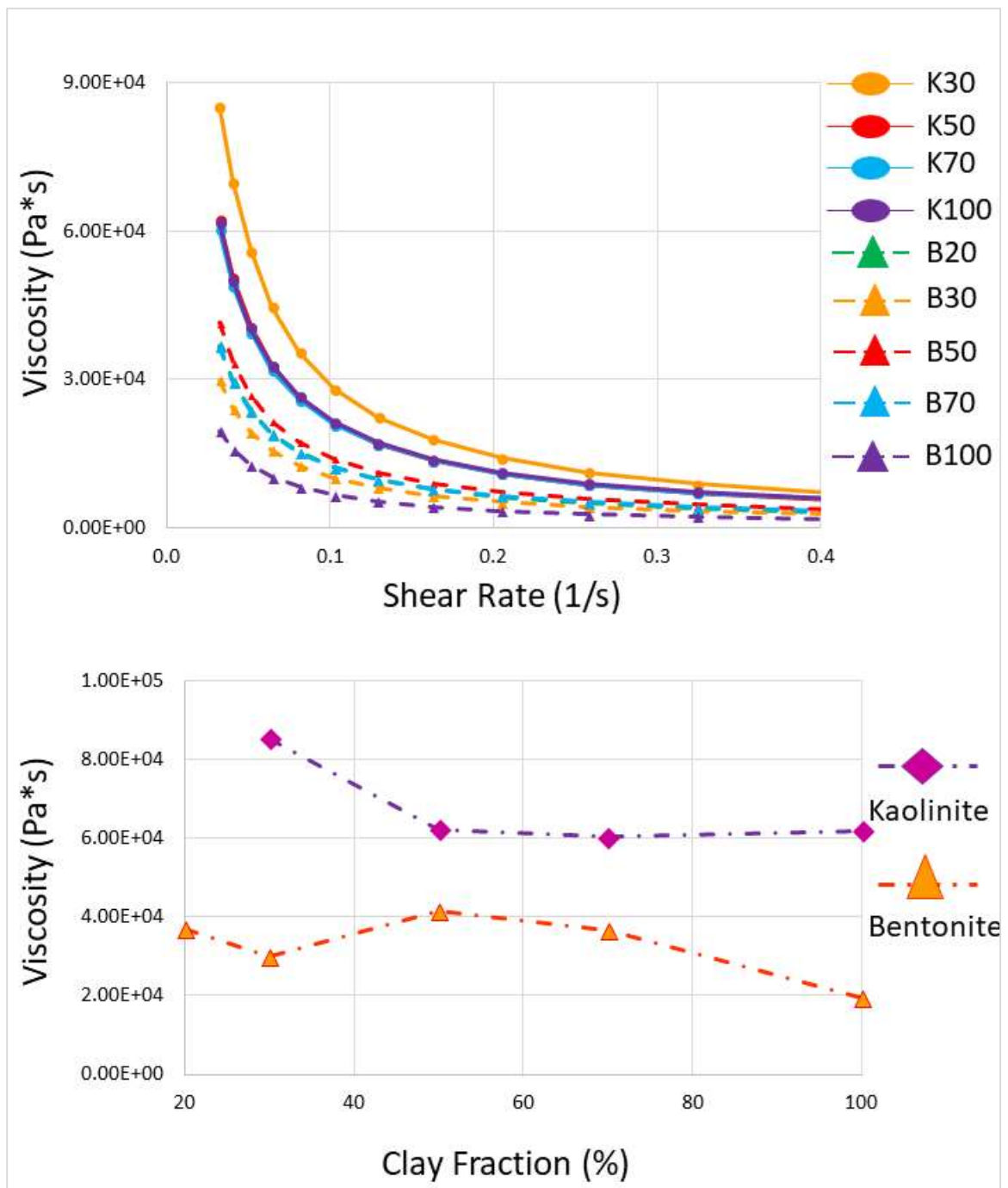


Figure 5-14. Viscosity versus shear rate for sheared conditions (top). Samples with straight line are kaolinite, and with dashed line are bentonite. In the figure at the bottom, the viscosity values for each sample were plotted against the clay fraction (kaolinite with diamonds and bentonite with triangles) for sheared conditions.

Rheological tests of samples with a higher consistency index should be conducted with another rheometer that allows higher yield stress values, and the samples could be then compared to an I_C closer to the EPB required conditions. Furthermore, tests with foam and polymers could improve the understanding of the behaviour of these mixed samples when conditioned. However, the methodology of mixing and testing the sample with enough time before the foam separates might be a challenge for the more cohesive samples.

Slump tests were also conducted to compare them with the results of the flow table. Initially, it was tried to measure the height of the cone for the flow table, as it was done with the slump test, but it was not feasible with the current device, at least with accuracy, and owing to the size of the sample, any error would have a much higher influence than any error with the slump test. This could be considered in the future for assembling of a modified flow table to provide accurate results of the slump height.

Then, the slump flow was considered and compared with the flow table results, which in this case must be done with 15 drops of the table, as the samples were too liquid to justify 40 drops. The values of the slump flow were transformed in percentage, comparing the initial slump diameter (200 mm) with the final diameter. The $Flow_{40}$ was also measured, but the samples were completely spread over the table, without showing any differences between them.

The samples tested had 10%, 20%, and 30% of clay and the results are shown in Figure 5-15. For the sample with 10% of clay, without any foam, there was excessive loss of water (Figure 5-15c), which affected the accuracy of the results, with bigger standard deviations and without a good correlation with the percentage of the slump flow (Figure 5-15a). This also shows the benefits of adding foam for low cohesion, or even cohesionless, soils, to decrease the permeability of the sample. If water is replaced by foam, not much liquid loss should be expected.

Only the results with 20% and 30% of clay were considered (Figure 5-15b). It is possible to see a good correlation between the flow of the slump and the flow of the table. For samples of 20% and 30% of clay, as the clay fraction increases, the flow diminishes, which was already observed in all the other tests. However, the consistency index of the sample with 30% of clay was minus 0.65 ($I_c = -0.65$) for the slump height of 10 cm, and minus 1.61 ($I_c = -1.61$), for the slump height of 20 cm.

The samples were touched, and a video was produced showing a compilation of the tested samples (see Video 5-1). By visualising that, it is possible to say that, even for 30% of clay, the samples were too liquid for a good muck consistency inside the EPB excavation chamber. However, for the consistencies of 0.4, for kaolinite mixed soils, the samples were too dry for an EPB operation, as showed in Figure 5-11. This brings back the question, for low cohesive soils, e.g., with 30% of kaolinite content mixed with sand, of how to assess the desired workability of the muck, with the slump test or by considering the consistency index?

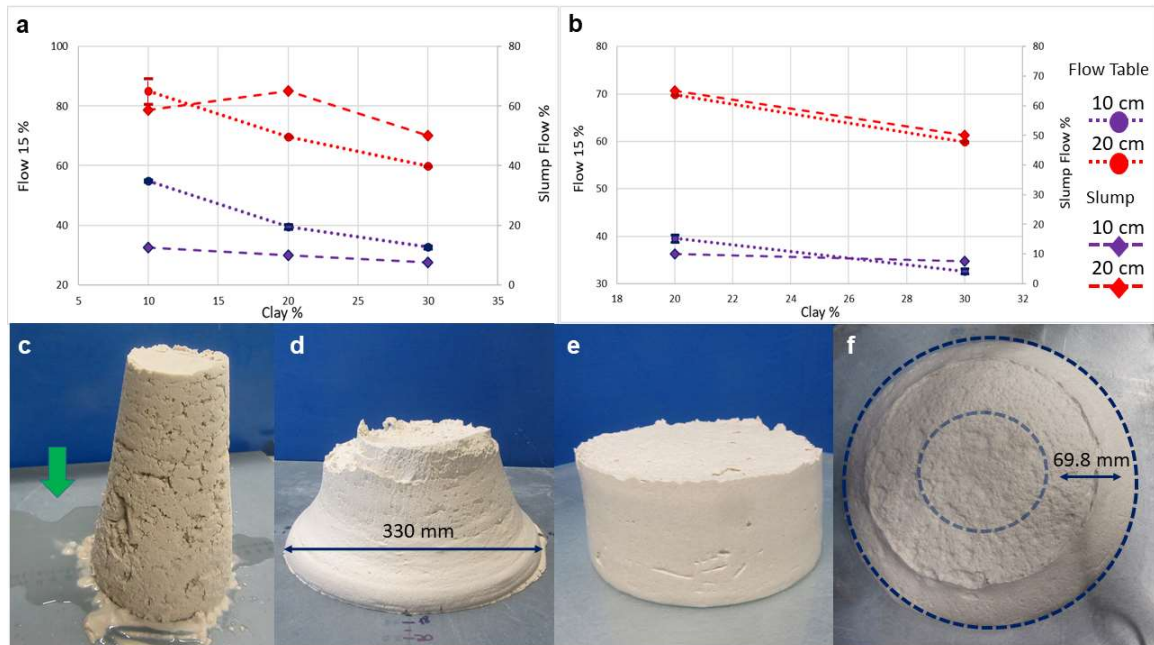


Figure 5-15. Comparison between slump and flow table for mixed clay–sand samples. (a) and (b) show the graph of flow 15 versus clay content (on the x axis) together with the slump flow, in percentage. The issue with water loss for samples with 10% of clay is shown in (c), which includes an example of slump with 20%, for slump height of 20 cm. Tests were conducted with the flow table as shown in (e), before jolting, with the same samples as in (d) (no changes in height without jolting), and after jolting in (f), showing the total spread for 15 drops of the table.

Finally, kaolinite samples mixed with “sfine” sand, with different clay percentages (30%, 50%, and 70%) were tested with three different conditioning products: products A, B, and C, as described in the section 5.3.2, and Tables 5-3 and 5-4. There was an increasing difficulty in conducting the tests as the clay percentage increased, owing to mixing issues with a more cohesive sample. Figures 5-16, 5-17 and 5-18 present all the results of the tested samples, always compared with a sample without any additive, only water (blue circle). All the samples were tested with the same consistency index (0.0), having similar water and foam content, and for the same clay percentage. Therefore, the conditions tested for each clay percentage were similar for each product, to assure a fair comparison between them.

Video 5-2 shows the compilation of these tests, which allows to have a better visualisation of the effect between the different products, along with an idea of the muck strength and its conditions when squeezed with the hands.

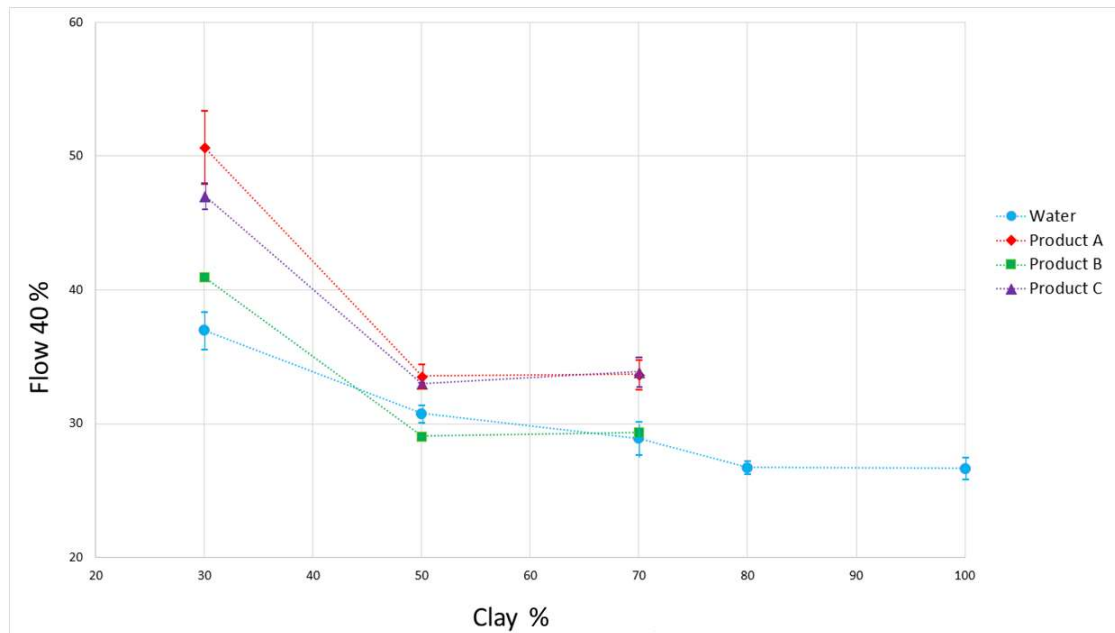


Figure 5-16. Flow₄₀ for Kaolinite with fine sand and different additives.

In Figure 5-16, it is observed a small change in flow for products A and C, which can surely improve the flow of a clayey soil along the conveyor belt. Product B, nevertheless, has a very similar flow when compared with the sample with only water, which does not necessarily imply that it has no other beneficial function when added to the soil, such as functioning as a clay aggregate dispersant, for example, diminishing clogging occurrence.

Figure 5-17 shows the bulk density and void ratio of the tested samples. Density is the main information regarding the muck and filling grade inside the excavation chamber. The sensors inside an excavation chamber in a shield machine do not measure consistencies, but pressures. The difference between the measured pressures, in different levels, allows apparent

densities to be calculated. Therefore, densities are the guidance for any change in the properties of the muck inside the excavation chamber. Surely, the homogeneity, or the lack of it, cannot be detected but, in general, the adjustments to achieve a desired muck state inside an excavation chamber will be performed based on pressure changes, and consequently, based on apparent densities.

The void ratio, as expected, increases with the inclusion of foam and foam with anti-clay polymers, as more air is inserted in the material, which is one of the benefits to add foam to the excavated material, as it increases the compressibility properties of the muck. As mentioned by Zumsteg et al. 2012, the higher the compressibility is, the smaller the pressure changes in the excavation chamber, thus assuring proper control of the excavation. Moreover, in terms of void ratio increase, product A seemed to be slightly more effective, for 30% and 70% of clay fraction, than the other two products.

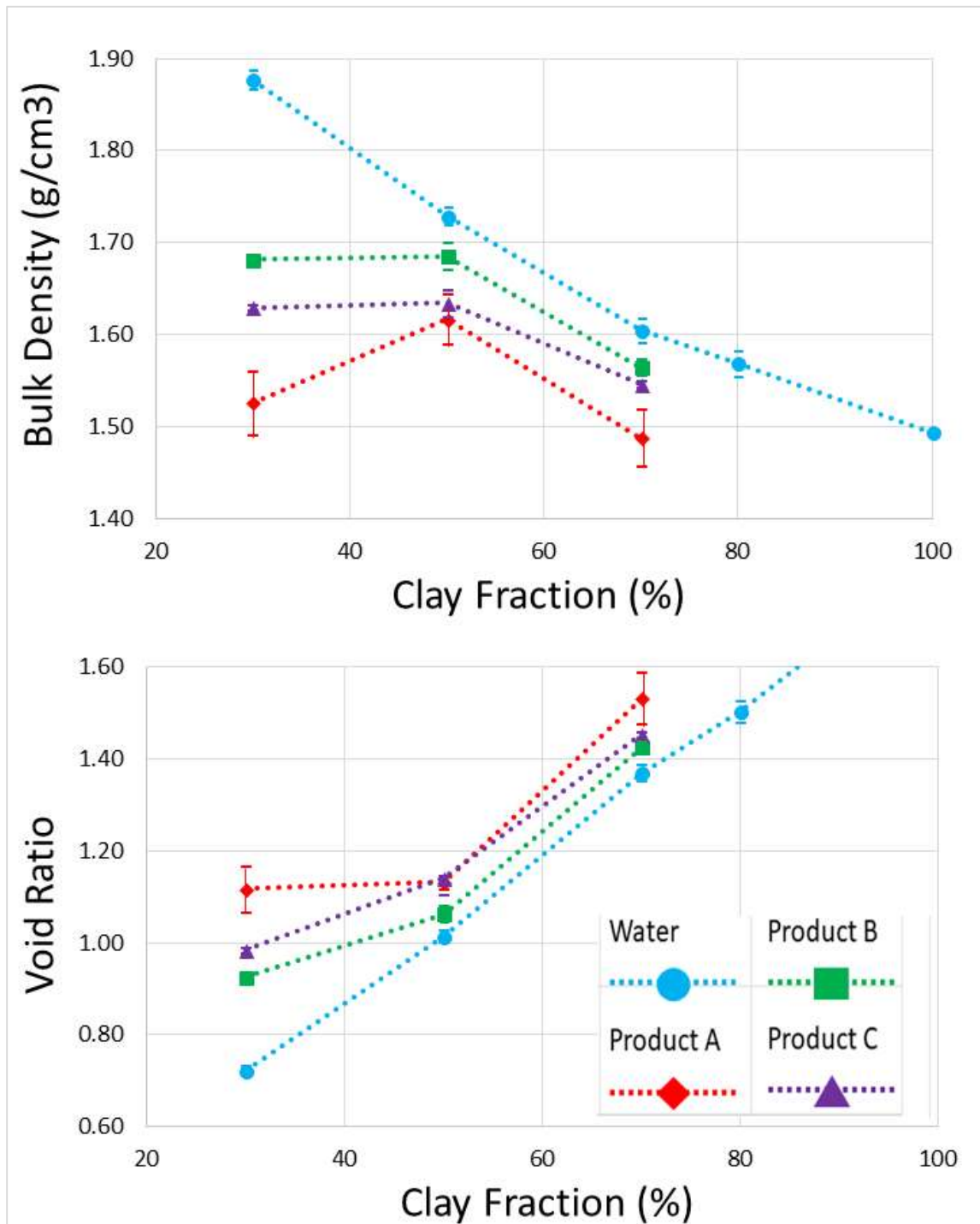


Figure 5-17. Physical indexes for the samples tested with water, then compared with Products A, B and C: top – bulk density; bottom – void ratio.

In Figure 5-18, with 30% of kaolinite, a higher FER seemed to produce a “better quality” of soil mixture (see Figure 5-18c and d), but it is difficult to say if it is better or worse for EPB operation, as the effect of the pressure inside the excavation chamber needs to be considered. This observation can be better visualised in the complementary video (Video 5-2).

The addition of product C in all the samples leads to pieces of material being stuck in the mould (Figure 5-18e), even with a high amount of oil coating in the cone surface, as described in the procedure (section 5.3.3). Product C is supposed to have anti-clay properties (polymer with dispersant functions of the clay aggregates). Thus, and most probably for kaolinite, this product would be functioning in quite the opposite way, increasing the amount of soil stuck in the metal surface. Even for samples with only water, it was not observed any stuck material in the cone, as it happened with product C.

In terms of FER values, the main difference was visual for the 30% sample. The values of void ratio did not have significant differences: the higher FER with product A resulted in a smaller void ratio, whereas for product B, with anti-clay, it was the opposite, and the same for the flow. The flow seems to go along with the void ratio, and the higher the void ratio is, the higher the flow. In terms of density, the highest values of bulk density corresponded to the lowest flows (sample with water), while the lowest values were related to the highest flows (sample with product A, only foam). Again, it is noted how useful it is to include the values of densities to guide not only the laboratory tests but the machine operation in order to face any changes in the properties of the muck.

The samples with the lowest clay fraction, 30%, were the ones with the highest influence of the foam products and with the most significant changes, showing that the larger the sand fraction of a material, the bigger the influence of adding foam, and vice-versa.

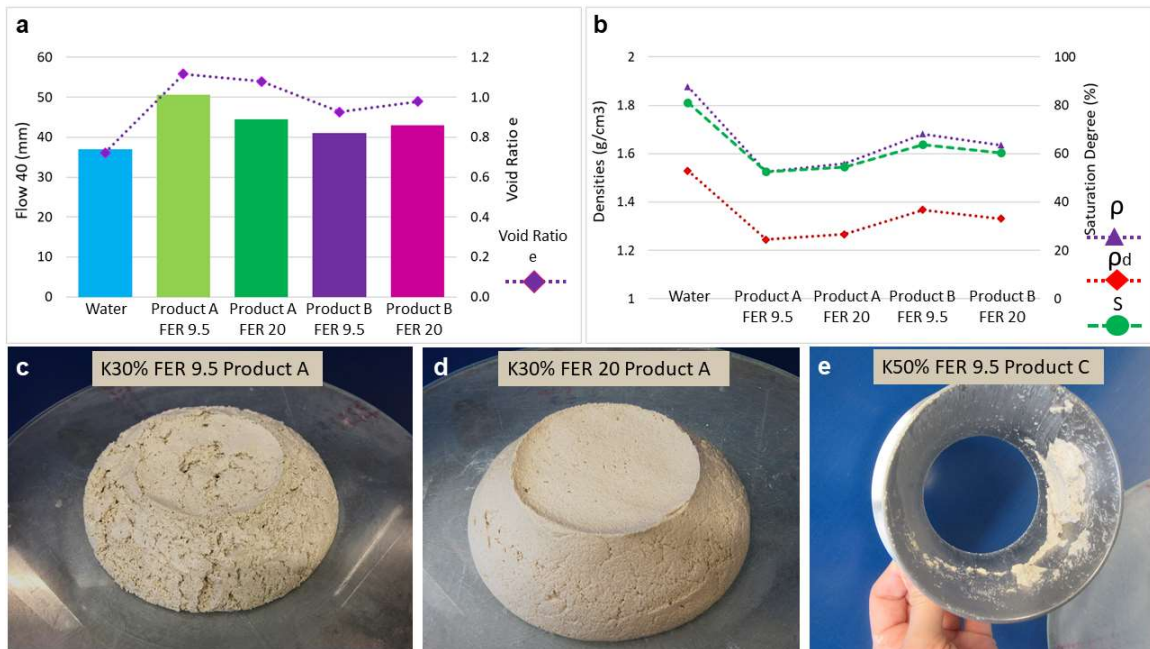


Figure 5-18. Flow of samples with 30% of kaolinite and two different target FERs (9.5 and 20) compared with their void ratio (a); bulk (ρ) and dry (ρ_d) densities compared with saturation degree (S) in (b); (c) and (d) compare the differences between samples for different FER values; and (e) shows the soil stuck in the mould after lifting it, using product C.

5.5 Final Considerations

Several considerations can be drawn from all the conducted tests, regarding the flow behaviour, the implications in EPB excavation and operation, and the methodology to evaluate these characteristics for mixed soils in the laboratory, including the tests limitations. It is essential to consider that any observation made here must be verified in the future with real tunnel drives. Ideally, tests should be conducted before a project is started and later, as the excavation proceeds, more tests should be conducted to elaborate a back-analysis of the results. Thus, it would be feasible to optimise the entire process of EPB excavation and soil conditioning, undoubtedly raising its efficiency and diminishing the operational costs.

Even though the intention in this study was primarily to focus on the soil matrix of the mixed transitional ground and on the residual soil, and to model its two main simplified components, i.e. quartz sand and clay mineral (kaolinite and bentonite), the observations here presented can be extended not only to most of the mixed soils, but to homogenous soil types in terms of laboratory testing for EPB soil conditioning and flow behaviour.

As explained by Trask and Close (1957), for cohesionless soils, the strength will depend on the friction of the clastic particles, whereas for cohesive soils, the strength depends mainly in the forces between the clay particles. Relating this to the results, it is possible to conclude that pure bentonite samples, with lower force between the clay particles (weak Van-der-Waals bonding), have the lowest shear stress versus shear rate values, when compared to pure kaolinite, which has the strongest bonding between the clay particles (O-OH hydrogen bonding).

In mixed samples, not only this clay particle bonding will influence the strength, but also the sand grain friction. With low kaolinite values, the friction of the sand added to the strong bonding of the kaolinite clay particles raised the yield stress when compared to the same clay fractions for bentonite, which has a much lower clay bonding. For all the samples, independently of the clay fraction, kaolinite always had higher flow values and higher yield stresses than bentonite.

In terms of consistency index and the role it plays in an EPB excavation and operation, it seems that the idealised consistency range is appropriate for pure clays, kaolinite and bentonite, but it seems to deviate for the samples with mixed kaolinite and sand. This should also be investigated further, especially testing samples obtained directly from the screw conveyor.

The results obtained from the slump test did not seem realistic for the EPB desired muck. For the tested mixed soils, it seems that neither of the options (10 and 20 cm) looks like a suitable material. If any consistency index would be guessed, at least for the samples with kaolinite mixed

with sand, but not for the pure clay sample, it would be around its liquid limit, as the consistency of 0.4 already seemed too dry for EPB operation (see Video 5-2). Therefore, to validate these assumptions concerning the mixed soils, tests with a flow table should be conducted with the muck extracted from the conveyor belt of an EPB machine; then, a realistic ideal consistency range could be drawn for the excavated ground.

In terms of laboratory methodology, after conducting over hundreds of tests, it is possible to conclude that the flow table is an easy and practical method to test samples with cohesive soil, mixed or not with other grains and with grain sizes up to 2 mm (up to coarse sand), and with conditioned soil. The required amount of soil is significantly lower (between 400-600 g) than the tests conducted with the slump test (between 10-13 kg), and the methodology is less physically demanding. The tests can easily be conducted even before the tunnel is excavated, just by testing borehole samples. Moreover, it would be easier to install a flow table inside the EPB machine and conduct tests with the excavated muck, and then proceeding to a back-analysis of the preliminary results, before the excavation begins.

Surely, limitations have been found in the tests with the flow table, and the first one was the challenge to conduct tests with consistencies above 0.4, which is the ideal EPB range, as stated by Maidl (1995), especially tests with pure clay in this consistency. Nonetheless, mixed soils seemed not to be a problem for conducting tests with this consistency index, or even higher, making the flow table a good asset for the characterisation of cohesive mixed soils, especially to test different conditioning systems and additives. It should also be considered the EPB muck pumping system, which requires more liquid materials; therefore, the flow table is an effective evaluation method.

Another issue with the flow table is how to conduct tests with samples containing grains larger than 2 mm. In this case, slump tests could be conducted instead. It is also necessary to look

for correlations between the slump flow and the flow table, even for cohesionless conditioned soils, which could improve and simplify the testing methodologies of materials and conditioned materials for a shield excavation, providing a non-static assessment of the flow behaviour of soils.

The used rheometer does not address the real needs of a cohesive soil in the context of an EPB operation. Another device should be tested to analyse the rheological behaviour of those samples. Besides the EPB muck pumping system, the results obtained with this rheometer are not relevant to a regular EPB operation, as the samples have to be investigated in an excessively liquid consistency.

Regarding soil additives and soil conditioning, preliminary tests should always be conducted before any additive or polymer supplier is chosen, to serve as a guidance of the effectiveness of a soil conditioner. As it can be observed in the test results, if clogging could be a problem along the project, product C would probably increase the clogging issues. No perfect or ideal products will function for all the types of ground.

Ideally, a preliminary testing campaign should be conducted to test the flow behaviour of the soils to be excavated and its interaction with conditioning additives, avoiding the trial-and-error approaches. The flow table can be a useful asset to cover this need, making it feasible to investigate the material obtained from boreholes or shafts, prior to the excavation. The densities should always be included, which will later be the most useful data to adjust the muck conditioning inside the excavation chamber.

Another aspect is that all the tests have been conducted at atmospheric pressure, which is not the actual condition in tunnel drives. This is still a subject for further investigation, and it has already been noted by Mooney et al. (2016).

Finally, concerning the tropical residual ground, it should be taken in consideration that this ground behaves differently than the regular sedimentary pure, or close to pure, clay layers,

especially concerning the flow behaviour when excavated by EPB machines. Many of the earlier assumptions about the properties of a cohesive material when excavated by shield machines had as case studies sedimentary soils, not the tropical residual ground. Nowadays, with the increase in mechanised tunnelling projects in tropical areas, e.g., Singapore, Brazil, China, Hong Kong, Malaysia, and other locations, it is essential to extend the investigation into this particular type of ground and question which of the earlier assumptions can, or cannot, be extended to the tropical ground.

5.6 Acknowledgements

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Chapter 6

Clogging and flow assessment of cohesive soils for EPB tunnelling: laboratory methodology and soil characterisation

Abstract

For earth pressure balance (EPB) machines, it is fundamental to characterise the excavated ground in its natural and conditioned states, as the material characteristics have a direct influence on machine operation, impacting the functions as a support medium and maintaining the support pressure ahead of the face. Some proposals for the assessment of workability and clogging potential of cohesive soils already exist; however, they do not focus on mixed soils, but mainly on pure sedimentary clayey soils. Mixed soils exist widely in nature, including a wide variety of residual soils, which are by-products of intense tropical weathering. This paper presents a combined laboratory routine to characterise and evaluate the clogging and fluidity of soils and conditioned soils, including mixed soils by considering different clay fractions. By testing clay–sand mixed soils assembled in the laboratory and investigating the influence of different contents of clay, water, and additives, for two different clay minerals and grain sizes, conclusions regarding cohesive mixed soils were achieved. The focus of this study is on mixed cohesive soils, representing residual soils; however, the work can be also applied to any type of soil that has any clay particle content, implying a cohesive behaviour. More than looking into the conclusions about the soil mixture itself, the main intention here is to recommend a laboratory routine to characterise these soils, correlating the results with an EPB machine excavation.

6.1 Introduction

The characteristics and behaviour of a conditioned ground play a fundamental role in the entire operation of earth pressure balance (EPB) machines. This conditioned material can be characterised by the excavated ground itself combined with any form of additives, such as water, foam, polymers, and fines. This material, called muck, has to contribute to sustain the pressure ahead of the machine, combined with the advance and retraction mechanisms of the hydraulic jacks and the rotation speed of the screw conveyor. Additionally, it needs to hold certain characteristics to be successfully transported and disposed. Several characteristics of muck that might be beneficial for face support or flow inside the chamber and screw conveyor can result in challenges for its transportation and disposal (EFNARC 2005, Herrenknecht et al. 2011, Maidl et al. 2012, Thewes et al. 2012, Galli and Thewes 2014).

Even though cohesive soils lie within the ideal operation field of EPB machines (BTS 2005, Maidl et al. 2012), some particularities of this soil type still cause certain challenges for tunnel operation, clogging being one of them (Thewes 1999, 2004, Sass and Burbaum 2009, Feinendegen et al. 2011, Hollman and Thewes 2012, 2013, Peila et al. 2016, Basmenj et al. 2016, Khabbazi et al. 2017). Moreover, the flow behaviour of soils should be available for any EPB excavation, as it will have a great influence on the operation, affecting the support pressure, machine parameters, such as the torque of machine and screw conveyor, as well as transportation and disposal (Galli and Thewes 2014, Galli 2016, Galli and Thewes 2018).

Both characteristics, i.e. clogging potential and flow behaviour, should be examined in combination, as they could influence each other. An adopted solution to decrease clogging should not turn into an issue for the flow of a certain excavated material, for example, being extremely liquid to cause transportation problems later.

Several laboratory tests and methodologies have been proposed for evaluating, directly or indirectly, the clogging potential (Thewes 1999, 2004, Sass and Burbaum 2009, Spagnoli et al. 2010, Feinendegen et al. 2011, Hollman and Thewes 2012, 2013, Zumsteg and Puzrin 2012, 2013, 2016) and flow behaviour by means of slump and rheometer devices (Quebaud et al. 1998, Leinala et al. 2000, EFNARC 2005, Peila et al. 2008, Borio 2010, Peila et al. 2013, Galli and Thewes 2014, Peila et al. 2016, Galli 2016, Galli and Thewes 2018).

Most of the research, as pointed out by Oliveira et al. (2018a, b), focused on either sand or clayey soils mainly originating from sedimentary deposition. However, mixed soils also exist frequently in nature, such as the tropical residual ground, which can be found at great depths in tropical areas (Vargas 1980, Rahardjo et al. 2004, Viana da Fonseca and Topa Gomes 2010, Huat et al. 2012, Oliveira et al. 2017a, b). EPB excavation along this type of ground, especially when including mixed transitional ground, as defined by Oliveira and Diederichs (2016), can lead into major excavation issues, as pointed out by several authors (Della Valle 2001, Thewes 2004, Zhao et al. 2007, Shirlaw 2015, 2016, Comulada et al. 2016, Gong et al. 2016, Peixoto da Silva et al. 2017, Oliveira and Diederichs 2016, Oliveira et al. 2017a, b).

Oliveira et al. (2018a), after testing assembled soils by mixing clay and sand in different proportions to simulate the basis of a residual soil in the laboratory, proposed tests to evaluate the clogging potential and flow behaviour of those mixed soils. The clogging potential was a modification from the original method proposed by Zumsteg and Puzrin (2012). In the original methodology, Zumsteg and Puzrin (2012) used a HOBART mixer and defined a clogging parameter (λ) by comparing the amount of soil stuck in the mixing tool (MT) and the total amount of soil mass.

Oliveira et al. (2018b) demonstrated that this method, as originally proposed, would lead into an overestimation of the clogging potential, especially for the case of mixed soils. In

response to this issue, the authors included an additional step done by dropping the MT from a certain height, to a certain surface, by means of a newly assembled device, ATUR. Subsequently, a flow table was used to evaluate the flow behaviour of mixed cohesive soils, resulting in the proposed method described by Oliveira et al. (2018c). Results obtained from the flow table, slump tests, and rheometer were compared and some assumptions about flow behaviour of soils, depending on the clay percentage and mineral, water content, grain size, and additives, were made.

This present study combines all the above-mentioned methods, the clogging and flow evaluation, adding more stages to it to characterise cohesive soils, mixed or not, and correlate the results with the expected behaviour ahead of an EPB machine excavation. This laboratory routine intends to provide a method of evaluating tendencies for clogging and changes in flow behaviour of any cohesive soil, with or without conditioning, offering a possibility for laboratory investigation prior and during an EPB tunnel excavation.

Considering that some of the tests included foam, the main foam parameters, namely foam concentration (C_f), foam expansion ratio (FER), and foam injection ratio (FIR), are defined here according to EFNARC (2005).

Foam concentration (C_f) – Equation 6-1:

$$C_f (\%) = 100 \times (m_{Surfactant} / m_{Foam Solution}) \quad (6-1)$$

Where:

$m_{Surfactant}$	mass of surfactant in foaming solution
$m_{Foam Solution}$	mass of foaming solution

Foam expansion ratio (FER) – Equation 6-2:

$$FER = V_{Foam} / V_{Foam\ Solution} \quad (6-2)$$

Where:

V_{Foam} volume of foam

$V_{Foam\ Solution}$ volume of foaming solution

Foam injection ratio (FIR) – Equation 6-3:

$$FIR (\%) = 100 \times (V_{Foam} / V_{Soil}) \quad (6-3)$$

Where:

V_{Foam} volume of foam

V_{Soil} volume of tested soil

6.2 Material and Methods

In this section, the authors describe the assembled soils and the laboratory methodology sequence. Some testing limitations are also pointed out.

6.2.1 Tested Material

To assure reproducibility of the tests and control of all variables, the soils were prepared in the laboratory, including three main mineral products: quartz sand, kaolinite, and bentonite. All the main characteristics of these three elements are detailed in Table 6-1.

There were a couple of reasons for choosing kaolinite and bentonite as tested clay minerals. First, they represent the extreme spectra of plasticity index for clay minerals, kaolinite being a considerably low plasticity clay, and bentonite, a highly plastic one. Second, kaolinite is the main clay mineral present in tropical residual ground (Vargas 1980, Huat et al. 2012), justifying the choice of this mineral and a slightly higher number of tests with it. Bentonite, because of its swelling properties, is expected to be a bigger challenge for shield tunnel drives.

Table 6-1. Main characteristics of each component of assembled soils.

Product Specification	Supplier	Spec. Grav.	Chemical Comp.	Specific Surface (m ² /g)	Cation Exc. Capacity (meq./100 g)	pH	Moisture (%)
Silica Sand	Euroquarz	2.65	SiO ₂	-	-	6.5-7	-
Kaolinite GHL KS 80*2	Georg H. Luh GmbH	2.62	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃	18.2	-	4.5-9.5	0.6-2
Active Bentonite IBECO B1	Imerys Civil Engineering	2.65	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃ , MgO, CaO, K ₂ O, Na ₂ O	600-800	70±10	10	11±3

A total of 9 sample combinations were assembled, with different water contents and, for some of the tests, with different conditioning products. Table 6-2 presents all the sample combinations with proportions of minerals and their Atterberg values. The sand portion of the samples were assembled with two different sand gradations, as illustrated in Figure 6-1. Each sample combination, for each specific water content, was tested for a minimum of three times. Most of the time, a high amount of soil was prepared (approximately 2 kg of wet soil), using the same specimen for several tests, which guaranteed testing the same consistency always.

Regarding the conditioning additives, it was chosen to keep the details partially undisclosed, replacing the names of products and their respective suppliers by letters (A, B, and C), specifying if it is only foam, or foam with anti-clay polymer, as indicated in Table 6-3. For all tests, Cf was always 3%, the FER target was always 9.5±0.3, and FIR was calculated considering

the moisture content necessary to bring the consistency of a soil down from a certain IC (consistency index) to another.

Once the first test was performed with a certain foam parameter, the variables were kept practically the same between all the tested products, to assure a fair comparison. All the tests with conditioned soils were performed with the samples containing 50% of clay (Ksfine50, Bsfine50) and all the variables can be accessed using Table 6-4. The final tests with Kmedium50 were not included in this table and will be described directly in section 6.3.4.

Table 6-2. Tested samples with proportions of minerals and Atterberg limits values.

Sample ID	Type of sand	Sand (%)	Kaolinite (%)	Bentonite (%)	W _L (%)	W _P (%)	IP (%)
Ksfine30	Sfine	70	30	-	22	16	6
Ksfine50	Sfine	50	50	-	32	21	11
Ksfine70	Sfine	30	70	-	44	29	16
K100	-	0	100	-	62	40	22
Kmedium50	Medium	50	50	-	31	23	8
Bsfine20	Sfine	80	-	20	86	27	59
Bsfine50	Sfine	50	-	50	171	31	140
Bsfine70	Sfine	30	-	70	245	34	211
B100	-	0	-	100	486	53	433

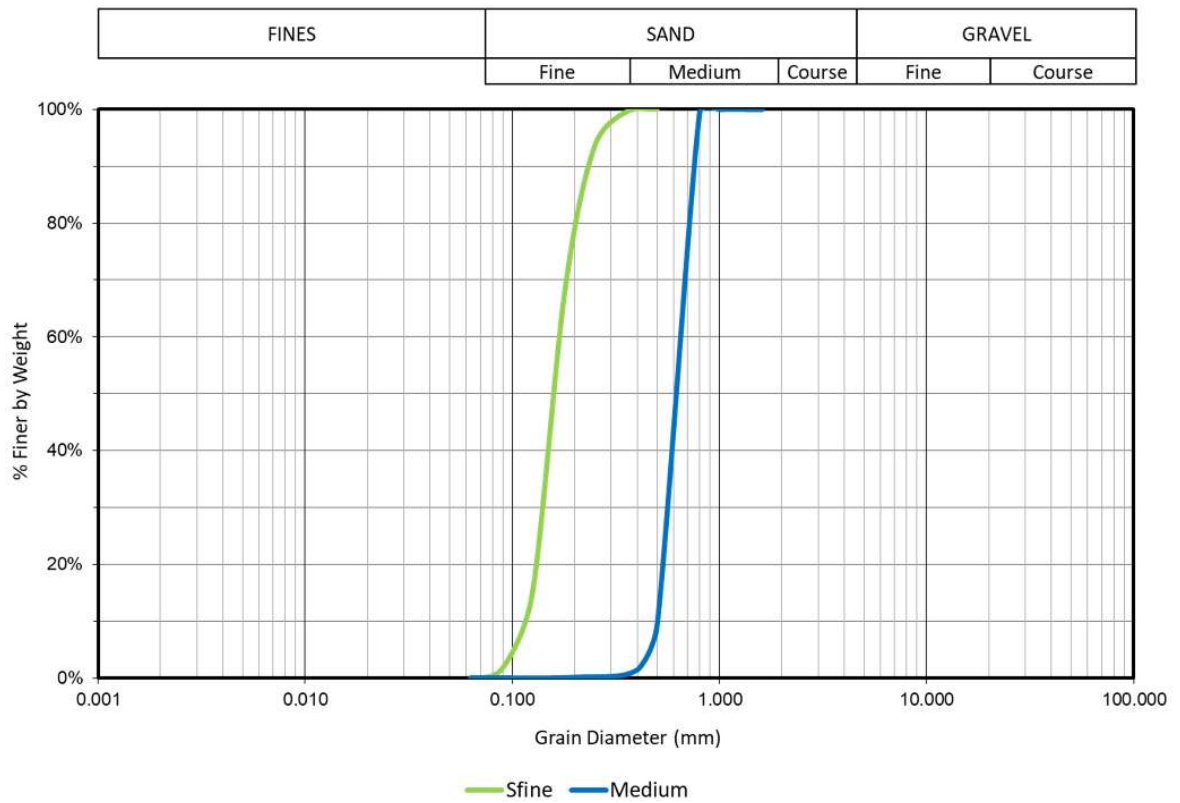


Figure 6-1. Grain size distribution for sand portion of soil mixture presented in Table 6-2.

Table 6-3. General description of additives used for tests with conditioned samples.

Product Name	Description
Product A	Foam
Product B	Foam with Anti-Clay Polymer
Product C	Foam with Anti-Clay Polymer

Table 6-4. Variables for tests with soil additives, including main foam parameters.

Soil Sample	Test ID	I _c Initial (only soil + water)	I _c Final ⁸ (soil+water+additives)	FER	FIR (%)
Ksfine50	1	1.33	0.81	9.6	91.5
	2	1.16	0.68	9.3	77.7
	3	1.05	0.54	9.4	84.5
	4	0.98	0.43	9.5	90.0
	5	0.50	0.02	9.6	86.3
Bsfine50	1	1.03	0.99	9.4	88.0
	2	0.94	0.88	9.4	114.5
	3	0.77	0.73	9.3	103.5
	4	0.71	0.68	9.3	80.5
	5	0.54	0.44	9.2	118.3

6.2.2 Proposed Methodology

The methodology routine is a product of an extensive testing campaign and has been modified to achieve better results with an effective procedure. Undoubtedly, there is still room for improvements, especially when tests are conducted with natural soils for tunnel projects, by crosschecking the results of preliminary laboratory investigation with the results obtained from tunnel drives.

In summary, the methodology combines a sequence of tests, starting with the preparation of samples, verification of the target consistency index, and test with the HOBART mixing methodology; this is followed by the sequential dropping of the beater with the ATUR device, then cleaning and washing the beater and bowl, the flow table test, and finally, the assessment of samples by squeezing them. Figure 6-2 illustrates the stages of this methodology and Figure 6-3

⁸ Values are an average from all the tests. The standard deviations are lower than 0.1.

provides a flowchart of the methodology sequence. Each stage now will be described in detail, allowing the work to be reproduced easily.

All samples were assembled initially as dried specimens. The target water content was added (deionised water) and after the mixing stage, soils were stored in closed containers (Figure 6-2a). For low plasticity specimens the moisture content was checked after 24 h of sample curing⁹ and for highly plastic samples ($IP > 40\%$), the samples were mixed a second time and we waited another 24 h to check the moisture content. If the target I_c was achieved, with a difference smaller than ± 0.04 , the tests were conducted. Otherwise, either dried soil or water was added to correct the consistency. The samples were mixed again, and we waited another 24 h before checking the moisture content. Some of the samples, such as the ones with higher content of bentonite, took longer than 72 h to be fully homogenised.

⁹ Sample curing means that the sample was given enough time for moisture homogenisation by simply waiting for a certain period.

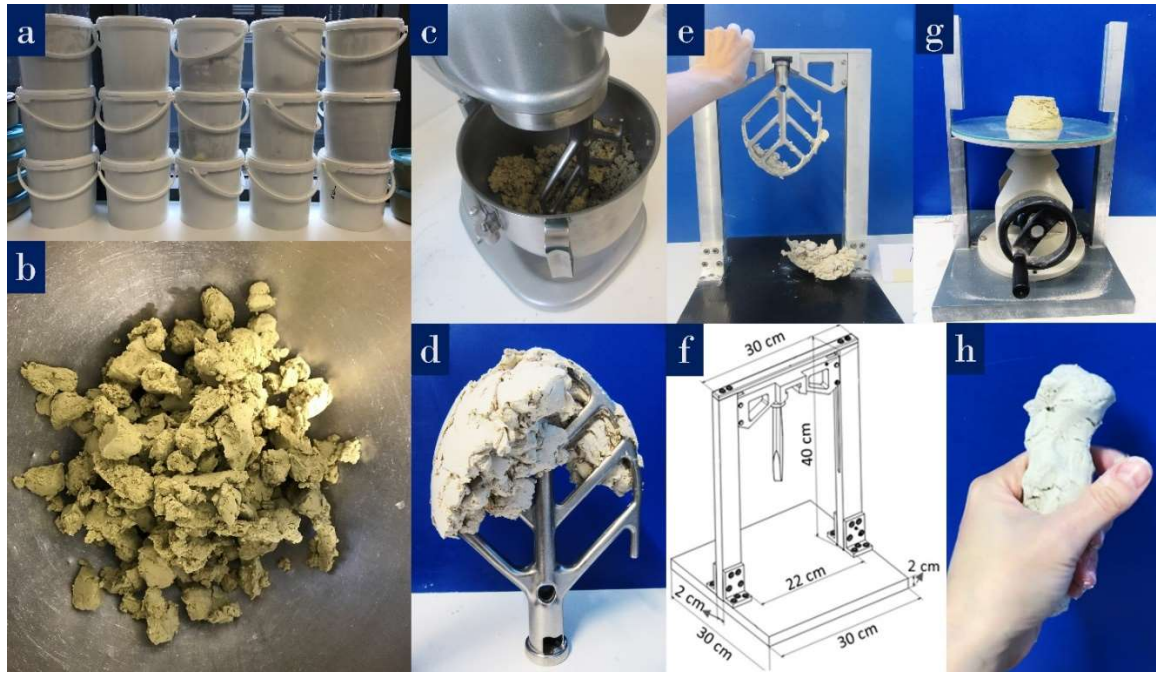


Figure 6-2. Illustration of parts of testing sequence: (a) soils were mixed with required mineral and water contents and stored in closed containers; (b) a reasonable quantity of sample was placed as lumps in the HOBART mixing bowl; (c) the HOBART mixer with soil specimen; (d) B-flat beater with stuck soil on it before any drop (λ_0); (e) B-flat beater placed in the device ATUR after one drop; (f) detail of the dimensions of the ATUR device; (g) flow table with soil specimen after 40 joltings of the table; (h) squeezing of the soil specimen to assess muck consistency and flow.

For calculating the consistency index, Equation (6-4) was applied, considering the values of plastic and liquid limits (W_p and W_L , respectively), together with the water content (W).

Equation 6-4:

$$I_c = (W_L - W) / (W_L - W_p) \quad (6-4)$$

Whenever a sample had grains with size above 0.425 mm, that portion was also included, which is not what is suggested by the Atterberg limits standard procedure (ASTM D4318 2017). The reasons for this modification were discussed in detail by Oliveira et al. (2018c).

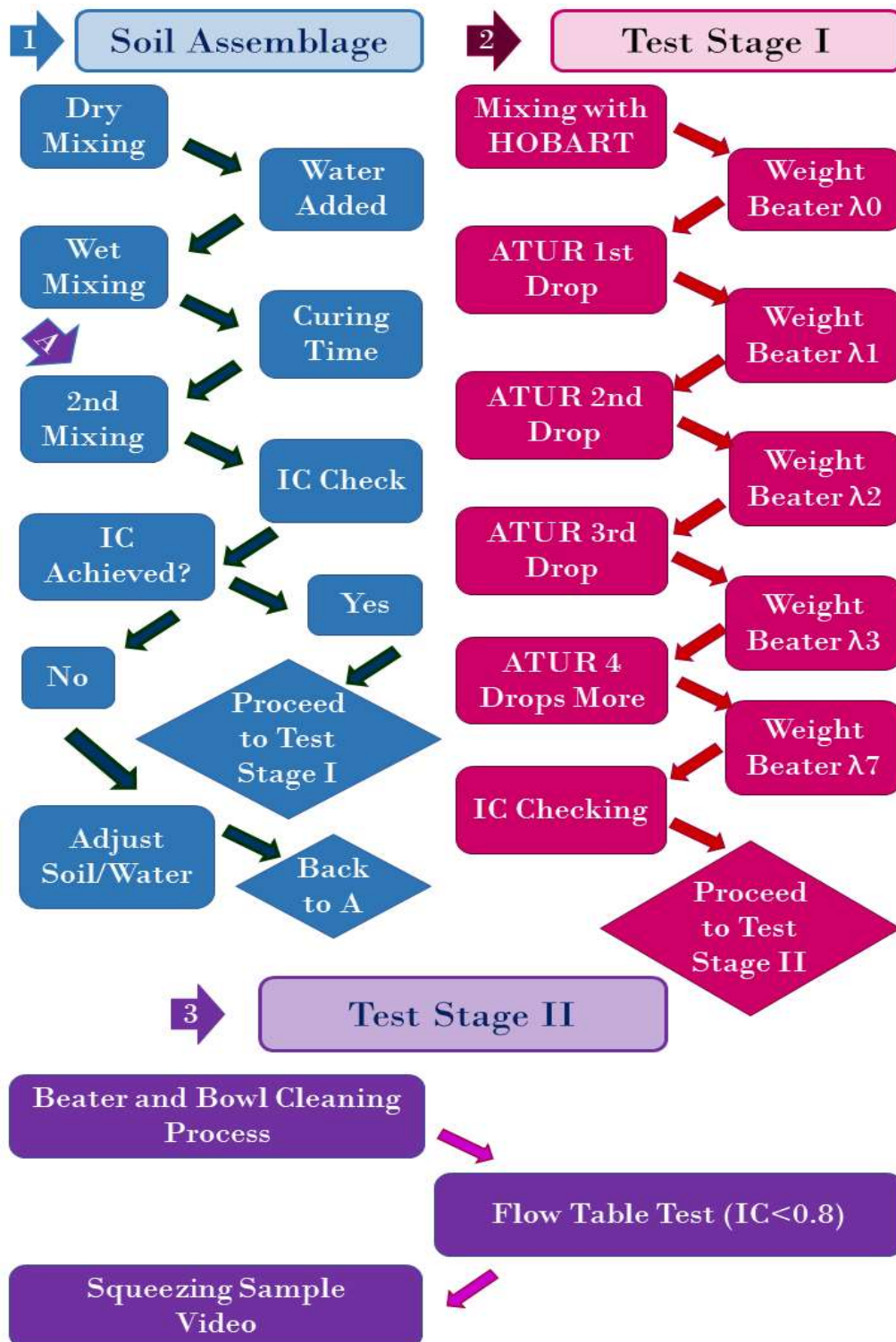


Figure 6-3. Flowchart of combined testing sequence suggested to assess clogging and conditioning of cohesive soils for EPB machine operation.

Once the target I_C was achieved, the next step was to proceed with the HOBART mixing. Figures 6-2b and c show that whenever possible (i.e. sample not too liquid), the soil was placed in the mixing bowl as lumps, which is closer to the reality of an EPB excavation that scrapes off the soil ahead of the cutterhead in sizes of lumps (Peila et al. 2016). The total soil mass was recorded, which represents the parameter G_{TOT} , as suggested by Zumsteg and Puzrin (2012).

In terms of the suggested total mass for the tested soil, it will basically depend on the shear resistance of each specimen and is also related to the consistency index. A stiff soil with I_C larger than 1.0 can easily break the mixing machine with only 700 g of total wet mass. For instance, tests with pure kaolinite, with $I_C = 1.04$, could only proceed with maximum of 400 g of wet soil. Usually, for lower consistencies, the wet total mass was approximately 800-1000 g. It was also noticed that changing the mass around these values did not significantly affect the results.

The B-flat beater (MT) was then placed in the machine, mixing with the HOBART in speed 1 (100 rpm) for 3 min. The beater was carefully removed and the mass of the stuck soil in the MT was weighed. The mass of the beater was subtracted, resulting in the parameter G_{MT0} , following the procedure as suggested by Zumsteg and Puzrin (2012) (illustrated in Figure 6-2d).

The next step was to place the beater in the ATUR device and let the beater drop, as described by Oliveira et al. (2018b) (shown in Figures 6-2e and f). The mass of the soil stuck at the beater was weighed again, obtaining G_{MT1} . The same procedure was performed until the third drop of the beater, always completely cleaning the surface base of ATUR and avoiding influence of the already dropped material stickiness, and then, obtaining G_{MT2} and G_{MT3} . The MT was then dropped four consecutive times and weighed to obtain G_{MT7} .

The next step was to clean the tools, which is a new stage not included previously by Oliveira et al. (2018b). This stage adjusts the final values of the clogging potential, detailed in

section 6.3.1. The cleaning process can also be correlated in the future with the intervention time in tunnel boring machines. Figure 6-4 illustrates the classification for the removing and cleaning of the MT and bowl, called the CB_{Factor} . The water used to clean the tools was always lukewarm tap water.

With numerous tests using different soil specimens, it was possible to observe that the cleaning process depended on the soil properties, as well as the beater metal/coating. Specimens with similar properties surely might be not as easy to differentiate between two sub-categories as soils with very different properties (i.e. low versus high plasticity). Surely, this is still a point to be further refined. However, extremes such as easiness in removing and cleaning the bowl after a test done with a specimen containing only 30% of kaolinite, against extreme difficultness of cleaning the beater and bowl after a 100% bentonite sample were extremely obvious. Besides recording the difficulties to remove and clean the soil by describing it in detail, pictures and even videos were recorded, making it easier later to compare between samples.

For samples with I_c equal to or smaller than 0.8 (not feasible for stiffer samples), the next step was to conduct the flow table test, as proposed by Oliveira et al. (2018c) and shown in Figure 6-2g. For that, a cone was greased with oil; the soil was moulded inside the cone, pressing it in layers and against the cone walls to assure complete filling. The cone was then lifted, and the diameter of the soil base was measured (m_0). The table was jolted 40 times and an average of three diameter measurements was taken, obtaining m_{40} . Pictures of the cone were also taken, so that whenever soil was left stuck in the cone even with an oil-greased surface, it could indicate some tendencies for clogging.

I _c – Consistency Index ↑	Main category	Sub-category	Description	Added Value	IP – Plasticity Index ↓
	St (stiff)	III	Practically, all stuck materials can be easily removed by hand; final cleaning requires only water (< 10 s).	3	
		II	Considerable portion of stuck material can be removed by hand; remaining portion is slightly stiff and difficult to remove without water. Bowl needs spooning and washing (< 30 s) to remove stuck material at the bottom.	2	
		I	Similar as above but requires longer than 30 s to clean beater and bowl, plus brushing and soap washing with lukewarm water.	1	IP – Plasticity Index ↓
	So (soft)	III	Part of the soil can be removed by hand but soil gets easily stuck on them. Remaining material removed with water.	3	
		II	Great portion of soil gets stuck on fingers; mainly, it can be cleaned with water but might require brushing for corners of beater. Bowl takes longer than 30 s to clean, maybe requiring sponging.	2	
		I	Similar as above but more sticky and requires brush and soap to completely remove material	1	IP – Plasticity Index ↓
	Vs (very soft)	III	Not possible to remove soil by hand; too soft, but requires only approximately 30 s to remove it only with water.	3	
		II	Similar as above, but it needs sponge to remove all soil. It takes longer than 30 s.	2	
		I	Similar as above, but needs brushing and soap washing to remove all soil. It takes longer than a minute	1	

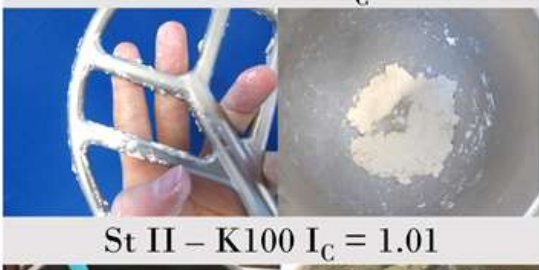
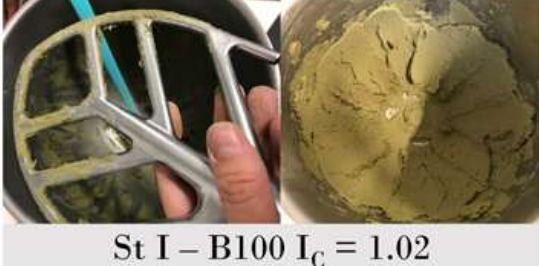
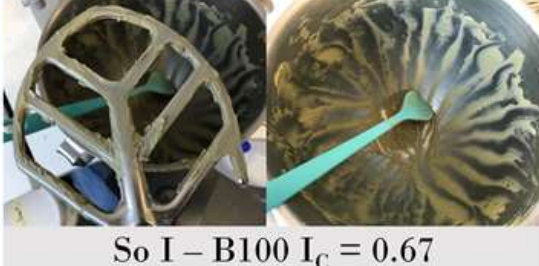
	
St III – Ksfine30 I _c = 1.18	So III – Ksfine50 I _c = 0.59
	
St II – K100 I _c = 1.01	So II – Ksfine70 I _c = 0.62
	
St I – B100 I _c = 1.02	So I – B100 I _c = 0.67

Figure 6-4. Cleaning beater and bowl factor (CB_{Factor}), with categories and sub-categories description on the top, including the respective values to be added at the Equation 6-7. On the bottom some examples of stiff and soft sub-categories.

The last stage of the test was to squeeze the sample, ideally recording a video (Figure 6-2h and Video 6-1). This procedure is similar to what is already done inside EPB machines, when a site worker touches and squeezes the material that comes out of the screw conveyor, feeling the muck. Even though it could be a slightly subjective procedure, it can still provide a useful information of the muck consistency and its properties, allowing some adjustments of the conditioning parameters.

Still to be correlated in the future with real tunnel drives, combined with successful excavation results, the ideal muck most probably is similar to a toothpaste being squeezed out of its tube. If this paste is extremely liquid, it will fall aside, or if it is extremely stiff, squeezing and flowing are not possible (Video 6-1).

Clogging tendencies can also be observed when squeezing the sample, or even excess in air due to high FER values (detailed in section 6.2.3). A compilation of several recorded videos from several tests can be accessed in Video 6-1.

One of the significant limitations of this methodology is conducting tests with conditioned samples because it is performed in atmospheric pressure in an open space. This ignores one important aspect for soil conditioning, that for some foam parameters, the entrained air could separate from the muck and would be restrained on top of the excavation chamber, which could represent a challenge for certain scenarios of face support (i.e. loose soil layers on top). In laboratory tests, this characteristic is lost and what could seem as a successful conditioning in the laboratory could result in an ineffective tunnel operation. The same goes for all the differences between tests conducted in atmospheric pressure, while tunnels are being driven with higher pressures, especially in terms of foam behaviour, as already pointed out by Mooney et al. (2016).

6.2.3 Calculations of Parameters

Parameters regarding tendencies for clogging and flow behaviour can be drawn from the above-mentioned methodology. The calculation for clogging potential obtained from weighing the mass of the soil stuck in the MT, as proposed by Zumsteg and Puzrin (2012), is calculated by Equation 6-5, obtaining λ , which is equivalent to λ_0 (no drop of the beater).

$$\lambda_0 = G_{MT} / G_{TOT} \quad (6-5)$$

Where G_{MT0} is the total mass of soil stuck in the MT (B-flat beater) and G_{TOT} the total mass of soil.

As the MT is dropped, the other parameters can be calculated similarly, with the same general equation (Equation 6-6), as initially proposed by Oliveira et al. (2018b).

$$\lambda_x = G_{MTx} / G_{TOT} \quad (6-6)$$

Where G_{MTx} is the total mass of soil stuck at the MT after x numbers of drops and G_{TOT} the total mass of soil.

The final clogging parameter is λ_F , which was adjusted after several tests with pure kaolinite samples, combined with the universal clogging potential diagram of Hollman and Thewes (2012), as explained in detail in section 6.3.1 and by Oliveira et al. (2018b). The final clogging parameter, λ_F is defined by Equation 6-7, where $(\overline{\lambda_x})$ is the average of all the obtained λ_x values, as indicated in Equation 6-8. The CB_{Factor} is calculated according the categories/sub-categories in Figure 6-4. Finally, Table 6-5 presents the main clogging potential categories. These categories were also defined with the calibration procedure, as detailed in section 6.3.1.

Equation 6-7:

$$\lambda_F = \overline{\lambda_x} / CB_{Factor} \quad (6.7)$$

Equation 6-8:

$$\overline{\lambda_x} = (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 + \lambda_7) / 5 \quad (6.8)$$

What is essential to mention is the flexibility of the CB_{Factor} , such as the possibility to include fractions to better represent two different sub-categories. That has been done with the samples containing pure bentonite, as described in section 6.3.1. This could be used as a type of a factor of safety, or for instance, for increasing the clogging effects of a sandblasted surface, which in this case, it is suggested to subtract 0.5 from the final CB_{Factor} , increasing the clogging potential. However, that can only be validated with several tunnel drives, as a back analysis of prior laboratory investigations.

Additionally, for all the samples classified as stiff (St in Figure 6-4; $I_C > 0.75$), a low clogging value would result in lumps, as the clogging category.

Table 6-5. Clogging potential categories defined by the parameter λ_F .

λ_F	Clogging Potential
≤ 0.27	Low
$0.27 - 0.48$	Medium
≥ 0.48	High

For the flow behaviour, as explained also by Oliveira et al. (2018c), the flow can be defined by Equation 6-9.

$$Flow_{40}(\%) = ((m_{40} - m_0)/m_0) \times 100 \quad (6-9)$$

Where:

m_0 - average of three measurements of the initial specimen diameter

m_{40} - average of three measurements of the specimen diameter after 40 times jolting

6.3 Results and Discussions

This section presents all the results obtained from the testing campaign with the combined methodology as described in section 6.2.2. It also considers some of the results gained from previous testing phases, which can be accessed in Oliveira et al. (2018b, c). Initially, it is demonstrated how a final calibration of the method was performed by comparing the results with the universal clogging graph from Hollman and Thewes (2012). Furthermore, aiming to investigate particularities of the cohesive mixed soil, tests with different proportions of clay and minerals, in diverse consistency ranges, were conducted. Tests were conducted as well with four different beaters, detailed in section 6.3.3. Finally, the influence of additives when added to these mixed soils was also investigated.

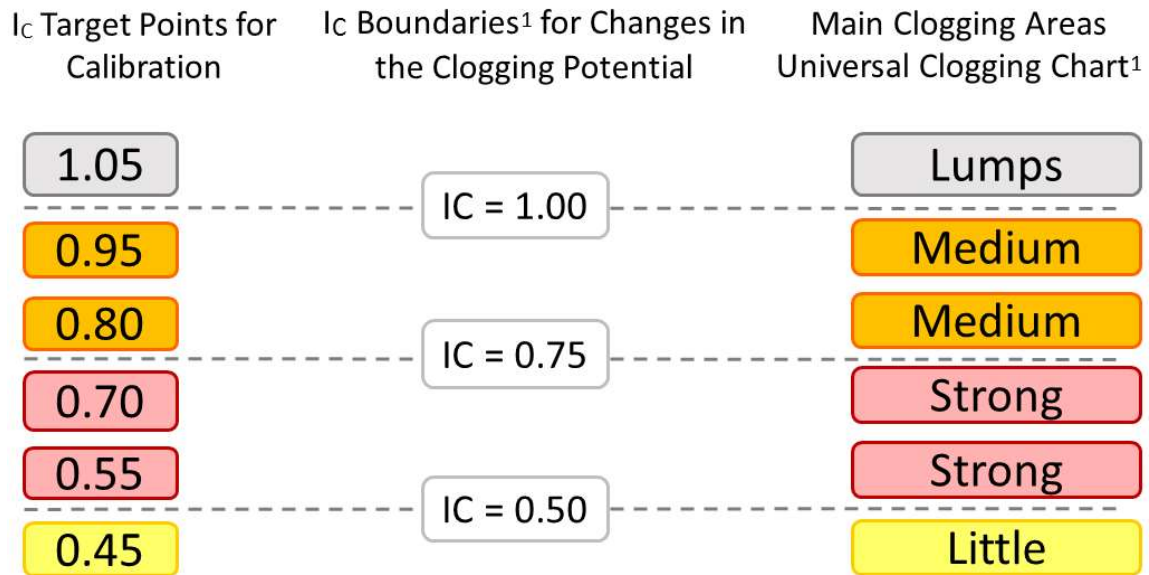
6.3.1 Calibration with Pure Clay

This calibration was done mainly to define how many drops of MT had to be performed to provide reasonable results of the clogging parameter (λ_F), which directly depends on the height of the drop of the beater and stiffness of the device base (ATUR), besides the properties of the soil. An initial calibration trial was presented by Oliveira et al. (2018b); however, these tests were conducted starting with dried mixtures, where water was added as the test went along, without curing time, which seemed to increase the standard deviation of the final results. By modifying the soil assemblage, the standard deviation values decreased, obtaining results that were more reliable.

This calibration was performed by comparing the results with the universal clogging chart of Hollman and Thewes (2012). Considering that this chart was elaborated based on real tunnel drives, indirectly, this calibration would have also a correlation with tunnel drives. The data for this universal chart, however, were obtained with excavations along sedimentary soils, mainly with pure clay layers of low to medium plasticity index (average IP of approximately 40%, with the highest IP = 72%) (Hollman and Thewes, 2012, 2013). The database did not include residual soils, neither very highly plastic clays, such as the bentonite tested in this research (IP = 433%; Table 6-2). Therefore, it was concluded that to guarantee a good comparison and, consequently, calibration, tests with pure kaolinite should be conducted and compared first.

The target I_c values were approximately 0.5 points above and below the clogging limits defined in the graph by Hollman and Thewes (2012), as illustrated in Figure 6-5, and also presented in the study by Oliveira et al. (2018a, b). Numerous tests for each I_c were conducted and the results for kaolinite can be found in Tables 6-6 and 6-7. To facilitate visualisation, the cells with the resulting λ and clogging category, as defined in the graph by Hollman and Thewes (2012), have been coloured as grey for lumps, yellow for little clogging, orange for medium clogging, and red for strong clogging.

After realising that the cleaning and removing of the stuck soil from the beater and the bowl were also providing valuable information, the CB_{Factor} was incorporated. The values of CB_{Factor} and the resulting equations that finally lead into the resulting clogging parameter λ_F have been defined by adjusting the results presented in Tables 6-6 and 6-7. To differentiate little clogging from lumps, a rule was applied that whenever the clogging was little, considering the values given in Table 6-5, and the CB_{Factor} category was St (stiff), then the final resulting clogging category was, logically, lumps (grey colour), and not little clogging.



¹ Obtained from the diagram by Hollman & Thewes (2012)

Figure 6-5. Main testing target points for the initial ATUR calibration considering clogging potential boundaries as defined in the chart by Hollman and Thewes (2012), with clogging categories presented on the right side.

As it can be observed in these tables, the results represented by the two last columns, ‘Clogging Category’ and ‘Clogging Potential H&T’, are consistent, except for only one point out of 49, in ID 6, which is relatively close to the limit of little/medium clogging category.

Considering this effective correlation with pure kaolinite samples, tests with bentonite were then conducted. The results are presented in Table 6-8 and it can be noticed that there is no close correlation, with the highest clogging values occurring for higher consistency values. Figure 6-6 illustrates better both cases, comparing the results with kaolinite and bentonite for all the λ values, from no drop (λ_0) to the final resulting value (λ_F).

It is interesting to note that from all the options (λ_0 to λ_F), bentonite always seems to hold a lower clogging potential than kaolinite, which is not entirely expected, considering its swelling behaviour. The CB_{Factor} value was changed from 1 to 0.7 and even 0.5, resulting in the last two graphs (λ_{F2} and λ_{F3}), which provided higher clogging values when compared to kaolinite.

Regardless of the alterations done with the CB_{Factor} , bentonite had always its clogging peak starting in higher consistency values than kaolinite.

One assumption that could be made for the different results between both clays is related to the differences between the molecular bindings in both clay minerals, where kaolinite has a strong O-OH hydrogen bonding and bentonite has a weak van der Waals O-O bonding, which could have a higher tendency to separate with the drops of the beater.

Again, it will be possible to investigate this matter further only by combining this laboratory routine with the results from real tunnel drives.

Table 6-6. Calibration table for kaolinite samples for I_C above 0.54 compared with clogging potential chart by Hollman and Thewes (2012; 2013), shown as H&T.

ID	I_C	λ_0	λ_1	λ_2	λ_3	λ_7	Cleaning Category	Final λ_F Value	Clogging Category	Clog. Potential H&T ¹⁰
1	1.02	0.83	0.81	0.20	0.02	0.02	St II	0.19	Lumps	Lumps
	1.02	0.81	0.55	0.01	0.01	0.01		0.14	Lumps	Lumps
	0.98	0.62	0.62	0.62	0.62	0.06		0.25	Lumps	Lumps
	0.99	0.62	0.62	0.62	0.27	0.08		0.22	Lumps	Lumps
	1.01	0.63	0.62	0.61	0.58	0.07		0.25	Lumps	Lumps
	1.01	0.64	0.64	0.64	0.64	0.09		0.27	Lumps	Lumps
	1.04	0.61	0.59	0.56	0.45	0.04		0.22	Lumps	Lumps
Ave 1	1.01	0.68	0.64	0.47	0.37	0.05		0.22	Lumps	Lumps
Std Dv	0.02	0.10	0.08	0.25	0.27	0.03		0.04		
2	0.86	0.74	0.74	0.74	0.74	0.63	St II	0.36	Medium	Medium
	0.86	0.76	0.76	0.76	0.75	0.62		0.36	Medium	Medium
	0.86	0.76	0.76	0.76	0.76	0.57		0.36	Medium	Medium
	0.86	0.78	0.77	0.77	0.77	0.66		0.38	Medium	Medium
	0.91	0.67	0.67	0.67	0.67	0.67		0.33	Medium	Medium
	0.90	0.65	0.65	0.65	0.39	0.39		0.27	Medium	Medium
	0.89	0.56	0.56	0.56	0.56	0.41		0.27	Medium	Medium
	0.93	0.59	0.59	0.59	0.59	0.56		0.29	Medium	Medium
	0.94	0.67	0.64	0.61	0.58	0.46		0.30	Medium	Medium
Ave 2	0.89	0.69	0.68	0.68	0.64	0.55		0.32	Medium	Medium
Std Dv	0.03	0.08	0.08	0.08	0.13	0.11		0.04		
3	0.80	0.66	0.66	0.66	0.66	0.66	St II	0.33	Medium	Medium
	0.80	0.68	0.68	0.59	0.59	0.59		0.31	Medium	Medium
	0.80	0.69	0.69	0.69	0.69	0.59		0.33	Medium	Medium
	0.80	0.73	0.73	0.73	0.73	0.72		0.36	Medium	Medium
	0.80	0.71	0.71	0.71	0.71	0.70		0.35	Medium	Medium
	0.78	0.67	0.67	0.67	0.67	0.62		0.33	Medium	Medium
	0.77	0.60	0.60	0.60	0.60	0.58		0.30	Medium	Medium
	0.77	0.60	0.60	0.60	0.60	0.58		0.30	Medium	Medium
Ave 3	0.79	0.68	0.68	0.66	0.66	0.64		0.33	Medium	Medium
Std Dv	0.01	0.04	0.04	0.05	0.05	0.06		0.02		
4	0.69	0.63	0.63	0.63	0.63	0.59	So I	0.62	Strong	Strong
	0.69	0.59	0.59	0.58	0.58	0.58		0.59	Strong	Strong
	0.67	0.60	0.60	0.60	0.60	0.60		0.60	Strong	Strong
	0.73	0.67	0.67	0.53	0.53	0.48		0.58	Strong	Strong
	0.55	0.56	0.56	0.56	0.56	0.56		0.56	Strong	Strong
	0.60	0.60	0.60	0.60	0.60	0.37		0.55	Strong	Strong
	0.68	0.66	0.66	0.66	0.66	0.66		0.66	Strong	Strong
	0.65	0.66	0.66	0.66	0.66	0.66		0.66	Strong	Strong
	0.65	0.66	0.66	0.66	0.66	0.66		0.66	Strong	Strong
Ave 4	0.66	0.62	0.62	0.60	0.60	0.56		0.60	Strong	Strong
Std Dv	0.06	0.04	0.04	0.05	0.05	0.10		0.04		

¹⁰Universal clogging potential diagram chart according to Hollman and Thewes (2012).

Table 6-7. Calibration table for kaolinite samples for I_C below 0.55 compared with clogging potential chart by Hollman and Thewes (2012; 2013), shown as H&T.

ID	I_C	λ_0	λ_1	λ_2	λ_3	λ_7	Cleaning Category	Final λ_F Value	Clogging Category	Clog. Potential H & T ¹⁰
5	0.54	0.62	0.62	0.62	0.62	0.62	So I	0.62	Strong	Strong
	0.54	0.66	0.66	0.66	0.66	0.66		0.66	Strong	Strong
	0.54	0.67	0.67	0.67	0.67	0.67		0.67	Strong	Strong
	0.54	0.67	0.67	0.67	0.67	0.67		0.67	Strong	Strong
	0.54	0.61	0.61	0.61	0.61	0.60		0.61	Strong	Strong
	0.52	0.71	0.71	0.71	0.71	0.62		0.69	Strong	Strong
	0.51	0.72	0.72	0.72	0.72	0.35		0.64	Strong	Strong
	0.51	0.72	0.72	0.72	0.72	0.35		0.65	Strong	Strong
Ave 5	0.53	0.66	0.66	0.66	0.66	0.60		0.65	Strong	Strong
Std Dv	0.01	0.04	0.04	0.04	0.04	0.11		0.03		
6	0.44	0.60	0.60	0.60	0.60	0.32	Vs II	0.27	Little	Little
	0.44	0.59	0.59	0.59	0.59	0.35		0.27	Little	Little
	0.24	0.54	0.54	0.53	0.30	0.13		0.20	Little	Little
	0.40	0.66	0.66	0.65	0.65	0.13		0.27	Little	Little
	0.39	0.61	0.61	0.61	0.60	0.10		0.25	Little	Little
	0.37	0.68	0.68	0.67	0.67	0.09		0.28	Medium	Little
	0.36	0.64	0.64	0.64	0.29	0.09		0.23	Little	Little
	0.36	0.64	0.64	0.64	0.29	0.09		0.23	Little	Little
Ave 6	0.37	0.62	0.62	0.61	0.53	0.17		0.26	Little	Little
Std Dv	0.07	0.05	0.05	0.05	0.16	0.11		0.03		
7	-0.05	0.52	0.51	0.25	0.25	0.06	Vs II	0.16	Little	Little
	-0.05	0.49	0.47	0.31	0.13	0.10		0.15	Little	Little
	0.02	0.53	0.52	0.16	0.12	0.06		0.14	Little	Little
	0.04	0.52	0.51	0.22	0.13	0.05		0.14	Little	Little
Ave 7	-0.01	0.51	0.50	0.23	0.16	0.06		0.15	Little	Little
Std Dv	0.05	0.02	0.02	0.06	0.06	0.02		0.01		

Table 6-8. Calibration table for bentonite samples compared with clogging potential chart by Hollman and Thewes (2012; 2013), shown as H&T.

ID	I _c	λ_0	λ_1	λ_2	λ_3	λ_7	Cleaning Category	Final λ_F Value	Clogging Category	Clog. Potential H & T ¹⁰
1	1.01	0.56	0.56	0.54	0.54	0.04	St I	0.45	Medium	Lumps
	1.01	0.51	0.51	0.39	0.39	0.05		0.37	Medium	Lumps
	1.01	0.42	0.42	0.42	0.42	0.01		0.34	Medium	Lumps
	1.02	0.57	0.56	0.56	0.55	0.09		0.47	Medium	Lumps
	1.02	0.51	0.51	0.48	0.48	0.05		0.40	Medium	Lumps
Ave 1	1.02	0.51	0.51	0.48	0.48	0.05		0.40	Medium	Lumps
Std Dv	0.01	0.07	0.07	0.08	0.08	0.03		0.06		
2	0.93	0.50	0.50	0.50	0.50	0.50	St I	0.50	Strong	Medium
	0.93	0.59	0.59	0.59	0.59	0.59		0.59	Strong	Medium
	0.93	0.46	0.46	0.46	0.46	0.46		0.46	Medium	Medium
	0.89	0.80	0.80	0.80	0.80	0.26		0.69	Strong	Medium
	0.92	0.59	0.59	0.59	0.59	0.45		0.56	Strong	Medium
Ave 2	0.92	0.59	0.59	0.59	0.59	0.45		0.56	Strong	Medium
Std Dv	0.02	0.15	0.15	0.15	0.15	0.14		0.10		
3	0.77	0.60	0.60	0.60	0.59	0.05	St I	0.49	Strong	Medium
	0.77	0.59	0.59	0.58	0.58	0.19		0.50	Strong	Medium
	0.77	0.61	0.61	0.60	0.60	0.16		0.52	Strong	Medium
	0.77	0.60	0.60	0.60	0.59	0.13		0.50	Strong	Medium
Ave 3	0.77	0.60	0.60	0.60	0.59	0.13		0.50	Strong	Medium
Std Dv	0.00	0.01	0.01	0.01	0.01	0.07		0.01		
4	0.67	0.70	0.69	0.49	0.20	0.03	So I	0.42	Medium	Strong
	0.67	0.62	0.61	0.61	0.41	0.10		0.47	Medium	Strong
	0.67	0.64	0.64	0.63	0.11	0.06		0.41	Medium	Strong
	0.67	0.65	0.65	0.57	0.24	0.06		0.43	Medium	Strong
Ave 4	0.67	0.65	0.65	0.57	0.24	0.06		0.43	Medium	Strong
Std Dv	0.00	0.04	0.04	0.08	0.15	0.04		0.03		
5	0.57	0.63	0.63	0.07	0.03	0.03	So I	0.28	Medium	Strong
	0.57	0.67	0.66	0.11	0.03	0.03		0.30	Medium	Strong
	0.57	0.65	0.65	0.14	0.03	0.03		0.30	Medium	Strong
	0.57	0.65	0.65	0.11	0.03	0.03		0.29	Medium	Strong
Ave 5	0.57	0.65	0.65	0.11	0.03	0.03		0.29	Medium	Strong
Std Dv	0.00	0.02	0.02	0.04	0.00	0.00		0.01		
6	0.43	0.61	0.45	0.10	0.04	0.03	Vs I	0.24	Little	Little
	0.43	0.52	0.45	0.05	0.03	0.03		0.22	Little	Little
	0.43	0.59	0.12	0.06	0.03	0.03		0.16	Little	Little
	0.43	0.57	0.34	0.07	0.03	0.03		0.21	Little	Little
Ave 6	0.43	0.57	0.34	0.07	0.03	0.03		0.21	Little	Little
Std Dv	0.00	0.05	0.19	0.03	0.00	0.00		0.04		
7	0.02	0.07	0.03	0.02	0.02	0.01	Vs II	0.02	Little	Little
	0.02	0.00	0.04	0.03	0.03	0.01		0.01	Little	Little
	0.02	0.06	0.04	0.03	0.02	0.01		0.02	Little	Little
	0.02	0.05	0.03	0.03	0.02	0.01		0.01	Little	Little
Ave 7	0.02	0.05	0.03	0.03	0.02	0.01		0.01	Little	Little
Std Dv	0.00	0.04	0.01	0.01	0.00	0.00		0.00		

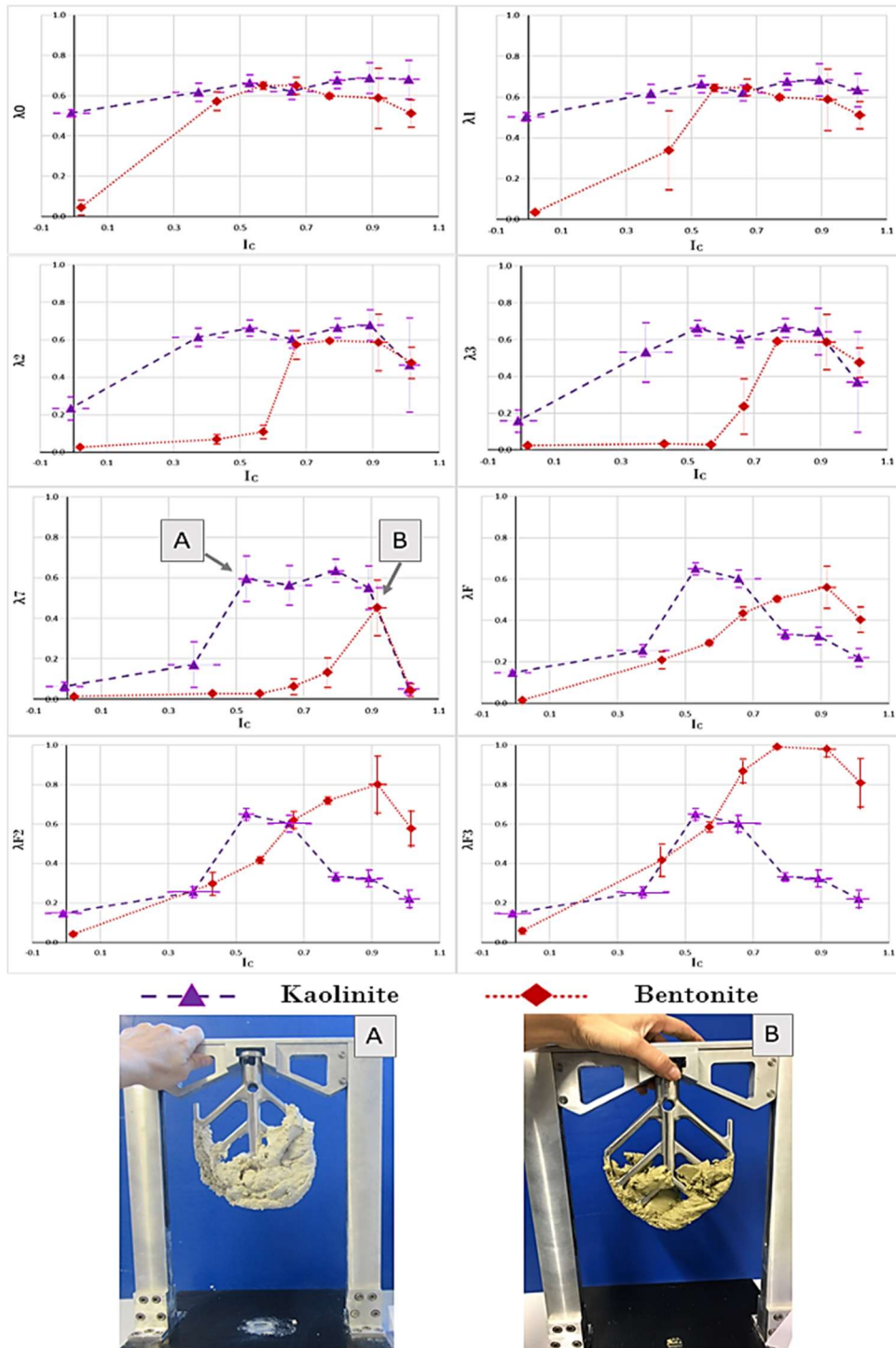


Figure 6-6. Clogging parameters λ from 0 to 7 and resulting λ_F , for kaolinite (purple triangle) and bentonite (red diamond). The last two bottom graphs had their CB_{Factor} modified from 1 to 0.7 (λ_{F2}) and 0.5 (λ_{F3}). Points A and B refer to the highest clogging values after 7 drops.

6.3.2 Tests Performed with Water

Figures 6-7, 6-8, and 6-9 show the results from tests for different clay fractions and I_C values. All the graphs show the original parameter proposed by Zumsteg and Puzrin (2012), labelled here as λ_0 , and the new clogging category, λ_F . On all graphs, labels represent the average values including their standard deviation, represented by a straight fine line.

There is an overestimation of the clogging values λ_0 for mixed soils, when compared to the final clogging category λ_F . As the clay content increases, the differences between λ_0 and λ_F decrease, especially for the consistencies between 0.5 and 0.7 for kaolinite, and 0.7 and 1.0 for bentonite, suggesting that when clogging is prominently present, the drops of the beater do not significantly decrease the resulting λ values.

The standard deviation values are higher for λ_0 , which means that in the testing procedure when moving the beater from the mixer to the scale, the soil that was not really stuck would occasionally fall off from the MT.

For kaolinite samples mixed with sand (Figure 6-7), only from 50% of clay content that medium clogging starts to occur for a consistency of approximately 0.8, being practically absent for 30% of kaolinite. Clogging becomes more evident with 70% of clay content, with medium clogging for I_C between 0.5 and 0.8. The highest clogging values seem to be slightly towards the right of the x-axis for mixed soils, when compared to pure clay.

For bentonite (Figure 6-8), medium clogging already occurs with 20% of clay, intensifying with increasing clay content. For most of the four graphs, except for 50% of clay content, the highest clogging values are at I_C between approximately 0.7 and 0.95. For 50% of clay content, there is a peak clogging value for a higher consistency around the plastic limit. High clogging values in this consistency also occur for 70% of clay content. In the test results with the flow table presented by Oliveira et al. (2018c), the lowest flow values for bentonite occurred at

approximately 50% and 70% of clay. Here, for I_c of approximately 1.0 (plastic limit), the highest clogging values are also at approximately 50% and 70% of bentonite.

A similar correlation between flow table test results from Oliveira et al. (2018c) can also be observed in Figure 6-9, where a larger grain size of sand fraction seemed to imply higher clogging values. For the flow table tests, the same results were found: larger sand grains implied lower flow values. This could indicate that for the same consistency index, lower flow values could indicate higher clogging tendencies.

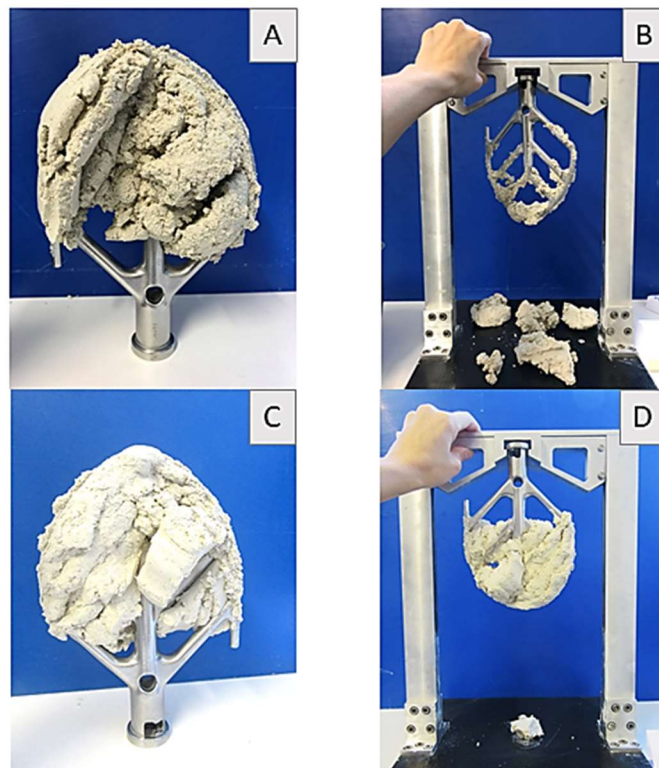
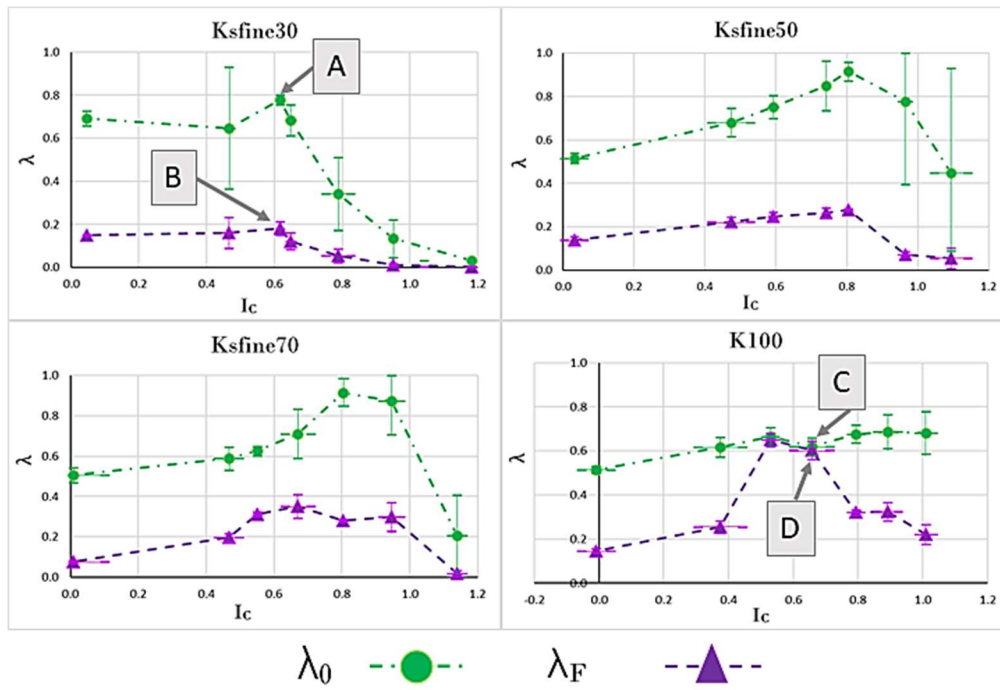


Figure 6-7. Comparison between tests with kaolinite and different clay contents, for fine sand, showing the resulting clogging parameters of λ_0 and λ_F . The letters A, B, C and D show points in the graphs.

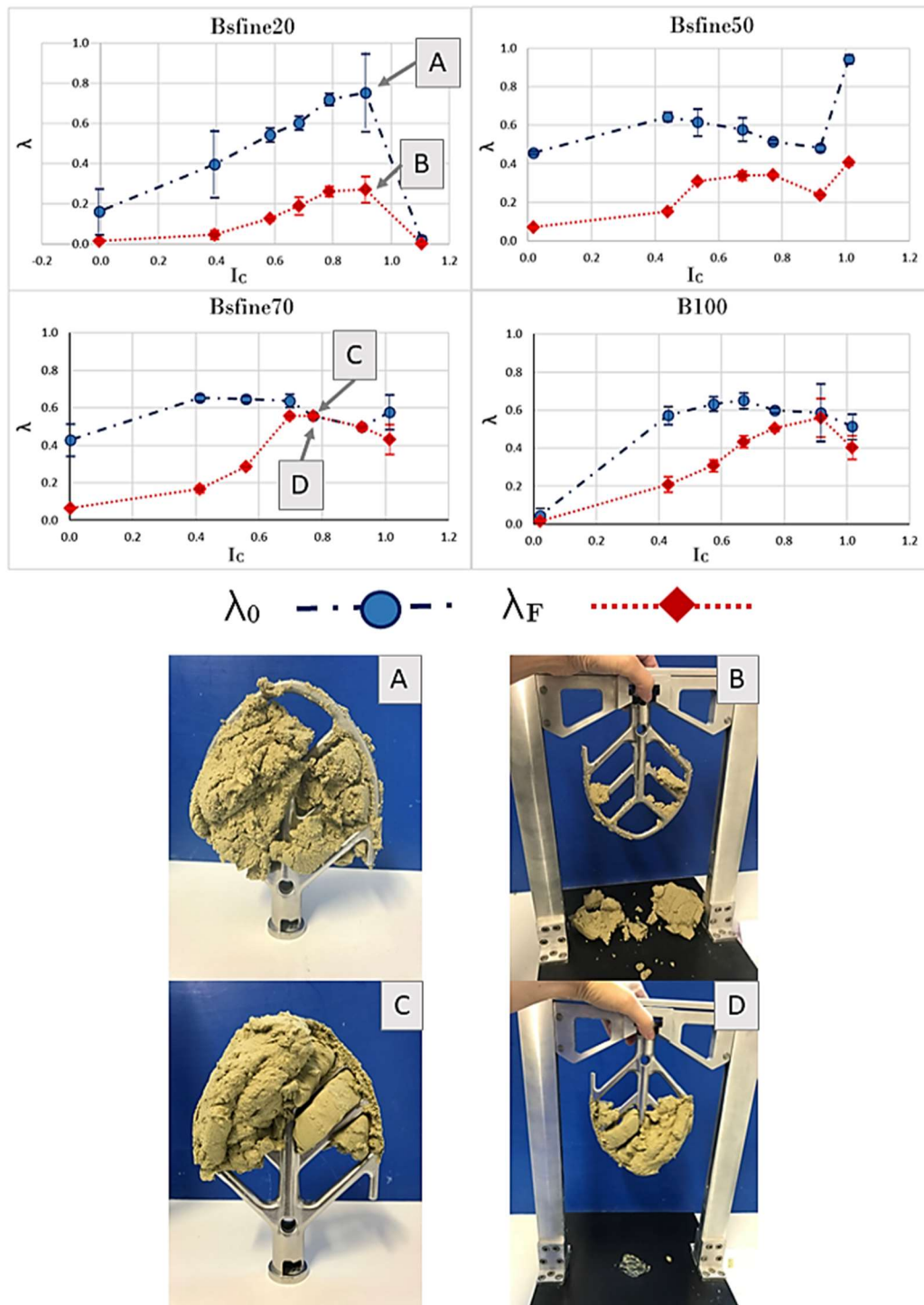


Figure 6-8. Comparison between tests with bentonite and different clay contents, for fine sand, showing the resulting clogging parameters of λ_0 and λ_F . The letters A, B, C and D show points in the graphs.

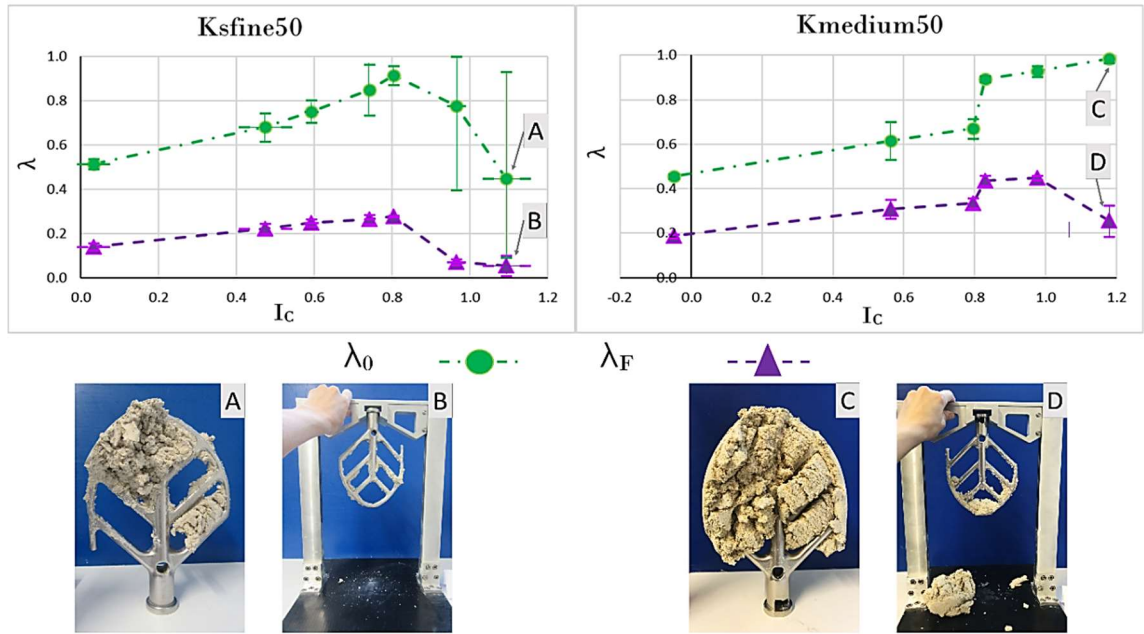


Figure 6-9. Comparison between tests with kaolinite and different sizes of sand, showing λ_0 and λ_F . The letters A, B, C and D show points in the graphs.

6.3.3 Tests with Different Beaters

Tests with mixed soils were also conducted with different beaters to investigate the influence of the metal or addition of a coating. Both clogging parameters (λ_0 and λ_F) were plotted. Again, λ_0 presented higher standard deviation values, as well as higher clogging (Figures 6-10 and 6-11). Tests were conducted with kaolinite and fine sand (50% each; Figure 6-10) and with 20% of bentonite and 80% of fine sand (Figure 6-11).

The results obtained with a stainless-steel beater, used also for all the other tests, were compared with three aluminium beaters, where two of them were covered with a sandblasted coating and a special red coating. Several machine cutterheads are sandblasted, justifying this choice. The sandblasted beater was the one that produced the highest clogging values. It was also more difficult to remove and clean the soil stuck at the MT for the sandblasted case, which justified the decrease of the CB_{Factor} added value to 0.5. Additionally, the special coating did not

produce a very significant reduction in clogging when compared to the aluminium or stainless-steel beater.

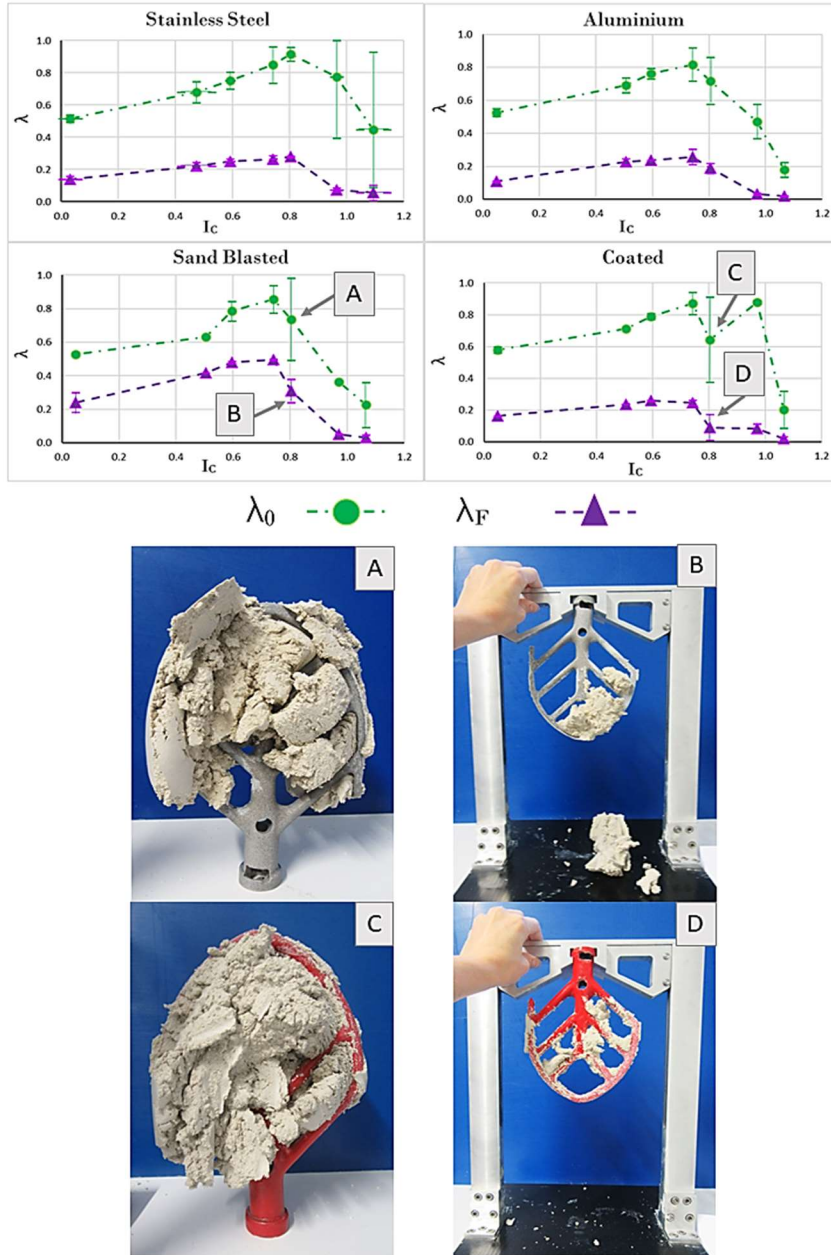


Figure 6-10. Comparison between tests with Ksfine50 and different beaters, showing the resulting clogging parameters of λ_0 and λ_F . The letters A, B, C and D show certain points in the graphs.

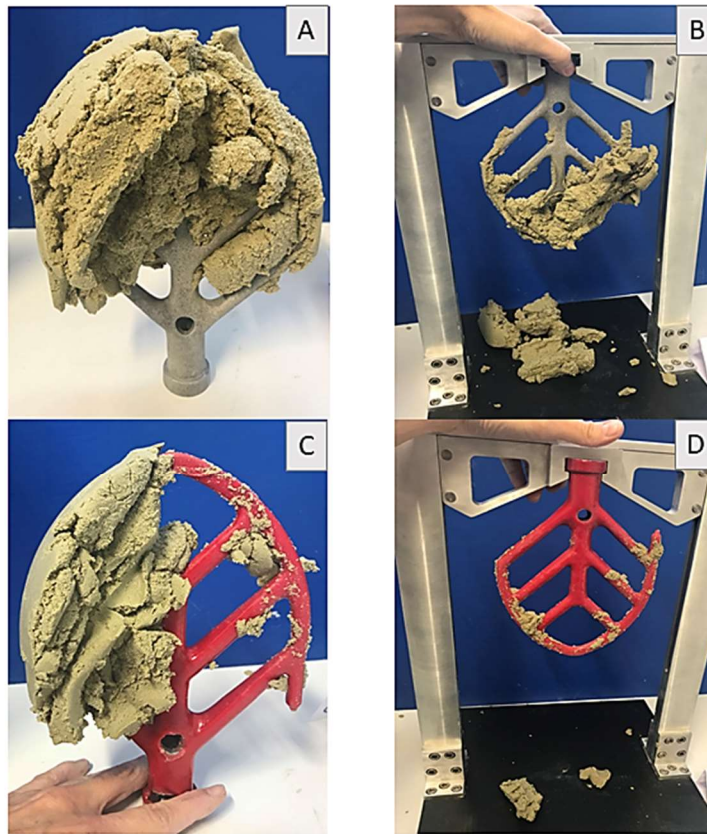
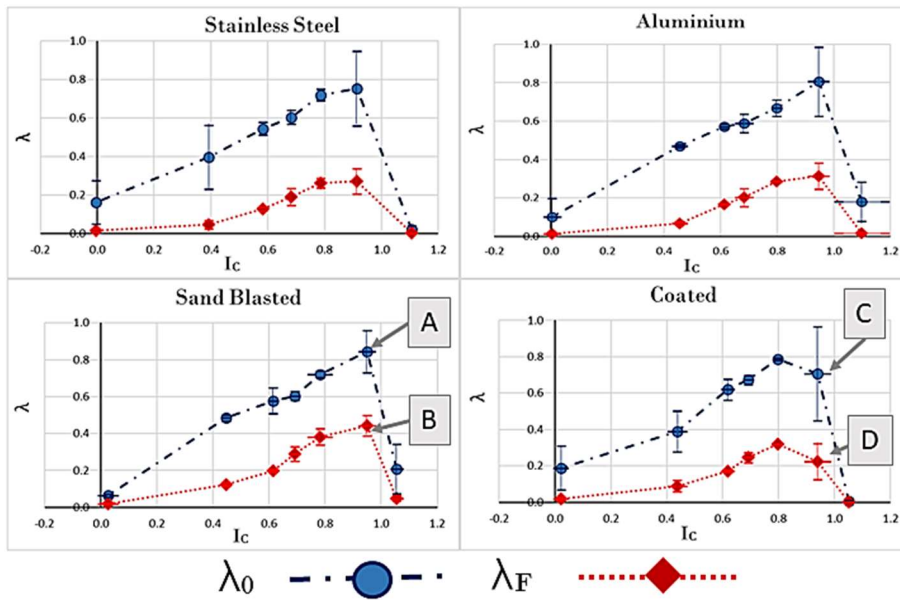


Figure 6-11. Comparison between tests with Bsfine20 and different beaters, showing the resulting clogging parameters of λ_0 and λ_F . The letters A, B, C and D show certain points in the graphs.

6.3.4 Tests Performed with Additives

The tests conducted with foam and polymers were the most laborious ones, if compared to the tests with only water, due to several reasons, such as achieving the target FER, the mixing procedure of the wetted soil with foam, and the choice of parameters for possible comparison between soils, especially without using unrealistic soil conditioning parameters.

The difficulties in producing a foam with the target FER were greater whenever the surfactant was changed from one product to another. It took several trials to achieve the desired FER with the used a foam generator. There was also a limitation in producing FER of small values, which is usually applied for cohesive soils (FER of approximately 5–7). The lowest feasible values were approximately 9.5–10, which allowed the testing procedure to be conducted within the time before the foam was separated.

The preliminary idea was to have an initial consistency of the wet soil and lower it with a foam (or foam plus anti-clay polymer), achieving a final consistency, which is 0.25 points lower than the initial one, thereby always comparing the same consistency index among all soil samples. However, the required amount of foam between kaolinite and bentonite was extremely different and bringing down the consistency to the final target I_C resulted in too little foam for kaolinite, on one side, and extremely high and unrealistic FIR values for bentonite. The solution was to focus on applying realistic values, which are already used in tunnel projects for a cohesive mixed soil (Maidl et al. 2015, Peixoto da Silva 2017). The FIR was then always approximately 80% and 120%, as indicated in Table 6-4.

The other difficulty was in mixing the wet soil homogeneously with the foam, which for many cases, was practically impossible (Figures 6-12 and 6-13, bottom pictures). This fact was already pointed out by Peila et al. (2016), showing that only the surface of lumps was really conditioned in the case of clayey soils. Thus, to force a complete mixing was also far from the

reality in tunnel drives. Several options of mixing were tried: from a higher speed of machine mixing to the use of a spoon. This is one of the limitations of the testing methodology when conducting tests with foam that still requires improvements. Obtaining a homogenous soil–foam mixture was most difficult in the tests done with bentonite.

The most useful part of the test was the manual squeezing stage, which provided valuable information, even though on a more descriptive basis, on the effectiveness of a certain soil conditioning. Moreover, in general, when the additive would help to reduce the clogging values, because there was enough quantity of it, the texture of the muck was far from ideal: either extremely spongy or too liquid. The results can be observed in Figure 6-12 for kaolinite and in Figure 6-13 for bentonite. In general, it appeared that adding any conditioning did not result in any significant improvements for the clogging itself; however, the foams with anti-clay polymers produced slightly better results than those with none.

For kaolinite, for example, on the second point from the right, at $I_c = 0.68$ approximately, only Product A foam resulted in more clogging than with only water, whereas the anti-clay products B and C, resulted in slightly fewer clogging effects. For bentonite as well, both products with anti-clay polymers had the best results, even though with very small differences, which most probably cannot justify the cost of these products for this type of soil.

The tests with the flow table also contributed to more information, which will be explained in section 6.3.5. In the bottom picture showing the cones in Figures 6-12 and 6-13, the stuck soil between Products A, B, and C can be compared. Product A was the one leaving a higher amount of soil specimen stuck on the mortar cone. B seemed to be effective for kaolinite, and C, even though slightly better than foam, did not work so well for bentonite.

An experimental trial was performed to provide other alternative testing possibilities for the effectiveness of anti-clay polymers, which can be visualised in Figure 6-14. After a test with

only water and with clogging occurrence for the sample Kmedium50 (Figure 6-9 and Figures 6-14a, b, g, and h), an anti-clay polymer (Product B) was mixed with water and directly spilled in the MT with soil, without foam production. In the first test, 10% of water content was added, and in the second, 20%. The surfactant concentration was 12% and 20%, respectively, instead of 3%.

For the first test, the liquid was spilled, and after waiting for 20 min, the liquid was mixed again, and the test was performed, resulting in a not very significant difference in the clogging (Figures 6-14c, d, e, and f). For the second case, the beater was turned several times inside the bowl with the liquid, waiting several minutes after each turn, which resulted in a significant lower clogging effect (Figures 6-14i, j, k, and l). This could suggest, for example, that in cases of major stops of the machine due to clogging occurrence, by filling the chamber with a highly concentrated anti-clay polymer mixed with water, the removal of the clogged soil could be facilitated.

A third trial was conducted by lubricating the MT with an anti-clay polymer, which did not result in any clogging decrease. It was even tested to add sugar to the sample and check if that could affect clogging, as an additional experiment. Sugar increased clogging and deteriorated the entire cleaning process. Therefore, this combined testing methodology could certainly provide a small-scale and easy-to-perform experiment, avoiding trying unorthodox solutions directly during a tunnel operation.

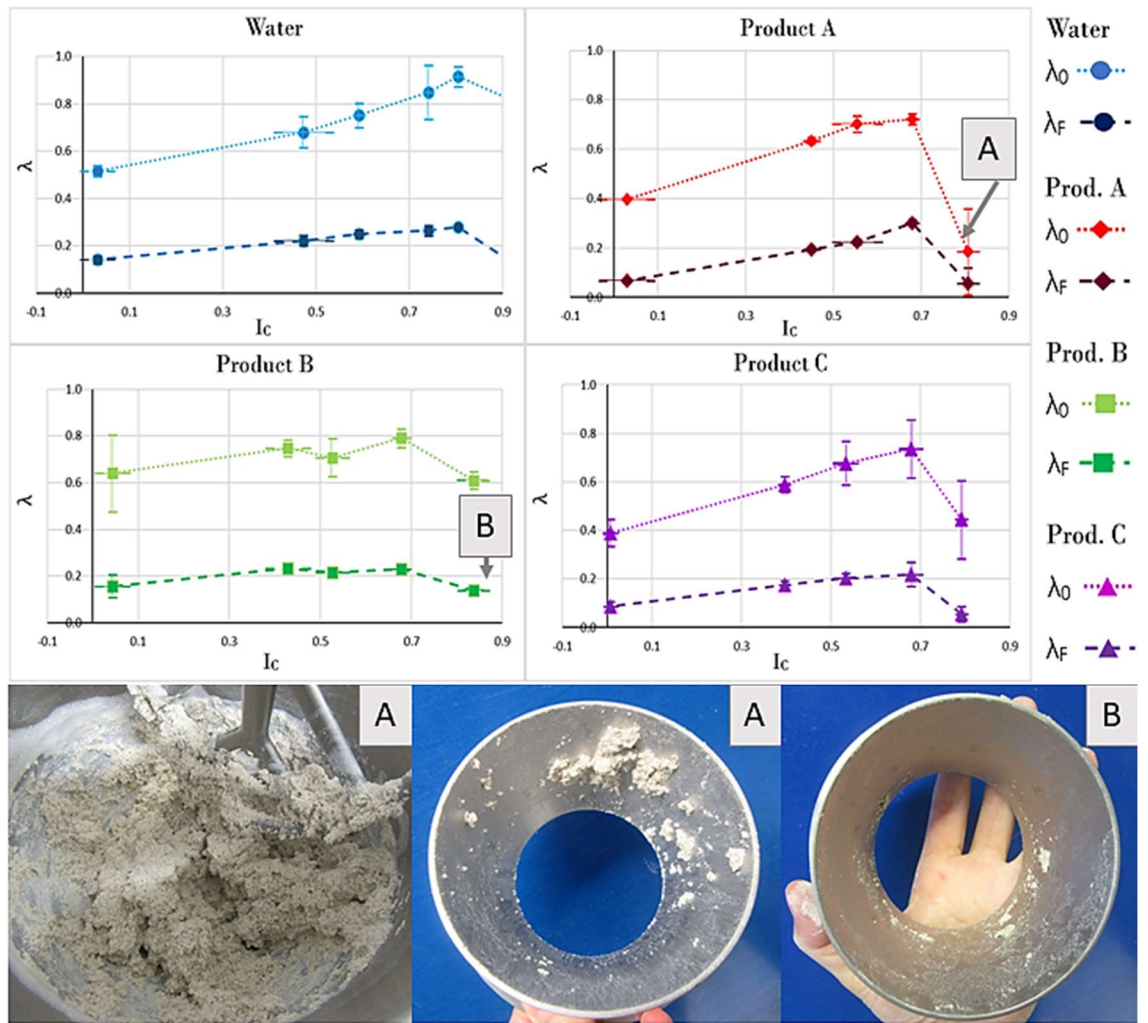


Figure 6-12. Results for Ksfine50 conditioned with only water, and with Products A, B and C. The pictures in the bottom are correlated with the letters in the graphs.

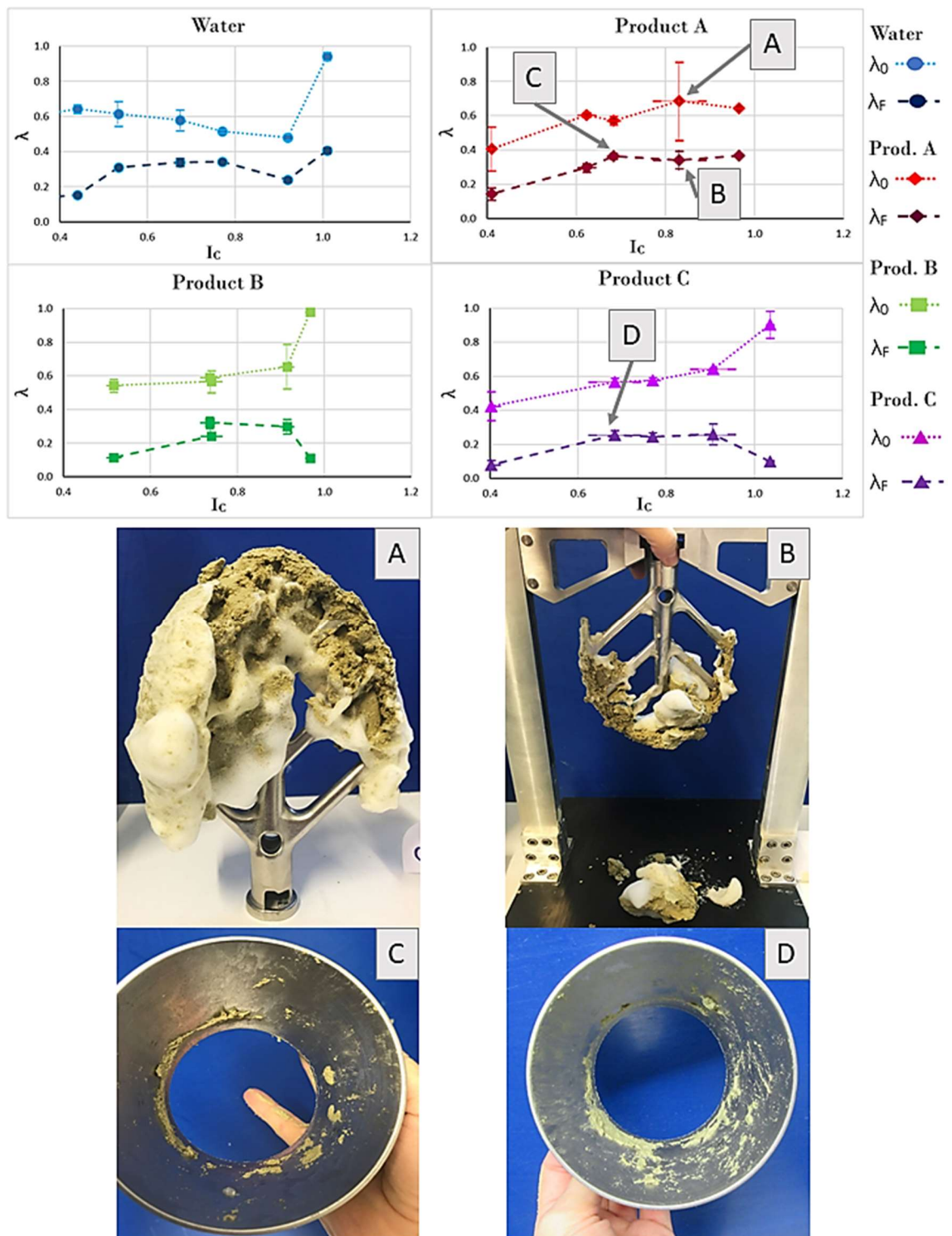


Figure 6-13. Results for Bsfine50 conditioned with only water, and with Products A, B and C. The pictures in the bottom are correlated with the letters in the graphs.



Figure 6-14. Alternative experimental trials with anti-clay polymer and Kmedium50 soil.

6.3.5 Flow table

The flow table, as detailed in the study by Oliveira et al. (2018c), was applied to evaluate the flow of cohesive soils, with and without conditioning. Several conclusions were already drawn from the flow behaviour of mixed clay–sand samples, such as the significant difference in the flow between samples containing kaolinite and bentonite, where the kaolinite samples had a higher flow than the ones with bentonite. The samples with bentonite also showed the lowest flow values for soils containing 50% and 70% of clay, and not 100%. The main concern was the relation between flow and clogging tendencies, if for the same consistency, soils that flow less would have a higher tendency for clogging.

In Figure 6-15, the results from tests conducted with the flow table for all tested samples with I_C below 0.8, after jolting the table 40 times ($Flow_{40}$), were plotted together with the changes in consistency and with the final clogging category, λ_F (reversed axis). The straight lines show the clogging λ_F values. As it is reversed, the points closer to the x-axis have the highest clogging potential, and the dotted lines represent the flow in mm. The top graph (Figure 6-15A) displays all the tested samples with only water, for different clay fractions and both clay minerals. The bottom graph (Figure 6-15B) shows the samples with 50% of either kaolinite or bentonite, mixed with either water, or additives A, B, or C.

It cannot be entirely assumed that in all cases, the lowest flow has also the highest clogging potential, but some conclusions can be drawn for the samples mixed with water, as follows:

1. In practically all consistencies, the lowest flow was always Bsfine70, followed by Bsfine50 and, then, B100, many times holding similar values.

2. Bsfine20, among the bentonite samples, always had the highest flow, and Ksfine30 had also the highest flow among kaolinite samples, except for consistencies close to the liquid limit ($IC = 0$).
3. Kaolinite samples were mainly the pure clay samples having the lowest flow, sometimes together with the 50% sample.
4. For $IC = 0.2$, the lowest flows were for the samples with bentonite, followed by the kaolinite ones. However, for the clogging, the highest values were among kaolinite samples, which is just opposite of the flow results. Additionally, by only looking at bentonite samples, the specimens with 70% of clay had the highest clogging values.
5. For $IC = 0.4$, pure bentonite samples had a higher clogging value than the other ones, except for 100% and 50% of kaolinite; however, the flow had similar results as $IC = 0.2$ for bentonite, and only kaolinite had a swap between Ksfine50 and K100.
6. For $IC = 0.7$, for instance, the bentonitic sample with 70% of clay had the second highest clogging potential, just after the pure kaolinite sample; it also had the lowest flow among all samples.

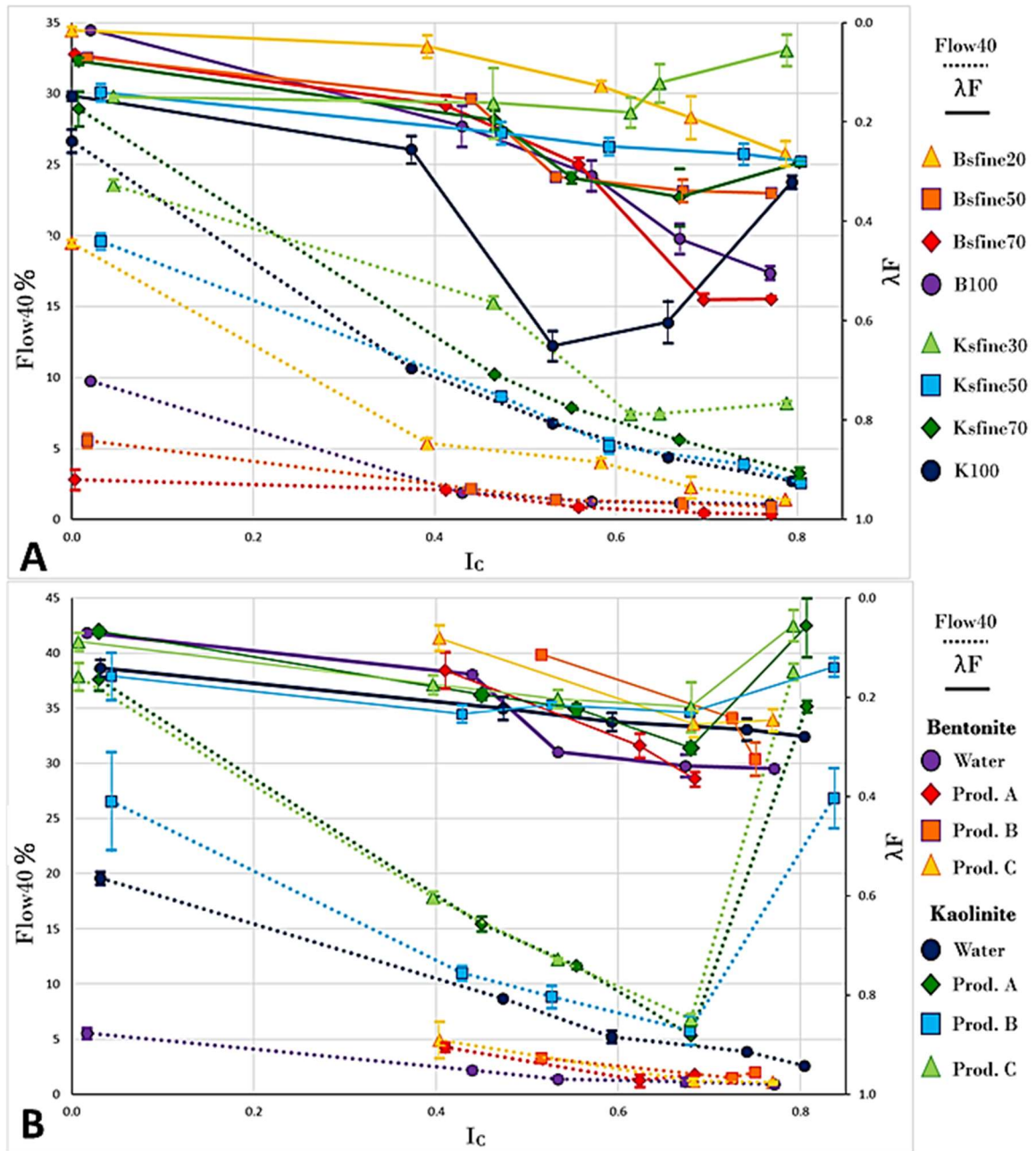


Figure 6-15. General comparison between Flow₄₀ percentage values (y-primary axis) and clogging values of λ_F (y-secondary axis, values reversed), for different I_c s: A for samples mixed with water, for different clay content and fine sand; B for samples with 50% of clay and fine sand, mixed with water and additives.

As illustrated in Figure 6-8, a change in the CB_{Factor} influenced greatly the results for pure bentonite. By assuming a lower CB_{Factor} value for all the samples with bentonite, a better

correlation could be obtained with the flow behaviour and clogging potential, and it is probably possible to assume that the lowest flow indicates the highest clogging. To validate this assumption, it is necessary to conduct more tests, including other clay minerals, such as illite and natural soils.

For the samples tested with additives, it can be easily noticed that the foam had a greater influence on the flow of the kaolinitic samples, depending on the FIR mainly, but it has not much influence on the clogging values.

The highest I_c point went from a flow of approximately 3 %, for the sample with only water, to a range of 27 to 40 %, for the ones with additives. The clogging values were also reduced. However, as it can be observed in the video (Video 6-1), these conditioned samples had a spongy aspect, with excessive air, and that could imply as having too much air inside the excavation chamber, for example, which could be an issue in case of mixed faces, with loose materials on top. In this case, the highest flow also implied the lowest clogging potential.

Besides this extreme example, the other samples did not have much significant differences in the flow/clogging behaviour. However, with slight differences, it can be said that for the clogging effect, Product B worked better for the bentonite samples and Product C, for the kaolinite ones.

In general, the addition of foam always implied higher flow values, for the same consistency, which confirms that foam, with or without polymers, functions as a lubricant, reducing the friction of soil surface, in this case of laboratory testing, the glass surface, or the cutterhead, in the case of a tunnel boring machine.

The samples with bentonite did not seem to change much, neither in flow or clogging, with the addition of the products, or, in the case of I_c of approximately 0.68, foam A even implied higher clogging values. For low plastic soils, such as the kaolinite samples, the addition of foam

had a higher influence, in terms of flow increase and friction decrease, than samples with high plasticity, such as the bentonite samples.

The video compilation is essential for further observations about clogging and flow behaviour. It seemed that an ideal muck for bentonite was not in the same I_C range as the kaolinite samples. If a guess could be made, the bentonitic samples would have an ideal muck range of approximately 0.4 to 0.75, as defined by Maidl (1995); however, the kaolinite samples, when mixed with sand, would have a lower I_C ideal range, around the liquid limit. This has also to be further crosschecked with real tunnel drives.

6.4 Conclusions

It was demonstrated here that by a simple yet complete laboratory routine, it is possible to characterise and classify cohesive samples, with or without sand content, making assumptions and forecasting the soil behaviour regarding EPB excavation. It is also feasible to include and compare changes in behaviour of soils after conditioning, which will not only avoid future issues along the excavation, but also reduce the costs of tunnel operation by avoiding the use of unnecessary products or adjusting the initial suggested quantities to the minimum required.

A proper and extensive description should always be the focus of any geological–geotechnical characterisation, not only by a detailed summary of the soil and muck characteristics, but also by complete photographic and video documentation. The classification may be altered or adjusted in the future, as it is still under development, requiring an extensive validation with real tunnel drives. However, with a precise and abundant description, the initial classifications can be adjusted afterwards, whenever necessary.

The conditioning of soil in an EPB operation depends on many factors, and not solely on the soil and conditioning characteristics, but also on the project logistics, such as details of the available water, or the price and regulations of the muck disposal, for example. Decreasing the

cost of additives while avoiding clogging by increasing the water content of the muck; however, this could create a major issue with the muck transportation and disposal, possibly justifying the initial costs with conditioning polymers.

Additionally, the wide range of soil/rock types, together with the influence of the local groundwater, makes it a challenge to define any ideal state of the excavated material in a broader scenario. Including, as well, the details of any project, it seems unrealistic to talk about ‘ideal mucks’, or ‘ideal consistencies’, as this will vary from point to point and from a tunnel project to another.

The ‘ideal’ must be found for each project and for that, a good characterisation methodology is essential. The greater the understanding about the natural soil and conditioned muck before the excavation occurs, the easier it will be to adjust afterwards any operation parameters, including making the best choices of soil conditioning.

In the case of different geological scenarios, such as soils originating from a sedimentary deposit versus a residual ground, for example, it will be necessary to identify the main differences in the properties of the ground that would affect the particularities of the excavation with an EPB machine, always questioning if a certain classification scheme would entirely fit a specific scenario.

Finally, this combined methodology provides a comprehensive choice for the characterisation of cohesive soils, from any geological environment, assessing as well as the influence of conditioning on the flow behaviour and on the clogging potential. To validate the assumptions made here, it is extremely necessary to conduct more tests by including other clay minerals, such as illite, for example, and, particularly natural soils.

Most of all, the methodology should be applied in real tunnel projects, before excavation, and, afterwards, as a back analysis of the assumptions made here, being then feasible to improve

the proposed methodology. A flow table, for example, could even be installed at the back of an EPB machine, where the muck can be directly checked and decisions about conditioning will not rely merely on the experience of an individual feeling the muck.

The greatest value of this laboratory routine is the easiness of reproduction in any other laboratory facility, making future comparisons possible, and, perhaps, allowing a standardisation that can be applied in tunnel projects worldwide, establishing the choices of soil conditioning and tunnel operation in more scientific-based conclusions.

6.5 Acknowledgements

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Chapter 7

Discussion and Conclusions

7.1 Discussion

This thesis presents the results from laboratory tests conducted with assembled soils, investigating the changes in certain variables. The final laboratory routine developed here, even though having as focus a cohesive residual soil, can be still applied for the investigation of any cohesive soil, including the conditioning of this material, evaluating its efficiency for an EPB excavation.

Several conclusions were already drawn in each preceding manuscript chapter. In this chapter, they are compiled, incorporating any missing points and indicating limitations. These limitations have been included in each sub-chapter to facilitate the unfolding of ideas, as well as some future recommendations. A summary of those contributions and recommendations for the research field of EPB excavation along cohesive soils is then provided.

Additionally, it includes some insights from a second calibration conducted with a new assembled device ATUR and some brief conclusions from tests conducted with natural soils. These results from the natural soils, as well as the location, cannot be fully disclosed here. However, the insights obtained from these tests provide a first impression of tests conducted with natural soils from a testing campaign to be applied in an actual tunnel project.

7.1.1 Combined Methodology for EPB Cohesive Soils Evaluation

Chapter 6 provided the final setup of a complete laboratory routine to empirically evaluate the clogging potential of any cohesive soil, assessing the effectiveness of additives used

to decrease clogging in an EPB excavation, and a general characterisation of the flow of the tested materials.

The major positive aspect of this routine is its simplicity, as all the used devices can be relatively easily acquired or assembled. The test routine, however time-demanding, is rather simple, being feasible to conduct tests in any soil laboratory or even at the tunnel project site. It is a tool which can be used not only for academic purposes, but also in the tunnel industry.

Tunnel projects can gain a lot by including a preliminary phase of soil characterisation to evaluate clogging potential and flow behaviour before any excavation is started, only by using the material obtained from borehole investigations. Testing the material obtained from the conveyor belt, at the job site, provides also a guidance in the adjustments of the excavation and soil conditioning parameters. At the tunnel site, the tests could, for example, be conducted in the backup area of the tunnel boring machine, or even in a container set up just outside the tunnel entrance.

The rather small quantity required for the tests, between 600-1200 g of moisturised soil per test, is much lower than the required for a slump test, which requires around 8-12 kg per test. Obtaining these quantities of soil directly from the material cored from the drilling phase allows for an early characterisation of the ground and the possibility of anticipating possible risks of clogging occurrence, or any issues related to the material which will be later excavated. It would also permit to investigate the effectiveness of any additive, such as foams and polymers, which could result in a considerable optimization of costs for the tunnel project.

Conducting slump tests with a clayey soil can be rather challenging because of the mixing procedure. When mixing the dry components with a concrete mixer used for the slump test (Vinai 2006, DIN EN12350-2 2009, Budach 2012, Galli 2016), there will be a considerable loss of the mass of the clay portion, as well as a formation of a cloud of dust. The mixing

procedure with water is also another issue, as the concrete mixer would not be able to completely break the clay lumps. This was solved, as explained in Chapter 5, by mixing the wet soil by hand, which requires a very long time (around 2 hours per sample) for a quantity of mass between 8-12 kg. Mixing the sample for a test with a flow table, on the other hand, takes no longer than 20 minutes.

It is essential to validate this entire methodology with real tunnel drives. This validation was not done in this thesis due to the main limitations related to the academic research in the field of tunnelling, even more in the mechanised area and civil tunnelling. As mentioned earlier in section 1.2, civil tunnel projects usually involve many stakeholders, with different contractual arrangements and most of the time claims happen during the project, extended even several years after the project is already completed.

Still, a validation with real tunnel drives must be made in the future, which might require a longer research period and more parties involved. It would be ideal to first conduct a preliminary investigation and characterisation of the soils to be excavated, including a conditioning assessment. Then the results can be later cross-referenced with a testing phase conducted while the excavation happens. Therefore, real clogging situations can be effectively characterised by a proper calibration of this proposed laboratory routine. And, the higher the amount of compared data, with more tunnels being excavated and analysed, the more confidence the researchers and technicians can have in the results.

The routine should also be tested with other types of soils assembled in the laboratory, such as clay with illite or other types of clay minerals as the main component. Other soil components such as mica or carbonates, for example, can be included in the investigation. Natural soils should be also tested, considering the limitations of its variances and the challenges that would be incurred to control the variables throughout the tests.

An initial characterisation of natural soils which are going to be excavated by tunnel boring machines in actual projects was conducted. The location and projects cannot be disclosed here. However, great results were achieved resulting in valuable insights, especially concerning the application of the device ATUR. It was clear the difference between certain soil specimens, also for different I_{Cs} , along the tests conducted with the combined testing methodology.

For one soil specimen, for example, with high plasticity and with expected strong potential for clogging, the beater had to be dropped forty times with ATUR until a very small portion of the soil would be released from the beater. Other soil specimen, differently, would had practically the entire mass of soil released after a single drop. Both specimens had the same λ_0 , for similar I_C , showing how the additional step with ATUR can be extremely useful in characterising the potential for clogging.

This combined laboratory routine adopted in this body of research suffered several alterations along its evolution, and what is considered as the final procedure (described in Chapter 6), still could be improved. One of these improvements is the base of the device ATUR, which would directly affect the calibration stage and the final value for the stickiness parameter.

Figure 7-1 shows what is here considered as last in this research, in terms of the parameter λ_F , the final resulting stickiness parameter, considering the average of all the λ s (λ_0 ; λ_1 ; λ_2 ; λ_3 ; and λ_7) and the CB_{Factor} , which refers to the procedures for cleaning the beater and bowl, as detailed in Chapter 6.

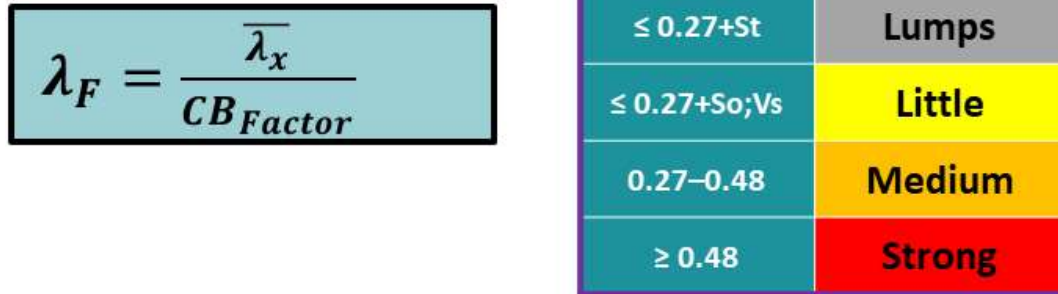


Figure 7-1. Final stickiness parameter calculation and the clogging potential resulting ranges.

Initially, when the idea of dropping the beater started, as explained in Chapter 3, it was dropped onto different surfaces, and one of the last trials was to drop on the Casagrande cup base, which has a standardised stiffness (ASTM D4318 2017). Casagrande (1932, 1958) spent several years defining the specifications of this base, as it had a great influence on the results, as revealed in Chapter 4. The same challenge is expected for the base of ATUR. Undoubtedly, this will require alterations in the future, including a new calibration stage where the relations and ranges of clogging potential are adjusted.

A second ATUR device was assembled for testing the natural soils, in this undisclosed location. It was tried to provide a very similar version of ATUR. However, with lack of specific standardisations of this device, defining precisely the quality and weight of the metal parts and the wooden base, it was a challenge to achieve a perfect similar assemblage. Therefore, a preliminary calibration stage was conducted with pure kaolinite, prior the tests with the natural soils, using the same methodology as detailed in the Chapter 6.

The second calibration was a great chance to verify in detail all the assumptions exposed in the Chapter 6, achieving satisfactory results and a successful calibration, with very similar

results than the ones detailed in the Chapter 6. Once the device is standardised, then the calibration stage might not be required.

The HOBART mixer was chosen as a brand because it is the mixer used in most of the concrete and soil laboratory facilities. It is an industrial machine with a good quality and greater resistance than a regular bakery mixer acquired in any supermarket. Still, other mixers brands could be tested in the future, making it even easier to conduct the tests anywhere, perhaps with even a lower cost. For that, it would be necessary to compare the results with the original used mixer.

The CB_{Factor} was a great add to the entire methodology, allowing more flexibility in the final values, considering the observations of the cleaning procedure. In the case of pure bentonite, for example, it was clear that the difficulties encountered when removing the remaining stuck soil in the beater and bowl were bigger than any other option with mixed bentonite soils or with the kaolinite samples.

The difficulties in cleaning the beater and the bowl decreased as the amount of sand increased. It is expected that the increase in sand would reduce the difficulties in removing the stuck soil at the TBM excavation tools, as well as the clogging occurrence. Nevertheless, due to the abrasiveness of the sand portion, it is expected that there will be an increase in the wear of the excavation tools with the increase of sand in the excavated soil.

The benefits of adding this factor were even clearer when characterising and testing the natural soils. The clay that had shown the highest clogging potential, which stayed stuck longer at the beater, even after forty drops, presented difficulties to be removed from the bowl and beater, in the cleaning process, requiring a brush and soap. Other specimens of mixed clay-sand soils, as well as residual soil with a clay component, were tested. All of them did not present the same

challenges in the cleaning procedure, providing as well lower values of the final clogging parameter λ_F .

The CB_{Factor} could be improved with some new techniques for washing and removing the soil in a methodical way. For example, for a certain period the bowl and beater being washed in a regular washing machine, checking the outcome, if after a certain time there was still soil left stuck in the beater and/or bowl.

In the absence of a precise methodology to calculate the CB_{Factor} , a detailed characterisation of the tested soils should be enough to provide comparative results of the potential for clogging between different tested soils. Moreover, in this research, the CB_{Factor} was divided between whole numbers, to be chosen between 1, 2 or 3 (Chapter 6). However, when testing the natural soils, the choice of fractional numbers seemed to detail better the CB_{Factor} , and, consequently, the results.

Table 7-1 shows how successful the method and proposed calculations were to correlate the final clogging potential results for pure kaolinite, with the results obtained with the clogging assessment chart from Hollman and Thewes (2012). For each I_C , at least 4 tests were conducted, with an average of 7 tests per each target I_C , providing low standard deviation values, showing how consistent the results are.

It is advisable to go through a similar calibration procedure with pure clay, with IP ideally between 10 % and 40%, to calibrate the base of ATUR, whenever this device is newly assembled, as it was conducted to assemble the second device for testing the natural soils.

Table 7-1. Resume of calibration for pure kaolinite samples, showing average values for λ_0 and λ_F , the standard deviation and comparing with clogging potential from Hollman and Thewes (2012).

TEST ID	N° of Points	I _C	λ_0 – no drop	CB Factor	Final λ_F	Clogging Category	Hollman and Thewes Clogging Potential
1	7	1.01	0.68	St II	0.22	Lumps	Lumps
SD		0.02	0.10		0.04		
2	9	0.89	0.69	St II	0.32	Medium	Medium
SD		0.03	0.08		0.04		
3	7	0.79	0.68	St II	0.33	Medium	Medium
SD		0.01	0.04		0.02		
4	8	0.66	0.62	So II	0.60	Strong	Strong
SD		0.06	0.04		0.04		
5	7	0.53	0.66	So I	0.65	Strong	Strong
SD		0.01	0.04		0.03		
6	7	0.37	0.62	Vs II	0.26	Little	Little
SD		0.07	0.05		0.03		
7	4	0.01	0.51	Vs II	0.15	Little	Little
SD		0.05	0.02		0.01		

Figure 7-2 illustrates why this plasticity range is recommended to calibrate ATUR with a pure clay sample. It is suggested to consider a soil with a plasticity index value higher than 10%, to assure that it is a plastic material, and up to 40%. Most samples considered in the universal diagram for clogging (Hollman and Thewes 2012) are between this plasticity range of 10-40% (samples represented by blue diamonds, red diamonds, and black triangles).

The sample for calibration should ideally contain 100% clay mineral fraction as the graph from Hollman and Thewes (2012) did not consider mixed soils. In mixed soils, here the sand contribution would add a friction component, not covered by this universal clogging diagram.

The calibration should not be conducted using soils with IP values much above than 60%, as there is not an assurance of validation with the universal clogging diagram from Hollman and Thewes (2012), considering that the highest IP used in this diagram lies around 70% (Figure 7-2).

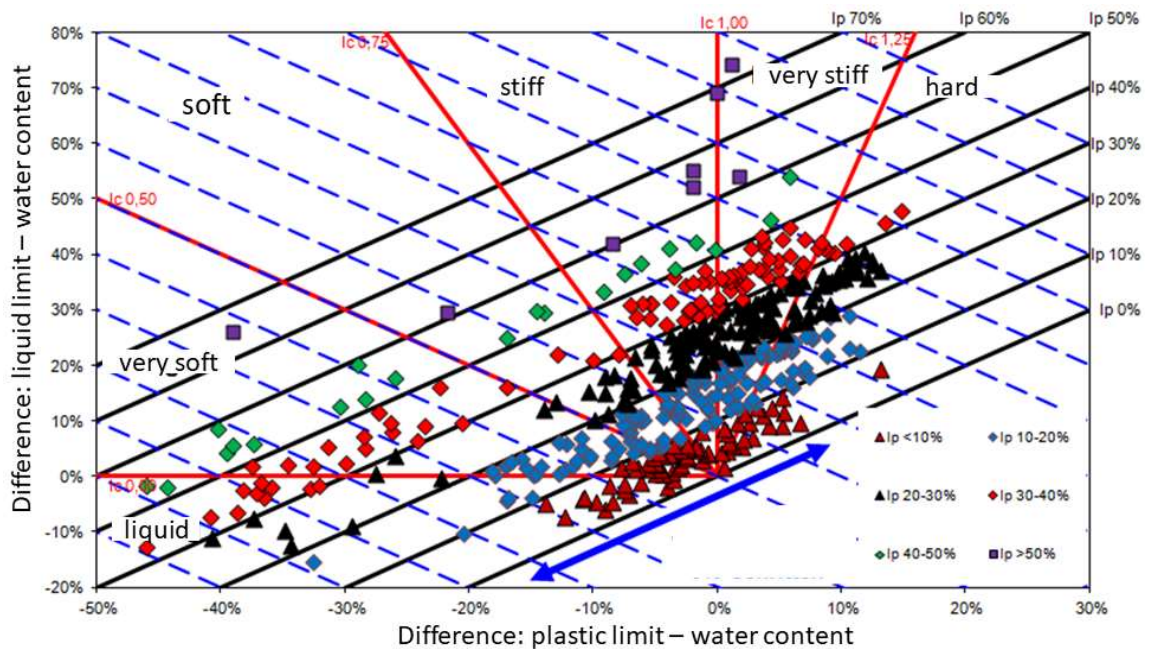


Figure 7-2. Universal clogging diagram sorted by plasticity index of the considered data samples from tunnel drives (translated from Hollman and Thewes 2017).

Table 7-2, for example, shows test results conducted with pure bentonite with an unsuccessful correlation between the obtained values of λ_F and the clogging potential fields as defined in the universal diagram from Hollman and Thewes (2012).

The plasticity index (IP) of this bentonite is 435%, as shown in Table B-4 (Appendix B), considerably greater than any data considered for the development of the universal clogging diagram as seen in Figure 7-2, where this plasticity is not even represented. Another aspect to consider is the activation nature of this bentonite. Activation of bentonite is not a naturally-occurring phenomenon and pure bentonite occurrence in nature is restricted to few locations which are unlikely to be excavated by an EPB machine or any TBM.

Table 7-2 shows that bentonite clogging is already occurring close to the plastic limit values at I_c around 1, where in the Hollman and Thewes (2012) graph, the material should be occurring as lumps. The strongest clogging occurrence is concentrated in ranges of higher I_c

values (something in between 0.75 and 0.95) when compared to Hollman and Thewes (2012) predictions, where high clogging potential would be occurring between lower I_C ranges (between 0.55 and 0.75).

Table 7-2. Resume of calibration for pure bentonite samples, showing average values for λ_0 and λ_F , the standard deviation and the clogging category from Hollman and Thewes (2012).

TEST ID	N° of Points	I_C	λ_0 – no drop	CB Factor	Final λ_F	Clogging Category	Hollman and Thewes Clogging Potential
1	4	1.02	0.51	St I	0.40	Medium	Lumps
SD		0.01	0.07		0.06		
2	4	0.92	0.59	St I	0.56	Strong	Medium
SD		0.02	0.15		0.10		
3	3	0.77	0.60	St I	0.50	Strong	Medium
SD		0.00	0.01		0.01		
4	3	0.67	0.65	So I	0.43	Medium	Strong
SD		0.00	0.04		0.03		
5	3	0.57	0.65	So I	0.29	Medium	Strong
SD		0.00	0.02		0.01		
6	3	0.43	0.57	Vs I	0.21	Light	Light
SD		0.00	0.05		0.04		
7	3	0.02	0.05	Vs II	0.01	Light	Light
SD		0.00	0.04		0.00		

When the tests were conducted, the feeling while manipulating the samples was that the material was indeed stickier when closer to the plastic limit. When this material was mixed with the HOBART, the leftover in the bottom of the mixing bowl could not be removed easily, and that was one of the causes that led to the differentiation of categories within the CB_{Factor} . Unless this fact is cross-checked with real tunnel drives, there is no possibility yet to validate any assumption made here regarding this aspect of the bentonite samples.

Another point which needs to be discussed is the comparison between the final empirical stickiness values of λ_F of pure kaolinite (Table 7-1) and pure bentonite (Table 7-2). Figure 7-3 facilitates the visualization of this contrast.

Both tables and Figure 7-3A show higher clogging values for the kaolinite, when using a CB_{Factor} of 1, than for the bentonite. This is opposite to what is expected, considering that bentonite is a swelling clay and as according to Thewes (1999, 2005), higher plasticity and swelling clays would have more tendencies for clogging during a shield excavation.

This CB_{Factor} was adjusted to 0.5 instead, providing a correlation which seems to be more realistic, showing higher clogging values for bentonite, as presented in Figure 7-3B and, as well as, in the Table7-3. Again, these are just suggested adjustments and it would only be feasible to reach a conclusion when these tests can be conducted with real tunnel drives, applying this methodology.

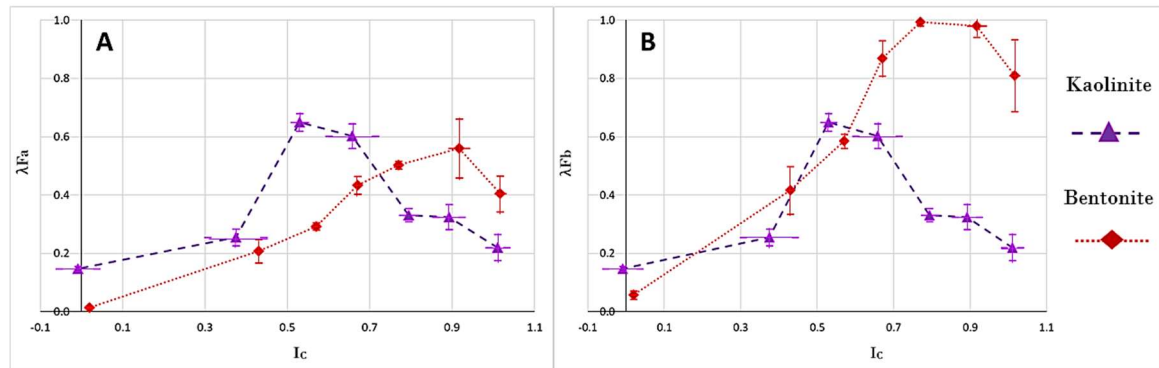


Figure 7-3. Comparison between the results for the final empirical stickiness parameter (λ_F) between pure kaolinite and pure bentonite. On A, it is considered the CB_{Factor} value as 1, and on B, as 0.5.

However, as mentioned in the section 2.5, especially concerning the occurrence of clogging along excavations in the residual soils of São Paulo and Singapore, and crossed with the results from all tests, clogging occurrence can also be expected when excavating soils having kaolinite as the main clay mineral component.

Table 7-3. Resume of calibration for pure bentonite samples, with a modified CB_{Factor} to 0.5.

TEST ID	N° of Points	I_c	λ_0 – no drop	CB Factor	Final λ_F	Clogging Category	Hollman and Thewes Clogging Potential
1	4	1.02	0.51	St I	0.81	Strong	Lumps
SD		0.01	0.07		0.12		
2	4	0.92	0.59	St I	0.98	Strong	Medium
SD		0.02	0.15		0.04		
3	3	0.77	0.60	St I	0.99	Strong	Medium
SD		0.00	0.01		0.01		
4	3	0.67	0.65	So I	0.87	Strong	Strong
SD		0.00	0.04		0.06		
5	3	0.57	0.65	So I	0.59	Strong	Strong
SD		0.00	0.02		0.02		
6	3	0.43	0.57	Vs I	0.42	Light	Light
SD		0.00	0.05		0.08		
7	3	0.02	0.05	Vs II	0.06	Light	Light
SD		0.00	0.04		0.01		

Another interesting aspect of the testing routine is the modification done for the sample preparation. In Chapter 3, the sample was initially dry, water was added to achieve the target consistency while the test was being conducted. The consistency was checked after the test, not always successfully achieving the target I_c . In Chapter 6, the water content was added to the sample, and the target I_c value was checked before conducting the test. The latter option, even though it required a longer initial preparation time, not only reduced the total time of the test but also produced lower standard deviation values. It was also possible to start the test with small lumps, reproducing the reality of an EPB excavation, as explained by Peila et al. (2016).

The quantity of the wet soil to be tested, and as well as the quantity that got stuck initially at the beater, did not seem to considerably affect the end results. The same test was conducted with different quantities of the initial tested soil. The soil quantity did not overload the machine and a minimum amount capable of covering at least 1/3 of the flat beater when already placed in the mixer was maintained. It was not expected that there would be great differences in the results if the moisturised soil is kept between 500-1200 g. Nevertheless, it is prudent to always repeat the same test, for the same I_c , at least three times.

The biggest limitation of this test was encountered when testing soils with foam, as the biggest challenge was to properly mix the foam and soil, without forcing an unrealistic soil mixture, using higher speeds of the mixer for a longer period than the suggested 3 minutes. In an EPB excavation, the mixing process occurs mainly ahead of the cutterhead, where the foam pipes are injecting the foam directly to the ground in contact with the face of the machine. Once the excavated and conditioned soil (the muck) goes inside the excavation chamber, there is, most probably, no real further mixing.

It is not easy to exactly reproduce how much mixing should be realistic or not in these laboratory tests to reproduce the reality of an EPB excavation. It is important to consider the differences in the RPM between the HOBART machine and the EPB, while the first is around 100 RPM, the second unlikely exceeds 5 RPM.

This fact was already pointed out by Peila et al. (2016), where the authors explain that the soil is scrapped by the machine in lumps, and the additives are kept only around the lump surface, as illustrated in Figure 7-4. To evaluate the precise conditioning of a clayey soil, considering that the mixture will not occur homogenously in the EPB operation, will continue to be a limitation for any EPB laboratory test and this should be always considered in the interpretation of the results.



Figure 7-4. Lumps of clayey soil which were excavated by an EPB machine and showing the conditioned only around its surface (Peila et al. 2016).

The fact that the tests were always conducted in atmospheric pressure is another limitation, which cannot be easily solved. As demonstrated by Mooney et al. (2016), the changes in pressure have a great effect on the foam behaviour and, consequently, it could also affect the results of tests with the conditioned material. This fact should be taken into consideration, though it is not yet clear how. When certain foam parameters, such as FER and FIR, produce a certain result in an atmospheric pressure, they might produce different results inside a pressurised chamber.

For example, a certain FER which seems to produce reasonable results in the laboratory, could produce a significant quantity of air which is held in the top of the excavation chamber. If a loose sand layer is exactly in the same level as this contained air, it could lead into loss of face support. Therefore, not necessarily ideal target conditions achieved in laboratory tests would be ideal as well in the tunnel excavation.

Regarding the flow table, an automatic machine could contribute to the testing routine, avoiding exhaustion of the testing personnel, as the number of drops, combined with the high quantity of required tests, could lead into ergonomic injuries. The same ergonomic issue could be applicable to the slump test, adding one more advantage to conducting tests with cohesive soils using an automated flow table. However, these automated flow tables are adjusted for 15 drops, instead of 40, not always allowing adjustments to a different number of drops.

A correlation between the results from the flow table and the clogging evaluation was also attempted, as mentioned in Chapter 6. It seems that it can be concluded that for a same consistency of a cohesive soil, the decrease of flow values would somehow be indicative of a higher tendency for clogging occurrence. Again, this should be also validated with real tunnel data, besides further tests with other clay minerals.

Nevertheless, there were limitations to run tests on the flow table with soils with $I_C > 0.4$. Also, the higher the clay fraction, for these higher I_{Cs} (>0.4), higher the difficulty to conduct properly a test, as the soil could not be easily placed in the cone, or, in certain cases, would not flow even after 40 drops. As mentioned in the Chapter 5, this would imply in a considerable limitation for pure clay soils, as the ideal consistency, as stated by Maidl (1995), lies between 0.4 and 0.75. However, that seemed not to be the case of mixed soils, with probably “ideal” ranges below 0.4, still to be verified in tunnel projects.

There are also other methods to remove the muck from an EPB screw conveyor than only the muck cars or conveyor belt. A muck pumping system could be attached to the end of the screw conveyor to remove the muck. In this case, a more liquid muck would be ideal, which would justify the preliminary characterisation of the muck using the flow table. This, as well, must be verified with real tunnel excavation scenarios.

Another limitation of the flow table is to include tests which contain grain sizes greater than 2 mm. Because of the size of the cone, soils containing a considerable amount of a coarser fraction should be tested with the slump test, consequently, a larger cone.

The results of the flow table with bentonite showing that the lowest obtained flows were between 50% and 70% of clay fraction, and not with 100% of bentonite, could also be applied in other fields of applications. For example, the possibility of adding a mixture of bentonite and sand, instead of only bentonite, as an additive for the cases of mixed ground conditions, as illustrated in Figure 2-15B, could produce a successful workability of the muck. The mixture of 50-70% of bentonite with sand, could perhaps be more effective in obtaining the ideal slump and the target ranges of muck workability than adding 100% bentonite.

This is similar to one of the principles used in the TBM Variable Density[®], from Herrenknecht (Burger 2013, Schaub and Duhme 2014, Herrenknecht Website), with the high-density slurry, when the bentonite is mixed with the own excavated soil, implying in a decrease of the flow, ideal for highly permeable soils, such as gravels, or mixed faces with rock chips.

Considering the higher cost of bentonite when compared with the cost of sand, the same flow results could be obtained with only 40% of clay fraction as was obtained with the 100% clay fraction, as illustrated by Figure 5-6. This outcome suggests a possible lower-cost alternative that could be applied in practice.

This is also added as a suggestion for future research, where other parameters of this bentonite-sand mixture could be investigated, such as the changes in the permeability and the influence in the required conditions for a certain application (e.g. diaphragm walls, artificial soil used for the launch of TBMs).

7.1.2 EPB Excavation in Mixed Transitional Ground

As explained in section 1.2, the main intention of this research was initially to investigate tropical residual ground due to the lack of any basic research on the soil conditioning of mixed clay-sand soils.

Most researches in EPB excavation, soil conditioning, and clogging potential has been conducted in the northern hemisphere, which has led to a focus on cohesive soils from sedimentary origins. Consequently, the universal clogging diagram from Hollman and Thewes (2012), or the clogging evaluation tests from Zumsteg and Puzrin (2012), besides others, did not consider the effect of having different clay percentages in an intrinsically mixed cohesive soil, such as the residual soil.

Chapter 4 brought to light these questions, especially highlighting queries concerning the applicability of the Atterberg limits, and, consequently, the consistency index when dealing with those mixed cohesive tropical soils. It seems more realistic to apply the proposed modification of the Atterberg method, as suggested in Chapter 4, which could in part offer a solution for the case of the residual ground, especially when characterising materials to be excavated by an EPB machine.

The universal clogging graph from Hollman and Thewes (2012) did not consider those mixed cohesive soils, therefore, the combined methodology here proposed looks as a feasible alternative to classify the clogging potential of clay-sand mixed materials.

It was also suggested in Chapter 4 to apply the equation defined by Polidori (2007) to calculate the plastic limit, considering the liquid limit and clay fraction values. Polidori has used several samples to develop this equation, determining a relationship between the liquid limit and plastic limit values and comparing the slope of the regression lines for different clay fractions, as illustrated in Figure 7-5. The slope value found was 0.04, considering these regression lines.

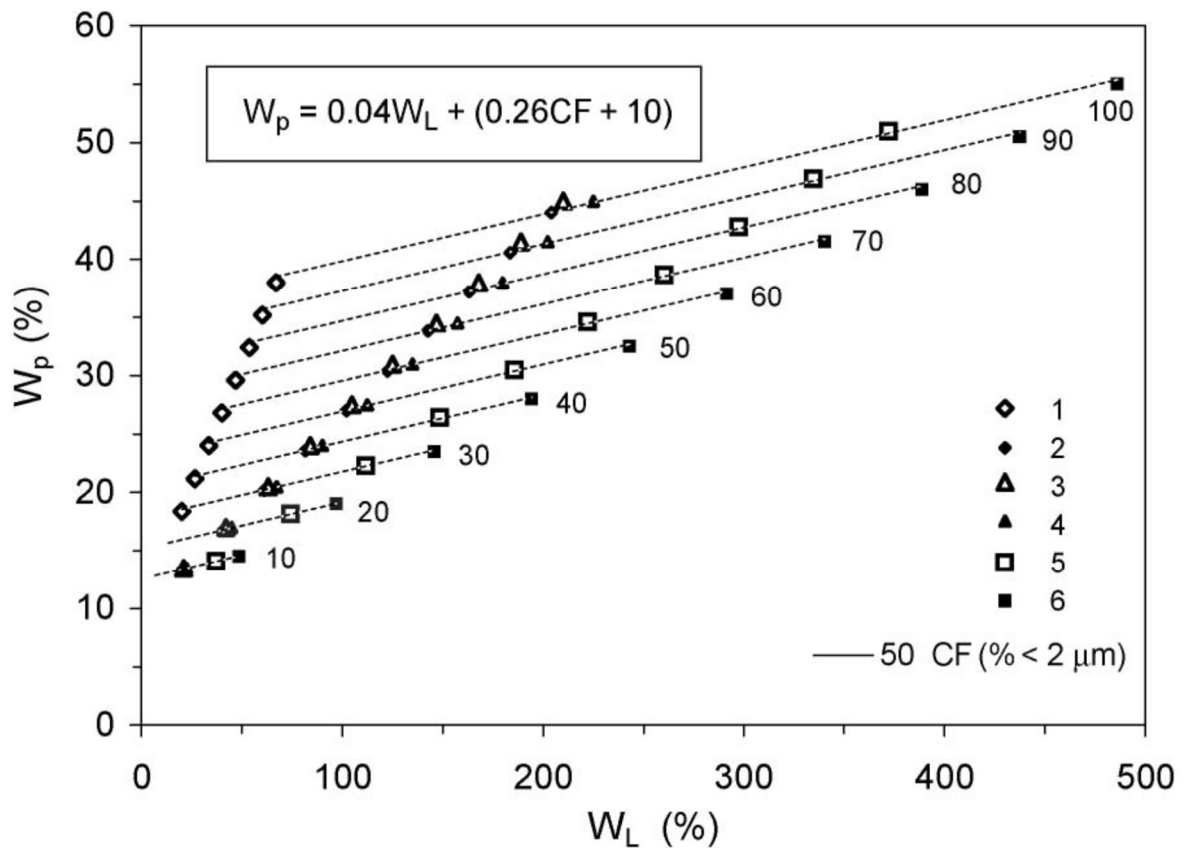


Figure 7-5. Relationship among plastic limit (W_p), liquid limit (W_L) and clay fraction (CF). The regression lines define variation of W_p as function of W_L in mixtures with the same CF (Polidori 2007).

The same comparison between the W_p and W_L values was done for the tested samples, separating them by the CF, including regression lines as shown in Figure 7-6. There was not a good fit as observed in the graph from Polidori (2007), presented in Figure 7-5, for several reasons. One is that the different grain sizes considered for the sand fraction could be influencing the low R-square values (for the samples with CF 30%, 50% and 70%). This same phenomenon is not happening for the pure clay line.

The samples with Friedland clay also showed a higher anomaly, being far from the regression line when compared to the kaolinite samples. This happens due to the low quality of

the product, not being advisable to be used for tests which require precision and control of the variables, especially in terms of controlling the amount of clay as a grain fraction.

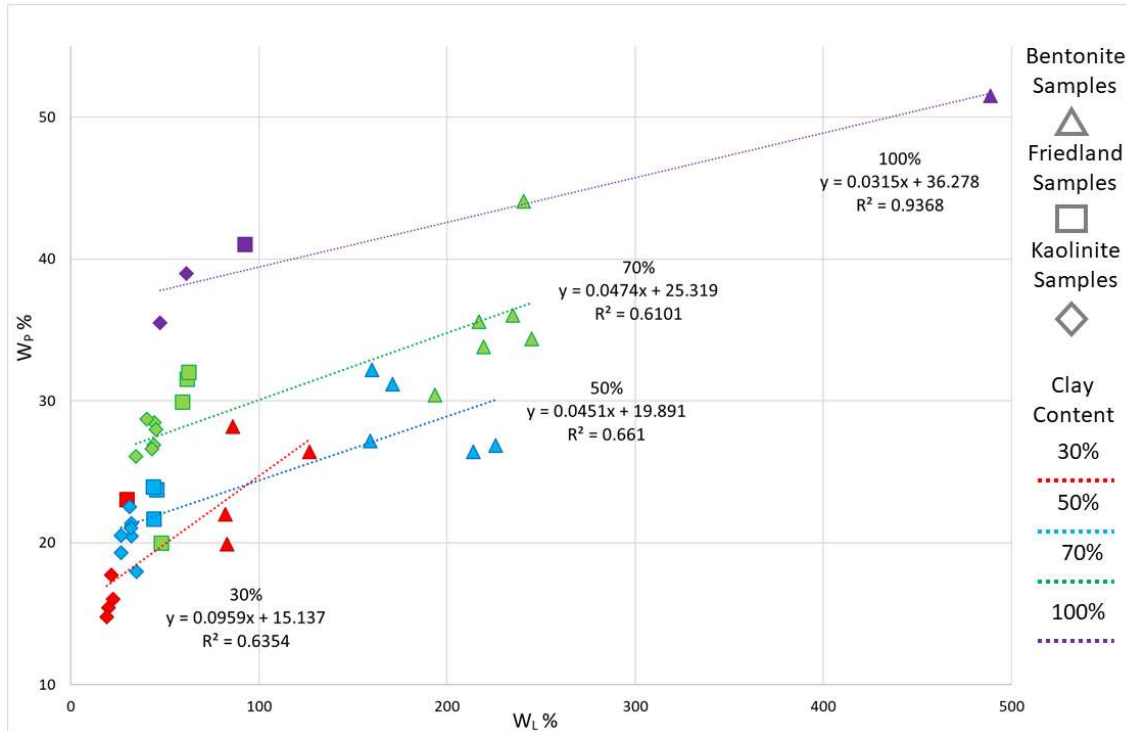


Figure 7-6. Relationship between W_P , W_L and CF, for the tested samples in this thesis. It was only considered samples with 30% (red), 50% (blue), 70% (green) and 100% (purple) of CF, as the other fractions were not tested for all the three main tested clays (kaolinite, bentonite and Friedland).

The samples with bentonite also are not so close to the regression line, and that is a consequence of its high plasticity and swelling behavior, which brings a higher challenge to precisely define the Atterberg limits. Samples with bentonite should be tested several times to obtain its Atterberg limits, which was not done in this research and is included here as a future recommendation.

The regression line of 30% is the one with a completely different slope than the other three lines. This is consistent with what other authors have mentioned; specifically, it has been

stated that for CF above 40% there is a linear relation between the W_P and the W_L as the CF increases, not happening for lower values of CF (in this case, 30% of CF) (Seed et al. 1964, Sivapullaiah and Sridharan 1985, Polidori 2007). However, the other two regression lines from the mixed clay-sand samples (CF 50% and 70%) were found to have an average slope of 0.04, exactly the same slope value as defined by Polidori (2007). The slope value of pure clay, however, is 0.03. For this comparison, more soils should be tested, providing a higher number of data points.

Polidori (2007) also tested other clay fractions in increments increasing by 10%. This iterative approach was not done here, only done with 30%, 50%, 70% and 100% of CF. Besides that, the author has included clays with very different IP values. The same is suggested as a future recommendation to include a middle range value of IP, such as illite clay. Moreover, if bentonite and Friedland clay were not considered for the regression lines presented in Figure 7-6, then the data would be insufficient to draw a new fitting equation for the plastic limit, as done by Polidori (2007). It is suggested that more tests with different sand grain sizes and clays with different IPs should be conducted to validate the suggested equation by Polidori (2007).

Another aspect to be debated is the ideal consistency for an EPB machine, which, as stated by Maidl (1995), would lie between I_{Cs} 0.4 and 0.75. This ideal consistency has been defined using sedimentary soils found in the northern hemisphere. Observing the video material mentioned in Chapters 5 and 6, the soils with I_C 0.4 for bentonite does not seem to have the same “squeezing” outcomes as the kaolinite soils with I_C 0.4. The second case seems a rather rigid consistency to be ideal for an EPB excavation.

For the case of mixed soils, the differences become even more clear. The sample with 30% of sand, in the I_C 0.4, looked too dry to be considered as a proper muck. Where consistencies

around the liquid limit ($I_C = 0$) seemed too liquid for the bentonite case, it seems to be an ideal state for the kaolinite samples, especially with the sand fraction.

It is also very unlikely to be feasible to define ideal consistencies of the muck, as a rule for all the clayey soils, considering the great variety of clay minerals, with diverse properties, as summarized in section 2.3, together with the fact that each tunnel project has different logistics in reality.

For example, even though the muck should be around 0.4 and 0.75 for one excavation case, if there is an issue with availability of water, it may be easier or even cheaper to excavate in drier conditions using a greater quantity of anti-clogging polymers. As another example, if the price of the muck disposal is the main issue, a drier muck would make more sense than trying to achieve a softer consistency, which would increase the volume of the muck to be disposed, and, consequently, the cost. In the other range, if a muck would be removed by means of drains, in an EPB with muck pumping system, the “ideal” muck would be more liquid, with higher fluidity, facilitating its removal.

The concept of an “ideal muck” must be checked and cross-referenced with data from real tunnel drives. The consistency of the excavated soil must be defined by sampling a small portion of the soil, conducting the moisture content verification (ASTM D2216 2010), and finally taking into account the excavation parameters (torque, thrust, excavation speed and rate, soil loss, face pressure).

It is suggested that a flow table is installed inside an EPB machine, in the backup, for example, or even outside the tunnel at the job site, where the flow of any excavated and conditioned soil can be analysed and, subsequently, compared with the main parameters of the tunnel drive. This would bring about a better understanding of the “ideal” conditions of the excavated material for an EPB excavation.

Still considering the matters of the tropical residual ground, this research focused only on its soil portion, ignoring the mixed face particularities. For example, this research did not include a study of the residual ground occurring as a portion of rock or even weathered rock. This part still needs to be investigated and modelled to bring to light the implications of this type of ground when excavated by an EPBM.

The results of flow obtained for different clay percentages, with great differences between bentonite and kaolinite, should also be checked with other testing methodologies, such as unconfined compression and triaxial tests. Tests conducted with another rheometer, which allows a greater shear stress than the rheometer used here, and detailed in Chapter 5, could also provide complementary information regarding the flow behaviour of cohesive mixed soils. These tests could also be conducted including conditioning additives.

7.1.3 Preliminary Investigations in EPB Tunnel Projects

Several studies demonstrated that a preliminary investigation and characterisation of the ground to be excavated by an EPB machine can contribute to the reduction of excavation costs and risks, and an increase in the efficiency of the entire process (BTS 2005, Thewes 2005, Thewes et al. 2012, Galli 2016, Ye et al. 2016, and others). In some countries mechanised tunnelling is a long-time tradition, and this practice of a preliminary investigation of ground conditioning is already happening, though this is not always the reality.

Often, ground conditioning characterisation tests are conducted only by the chemical suppliers, which could perhaps provide a biased result. These investigations and characterisation of the conditioned ground, as well as the quality assessment of the foam and additives, should always be conducted by other parties than only the supplier. For instance, the contractor or the client, depending on the type of contract of a tunnel project, should always provide a comparison between different suppliers.

A guideline for EPBM ground conditioning should be standardised worldwide. Currently, the EFNARC guidelines, which were last updated on 2005, provide the most recent set of guidelines that can be used. Therefore, there is an urgent need to provide updated standardised guidance to projects regarding EPB ground conditioning.

An example of what could be a resulting product of this preliminary analysis of EPB conditioning, including the aspects of clogging potential and mixed transitional ground (MTG), is illustrated in Figure 7-7. This figure did not use any real data from a project but is meant to provide an imaginary scenario for illustration purposes using the geological-geotechnical profile from the São Paulo metro line 4, phase 3, as presented in Oliveira et al. (2017c), as a base.

Something like this could always be implemented in the basic design stage of a tunnel project as soon as the borehole material is already available. A second stage can be later conducted including the material from the excavation of the tunnel. By doing that, we will be able to design, and construct tunnels excavated by EPB machines more efficiently, and it will be easier to overcome challenges such as clogging and mixed faces.

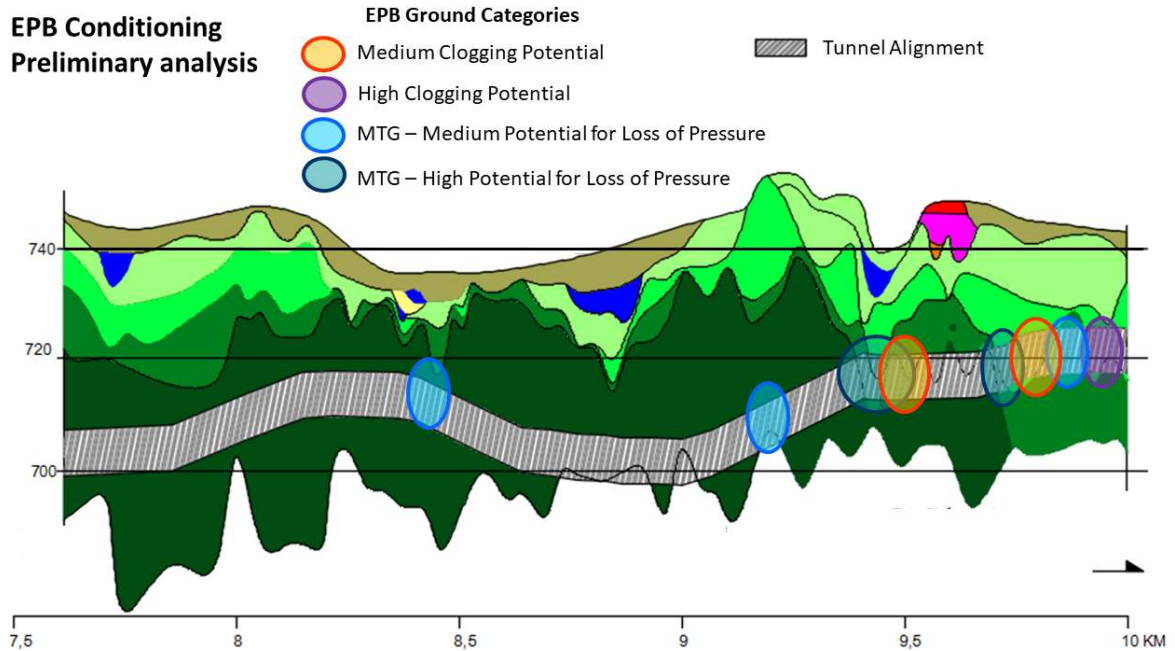


Figure 7-7. Example of an imaginary representation of a preliminary analysis regarding EPB flow behaviour and clogging potential in a longitudinal geological profile (modified from Oliveira et al. 2017c).

7.2 Summary of Conclusions

This thesis has achieved several conclusions, summarised below:

1. This new combined laboratory routine can provide a tool to characterise the clogging potential and flow behaviour of cohesive soil, mixed or not, conditioned or not, correlating that to an EPB excavation.
2. This methodology is rather simple and can be easily reproduced in any laboratory facility, or even at a tunnel site.
3. The methodology here proposed is so far the only procedure that addresses and evaluates the aspects of clogging occurrence in a residual mixed soil, with a contribution of a non-clay portion (e.g. sand).
4. Cohesive soils with the same consistency index do not flow necessarily the same.

5. There is a possible correlation between flow values and clogging potential, still to be entirely understood; however, it seems that for the same I_C , a decrease of flow could imply an increase in the tendencies for clogging. Clogging, in a simplified approach, could also be considered as an absence of flow.
6. The flow table can provide useful information regarding the flow behaviour of cohesive soils, requiring less quantity of material than a slump test, however, for materials with grain size larger than 2 mm, the slump test is more suitable.
7. Some considerations regarding the effectiveness of certain additives concerning clogging and flow have been given as an example of how the methodology could be applied for the assessment of soil conditioning.
8. Alternative solutions for the applicability of the Atterberg limits and the consistency index for cohesive mixed soils, especially for the residual tropical ground, have been proposed.
9. It is suggested that the equation from Polidori (2007) be used to calculate the W_p , considering the W_L and the CF for samples tested with larger grains than the standardized by the ASTM D4318 (2017).
10. The aspects regarding the ideal consistency of a cohesive soil for an EPB excavation have been brought into question. This ideal consistency very likely will not always be the same for all the tunnel projects excavated with an EPB machine along all the cohesive soils, especially when having a contribution of a sand portion. This needs to be still confirmed by testing the material obtained from the screw conveyor.
11. The preliminary use of this methodology for assessing the flow behaviour and the clogging potential before the tunnel is excavated, can provide the possibility to

increase efficiency in the tunnel excavation, especially in terms of soil conditioning optimization.

12. Even though the tests with rheometer, due to the device limitations, have been conducted with $I_c = -0.8$, which is too liquid to be relevant for a regular EPB operation, they could still be relevant to another tunnel operation, such as the muck pumpability.
13. The results of this research could be applied in fields and applications other than exclusively EPB excavation.

7.3 Summary of Recommendations for Future Work

The following list indicates recommendations for future work:

1. All the assumptions here made should be validated with real tunnel drives, especially concerning the clogging potential and the ideal consistency of the muck for EPB machines. A back analysis can be conducted between a preliminary investigation phase and during the tunnel excavation.
2. More tests should be conducted with other clay minerals, such as illite.
3. Tests should also be conducted with natural soils, including undeformed samples, to verify the effect of soil cementation, especially relevant in the case of the tropical soils.
4. All these tests could include a stage for inquiring the relationship among W_p , W_L and CF , for different sizes of sand grain, and different clay minerals.
5. Other laboratory tests can be conducted with similar samples, considering the same proportions of the variables here tested, such as unconfined compression and triaxial tests, besides other rheometers, vane shear and other laboratory tests.

6. A flow table should be installed inside an EPB machine, capable of checking the flow behaviour and as well as the material consistency during excavation, while also comparing these values with the excavation parameters.
7. More tests with foam, and other additives, varying foam parameters, such as FER and FIR, should be conducted to obtain more data on this aspect.
8. More slump tests with cohesive soils can be conducted, trying to find a correlation between the slump flow and the flow values obtained with the flow table.
9. Other alternatives for reducing clogging could be tested, such as the addition of mica flakes.
10. High plasticity and swelling clays should be further investigated in terms of clogging for an EPB excavation, trying to find a correlation in the Hollman and Thewes (2012) graph and the testing methodology proposed here.
11. Many aspects of the testing methodology could still be improved, such as: the base and the number of drops, checking the applicability of other mixer brands, or any other aspect still missing and not considered here.
12. The CB_{Factor} should be further investigated and, perhaps, to be included a laboratory methodology to measure quantitatively the cleaning of the beater and bowl.
13. All the methodology could be considered in a new assemblage which allows changes of pressure, instead of only being feasible in the atmospheric pressure, unrealistic for the majority of the EPB drives.

7.4 Contributions

Articles published in refereed journals

- 1- **Oliveira, D.G.G. de**, Thewes, M., Diederichs, M.S., Langmaack, L. 2018. EPB Tunnelling Through Clay-Sand Mixed Soils: Proposed Methodology for Clogging Evaluation. *Geomechanics & Tunnelling* **11** (4): 375-387. doi: 10.1002/geot.201800009
- 2- **Oliveira, D.G.G. de**, Thewes, M., Diederichs, M.S., Langmaack, L. 2018. Consistency Index and its Correlation with EPB Excavation of Mixed Clay-Sand Soils. *Geotechnical & Geological Engineering*: 4-11. doi: 10.1007/s10706-018-0612-x

Articles submitted to refereed journals

- 3- **Oliveira, D.G.G. de**, Thewes, M., Diederichs, M.S. Flow Behaviour of Mixed Clay-Sand Soils for EPB Machine Excavation. Submitted to *Acta Geotechnica*.
- 4- **Oliveira, D.G.G. de**, Thewes, M., Diederichs, M.S. Clogging and Flow Assessment of Cohesive Soils for EPB Tunnelling: Laboratory Methodology and Soil Characterisation. Submitted to *Tunnelling & Underground Space Technology*.

Conference Contributions with Oral Presentations

- 5- **Oliveira, D.G.G. de**, and Diederichs, M.S. 2016. TBM interaction with soil-rock transitional ground. TAC 2016 Annual Conference, Ottawa.
- 6- **Oliveira, D.G.G. de**, Diederichs, M.S., Rasmussen, L.L., and Cecílio Jr., M.O. 2017. Tropical residual soil data compilation as guidance for laboratory tests and EPB excavation process simulation. 9th International Symposium on Geotechnical Aspects of Underground Construction in Soft Ground. IS – São Paulo.

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APPENDIX A

Tropical Residual Soil Compilation

This appendix presents the data of several publications and technical reports, compiled for the work presented at the conference IS-Sao Paulo 2017 (Oliveira et al. 2017b). The original table with the complete data was made accessible at the research gate link: https://www.researchgate.net/publication/309564759_ResidualSoilCompilation_Final/overview, and is part of this compilation work. The password for opening the table from the research gate is “MiltonVargas”.

The reference sources are provided after the tables with the compiled data, cited by numbers in the tables. The tables were split according to the parent rock, defining separate tables for the parent rocks more frequent: granite, gneiss and migmatite/migmatite gneiss; then separating the remaining data by parent rock type: igneous, sedimentary and metamorphic. Tables A-1 and A-2 present, respectively, the color used on the above-mentioned tables and the country code cited at those. This compilation includes a total of 118 references, with 519 data points, from several countries. Some of the data is unpublished, obtained from technical reports. Whenever applicable, the original source, for these unpublished data, is also included.

Table A- 1. Explanation of the colors presented at the compilation of the residual soil.

Color	Meaning
	Calculated using the formula $GW=Se$
	Average of given values
	Considering the informed total granulometry for fines, considering 60% silt and 40% clay
	Calculated considering the solid density and water content $(\lambda s*(1+(W/100)))/(1+e)$
	Assumed value considering similar values for similar litotypes and saturated
	Calculated considering porosity value
	Calculated considering natural density and water content

Table A- 2. Country codes for the residual soil compilation tables.

Country	Code	Country	Code
Brazil	1	Tanzania	10
Colombia	2	UK	11
USA	3	China	12
Singapore	4	Hong Kong	13
Malaysia	5	Japan	14
Indonesia	6	India	15
Zimbabwe	7	Thailand	16
Portugal	8	Georgia	17
Hawaii, USA ¹¹	9	South Africa	18

Table A- 3. Compilation data with residual soil having as parent rock Gneiss, Part 1.

Soil Name/Local	Country	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Caxingui Gneiss	1	2	2	35.1	49.1	13.8	27.2	-	-	28.4	9.4	-	-	10	27	2.72	80.6
Caxingui Gneiss	1	2	-	36.8	63.2	0	27.2	-	-	-	-	-	-	24	28.5	2.72	80.6
Minas Gerais AM-01	1	7	2	62	17	19	26.6	16.4	0.48	40	19	14.7	-	0	28	2.66	80.6
Minas Gerais AM-02	1	7	3	57	23	17	27.4	16.7	0.25	31	13	7.3	-	12	21	2.74	80.6
Minas Gerais AM-03	1	7	6	65	21	8	27.5	20.8	0.32	NL	NP	9.4	-	-	-	2.75	80.6
Minas Gerais AM-04	1	7	0	60	35	5	26.8	17.9	0.29	NL	NP	8.8	-	50	25	2.68	80.6
Minas Gerais AM-05	1	7	1	72	18	9	26.7	19.5	0.37	NL	NP	11.1	-	-	-	2.67	80.6
Minas Gerais AM-06	1	7	2	58	31	9	26.8	16	0.58	32	15	17.4	-	11	26	2.68	80.6
Minas Gerais AM-07	1	7	2	57	30	11	27.0	18.6	0.38	27	12	11.3	-	60	25	2.70	80.6
Minas Gerais AM-08	1	7	1	72	18	9	26.6	19.3	0.37	NL	NP	11.3	-	-	-	2.66	80.6
Minas Gerais AM-09	1	7	0	71	26	3	26.6	18.5	0.44	24	12	13.2	-	-	-	2.66	80.6
Minas Gerais AM-10	1	7	0	73	21	6	26.4	17.9	0.27	NL	NP	8.3	-	71	20	2.64	80.6
Minas Gerais AM-11	1	7	1	73	20	6	25.9	19.4	0.17	NL	NP	5.4	-	83	24	2.59	80.6
Ribeirao Guaratingueta Slope 1	1	9	-	75	15	10	26.3	13.8	0.9	40	11	8.9	-	9	31	2.63	26.0
Ribeirao Guaratingueta Slope 2	1	9	-	55	27	18	27.3	11.0	1.5	61	15	26.9	-	21.5	24.0	2.73	49.0
III Perimetral/Porto Alegre 3m	1	10	-	45	53	2	26.2	14.0	0.88	47	15	7.3	-	15.8	27.6	2.62	20.9
III Perimetral/Porto Alegre 11m	1	10	-	50	50	0	26.4	15.2	0.75	40	11	19.9	-	20.4	28.5	2.64	63.3
UHE Chapeu D'Uvas	1	16	-	34	38.2	27.8	27.1	11.4	1.38	54	39	41.0	-	-	-	2.71	80.6
UHE Chapeu D'Uvas 5	1	16	1	28	5	66	27.7	14.45	0.90	53	20	26.2	-	-	-	2.77	80.6
UHE Chapeu D'Uvas 7	1	16	7	42	25	26	28.7	12.9	1.07	39	20	29.9	-	-	-	2.87	80.6
UHE Chapeu D'Uvas 9	1	16	2	65	28	5	27.2	13.5	1.04	-	NP	30.7	-	-	-	2.72	80.6
UHE Chapeu D'Uvas 11	1	16	1	59	30	10	27.5	13.3	1.08	-	NP	31.5	-	-	-	2.75	80.6
Caxingui Gneiss	1	19	8	40	42	10	26.8	16.7	0.6	36	10	21.9	1.70E-07	37.3	26.2	2.68	97.5
Caxingui Gneiss	1	19	7	41	42	10	27.2	17.0	0.6	32	9	19.2	-	-	-	2.72	97.5
Caxingui Gneiss	1	19	4	42	42	12	27.6	17.3	0.6	33	9	18.1	-	-	-	2.76	97.5
Caxingui Gneiss	1	19	8	40	42	10	26.8	16.8	0.6	36	10	21.5	-	-	-	2.68	97.5
Euclides da Cunha Dam	1	25	-	36	50	14	27.1	17.14	0.55	48	19	16.4	-	-	-	2.71	80.6

¹¹ Separated from USA, as it belongs to a very particular geological environment.

Table A- 4. Compilation data with residual soil having as parent rock Gneiss, Part 2.

Soil Name/Local	Country	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Morumbi Gneiss	1	28	-	-	-	-	27.2	16.4	0.53	-	-	15.7	-	34	36.5	2.72	80.6
Rodoanel-Sul Gneiss	1	29	0.2	47.6	34.2	18	27.2	13.6	0.84	49.5	24	24.9	-	30	31.7	2.72	80.6
Santo Amaro Gneiss	1	30	-	-	-	-	27.2	15.0	0.81	-	-	24.0	-	45	31.1	2.72	80.6
Jandira Gneiss	1	34	-	32	53	15	27.2	11.1	1.18	68	26	35.0	-	24	29	2.72	80.6
Pinheiros Gneiss	1	35	-	50.3	44.4	5.3	27.2	15.7	0.68	35.3	5.4	20.2	-	14	32.9	2.72	80.6
Ouro Preto	1	36	-	47	5	48	26.3	11.9	1.5	58	29	28.0	-	7	28.1	2.63	80.0
Ouro Preto	1	36	-	51	14	35	26.3	12.6	1.1	50	22	31.0	-	5	29.1	2.63	82.0
Ouro Preto	1	36	-	45	37	18	26.8	13.8	0.75	44	24	34.0	-	8	29.6	2.68	84.0
Ouro Preto	1	36	-	48	37	15	26.8	13.5	0.8	42	21	37.0	-	3	33.9	2.68	85.0
Ouro Preto	1	36	-	45	43	12	26.8	12.6	0.95	41	22	40.0	-	15	31	2.68	90.0
Ouro Preto	1	36	-	47	43	10	26.8	12.4	1	41	19	43.0	-	10	30.5	2.68	93.0
Ouro Preto	1	36	-	49	39	12	26.8	12.3	0.95	42	16	46.0	-	20	25.4	2.68	96.0
Rodrigo de Freitas Lagoon Slope, Sample Slip Surface Saprolitic Soil	1	37	-	63	27.5	9.5	26.4	11.8	1.19	38.2	NP	19.0	-	-	-	2.64	42.2
Rodrigo de Freitas Lagoon Slope, Sample Slip Surface Mature Residual Soil	1	37	-	56	34	10	26.3	12.0	1.14	39.5	15	21.2	-	-	-	2.63	48.9
Rodrigo de Freitas Lagoon Slope, Sample Trench Saprolitic Soil	1	37	-	82	9.8	8.2	26.6	16.1	0.62	NL	NP	6.4	-	-	-	2.66	27.5
Poli/USP/SP Campo Experimental	1	41	-	31.5	45.5	23	27.1	16.3	0.66	48	19	21.5	-	22.9	29	2.71	88.6
North Caroline State University test site	3	50	-	-	-	-	28.0	13.0	1.15	80	35	30.8	1.00E-04	-	-	2.80	75.0
Santo Andre Gneiss/Soil1	1	53	-	35	18	47	28.0	13.1	1.13	48	21	31.0	-	29	32	2.80	76.8
Osasco Gneiss/Soil 2	1	53	-	45	13	42	26.3	13.9	0.89	54	23	30.0	-	27	39	2.63	88.7
Santo Andre Gneiss/Soil2	1	53	-	45	16	39	28.2	16.2	0.74	56	24	20.3	-	35	35	2.82	77.4
Osasco Gneiss/Soil 1	1	53	0	40	15	45	26.5	11.7	1.26	56	23	23.0	-	13	39	2.65	-
Espinhaço da Gata 1 m	1	54	1	37	20	42	26.8	12.2	1.2	50	21	12.5	1.10E-05	-	-	2.68	27.9
Espinhaço da Gata 2 m	1	54	1	47	28	24	27.3	13.3	1.05	52	18	12.5	1.10E-05	-	-	2.73	32.5
Espinhaço da Gata 3	1	54	1	71	20	8	27.4	15.2	0.8	36	7	12.5	4.10E-06	-	-	2.74	42.8
Espinhaço da Gata 4	1	54	1	71	22	6	27.3	14.4	0.9	35	7	12.5	4.10E-06	-	-	2.73	37.9
Espinhaço da Gata 5	1	54	1	79	15	5	27.4	14.7	0.86	32	5	12.5	4.10E-06	-	-	2.74	39.9
Espinhaço da Gata 6	1	54	0	81	14	5	27.3	15.6	0.75	34	8	12.5	4.10E-06	-	-	2.73	45.5
Espinhaço da Gata 7	1	54	1	87	8	4	27.5	-	-	35	6	-	-	-	-	2.75	80.6
Espinhaço da Gata 8	1	54	8	72	15	5	28.0	-	-	40	7	-	-	-	-	2.80	80.6
Espinhaço da Gata 9	1	54	1	91	7	1	27.4	-	-	31	4	-	-	-	-	2.74	80.6
Espinhaço da Gata 10	1	54	8	74	14	4	27.7	-	-	31	7	-	-	-	-	2.77	80.6
Espinhaço da Gata 11	1	54	3	78	15	4	27.7	-	-	31	7	-	-	-	-	2.77	80.6
Espinhaço da Gata 12	1	54	2	79	14	5	27.7	-	-	32	6	-	-	-	-	2.77	80.6
Espinhaço da Gata 13	1	54	1	80	14	5	27.7	-	-	32	6	-	-	-	-	2.77	80.6
Viosa	1	56	0	28	14	58	27.7	12.87	1.16	66	34	24.5	3.78E-06	-	-	2.77	58.4
Sao Paulo Upper	1	57	-	-	-	-	27.2	-	-	60	33	-	-	-	-	2.72	80.6
Sao Paulo Intermediate	1	57	-	-	-	-	27.2	-	-	45	25	-	-	-	-	2.72	80.6
Sao Paulo Saprolite	1	57	-	-	-	-	27.2	-	-	49	23	-	-	-	-	2.72	80.6
Urubu's Hill, Rio de Janeiro, Residual Clayey	1	61	-	35	25	40	27.2	-	-	58	24	-	-	29	30	2.72	-
Urubu's Hill, Rio de Janeiro, Saprolitic	1	61	-	75	11	14	27.2	-	-	48	16	-	-	19	32	2.72	-
Machados, Pernambuco A1 0.75m	1	62	2	32	14	52	27.4	-	-	55	20	-	-	-	-	2.74	80.6
Machados, Pernambuco A1 1.14m	1	62	2	25	25	48	28.3	-	-	50	18	-	-	-	-	2.83	80.6
Machados, Pernambuco A1 8.7m	1	62	2	83	11	4	27.5	-	-	36	10	-	-	-	-	2.75	80.6
Machados, Pernambuco A3 1.9m	1	62	4	33	20	43	27.5	-	-	52	17	-	-	-	-	2.75	80.6
Machados, Pernambuco A3 4.6m	1	62	2	51	36	11	27.9	-	-	44	13	-	-	-	-	2.79	80.6
Machados, Pernambuco A3 5.6m	1	62	1	65	22	12	28.1	-	-	39	10	-	-	-	-	2.81	80.6

Table A- 5. Compilation data with residual soil having as parent rock Gneiss, Part 3.

Soil Name/Local	Country	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
USP/SP	1	66	-	34	46	20	27.1	15.3	0.77	47	13	19.0	-	12	30	2.71	88.9
Real Parque São Paulo 1A	1	69	-	-	-	-	27.2	14.3	0.98	52.4	29	29.8	-	27	20	2.72	82.7
Real Parque São Paulo 1B	1	69	-	-	-	-	27.2	-	-	-	-	-	-	1	33	2.72	80.6
Real Parque São Paulo 2	1	69	-	-	-	-	27.2	17.4	0.54	28.7	9.5	13.7	-	21	31	2.72	69.0
Real Parque São Paulo 3	1	69	-	-	-	-	27.2	14.9	0.86	45.1	20	25.3	-	34	20	2.72	80.0
Real Parque São Paulo 4	1	69	-	-	-	-	27.2	21.4	0.7	62.7	28	16.7	-	4	26	2.72	64.9
Real Parque São Paulo 5	1	69	-	-	-	-	27.2	15.0	0.96	65.2	30	28.3	-	2	32	2.72	80.2
Real Parque São Paulo 6	1	69	-	-	-	-	27.2	18.0	0.67	33.8	4.6	19.3	-	71	21	2.72	78.4
Real Parque São Paulo 7	1	69	-	-	-	-	27.2	-	-	-	-	-	-	12	22	2.72	80.6
Real Parque São Paulo 8	1	69	-	-	-	-	27.2	-	-	-	-	-	-	8	24	2.72	80.6
Real Parque São Paulo 9	1	69	-	-	-	-	27.2	14.8	0.77	45.4	12	25.0	-	13	15	2.72	88.3
Real Parque São Paulo 10	1	69	-	-	-	-	27.2	15.5	0.77	-	-	21.3	-	5	19	2.72	80.6
County of Pacatuba, Ceara, 1	1	70	-	52	35	13	26.4	24.8	0.07	29	12	2.0	-	-	-	2.64	80.6
SP countryside Granitic Gneiss	1	72	-	38	42	20	27.5	13.4	1.05	47.5	22	19.8	5.70E-03	50	32	2.75	50.0
Viçosa	1	79	0	53	30	27	27.9	-	-	48	17	-	-	-	-	2.79	80.6
Viçosa Maduro	1	80	-	27	15	58	27.2	13.1	1.07	68	29	42.0	-	19.2	31	2.72	106.8
Viçosa Jovem	1	80	-	50	45	5	26.7	15.5	0.72	38	15	28.8	-	17	28	2.67	106.8
Canguçu Complex, Porto Alegre, Sample A	1	81	-	-	-	6	27.2	11.6	1.35	-	11	40.0	-	-	-	2.72	80.6
Canguçu Complex, Porto Alegre, Sample B	1	81	-	-	-	4	27.2	10.5	1.59	-	15	47.0	-	-	-	2.72	80.6
Ribeirão Taboão/Lorena/Solo 01	1	85	-	85	6	9	27.2	17.3	0.57	23	8	11.0	-	-	-	2.72	52.5
Ribeirão Taboão/Lorena/Solo 02	1	85	-	64	3	33	27.2	15	0.87	69	35	21.0	-	-	-	2.72	65.7
Ribeirão Taboão/Lorena/Solo 03	1	85	-	79	5	16	27.2	13.5	0.96	79	38	30.0	-	-	-	2.72	85.0
Ribeirão Taboão/Lorena/Solo 04	1	85	-	86	8	6	27.2	15.5	0.69	27	10	12.0	-	-	-	2.72	47.3
Ribeirão Taboão/Lorena/Solo 05	1	85	-	66	16	18	27.2	13.3	1.07	34	15	19.0	-	-	-	2.72	48.3
Ribeirão Taboão/Lorena/Solo 06	1	85	-	71	14	15	27.2	15	0.78	39	8	11.0	-	-	-	2.72	38.4
Line 5/SP Sample 1	1	87	0	40	28	32	27.2	-	-	32	9	19.2	-	-	-	2.72	-
Line 5/SP Sample 2	1	87	0	37	34	29	27.2	-	-	33	9	18.1	-	-	-	2.72	-
Line 5/SP Sample 3	1	87	36	51	10	3	27.2	-	-	28.4	9.4	20.0	-	-	-	2.72	-
Line 5/SP Sample 4	1	87	1	79	9	11	27.2	-	-	26	10	21.5	-	-	-	2.72	-
Line 5/SP Sample 5	1	87	22	66	8	4	27.2	-	-	44.4	17	20.0	-	-	-	2.72	-
Line 5/SP Sample 6	1	87	1	53	30	16	27.2	-	-	47.6	17	20.0	-	-	-	2.72	-
Line 5/SP Sample 7	1	87	0	65	20	15	27.2	-	-	47.3	20	20.0	-	-	-	2.72	-
Line 5/SP Sample 8	1	87	0	57	21	22	27.2	-	-	52	22	20.0	-	-	-	2.72	-
Line 5/SP Sample 9	1	87	0	68	22	10	27.2	-	-	55	23	20.0	-	-	-	2.72	-
Line 5/SP Sample 10	1	87	0	56	32	12	27.2	-	-	49.5	24	20.0	-	-	-	2.72	-
Line 5/SP Sample 11	1	87	0	38	41	21	27.2	-	-	68	26	20.0	-	-	-	2.72	-
São Paulo Gneiss	1	91	-	-	-	5	27.2	12.9	1.11	-	6	33.0	-	0	28	2.72	80.6
São Paulo Gneiss	1	91	-	-	-	12	27.2	11.7	1.32	-	11	39.0	-	0	30	2.72	80.6
Rio de Janeiro Upper	1	99	-	-	-	-	27.2	-	-	-	-	-	-	-	-	2.72	80.6
Rio de Janeiro Intermediate	1	99	-	-	-	-	27.2	10.7	1.55	37.5	15	45.9	-	0.2	33	2.72	80.6
Rio de Janeiro Saprolite	1	99	-	-	-	-	27.2	-	-	-	-	-	-	0.35	30.5	2.72	80.6
Iguana 1	2	103	2	10	56	32	26.6	11.1	1.4	61	16	14.0	-	-	-	2.66	26.6
Iguana 2	2	103	1	31	61	7	26.4	12.0	1.2	44	7	10.0	-	-	-	2.64	22.0
Iguana 3	2	103	4	38	56	2	26.4	14.8	0.78	35	-	6.0	-	-	-	2.64	80.6
Vila Anastacio Gneiss	1	105	4	39	41	16	26.5	12.0	1.04	53	17	39.0	-	34	29.3	2.65	99.5
Coastal Mountains	1	106	-	-	-	-	27.2	12.4	1.2	45	18	35.6	-	0.5	28	2.72	80.6
Vigário Stream, Sample H	1	107	-	44	17	39	27.2	11.2	1.44	59	27	42.6	-	-	-	2.72	80.6
Vigário Stream, Sample M	1	107	-	44	17	39	27.2	11.2	1.44	59	24	42.6	-	-	-	2.72	80.6
Ribeirão do Campo, Serra do Mar	1	107	18	65	14	3	27.2	18.5	0.47	32	0	13.9	-	-	-	2.72	80.6
Vila Anastacio Gneiss	1	107	5	33	17	45	27.2	12.8	1.13	53	17	33.4	-	-	-	2.72	80.6
Itapevi	1	107	27	51	17	5	27.2	18.5	0.47	32	0	13.9	-	-	-	2.72	80.6
São Paulo, 5 meters	1	108	0	26	49	25	27.2	13.6	1.00	70	38	29.6	-	-	-	2.72	80.6
São Paulo, 10 meters	1	108	0	14	71	15	27.2	13.1	1.08	47	18	32.1	-	-	-	2.72	80.6
São Paulo, 15 meters	1	108	0	40	39	21	27.2	11.2	1.44	45	10	42.6	-	-	-	2.72	80.6
São Paulo, 20 meters	1	108	-	48	40	12	27.2	10.1	1.70	50	18	50.5	-	-	-	2.72	80.6
São Paulo, 25 meters	1	108	-	48	38	14	27.2	13.6	1.00	48	13	29.6	-	-	-	2.72	80.6

Table A- 6. Compilation data with residual soil having as parent rock Migmatite/ Migmatite Gneiss, Part 1.

Soil Name/Local	Count ry	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	ϵ	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Poli/USP/SP Campo Experimental/Site I/Sample I.1	1	1	2	28	68	2	27.1	15.8	0.68	36	11	21.2	4.80E-05	-	-	2.71	83.0
Poli/USP/SP Campo Experimental/Site I/Sample I.2	1	1	1	30	51	18	27.1	14.7	0.8	35	11	23.6	1.00E-05	-	-	2.71	78.0
Poli/USP/SP Campo Experimental/Site I/Sample I.3	1	1	0	24	66	10	27.1	14.9	0.77	34	10	22.2	-	-	-	2.71	76.0
Poli/USP/SP Campo Experimental/Site I/Sample I.4	1	1	0	23	73	4	27.1	14.3	0.85	49	18	26.1	1.10E-05	60	25	2.71	81.0
Poli/USP/SP Campo Experimental/Site I/Sample I.5	1	1	0	30	65	5	27.1	16.6	0.63	46	16	17.9	2.00E-06	-	-	2.71	77.0
Poli/USP/SP Campo Experimental/Site I/Sample I.6	1	1	0	20	72	8	27.1	16.6	0.65	48	18	18.1	2.00E-06	-	-	2.71	76.0
Poli/USP/SP Campo Experimental/Site II/Sample II.1	1	1	4	17	68	11	27.1	13.4	1.03	51	20	29.2	-	-	-	2.71	77.0
Poli/USP/SP Campo Experimental/Site II/Sample II.2	1	1	3	17	61	19	27.1	13.2	1.1	49	20	28.5	2.70E-04	-	-	2.71	72.0
Poli/USP/SP Campo Experimental/Site II/Sample II.3	1	1	1	27	62	10	27.1	14.1	0.91	48	22	24.6	-	-	-	2.71	73.0
Poli/USP/SP Campo Experimental/Site II/Sample II.4	1	1	0	30	60	10	27.1	14.9	0.81	40	16	23.9	2.00E-05	-	-	2.71	79.0
Poli/USP/SP Campo Experimental/Site II/Sample II.5	1	1	0	17	74	9	27.1	14.7	0.81	48	20	25.5	-	-	-	2.71	83.0
Poli/USP/SP Campo Experimental/Site II/Sample II.6	1	1	2	16	66	16	27.1	14.4	0.86	44	19	28.5	2.50E-06	-	-	2.71	89.0
Poli/USP/SP Campo Experimental/Site III/Sample III.1	1		1	31	64	4	27.1	11.9	1.43	35	8	22.4	-	-	-	2.71	64.0
Poli/USP/SP Campo Experimental/Site III/Sample III.2	1	1	1	30	65	4	27.1	14.0	1.05	34	8	24.6	-	-	-	2.71	88.0
Poli/USP/SP Campo Experimental/Site III/Sample III.3	1	1	0	25	72	3	27.1	14.7	0.81	36	11	21.4	-	-	-	2.71	70.0
Poli/USP/SP Campo Experimental/Site III/Sample III.4	1	1	6	27	64	3	27.1	15.2	0.8	45	16	23.6	-	-	-	2.71	92.0
Poli/USP/SP Campo Experimental/Site III/Sample III.5	1	1	10	30	52	8	27.1	15.3	0.8	42	21	24.6	-	-	-	2.71	85.0
Poli/USP/SP Campo Experimental/Site III/Sample III.6	1	1	1	11	83	5	27.1	16.4	0.68	45	21	21.3	-	-	-	2.71	87.0
Poli/USP/SP Campo Experimental	1	18	-	20	70	10	27.1	14.8	0.925	46	16	25.0	2.00E-07	20	-	2.71	81.5
Rio de Janeiro Migmatite Red Soil	1	20	-	-	-	25	27.5	9.6	1.86	-	-	54.0	-	0	25	2.75	80.0
Rio de Janeiro Migmatite Variegated	1	20	-	-	-	20	27.5	9.0	2.06	-	-	60.0	-	0	17	2.75	80.0
Rio de Janeiro Migmatite Dark Grey	1	20	-	-	-	20	27.5	10.1	1.72	-	-	50.0	-	0	27	2.75	80.0
Passauna Dam	1	25	-	0	79	21	27.5	15.4	0.88	50	14	25.7	-	-	-	2.75	80.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4508a	1	67	0	32	59	9	27.1	14.1	0.94	40.9	18	33.9	2.90E-04	18	35.3	2.71	98.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4508b	1	67	0	43	50	7	27.4	14.5	0.86	38.8	15	30.4	2.90E-04	0	39.5	2.74	97.3

Table A- 7. Compilation data with residual soil having as parent rock Migmatite/ Migmatite Gneiss, Part 2.

Soil Name/Local	Count ry	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4508c	1	67	1	39	52	8	27.5	16.6	0.72	36.1	13	24.1	2.90E-04	26.8	35.3	2.75	92.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4527a	1	67	1	43	35	21	27.6	14.5	0.86	37.5	16	28.4	3.91E-05	3.4	39.6	2.76	91.8
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4527b	1	67	0	37	53	10	27.1	15.6	0.58	37.8	14	19.5	3.91E-05	-	-	2.71	90.9
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4527c	1	67	3	50	37	10	28.1	16.1	0.48	31.4	13	16.2	1.54E-03	41.9	40.5	2.81	94.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4527d	1	67	5	53	33	9	28.2	16.4	0.45	29.4	14	13.8	1.54E-03	0	42.5	2.82	87.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4553a	1	67	0	47	23	30	26.5	14.9	0.72	31	14	24.6	8.37E-05	-	-	2.65	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4553b	1	67	6	43	39	12	27.1	15.2	0.74	36.5	13	24.6	8.37E-05	0	37	2.71	90.3
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4553c	1	67	5	44	35	16	27.1	15.3	0.74	35.4	13	24.6	3.29E-03	-	-	2.71	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4553d	1	67	1	37	45	17	27.8	15.2	0.76	39.8	16	24.6	3.29E-03	-	-	2.78	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580a	1	67	0	31	20	49	26.7	13.5	0.71	52.3	24	23.8	-	-	-	2.67	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580b	1	67	0	22	20	58	27.2	13.5	0.72	57.3	28	23.8	-	-	-	2.72	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580c	1	67	1	40	51	8	27.4	13.5	0.73	36.8	12	23.8	-	-	-	2.74	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580d	1	67	2	43	48	7	26.9	13.5	0.71	NL	NP	23.8	-	-	-	2.69	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580e	1	67	4	48	42	6	27.5	13.5	0.73	NL	NP	23.8	-	-	-	2.75	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4580f	1	67	3	44	47	6	26.9	13.5	0.71	NL	NP	23.8	-	-	-	2.69	90.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4594a	1	67	7	52	29	12	27.6	15.1	0.69	32.5	10	23.8	4.64E-03	-	-	2.76	95.0
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4594b	1	67	1	54	35	10	27.6	14.9	0.73	35.7	12	25.4	1.18E-04	0	37.8	2.76	95.8
Line 4-Phase 3 São Paulo, Embu Complex, Borehole 4594c	1	67	2	57	30	11	27.5	15.6	0.62	30.3	12	19.6	1.18E-04	11.5	35.3	2.75	87.3
SP Countryside Migmatite/Sample I	1	72	0	10	66	24	27.5	13.9	1.01	85	38	26.5	1.10E-04	36	25.5	2.75	74.0
SP Countryside Migmatite/Sample II	1	72	1	31	53	15	27.5	13.9	1.01	50	23	26.5	2.60E-04	26	29	2.75	74.0
SP Countryside Migmatite/Sample III	1	72	10	52	33	5	27.5	13.9	1.01	29	10	26.5	2.60E-04	67	32	2.75	74.0
Rio de Janeiro Migmatite Light Grey	1	86	-	-	-	4	27.5	13.8	1.00	-	-	29.0	-	0	33.5	2.75	80.0
Rio de Janeiro Migmatite Dark Grey	1	86	-	-	-	2	27.5	13.1	1.10	-	-	32.0	-	4	27	2.75	80.0
São Paulo Migmatite	1	91	-	-	-	20	27.5	10.4	1.65	-	20	48.0	-	0	17	2.75	80.0
São Paulo Migmatite	1	91	-	-	-	10	27.5	11.6	1.38	-	18	40.0	-	0	22	2.75	80.0
São Paulo Migmatite	1	91	-	-	-	8	27.5	10.4	1.65	-	21	48.0	-	0	21	2.75	80.0

Table A- 8. Compilation data with residual soil having as parent rock Granite, Part 1.

Soil Name/Local	Coun try	Source No	Gravel %	Sand %	Silt %	Clay %	λs kN/m3	λd kN/m3	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S
4 Bukit Timah Form. Sample SLR-1a	4	3	-	61	23.4	15.6	26.0	-	-	108	61	-	2.50E-08	-	-	2.6	80.0
4 Bukit Timah Form. Sample SLR-1b	4	3	-	72	16.8	11.2	26.1	-	-	64	30	-	7.36E-08	-	-	2.61	80.0
4 Bukit Timah Form. Sample SLR-1c	4	3	-	75	15	10	25.5	-	-	60	25	-	2.88E-07	-	-	2.55	80.0
4 Bukit Timah Form. Sample SLR-2a	4	3	-	38	37.2	24.8	26.0	-	-	105	58	-	2.78E-08	-	-	2.6	80.0
4 Bukit Timah Form. Sample SLR-2b	4	3	-	38	37.2	24.8	26.1	-	-	116	68	-	3.00E-08	-	-	2.61	80.0
4 Bukit Timah Form. Sample SLR-2c	4	3	-	43	34.2	22.8	25.5	-	-	61	25	-	1.36E-08	-	-	2.55	80.0
4 Bukit Timah Form. Sample YS-a	4	3	-	44	33.6	22.4	26.1	-	-	54	21	-	7.27E-08	-	-	2.61	80.0
4 Bukit Timah Form. Sample YS-b	4	3	-	32	40.8	27.2	25.3	-	-	68	19	-	4.80E-07	-	-	2.53	80.0
4 Bukit Timah Form. Sample LA-a	4	3	-	66	20.4	13.6	25.3	-	-	48	11	-	9.51E-07	-	-	2.53	80.0
4 Bukit Timah Form. Sample LA-b	4	3	-	72	16.8	11.2	26.0	-	-	97	59	-	5.37E-08	-	-	2.6	80.0
4 Bukit Timah Form. Sample LA-c	4	3	-	62	22.8	15.2	25.4	-	-	61	57	-	1.86E-08	-	-	2.54	80.0
4 Bukit Timah Form. Sample LA-d	4	3	-	75	15	10	26.0	-	-	97	59	-	3.57E-08	-	-	2.6	80.0
Penang Island, Tanjung Bungah, TB	5	4	28	37	23	12	26.8	16.8	0.77	51	19	23.0	-	21	17.5	2.68	80.0
Penang Island, Paya Terubong, PT	5	4	14	55	18	13	26.3	16	0.76	73	48	23.0	-	39	8.9	2.63	80.0
Penang Island, Batu Ferringhi, BF	5	4	27	40	26	7	27.1	17	0.81	37	9	24.0	-	30	27	2.71	80.0
Florianopolis 112 A	1	5	-	39.32	40	20.6	27.6	13.1	1.07	46	16	37.9	-	35	22	2.76	97.8
Florianopolis 112 B	1	5	7	32.36	43.1	17.58	26.2	13.5	0.8	37	17	32.9	-	20	27	2.62	107.7
Florianopolis 112 C	1	5	1.8	36	43.3	18.82	26.7	13.4	0.75	43	20	45.2	-	30	27	2.67	160.9
Florianopolis 120 A	1	5	-	26.67	38.3	35.06	27.1	13.5	1	56	26	36.0	-	10	19	2.71	97.6
Florianopolis 120 B	1	5	-	46.09	44.3	9.6	27.2	15.6	0.85	35	18	15.1	-	20	25	2.72	48.3
Florianopolis 120 C	1	5	-	48.3	41.9	9.81	26.7	15.5	0.7	32	14	23.5	-	0	40	2.67	89.6
Nilai, Negeri Sembilan	5	6	34.6	31.1	20.6	13.72	26.6	-	-	25.5	5.5	-	-	-	-	2.66	80.0
Polo da Asprela, FEUP, Sample 1	8	13	2	74	16	8	26.7	16.39	0.58	-	-	-	-	9.34	40.33	2.67	
Polo da Asprela, FEUP, Sample 2	8	13	11	77	8	4	26.3	16.39	0.58	-	-	-	-	9.34	40.33	2.63	
Polo da Asprela, FEUP, Sample 3	8	13	2	70	23	5	26.0	16.39	0.58	-	-	-	-	9.34	40.33	2.6	
Federal H'Way/Kuala Lumpur	5	21	17	34	29.4	19.6	26.8	13.9	0.92	41.5	14	27.5	-	23.5	6	2.68	80.0
Carta de Zoneamento Geotecnico, G4-G	8	23	8.4	59.6	25.6	6.4	26.1	15.5	0.66	28	9	11.4	1.90E-06	27	31	2.61	45.1
Carta de Zoneamento Geotecnico, G4-C	8	23	9.4	46.6	35.2	8.8	26.3	15	0.79	36	12	-	7.58E-07	2	36	2.63	
Carta de Zoneamento Geotecnico, G8-A	8	23	14.4	57.6	22.4	5.6	26.1	15.8	0.75	30	8	12.7	6.79E-07	6	33	2.61	44.2
Carta de Zoneamento Geotecnico, G8-B	8	23	14	54	25.6	6.4	26.2	13.4	0.18	31	7	11.5	2.10E-06	-	-	2.62	170.7
Itaparica Dam	1	25	-	95	3	2	27.0	18.79	0.49	-	-	14.5	-	-	-	2.7	80.0
Porto, Sample G4-k	8	26	12	46	42	0	26.2	15.2	0.77	35	13	20.0	1.00E-07	-	-	2.62	80.0
Porto, Sample G4-G	8	26	12	50	38	0	26.3	15.7	0.65	32	8	20.0	1.00E-06	-	-	2.63	80.0
Porto, Sample G8-A	8	26	16	58	26	0	26.0	20	0.78	28	4	20.0	1.00E-06	-	-	2.6	80.0
Porto, Sample W5	8	26	18	58	24	0	26.4	23.7	0.18	32	7	20.0	1.00E-06	20	38	2.64	80.0
Shenzhen Metro Project	12	32	27.5	17.5	33	22	26.6	14.0	0.9	-	19	28.6	1.00E-05	-	-	2.66	84.7
Vila Albertina Granite/Sample 4	1	38	1	35	18	46	27.1	14.9	0.81	63	41	26.5	-	16.45	30.1	2.71	88.7
5 Sample a	5	42	25	61	8.4	5.6	23.9	14.8	0.61	59	29	20.4	-	-	-	2.39	80.0
5 Sample b	5	42	46	45	5.4	3.6	25.7	13.4	0.92	57.2	23	28.5	-	-	-	2.57	80.0
5 Sample c	5	42	43	50	4.2	2.8	25.4	12.8	0.98	49	24	31.0	-	-	-	2.54	80.0
4 Bukit Timah Formation	4	43	1	61	22	16	25.9	14.7	0.76	-	-	23.5	-	-	-	2.59	80.0
HS4	13	44	55	41	3	1	26.3	-	-	-	-	-	-	-	-	2.63	80.0
SH2	13	44	40	33	17	10	26.3	12.1	1.17	-	-	14.6	-	-	-	2.63	33.5
SH8	13	44	41	45	11	3	26.3	12.1	1.17	-	-	18.7	-	-	-	2.63	41.7
SH6	13	44	42	40	14	4	26.3	12.1	1.17	-	-	18.6	-	-	-	2.63	42.0
SH7	13	44	40	45	11	4	26.2	12.1	1.16	-	-	17.1	-	-	-	2.62	38.6
SH3	13	44	31	42	18	9	26.3	13	1.09	-	-	18.7	-	-	-	2.63	45.0
SH5(w)	13	44	35	38	22	5	26.3	12.9	1.05	-	-	20.6	-	-	-	2.63	52.0
SH5(a)	13	44	34	34	26	6	26.2	13.1	1.01	-	-	20.6	-	-	-	2.62	54.0
HS5	13	44	6	30	56	8	26.3	-	-	-	-	-	-	-	-	2.63	80.0
Melaka, 5	5	46	0	31.5	41	27.5	25.9	9.3	1.79	71.8	40	55.2	-	18.75	12.75	2.59	80.0
Harak Highway/Selangor/01	5	47	6	59	17	18	26.3	16.2	0.87	60	24	24.0	-	7.9	45	2.63	72.6
Harak Highway/Selangor/02	5	47	45	33	10	12	26.3	16.4	0.86	52	16	22.0	-	2.5	50	2.63	67.3
Harak Highway/Selangor/03	5	47	44	39	8	9	26.3	15.8	0.81	47	11	24.0	-	3.6	40	2.63	77.9
Harak Highway/Selangor/04	5	47	40	48	5	7	26.3	15.7	0.7	46	14	16.0	-	2.7	46	2.63	60.1
Harak Highway/Selangor/05	5	47	12	64	16	8	26.3	15.4	0.64	40	7	20.0	-	2	42	2.63	82.2
Harak Highway/Selangor/06	5	47	44	46	6	4	26.3	14.8	0.63	36	12	19.0	-	4.3	32	2.63	79.3
Harak Highway/Selangor/07	5	47	55	40	5	0	26.3	13.8	0.57	24	5	14.0	-	-	-	2.63	64.6
Petaling Jaya/Selangor	5	48	15	32.5	31.5	21	26.3	-	-	-	-	-	-	-	-	2.63	80.0
Cheras/Selangor	5	48	20	47.5	19.5	13	26.3	9.3	1.82	52.5	17	55.5	-	-	-	2.63	80.0
Hill side slope at the crest	13	52	17.35	46.8	20.6	15.25	26.4	15.2	0.76	60.7	20	23.1	1.00E-06	-	-	2.635	80.0
Hill side slope near the berm	13	52	17.3	68.7	13.7	0	26.4	14.6	0.82	47.6	15	24.7	1.00E-05	-	-	2.64	80.0
Mt Beacon, Kowloon Tong	13	55	22	61	15	2	26.3	18.9	0.36	33.5	18	11.0	-	-	-	2.63	80.0
Bukit Timah	4	58	-	49	25	26	26.6	13.0	0.52	47	18	15.6	-	13	29.7	2.66	80.0
Nova Friburgo	1	59	-	66	29	5	26.3	12.98	1.05	36.4	5.5	13.1	-	-	-	2.63	33.0

Table A- 9. Compilation data with residual soil having as parent rock Granite, Part 2.

Soil Name/Local	Country	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Maramduba/Ubatuba M2A30	1	60	4.5	41.5	10.5	43.5	27.6	13.08	1.11	66.4	27	28.0	-	-	-	2.76	66.0
Maramduba/Ubatuba M2A50	1	60	0	35	8.5	56.5	27.6	14.28	0.93	89.5	32	36.0	-	-	-	2.76	72.0
Maramduba/Ubatuba Block2	1	60	0	64	30.5	5.5	27.8	13.06	1.13	51.2	15	27.0	2.30E-06	-	-	2.78	68.0
Maramduba/Ubatuba M2B130	1	60	0	44.5	37	18.5	28.2	14.21	0.98	49	7.6	23.0	1.60E-07	-	-	2.82	60.0
Maramduba/Ubatuba M2B335	1	60	0	46	34	20	27.6	14.46	0.9	49.4	11	24.0	3.00E-07	-	-	2.76	58.0
Maramduba/Ubatuba M2B500	1	60	0	61	35.5	3.5	27.5	13.39	1.05	49.1	6.2	22.0	8.00E-07	-	-	2.75	52.0
Maramduba/Ubatuba M1C600	1	60	0	69.5	28	2.5	27.5	15.62	0.76	39.8	15	29.0	-	-	-	2.75	74.0
Maramduba/Ubatuba M1C800	1	60	0	57.5	38.5	4	27.6	10.83	1.55	42.9	8.1	57.0	-	-	-	2.76	95.0
Maramduba/Ubatuba Block1	1	60	0	48	46	6	28.2	12.44	1.27	53.8	12	31.0	2.30E-06	-	-	2.82	63.0
Maramduba/Ubatuba M1C1000	1	60	0	25	53.5	21.5	32.2	11.03	1.92	78.1	26	48.0	-	-	-	3.22	82.0
Kowloon Bay	13	63	7	39	35	19	26.3	13.95	0.89	-	-	27.0	-	-	-	2.63	80.0
Ikoma Mountain, Osaka, Sample a	14	65	-	-	-	-	27.1	11.4	1.38	-	-	22.9	-	-	-	2.71	54.4
Ikoma Mountain, Osaka, Sample b	14	65	-	-	-	-	27.3	12.5	1.19	-	-	19.9	-	-	-	2.73	58.9
Ikoma Mountain, Osaka, Sample c	14	65	-	-	-	-	27.4	13.8	0.99	-	-	17.7	-	-	-	2.74	48.0
Ikoma Mountain, Osaka, Sample d	14	65	-	-	-	-	27.4	16.5	0.66	-	-	11.5	-	-	-	2.74	49.6
SP Countryside Granite/Average Sample I	1	72	2	26	22	50	27.1	12.5	1.18	54.2	20	24.2	7.00E-04	10	27	2.71	56.0
SP Countryside Granite/Average Sample II	1	72	6	54	20	20	27.1	12.5	1.18	54.2	22	24.2	3.50E-03	11	31	2.71	56.0
Yishun Slope, 4, Sample 1	4	73	0	26	66	8	27.0	11.6	1.4	62	22	38.0	-	26	27	2.7	73.3
Yishun Slope, 4, Sample 2	4	73	0	26	68	6	26.0	11.2	1.3	64	24	52.0	4.00E-09	26	27	2.6	104.0
Yishun Slope, 4, Sample 3	4	73	0	22	69	9	27.5	14.5	0.85	54	22	24.0	5.90E-09	13	35	2.75	77.6
Yishun Slope, 4, Sample 4	4	73	4	33	57	6	25.5	15.1	0.55	42	NP	26.0	7.00E-09	12	38	2.55	120.5
Yishun Slope, 4, Layer 1	4	74	-	42	34.8	23.2	26.8	13.9	1.17	45	14	35.0	7.20E-07	12	33	2.68	80.0
Yishun Slope, 4, Layer 2	4	74	-	22	46.8	31.2	27.1	11.6	1.76	57	17	52.0	-	13	29	2.71	80.0
Yishun Slope, 4, Layer 3	4	74	-	66	20.4	13.6	26.7	16.8	0.63	42	16	19.0	-	35	31	2.67	80.0
4 Bukit Timah Form. Sample MR1	4	75	-	53	28.2	18.8	26.6	-	-	52	21	-	6.00E-06	-	-	2.66	80.0
4 Bukit Timah Form. Sample MR2	4	75	-	40	36	24	25.8	-	-	54	15	-	3.30E-05	-	-	2.58	80.0
4 Bukit Timah Form. Sample TR1	4	76	-	27	43.8	29.2	25.9	-	-	71	37	-	3.19E-08	-	-	2.59	80.0
4 Bukit Timah Form. Sample TR2	4	76	-	30	42	28	26.6	-	-	51	22	-	5.65E-05	-	-	2.66	80.0
4 Bukit Timah Form. Sample BBA1	4	76	-	43	34.2	22.8	28.2	-	-	31	11	-	5.21E-08	-	-	2.82	80.0
4 Bukit Timah Form. Sample BBA2	4	76	-	20	48	32	26.4	-	-	34	14	-	8.28E-08	-	-	2.64	80.0
4 Bukit Timah Form. Sample BB1	4	76	-	3	58.2	38.8	26.5	-	-	55	23	-	1.20E-08	-	-	2.65	80.0
4 Bukit Timah Form. Sample BB2	4	76	-	51	29.4	19.6	27.1	-	-	49	19	-	2.15E-08	-	-	2.71	80.0
4 Bukit Timah Form. Sample AMK1	4	76	-	58	25.2	16.8	25.4	-	-	56	18	-	8.52E-06	-	-	2.54	80.0
4 Bukit Timah Form. Sample AMK2	4	76	-	53	28.2	18.8	25.8	-	-	58	22	-	3.96E-06	-	-	2.58	80.0
4 Bukit Timah Form. Sample AMA	4	76	-	29	42.6	28.4	27.0	-	-	54	25	-	2.52E-07	-	-	2.7	80.0
4 Bukit Timah, Sample 1	4	76	-	43	35	22	26.3	16.3	0.61	54	9	43.0	-	15	40	2.63	185.4
4 Bukit Timah, Sample 2	4	76	-	44	35	21	26.3	18.5	0.42	35	12	21.0	-	5	31	2.63	131.5
4 Bukit Timah, Sample 3	4	76	-	43	34	23	26.3	17.0	0.55	45	9	34.0	-	8	35	2.63	162.6
4 Bukit Timah, Sample 4	4	76	-	52	23	25	26.3	16.6	0.58	52	10	41.0	-	14	41	2.63	185.9
4 Bukit Timah, Sample 5	4	76	-	51	26	23	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 6	4	76	-	43	33	24	26.3	17.3	0.52	-	-	15.8	-	9	36	2.63	80.0
4 Bukit Timah, Sample 7	4	76	-	51	30	19	26.3	12.8	1.05	43.6	10	32.0	-	-	-	2.63	80.0
4 Bukit Timah, Sample 8	4	76	-	53	25	22	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 9	4	76	-	49	30	21	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 10	4	76	-	54	30	16	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 11	4	76	-	55	25	20	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 12	4	76	-	45	34	21	26.3	17.4	0.51	43	11	31.0	-	9	36	2.63	159.9
4 Bukit Timah, Sample 13	4	76	-	59	30	11	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 14	4	76	-	55	27	18	26.3	17.7	0.49	42	8	31.0	-	8	37	2.63	166.4
4 Bukit Timah, Sample 15	4	76	-	60	32	8	26.3	19.2	0.37	37	11	25.0	-	4	34	2.63	177.7
4 Bukit Timah, Sample 16	4	76	-	56	31	13	26.3	16.5	0.59	46	8	36.0	-	12	40	2.63	160.5
4 Bukit Timah, Sample 17	4	76	-	61	31	8	26.3	19.9	0.32	28	6	16.0	-	2	32	2.63	131.5
4 Bukit Timah, Sample 18	4	76	-	58	29	13	26.3	17.7	0.49	41	10	30.0	-	8	39	2.63	161.0
4 Bukit Timah, Sample 19	4	76	-	55	34	11	26.3	16.1	0.63	54	12	44.0	-	12	41	2.63	183.7
4 Bukit Timah, Sample 20	4	76	-	43	35	22	26.3	16.3	0.61	54	9	43.0	-	15	40	2.63	185.4
4 Bukit Timah, Sample 21	4	76	-	44	35	21	26.3	18.5	0.42	35	12	21.0	-	5	31	2.63	131.5
4 Bukit Timah, Sample 22	4	76	-	43	34	23	26.3	17.0	0.55	45	9	34.0	-	8	35	2.63	162.6
4 Bukit Timah, Sample 23	4	76	-	52	23	25	26.3	16.6	0.58	52	10	41.0	-	14	41	2.63	185.9
4 Bukit Timah, Sample 24	4	76	-	51	26	23	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 25	4	76	-	43	33	24	26.3	17.3	0.52	-	-	15.8	-	9	36	2.63	80.0
4 Bukit Timah, Sample 26	4	76	-	51	30	19	26.3	12.8	1.05	43.6	10	32.0	-	-	-	2.63	80.0
4 Bukit Timah, Sample 27	4	76	-	53	25	22	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 28	4	76	-	49	30	21	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 29	4	76	-	54	30	16	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 30	4	76	-	55	25	20	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 31	4	76	-	45	34	21	26.3	17.4	0.51	43	11	31.0	-	9	36	2.63	159.9
4 Bukit Timah, Sample 32	4	76	-	59	30	11	26.3	-	-	-	-	-	-	-	-	2.63	80.0
4 Bukit Timah, Sample 33	4	76	-	55	27	18	26.3	17.7	0.49	42	8	31.0	-	8	37	2.63	166.4
4 Bukit Timah, Sample 34	4	76	-	60	32	8	26.3	19.2	0.37	37	11	25.0	-	4	34	2.63	177.7
4 Bukit Timah, Sample 35	4	76	-	56	31	13	26.3	16.5	0.59	46	8	36.0	-	12	40	2.63	160.5

Table A- 10. Compilation data with residual soil having as parent rock Granite, Part 3.

Soil Name/Local	Country	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Maramduba/Ubatuba M2A30	1	60	4.5	41.5	10.5	43.5	27.6	13.08	1.11	66.4	27	28.0	-	-	-	2.76	66.0
Maramduba/Ubatuba M2A50	1	60	0	35	8.5	56.5	27.6	14.28	0.93	89.5	32	36.0	-	-	-	2.76	72.0
Maramduba/Ubatuba Block2	1	60	0	64	30.5	5.5	27.8	13.06	1.13	51.2	15	27.0	2.30E-06	-	-	2.78	68.0
Bukit Timah, Sample 36	4	76	-	61	31	8	26.3	19.9	0.32	28	6	16.0	-	2	32	2.63	131.5
Bukit Timah, Sample 37	4	76	-	58	29	13	26.3	17.7	0.49	41	10	30.0	-	8	39	2.63	161.0
Bukit Timah, Sample 38	4	76	-	55	34	11	26.3	16.1	0.63	54	12	44.0	-	12	41	2.63	183.7
Bukit Timah Formation Ave Val	4	76	-	-	-	-	26.3	17.4	0.51	43.4	9.7	32.0	-	8.83	36.83	2.63	165.0
Kampar Perak/W.Grade VI	5	77	8	43.5	24	24.5	26.4	14.2	1.1	31.5	17	16.5	-	-	-	2.635	39.5
Kampar Perak/W.Grade V	5	77	16	57	18	9	26.4	15.3	0.925	21.5	17	18.0	-	29	18.25	2.635	51.3
Kampar Perak/W.Grade IV	5	77	32.5	40	9.5	18	26.4	14.5	0.85	18.5	15	17.0	-	-	-	2.64	52.8
Adilabad, Yellowish Brown Clay	15	78	-	-	-	-	18.5	11.1	0.67	69	47	32.5	-	-	-	1.85	90.0
Guarda 0-4	8	82	35.3	41.1	18.6	5	24.2	-	-	38.7	8.2	-	-	8	39	2.42	80.0
Guarda 4-9	8	82	34.2	41.54	18.5	5.7	30.1	-	-	40.1	9.5	-	-	15	40	3.01	80.0
16010	16	83	16	62	8	14	26.3	25	0.33	27	7	10.0	-	-	35.6	2.63	80.0
16020	16	83	4	52	24	20	26.3	18.8	1.22	56	28	37.0	-	-	-	2.63	80.0
16030	16	83	6	64	23	7	26.5	19.6	0.53	NL	NP	16.0	-	-	-	2.65	80.0
UTM, Johor, 5	5	84	35.26	11.93	31.69	21.12	25.9	13.9	0.86	68.4	26	26.5	-	10.5	30.5	2.59	80.0
Bukit Lajang/Kuala Lumpur	5	89	30	40	17.5	12.5	26.5	16.7	0.7	37.5	18	20.0	-	-	-	2.65	75.7
Johor Bahu Granite	5	90	0	57.5	22.5	20	25.5	15.15	0.60	73	32	18.9	-	-	-	2.55	80.0
Second Link Highway	5	90	0	45	37	18	24.9	14.8	1.08	72.5	35	34.7	-	-	-	2.485	80.0
Upper	17	93	-	-	-	-	26.7	15.3	0.75	40	17	22.5	-	-	-	2.67	80.0
Intermediate	17	93	-	-	-	-	26.7	13.4	1	30	2.5	30.0	-	-	-	2.67	80.0
Saprolite	17	93	-	-	-	-	26.7	16.7	0.6	-	-	18.0	-	-	-	2.67	80.0
ITM Shah Alam, 5	5	94	0	77	13.5	9.5	26.3	-	-	47.5	19	-	-	28	39	2.63	80.0
Cheras, Kuala Lumpur	5	95	0	38	13	49	25.8	12.9	1.00	67.8	34	31.0	-	-	-	2.58	80.0
4 Bukit Timah Form. Sample SS	4	96	-	61	23.4	15.6	25.9	-	-	48	21	-	2.20E-06	-	-	2.59	80.0
Sugai Ara, Penang	5	97	9.8	65.3	24.9	0.0	26.4	16.9	0.56	41.5	11	17.1	-	-	-	2.64	80.0
Bt. Bendera, Penang	5	97	10.3	67.3	22.5	0.0	25.6	17.9	0.43	47.5	12	13.4	-	-	-	2.56	80.0
Banjaran Kledang, Perak	5	97	21.7	32.9	20.3	25.2	26.5	-	-	70	34	-	-	-	-	2.65	80.0
Km 26.4 Lebu Raya KL-Karak	5	97	25.7	53.6	18.3	2.5	24.9	17.9	0.39	53.5	20	12.5	-	-	-	2.49	80.0
Km 39.9 Lebu Raya KL-Karak	5	97	4.4	44.3	41.3	10.1	25.9	14.8	0.74	54.5	25	23.0	-	-	-	2.59	80.0
Kuantan	5	97	28.0	47.5	17.5	7.0	26.1	9.3	1.80	57.5	26	55.2	-	-	-	2.61	80.0
Sg. Besi/Kuala Lumpur	5	100	0	57.5	25.5	17	26.3	16.0	0.645	-	-	25.0	-	81	4	2.63	101.9
Mandaqui Granite	1	105	0	24	36	40	27.0	12.8	1.1	68	26	30.0	-	34	29.3	2.7	73.5
Mandaqui	1	107	-	18	17	65	26.3	9.2	1.86	65	26	56.5	-	-	-	2.63	80.0
Porto typical saprolitic soil	8	109	8	62	25	5	26.1	16.1	0.55	-	-	20.0	1.00E-05	8.7	37.2	2.61	90.0
NW of 8, Borehole SP3, Sample 1	8	110	5	62	25	8	26.1	15.1	0.725	28	3	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 2	8	110	2	65	29	4	26.1	15.1	0.725	29	5	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 3	8	110	5	63	26	6	26.1	15.1	0.725	27	8	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 4	8	110	5	58	29	8	26.1	15.1	0.725	30	9	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 5	8	110	6	68	23	3	26.1	15.1	0.725	31	3	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 6	8	110	7	63	25	5	26.1	15.1	0.725	32	11	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 7	8	110	7	59	29	5	26.1	15.1	0.725	NL	NP	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 8	8	110	6	67	22	5	26.1	15.1	0.725	NL	NP	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 9	8	110	7	67	21	5	26.1	15.1	0.725	NL	NP	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP3, Sample 10	8	110	6	67	22	5	26.1	15.1	0.725	32	4	21.0	1.00E-05	-	-	2.61	85.0
NW of 8, Borehole SP4, Sample 1	8	110	10	64	19	7	26.1	16.1	0.625	NL	NP	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 2	8	110	7	55	31	7	26.1	16.1	0.625	32	7	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 3	8	110	5	65	24	6	26.1	16.1	0.625	30	8	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 4	8	110	9	63	22	6	26.1	16.1	0.625	NL	NP	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 5	8	110	11	59	24	6	26.1	16.1	0.625	28	4	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 6	8	110	13	60	22	5	26.1	16.1	0.625	28	3	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 7	8	110	11	64	20	5	26.1	16.1	0.625	28	6	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 8	8	110	10	65	21	4	26.1	16.1	0.625	28	7	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 9	8	110	10	64	21	5	26.1	16.1	0.625	NL	NP	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 10	8	110	13	58	24	5	26.1	16.1	0.625	31	6	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 11	8	110	12	62	22	4	26.1	16.1	0.625	NL	NP	20.0	1.00E-05	-	-	2.61	80.0
NW of 8, Borehole SP4, Sample 12	8	110	11	66	19	4	26.1	16.1	0.625	28	2	20.0	1.00E-05	-	-	2.61	80.0
Experimental Site FEUP S2/1	8	111	12	53	30	5	27.4	15.6	0.72	32	5	16.3	-	6	36.9	2.74	62.0
Experimental Site FEUP S2/2	8	111	12	53	30	5	27.7	14.0	0.93	-	-	18.8	-	6	36.9	2.77	56.0
Experimental Site FEUP S2/3	8	111	24	32	30	14	28.2	15.8	0.75	44	17	16.2	-	6	36.9	2.82	61.0
Experimental Site FEUP S2/4	8	111	24	32	30	14	28.3	15.1	0.83	-	-	21.7	-	6	36.9	2.83	74.0
Experimental Site FEUP S2/5	8	111	16	46	33	5	27.1	15.5	0.71	NL	NP	22.5	-	6	36.9	2.71	86.0
Experimental Site FEUP S2/6	8	111	11	53	32	4	27.3	15.8	0.68	NL	NP	20.2	-	6	36.9	2.73	81.0
Yen Chow Street/Kowloon Peninsula	13	112	-	50	32	18	26.4	14.1	0.86	NL	NP	33.1	-	-	-	2.64	101.6
Island Country Club	4	114	0.2	59.8	23	17	25.9	14.7	0.762	48	21	23.5	-	-	-	2.59	80.0
Taishan, Guangdong Prov Sample 1	12	115	13.5	25.5	29.6	31.4	27.7	14.3	0.993	69.9	32	30.2	-	-	-	2.77	84.2
Taishan, Guangdong Prov Sample 2	12	115	3	35.1	41.1	20.5	27.3	12.8	1.157	61	32	38.3	-	-	-	2.73	90.4
Taishan, Guangdong Prov Sample 3	12	115	4.6	33.4	45.9	16.1	27.3	12.6	1.175	57.1	26	40.2	-	-	-	2.73	93.4
Taishan, Guangdong Prov Sample 4	12	115	5	34.4	44.2	16.4	27.3	12.1	1.263	61.4	31	44.1	-	-	-	2.73	95.4
Taishan, Guangdong Prov Sample 5	12	115	6.8	31.7	44.6	16.9	27.4	12.8	1.164	56.9	28	37.4	-	-	-	2.74	88.0
Granite South Africa	18	117	-	30	40	30	27.1	15.1	0.66	39	24	19.5	-	-	-	2.71	80.0
Shenzhen Futian Dist, Guangdong	12	118	11	45	21	23	26.0	15	0.736	40	15	22.0	-	-	-	2.6	77.7

Table A- 11. Compilation data with residual soil having sedimentary rocks as parent rock.

Soil Name/Local	Country	Parent Rock	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Vila Scharlau	1	Sandstone	11	-	30	63	7	26.5	15.9	0.685	21	NP	13.0	1.00E-06	-	-	2.65	50.3
Ilima	10	Mudstone	15	-	-	-	-	26.5	-	-	48	26	-	-	-	22.2	2.65	80.0
Kings Norton	11	Marl	22	-	-	-	7	27.6	15.7	0.76	35	10	22.0	-	-	-	2.76	80.0
Erdington	11	Marl	22	-	-	-	11	27.8	16.9	0.64	36	10	18.5	-	-	-	2.78	80.0
Bells Lane	11	Marl	22	-	-	-	19	27.6	16.2	0.71	39	18	20.5	-	-	-	2.76	80.0
CAIC/Laranjeiras/Sergipe Sample 01 0-0.5 meters	1	Limestone	39	-	-	-	-	27.1	16.7	0.63	39	20	18.5	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 01 0.5-1 meters	1	Limestone	39	-	-	-	-	27.1	15.7	0.72	38	20	21.3	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 01 1-1.5 meters	1	Limestone	39	-	-	-	-	27.1	14.9	0.82	57	29	24.2	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 01 1.5-2 meters	1	Limestone	39	-	-	-	-	27.1	14.3	0.89	57	24	26.3	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 01 2-2.5 meters	1	Limestone	39	-	-	-	-	27.1	14.3	0.90	57	23	26.6	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 02 0-0.5 meters	1	Limestone	39	-	-	-	-	27.1	21.3	0.27	40	19	8.1	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 02 0.5-1 meters	1	Limestone	39	-	-	-	-	27.1	19.6	0.38	38	17	11.3	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 02 1-1.5 meters	1	Limestone	39	-	-	-	-	27.1	19.4	0.40	38	17	11.7	-	-	-	2.71	80.0
CAIC/Laranjeiras/Sergipe Sample 02 1.5-2 meters	1	Limestone	39	-	-	-	-	27.1	17.9	0.51	38	19	15.1	-	-	-	2.71	80.0
SP Countryside Siltstone/ Sample I	1	Siltstone	72	0	20	55	25	27.0	15.5	0.76	57	28	23.6	-	65	35	2.7	86.0
SP Countryside Siltstone/ Sample II	1	Siltstone	72	0	20	62	18	27.0	15.5	0.76	48	20	23.6	-	43	30	2.7	86.0
SP Countryside Siltstone/ Sample III	1	Siltstone	72	0	5	90	5	27.0	15.5	0.76	33	11	23.6	-	21	24	2.7	86.0
SP Countryside Sandstone/Sample Bauru	1	Sandstone	72	0	60	25	15	26.9	17.2	0.56	32	14	16.6	2.50E-04	5	31	2.69	79.0
SP Countryside Sandstone/Sample Caiua	1	Sandstone	72	0	85	7	8	26.8	18.3	0.47	22	8	8.4	2.50E-04	18	36	2.68	48.0
5n Shale	5	Shale	90	0	49	38.5	12.5	25.8	15.9	0.82	55.5	24	25.5	-	-	-	2.58	80.0
Viracopos, Campinas	1	Sandstone	107	-	44	25	31	27.0	13.8	0.96	49	19	28.5	-	-	-	2.7	80.0
Perus	1	Limestone	107	-	9	31	60	27.1	14.9	0.82	45	19	24.2	-	-	-	2.71	80.0
Campinas, 5 meters	1	Siltstone	108	-	55	23	22	26.5	14.6	0.82	40	8	24.7	-	-	-	2.65	80.0
Campinas, 10 meters	1	Siltstone	108	-	62	18	20	26.5	11.1	1.38	42	4	41.7	-	-	-	2.65	80.0
Campinas, 15 meters	1	Siltstone	108	-	51	21	28	26.5	8.0	2.33	38	10	70.4	-	-	-	2.65	80.0
Sandstone, 5 meters	1	Sandstone	108	-	80	4	16	27.0	12.2	1.22	23	9	36.2	-	-	-	2.7	80.0
Sandstone, 10 meters	1	Sandstone	108	-	78	6	16	27.0	13.5	1.00	26	13	29.6	-	-	-	2.7	80.0
Kuala Lumpur	5	Sandstone/Siltstone	113	-	-	39	19	27.0	18.7	0.51	-	15	15.0	-	117.5	-	2.7	80.0

Table A- 12. Compilation data with residual soil having igneous rocks as parent rock.

Soil Name/Local	Country	Parent Rock	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Bali, Test 1979	6	Volcanic Ash	8	-	-	-	44	26.3	11.7	1.74	79	43	53.0	-	7.3	11.6	2.63	80.0
Bali, Test 1983	6	Volcanic Ash	8	-	-	-	21	27.4	13.6	1.20	61	30	35.0	-	2.2	7.8	2.74	80.0
Bali, Test 1985	6	Volcanic Ash	8	-	-	-	39	26.5	11.7	1.56	69	37	47.0	-	11.8	-	2.65	80.0
Troutbeck	6	Dolerite	12	-	8	70	22	27.0	-	-	67	30	-	-	4	13.2	2.7	80.0
Saprolite 3 m	9	Basalt	14	-	12	52.8	35.2	31.5	7.1	3.44	77	18	77.0	-	0	33	3.15	70.5
Saprolite 5 m	9	Basalt	14	-	25	45	30	30.3	7.5	3.05	71	9	71.0	-	-	-	3.03	70.5
Ruagarents	10	Volcanic Ash	15	-	-	-	-	26.8	-	-	112	44	-	-	-	34	2.68	80.0
Arusha	10	Volcanic Ash	15	-	-	-	-	26.8	-	-	72	39	-	-	-	31	2.68	80.0
Poços de Caldas	1	Sienite	17	-	8.1	91.9	0	26.3	10.775	1.37	59.3	16	41.6	-	-	38	2.627	80.0
UFRGS	1	Basalt	24	-	44	32	24	30.3	14.4	0.82	43	20	25.0	-	17	26	3.03	78.0
Nova Avanhadava Dam	1	Basalt	25	-	9	41	50	30.2	12.8	1.65	86	31	38.5	-	-	-	3.02	70.5
Maringa, Sample 1	1	Basalt	40	-	10	14	76	29.8	10.94	1.76	58	18	32.7	-	-	-	2.98	56.6
Maringa, Sample 2	1	Basalt	40	-	7	17	76	30.3	10.18	1.97	57	15	31.8	-	-	-	3.03	49.4
Maringa, Sample 3	1	Basalt	40	-	5	19	76	30.7	9.50	2.2	58	16	31.6	-	-	-	3.07	43.5
Maringa, Sample 4	1	Basalt	40	-	8	23	69	30.3	11.67	1.61	57	18	30.4	-	-	-	3.03	57.9
Maringa, Sample 5	1	Basalt	40	-	7	26	67	30.8	12.12	1.54	59	19	30.1	-	-	-	3.08	60.1
Maringa, Sample 6	1	Basalt	40	-	8	32	60	30.8	11.63	1.63	61	19	31.6	-	-	-	3.08	59.0
Maringa, Sample 7	1	Basalt	40	-	10	31	59	30.2	11.24	1.67	73	24	35.7	-	-	-	3.02	63.7
Maringa	1	Basalt	40	0	4	29	67	30.1	10	2	54	17	29.4	-	-	-	3.01	44.2
Ilha Solteira, São Paulo	1	Basalt	45	-	-	-	66	30.3	5.2	4.81	-	74	112.0	-	0	10	3.03	70.5
Rio de Janeiro Dolerite	1	Dolerite	49	-	-	-	9	27.0	11.8	1.28	-	8	38.0	-	0	16	2.7	80.0
Lantau Island	13	Volcanic Ash	51	-	35	40	25	26.8	16	0.68	34	14	17.3	3.00E-06	-	-	2.68	68.2
Parana Upper	1	Basalt	57	-	-	-	-	30.3	-	-	56.5	22	-	-	-	-	3.03	70.5
Parana Intermediate	1	Basalt	57	-	-	-	-	30.3	-	-	55	14	-	-	0.4	27.5	3.03	70.5
Pereque-Mirim/Ubatuba/P1A30	1	Charnockite	60	1	54	16.5	28.5	26.6	13.9	0.909	41	11	19.0	1.00E-07	-	-	2.66	48.0
Pereque-Mirim/Ubatuba/BLOCK 3	1	Charnockite	60	0	53.5	15.5	31	27.1	14.3	0.896	50	13	23.0	2.60E-06	-	-	2.71	59.0
Pereque-Mirim/Ubatuba/P1C250	1	Charnockite	60	1	56.5	20.5	22	27.5	11.7	1.34	48	18	18.0	3.20E-07	-	-	2.75	43.0
Pereque-Mirim/Ubatuba/P1C350	1	Charnockite	60	5	59	21	15	27.4	11.9	1.3	42	9	27.0	-	-	-	2.74	70.0
Pereque-Mirim/Ubatuba/P1C500a	1	Charnockite	60	9.2	61.8	24	5	27.7	12	1.3	42	8	19.0	-	-	-	2.77	35.0
Pereque-Mirim/Ubatuba/P1C500b	1	Charnockite	60	0	55	26.5	18.5	27.6	15	0.845	NL	NP	19.0	-	-	-	2.76	37.0
Pereque-Mirim/Ubatuba/P1C650	1	Charnockite	60	1	51.5	33.3	14.2	27.4	14.3	0.91	36	3	23.0	-	-	-	2.74	81.0
Pereque-Mirim/Ubatuba/P1BLOCK4	1	Charnockite	60	0	50	40	10	27.6	11.5	1.41	43	4	35.0	1.20E-05	-	-	2.76	68.0
Pereque-Mirim/Ubatuba/P1C750	1	Charnockite	60	9	70	18.5	2.5	26.5	13.8	0.92	NL	NP	10.0	-	-	-	2.65	27.0
Sha Tin	13	Volcanic Ash	63	8	17	43	32	26.2	16.03	0.98	55.4	22	30.0	-	-	-	2.62	80.0
Fan Ling	13	Volcanic Ash	63	0	22	73	5	27.3	17.3	0.58	43	14	17.0	-	-	-	2.73	80.0
Halawa	9	Basalt	64	19	24	34.2	22.8	30.3	-	-	91	30	-	-	-	-	3.03	70.5
Granodiorite Antioquia Province	2	Granodiorite	71	0.5	65.5	20.4	13.6	26.9	12.5	0.91	28.2	3.7	29.2	-	32	36	2.69	80.1
SP Countryside Basalt/Sample I	1	Basalt	72	0	2	38	60	29.6	11.7	1.55	135	94	39.6	2.80E-04	35	20	2.96	77.0
SP Countryside Basalt/Sample II	1	Basalt	72	0	10	46	44	29.6	11.7	1.55	82	40	39.6	2.60E-03	19	29	2.96	77.0
SP Countryside Basalt/Sample III	1	Basalt	72	4	26	49	21	29.6	11.7	1.55	40	13	39.6	2.60E-03	13	31.5	2.96	77.0
SP Countryside Pegmatite	1	Pegmatite	72	0	56	34	10	26.6	13.0	1.05	61	24	32.4	2.20E-03	5	33	2.66	82.0
Segamat	5	Basalt	90	23.5	23.5	26.5	26.5	27.6	15.3	1.02	67	31	26.2	-	-	-	2.76	70.5
Kuantan	5	Basalt	90	17.5	21	36	25.5	27.4	14.1	1.39	58	19	35.8	-	-	-	2.74	70.5
5n Andesite	5	Andesite	90	11	21	27.5	40.5	28.3	15.55	1.18	78.3	15	33.5	-	-	-	2.83	80.0
5n Gabbro	5	Gabbro	90	0	4.5	53	42.5	24.4	13.3	1.45	84	40	47.5	-	-	-	2.44	80.0
São Paulo Basalt	1	Basalt	92	-	-	-	26	30.3	7.6	3.01	-	24	70.0	-	0	19	3.03	70.5
São Paulo Basalt	1	Basalt	92	-	-	-	54	30.3	4.7	5.50	-	93	128.0	-	0	10.5	3.03	70.5
São Paulo Basalt	1	Basalt	92	-	-	-	56	30.3	7.8	2.88	-	39	67.0	-	0	9	3.03	70.5
São Simão, Goiás, Dense Basalt	1	Basalt	98	-	-	-	30	30.3	8.5	2.58	-	21	60.0	-	8	30	3.03	70.5
São Simão, Goiás, Basalt, Vesicular	1	Basalt	98	-	-	-	25	30.3	7.6	2.97	-	36	69.0	-	8	22.7	3.03	70.5
Molokai, Oahu Island	9	Basalt	102	-	28.7	51	20.3	30.3	14.1	1.08	45.5	6.8	23.0	-	280	34	3.03	64.6
Lahaina, Oahu Island	9	Basalt	102	-	31	38.7	30.3	30.3	13.5	1.17	49.1	11	22.8	-	350	52	3.03	59.1
Wahiawa, Oahu Island	9	Basalt	102	-	10.7	45.5	43.8	30.3	14.8	1.02	51.3	15	27.3	-	175	27	3.03	81.0
Manana, Oahu Island	9	Basalt	102	-	69.3	16.7	14	30.3	14	1.14	67.1	17	33.9	-	210	57	3.03	90.0
Paaioa, Oahu Island	9	Basalt	102	-	27.8	39.2	33	30.3	13.2	1.35	57.1	18	39.6	-	184	38	3.03	88.8
Lihue, Kauai Island	9	Basalt	102	-	12.4	28.8	58.8	30.3	13.3	1.34	72.6	18	35.5	-	119	41	3.03	80.4
Puhi, Kauai Island	9	Basalt	102	-	24.5	39.9	35.6	30.3	12.9	1.58	44	3.7	41.3	-	84	52	3.03	79.3
Kapaa, Kauai Island	9	Basalt	102	-	24	41.7	34.3	30.3	11.7	1.9	52	10	45.8	-	70	44	3.03	73.0
Hali, Kauai Island	9	Basalt	102	-	45.1	34.4	20.5	30.3	12.3	1.75	40	4.6	43.7	-	49	35	3.03	75.7
Kilohana, Hawaii Island	9	Volcanic Ash	102	-	69	24.8	6.2	26.8	9.2	2.15	-	-	39.2	-	-	-	2.68	48.9
Waimea, Hawaii Island	9	Volcanic Ash	102	-	33	57.7	9.3	26.8	8.1	2.8	66.8	3.8	45.8	-	-	-	2.68	43.8
Kukiaiu, Hawaii Island	9	Volcanic Ash	102	-	89	10.4	0.6	26.8	4.4	5.99	93.3	4.7	164.1	-	70	37	2.68	73.4
Honokaa, Hawaii Island	9	Volcanic Ash	102	-	89.5	9	1.5	26.8	3	8.9	-	-	244.9	-	70	33	2.68	73.8
Hilo, Hawaii Island	9	Volcanic Ash	102	-	87.5	7.4	5.1	26.8	3.5	7.13	61.4	-	234.9	-	42	33	2.68	88.3
Porous Clay Upper	1	Basalt	106	-	-	-	-	30.3	14.8	1.05	55	28	24.4	-	0.15	29	3.03	70.5
Londrina	1	Basalt	107	-	-	-	19	30.3	10.6	1.86	65	19	43.2	-	-	-	3.03	70.5
Londrina, 5 meters	1	Basalt	108	-	40	23	37	30.3	21.2	0.43	55	27	10.0	-	-	-	3.03	70.5
Londrina, 10 meters	1	Basalt	108	-	45	35	20	30.3	13.6	1.22	62	16	28.4	-	-	-	3.03	70.5
Londrina, 13 meters	1	Basalt	108	-	53	32	15	30.3	13.6	1.22	52	10	28.4	-	-	-	3.03	70.5

Table A- 13. Compilation data with residual soil having metamorphic rocks as parent rock.

Soil Name/Local	Country	Parent Rock	Source No	Gravel %	Sand %	Silt %	Clay %	λ_s kN/m ³	λ_d kN/m ³	e	LL %	IP %	W %	K (cm/s)	Co (kPa)	θ°	G	S %
Harare	6	Phyllite	12	-	9	84	7	27.6	-	-	45	20	-	-	0	11.3	2.76	80.0
Carta de Zoneamento Geotecnico, G4-X	8	Schist	23	11	45	35.2	8.8	26.3	14.8	0.78	58	32	11.3	-	-	-	2.63	-
Carta de Zoneamento Geotecnico, G5-A	8	Schist	23	21.2	50.8	22.4	5.6	27.5	18.1	0.52	36	14	11.7	-	-	-	2.75	-
Emborcação Dam	1	Schist	25	-	50	40	10	26.7	16.05	0.63	46	19	18.9	-	-	-	2.67	80.0
Brasília	1	Slate	27	0	3	67	30	27.5	-	-	44	24	-	1.00E-05	13.6	19	2.75	80.0
Sao Mateus Quartz Mica Schist	1	Schist	31	0	35	52	13	27.3	15.5	0.76	40	4	15.5	9.00E-05	32	35	2.73	52.6
Cuiaba/Sample 1	1	Phyllite	33	-	43	38	19	26.4	20.0	0.4	32	12	10.0	-	-	-	2.64	66.0
Cuiaba/Sample 2	1	Phyllite	33	-	15	25	60	27.3	14.6	0.6	55	33	20.0	-	-	-	2.73	91.0
Cuiaba/Sample 3	1	Phyllite	33	14	23	47	16	27.6	17.4	0.59	21.6	8.1	15.5	-	-	-	2.76	69.0
Jalan Gurney/Kuala Lumpur	5	Schist	48	4	15	81	0	26.8	15.8	0.7	57.5	28	29.0	-	-	-	2.68	111.0
Ouro Preto	1	Phyllite	68	-	30	68	2	27.6	16.7	0.47	35	0	15.2	-	0	27	2.76	90.1
Castelo Branco Highway Km41	1	Phyllite	68	-	15	80	5	27.6	8.4	1.37	44	11	37.5	-	0	26	2.76	75.4
Tremembe Sao Paulo 11A	1	Phyllite	69	-	-	-	-	27.6	14.3	0.41	48	16	12.0	-	15	29	2.76	80.0
Tremembe Sao Paulo 11B	1	Phyllite	69	-	-	-	-	27.6	13.1	0.96	-	-	27.8	-	44	30	2.76	80.0
SP Countryside Phyllite/ Average Sample I	1	Phyllite	72	0	10	20	70	28.3	14.6	0.94	68.9	34	28.0	5.60E-05	60	24	2.83	84.0
SP Countryside Mica Schist	1	Schist	72	-	40	48	12	28.1	14.2	0.98	51.2	24	27.0	2.00E-03	20	30.5	2.81	77.0
SP Countryside Quartz Schist	1	Schist	72	2	76	17	5	27.3	17.6	0.55	NL	NP	28.0	1.10E-04	-	-	2.73	51.0
SP Countryside Phyllite/ Average Sample II	1	Phyllite	72	0	10	50	40	28.3	14.6	0.94	68.9	34	28.0	1.80E-04	46	27	2.83	84.0
SP Countryside Phyllite/ Average Sample III	1	Phyllite	72	5	13	64	18	28.3	14.6	0.94	68.9	34	28.0	1.80E-04	5	32	2.83	84.0
SP Countryside Metabasic/Sample I	1	Metabasic	72	0	5	60	35	28.9	11.6	1.49	76	40	36.5	6.40E-04	22	26	2.89	70.0
SP Countryside Metabasic/Sample II	1	Metabasic	72	0	10	70	20	28.9	11.6	1.49	61	22	36.5	2.90E-03	10	33	2.89	70.0
Kampar Perak/W.Grade VI	5	metasedimentary	77	29	24	21.5	25.5	26.3	13.3	1	46.5	18	28.0	-	26.5	24	2.63	73.6
Kampar Perak/W.Grade V	5	metasedimentary	77	19.5	41.5	28.5	10.5	26.3	13.8	1	44	15	30.0	-	62.5	19	2.625	78.8
Kampar Perak/W.Grade IV	5	metasedimentary	77	26	39.5	25.5	9	26.4	15.9	0.75	41	15	19.5	-	57.5	19.5	2.635	68.5
Gopeng Perak	5	Phyllite	88	-	-	-	-	27.6	11.2	1.64	-	-	47.5	-	-	-	2.76	80.0
Quartz-mica Schist 5	5	Schist	90	4	36.5	41.5	18	25.5	15.75	0.64	45	13	20.1	-	-	-	2.545	80.0
Graphitic Schist 5	5	Schist	90	5.5	17	60	17.5	26.2	14.75	0.66	52.5	21	20.1	-	-	-	2.62	80.0
5n Serpentine	5	Serpentine	90	3	27.5	33	36.5	32.1	14.75	1.91	66.5	16	47.6	-	-	-	3.21	80.0
Senawang/Negeri Sembilan	5	Schist	101	11.5	34	42.5	12	26.8	14.3	0.85	66	26	25.5	-	103	18.5	2.68	80.0
Medellin 1	2	Amphibolite	103	-	2	64	34	28.2	9.2	2.08	82	24	41.0	-	-	-	2.82	55.6
Medellin 2	2	Amphibolite	103	2	27	57	14	28.8	10.7	1.7	54	12	38.0	-	-	-	2.88	64.4
Medellin 3	2	Amphibolite	103	1	31	62	6	27.8	12.8	1.18	36	4	22.0	-	-	-	2.78	51.8
Guarne, Antioquia	2	Amphibolite	104	0	42	53	5	28.0	12.7	1.2	58	20	28.0	-	3	30	2.8	68.0
Perus	1	Schist	107	-	12	26	62	26.8	14.5	0.85	46	6	25.4	-	-	-	2.68	80.0
Cubatão	1	Schist	107	15	42	42.8	0.2	26.8	19.6	0.37	27	0	11.0	-	-	-	2.68	80.0
BR-116 Highway, Serra do Mar South	1	Schist	107	-	-	-	0.7	26.8	19.3	0.39	28	11	11.6	-	-	-	2.68	80.0
Agua Espraçada Avenue Mica Schist	1	Schist	116	-	70	24	6	26.8	12.6	0.97	-	-	55.0	-	15	30	2.68	152.0

Source references for Tables A-1 to A-13

- (1) ABEF 1989.
- (2) Abramento and Pinto 1998.
- (3) Zai et al. 2016.
- (4) Ahmad et al. 2006.
- (5) ANTT 2015.
- (6) Anuar and Ali 1997.
- (7) Araujo et al. 2014.
- (8) Belloni et al. 1988.
- (9) Benessiuti et al. 2011.
- (10) Bernardes 2003.
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APPENDIX B

Tested soils: resume of the main products and assembled samples

Table B- 1. Main characteristics of each component of the assembled soils.

Product Specification	Supplier	Spec. Grav.	Chemical Comp.	Specific Surface (m ² /g)	Cation Exc. Capacity (meq./100 g)	pH	Absorption (%)	Moisture (%)
Silica Sand ¹²	Euroquarz	2.65	SiO ₂	-	-	6.5-7	-	-
Silica sand ¹³	Bell & Mackenzie	2.65	SiO ₂	-	-	6-7	-	-
Friedland Clay	MRG Blautonwerk Friedland	2.71	Combination of several minerals ¹⁴	165.5	35-45	8.7	-	6-8
Kaolinite - EPK Kaolin	Edgar Minerals	2.65	Al ₂ O ₃ , 2SiO ₂ , 2H ₂ O	28.52	4.5	5.8	11.3	Max. 1
Kaolinite GHL KS 80	Georg H. Luh GMBH	2.62	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃	18.2	-	4.5-9.5	-	0.6-2
Active Bentonite IBECO B1	Imerys Civil Engineering	2.65	Al ₂ O ₃ , SiO ₂ , Fe ₂ O ₃ , MgO, CaO, K ₂ O, Na ₂ O	600-800	70±10	10	450	11±3
Na-Bentonite - Volclay® 325 mesh*1	American Colloid Company	2.35	(Na,Ca)0.33 (Al _{1.67} Mg _{0.33}) Si ₄ O ₁₀ (OH) ₂	0.09-1.8	>70	8.5-10.5	not informed	Max. 12
Phlogopite Mica PD900*1	LKAB	2.8	KMg ₃ (AlSi ₃ O ₁₀)(OH,F) ₂	0.3	No appreciable CEC	9	-	Max. 1

¹² For sand grain size: sfine, fine, fine2, medium, medium2 and coarse, as shown in Figure B-1.

¹³ For sand grain size: 11, 12, 13 and fine-medium, as shown in Figure B-1.

¹⁴ Montmorillonite (17%); Illite-Montmorillonite (12%); Illite-Mica (28%); Fireclay (13%); Chlorite (2%); Quartz (25%); Feldspar (2%); Pyrite (1%).

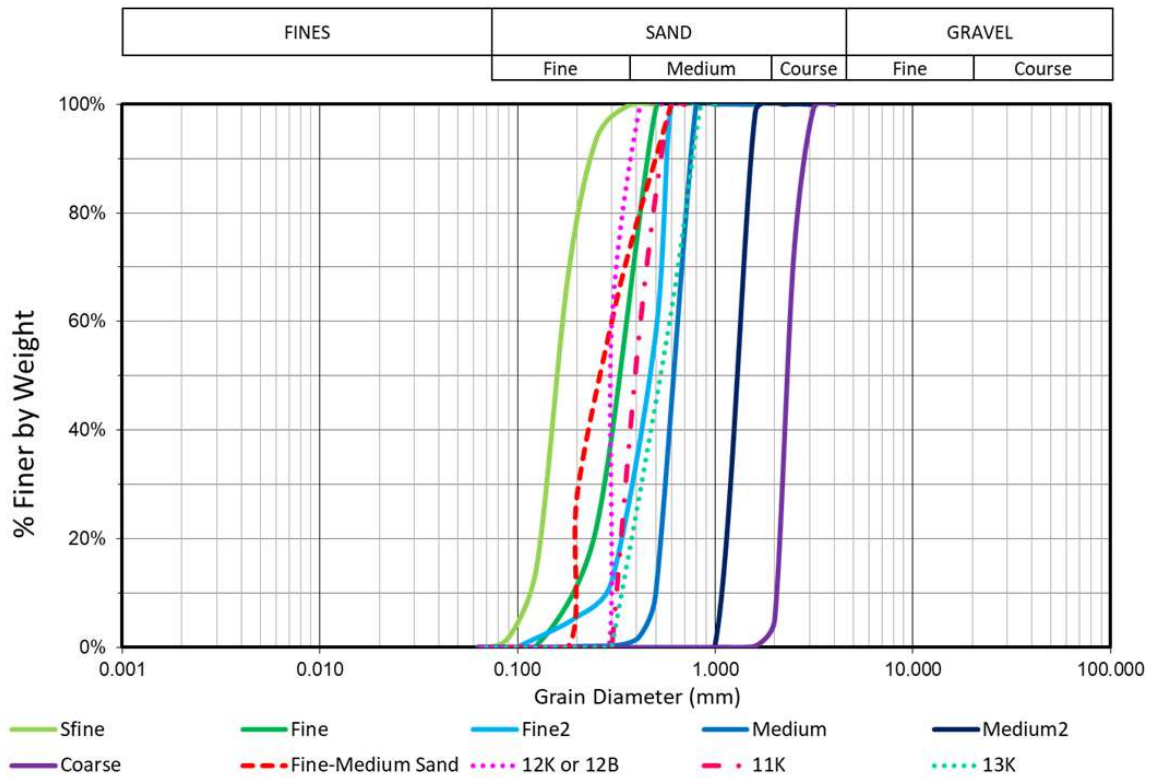


Figure B- 1. Sand grain distribution for all the samples.

Table B- 2. Soils assembled with EPK Kaolin, Volclay, Mica and Bell-Mackenzie Sand.

Sample ID	Sand type	Sand (%)	Kaolinite (%)	Bentonite (%)	Mica (%)	W _L (%)	W _P (%)	IP (%)
KFM20	Fine-medium	80	20	-	-	24	15	9
KFM25	Fine-medium	75	25	-	-	26	16	10
KFM35	Fine-medium	65	35	-	-	29	18	11
KFM40	Fine-medium	60	40	-	-	32	19	13
12K3	12	70	30	-	-	22	13	9
12B1	12	80	-	20	-	90	24	65
12B2	12	60	-	40	-	194	36	158
12KM1	12	25	60	-	15	41	24	17
12KM2	12	10	60	-	30	47	31	16

Table B- 3. Assembled soils with Friedland clay and Euroquarz sand.

Sample ID	Sand (%)	Clay (%)	W _L (%)	W _P (%)	IP (%)	W _{Pol} (%)
Fsfine20	80	20	26	NP	NP	16
Fsfine30	70	30	30	23	7	19
Fsfine50	50	50	46	24	22	25
Fsfine70	30	70	62	31	31	31
Ffine20	80	20	NP	NP	NP	-
Ffine30	70	30	26	NP	NP	19
Ffine50	50	50	35	18	17	24
Ffine70	30	70	48	20	28	30
Ffine2_20	80	20	NP	NP	NP	-
Ffine2_30	70	30	NP	NP	NP	-
Ffine2_50	50	50	45	22	23	25
Ffine2_70	30	70	60	30	30	31
Fmedium20	80	20	NP	NP	NP	-
Fmedium30	70	30	NP	NP	NP	-
Fmedium50	50	50	44	24	20	25
Fmedium70	30	70	63	32	31	31
F100	0	100	94	41	53	40

Table B- 4. Assembled soils with bentonite IBECO B1 and Euroquarz sand.

Sample ID	Sand (%)	Bentonite (%)	W _L (%)	W _P (%)	IP (%)	W _{Pol} (%)
Bsfine20	80	20	86	27	59	19
Bsfine 25	75	25	107	27	80	21
Bsfine30	70	30	127	26	101	23
Bsfine50	50	50	171	31	140	30
Bsfine70	30	70	245	34	211	38
Bfine20	80	20	59	24	35	18
Bfine30	70	30	86	28	58	21
Bfine50	50	50	139	27	112	29
Bfine70	30	70	217	36	181	37
Bfine2_20	80	20	54	NP	NP	17
Bfine2_30	70	30	83	20	63	21
Bfine2_50	50	50	157	26	131	29
Bfine2_70	30	70	219	34	185	37
Bmedium20	80	20	53.5	NP	NP	17
Bmedium30	70	30	82	22	60	21
Bmedium50	50	50	160	32	128	29
Bmedium70	30	70	194	30	163	36
Bmedium2_20	80	20	NP	NP	NP	-
Bmedium2_30	70	30	NP	NP	NP	-
Bmedium2_50	50	50	159	27	132	29
Bmedium2_70	30	70	241	44	197	38
Bcoarse20	80	20	NP	NP	NP	-
Bcoarse30	70	30	NP	NP	NP	-
Bcoarse50	50	50	NP	NP	NP	-
Bcoarse70	30	70	235	36	199	38
B100	0	100	486	52	435	55

Table B- 5. Soils assembled with EPK Kaolin and Bell-Mackenzie sand.

Sample ID	Sand (%)	Kaolinite (%)	W _L (%)	W _P (%)	IP (%)	W _{Pol} (%)
11K_2	80	20	11.4	NP	NP	16
11K_3	70	30	20	13	7	19
11K_4	60	40	24	15	9	21
11K_5	50	50	30	19	11	24
11K_6	40	60	38	23	15	27
11K_7	30	70	43	24	19	30
11K_8	20	80	49	28	21	33
11K_9	10	90	59	30	28	36
12K_2	80	20	14	NP	NP	16
12K_3	70	30	22	13	9	19
12K_4	60	40	27	16	11	21
12K_5	50	50	32	19	13	24
12K_6	40	60	38	23	15	27
12K_7	30	70	44	26	18	30
12K_8	20	80	52	28	23	33
12K_9	10	90	59	32	27	36
13K_2	80	20	12.6	NP	NP	16
13K_3	70	30	20	13	7	19
13K_4	60	40	26	15	11	21
13K_5	50	50	32	19	13	24
13K_6	40	60	39	22	17	27
13K_7	30	70	46	26	20	30
13K_8	20	80	52	29	23	33
13K_9	10	90	60	30	29	36

Table B- 6. Soils assembled with Kaolinite GHL and Euroquarz sand.

Sample ID	Sand (%)	Kaolinite (%)	W _L (%)	W _P (%)	IP (%)	W _{Pol} (%)
Ksfine10	90	10	NP	NP	NP	NP
Ksfine20	80	20	NP	NP	NP	NP
Ksfine30	70	30	22	16	6	19
Ksfine50	50	50	32	21	11	24
Ksfine70	30	70	44	29	16	30
Ksfine80	20	80	49	31	18	33
K100	0	100	62	40	22	38
Kfine30	70	30	22	18	4	19
Kfine50	50	50	32	21	11	24
Kfine70	30	70	45	28	17	30
Kfine2_30	70	30	20	15	5	19
Kfine2_50	50	50	32	21	11	24
Kfine2_70	30	70	44	27	17	30
Kmedium30	70	30	20	17	3	19
Kmedium50	50	50	31	23	8	24
Kmedium70	30	70	43	27	16	30
Kcoarse50	50	50	27	21	6	24
Kcoarse70	30	70	41	29	12	30